

Mission Systems Interface Volume

For

A-GRA

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1 MISSION SYSTEMS INTERFACE

The Mission Systems (MS) interface is the connection between MA and onboard MS such as sensor subsystems, payloads, and communications services. The major capabilities of the current L1 MS interface include the following:

- Data exchanges between A-GRA compliant, Mission Autonomy equipped platforms utilizing onboard Mission Systems
- Providing status data to MA such as position, navigation, and timing (PNT) data, and system status of the local and remote platforms
- Providing MS-generated operational data such as entities tracked by MS systems

The L1 MS Interface captures the UCI-based interactions that MA will use to control and interface with the both the local Mission Systems and other MA equipped ACP's. The design intent is to provide a succinct, service-based interface to capabilities provided by the MS that MA will interface with. The message set and content is intended to capture the minimal information exchange between MA and MS to support the capabilities required for the decomposed system use cases. Concerns that can be managed solely by MA, such as which ACP is the leader of an ACP team, are not exposed by MA in the L1 interface. Similarly, the L1 interface abstracts the detailed operation and implementation of MS subsystems such as network interfaces, and details such as the exact route messages take to reach the node which is the leader are not exposed by MS. The Mission Systems interface relies on OMS and A-GRA requires MA and MS to communicate via the platform hosted ASB. The A-GRA does not make any assumptions about the CAL used in MS subsystems except for the DMS.

1.1 Interface Description

Below we arrange the MS to MA interactions into a set of functional groups we refer to as services. These groupings are intended to help describe the capabilities the MS must provide and how they are accessed. They do not imply any requirements or restrictions on the internal architecture or implementation of MS. For each category, message definitions and interactions are detailed in the following section. We anticipate further refinement and extension will occur in future versions of the A-GRA.

Communications Service

The A-GRA L1 interface to MS seeks to enable communication services while isolating MA from the detailed management of the network, communication terminals and the like to the extent possible. The L1 interface defines message exchange with network endpoints currently visible to MS via publication/subscribe or a directed message. This is described below in Decentralized Message Service (DMS). To support that functionality, the Communication Service interactions provide insight into the network state. To connect to remote nodes on currently accessible network interfaces, MS may use a network discovery protocol, a pre-loaded configuration file, or other mechanism. Once connected, MS provides MA with a notification that a new node is accessible. MS provides a system identifier, analogous to a hostname in the case of an IP network, by which MA can direct messaging to the remote node. MS holds the detailed network and addressing information, making use of it when MA makes a request involving a particular system identifier. Once messaging is established between nodes, MA may exchange information with remote nodes about node type, capabilities, and other information as defined by the A-GRA MA L1 interfaces (C2, Peer). By abstracting addressing details

for MA, this enables Communications and DMS interactions to facilitate service across a set of heterogeneous networks. Methods that nodes use to exchange their system identifiers, or for consistent identifiers to be assigned for nodes on TDL or other networks that may not have the capability to provide a system identifier, has been considered and implemented. As that functionality is within the MS boundary, it is not currently included in the A-GRA interface. The comms terminal will automatically be configured based on a predetermined Comm Terminal Plan. The A-GRA does not currently require that an MA implementation make any use of the Comms Terminal Plan, though information such as any periods that lack comms and trigger specific configurations may be of use.

The "Additional Comms Capability" MUC specifies a more holistic approach to managing the OTA communications with a Comm Terminal Plan and other Comm Terminal Capability configuration/status messaging. The Comm Terminal Plan and Capability are similar to the Data Dissemination Plan described below which both add capability of updating the configurations at linked Mission Plans and other optional Triggers.

Decentralized Messaging Service

The L1 interface with MS defines interactions to send and receive messages between A-GRA compliant systems currently reachable by MS, either directly or via multiple network hops. Both publish/subscribe and directing a message to specific recipients are supported. The interface is designed to be consistent with the Gateway Capability message-passing paradigm of OMS while including additional information necessary to operate on over-the-air (OTA) networks. This includes message priority for bandwidth management and queuing as well as optional subscription filtering to eliminate unnecessary propagation of messages. The interface is designed to allow implementations to operate across heterogeneous network types (e.g. IP and non-IP) by utilizing generic asset IDs that can be mapped by the DMS onto multiple addressing schemes and by not defining interactions that require a connection-based network. The core method for messages to transfer between systems utilizes a UCI message wrapper (*MA_TxDataPayloadCommandMT*) to send messages to the DMS and another UCI message wrapper (*MA_RxDataPayloadMT*) the DMS publishes when messages are received. The prescribed UCI wrapper message require an *EncodedPayload* field for the message supplied by MA for transport. The datatype while crossing the L1 boundary between MA and MS is UTF-8 XML represented as hexBinary. The OTA wrapper message sent between the DMSs, *MA_DataPayloadWrapper* IDL, also requires an *EncodedPayload* field. The datatype while crossing between DMSs will be in EXI representation. Integration with assets that are not A-GRA compliant, and a linked send/respond message pair have been considered but are not included in the current ASK.

The "Additional Comms Capability" MUC specifies a more holistic approach to managing the OTA communications with a Data Dissemination Plan. The Data Dissemination Plan specifies how often and whether ingress/egress messages will be subscribed/published directly on the ASB or if the UCI wrappers can be used. The DMS, as a Gateway Capability, is managed by MS using the Data Dissemination Plan to configure the connection to other DMSs, the ingress/egress of messages, and how the ingress/egress occurs. The DMS can be configured to allow ingress/egress of UCI wrapped messages from/to topics and UCI wrapped messages that are intended to be sent only to specific destinations. Allowing direct message subscription from the local ASB is advantageous because it means MA does not need to manage how often and where messages need to be sent offboard, simplifying individual MA L2 service implementations.

Status Service

MS provides messages by which MA can request receipt of health, status, comms, or built-in test

(BIT) information for MS subsystems. MS will also report health, status, and BIT information of its subsystems to MA at a configurable periodic rate.

PNT Service

MS provides messages by which MA may request and receive platform position and current time. MA can also adjust the frequency PNT data is published.

Data Fusion Service

MS provides messages by which MA may receive processed tracks that are compiled from multiple sensor inputs.

1.2 Interface Interactions

The MS interactions with MA have been scoped as shown in the below figure. Within each interaction, there is a baseline set of functionalities that must be supported by MS and MA for a compliant system.

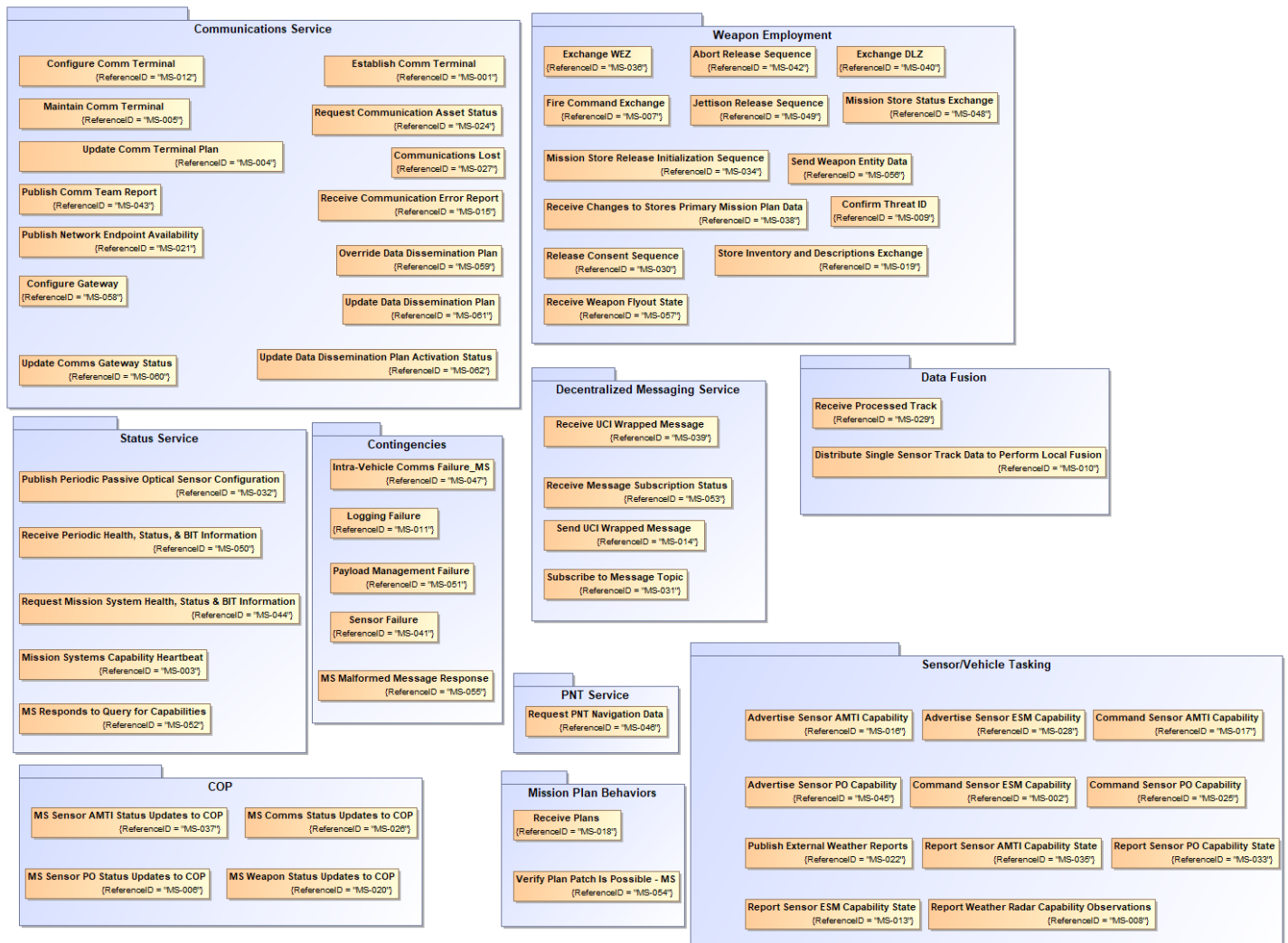


Figure 1-1: Mission Systems Interactions

1.2.1 Communications Service Package

MS is responsible for managing all parts of the communications systems that enable connecting to offboard systems, sending/receiving messages offboard, and other associated items. The parts of the communications system that will be covered in this section include the Comm Terminals, Gateways, and their plan management. The Comm Terminal Capability is used to manage the individual radio terminals that connect between systems. The Gateway Capability is used to manage what messages Egress/Ingress over those Comm Terminal systems. The core implementation includes MA monitoring specific aspects of the the comms system including loss of comms, state of the comms system, and available endpoints.

The "Additional Comms Capability" MUC includes the management of those capabilities where MA can use the associated Plan message set (*CommTerminalPlanMT* for Comm Terminals and *MA_DataDisseminationPlanMT* for Gateways). MS is responsible for using the Plans, when activated by their Activation Commands, to individually configure the Comm Terminal and Gateways. There is a method for MA to configure the Capabilities directly, but the Plans allow MS to manage and update the capabilities automatically. The Decentralized Messaging Service (DMS) is a specific type of Gateway Capability with a method for sending/receiving UCI wrapped messages, which is the core functionality, and the "Additional Comms Capability" MUC adds subscribing/publishing messages directly to/from the local ASB. The DMS adds some clarity on how communications connections are to be established using DDS with IP based Comm Terminal links. The DMS Package section will go into further detail for how it supports UCI wrapped messaging. With the "Additional Comms Capability" MUC, the DMS will support the Gateway Capability configuration that is setup by the *MA_DataDisseminationPlanMT* message (see the "Update Data Dissemination Plan" sequence for more details).

1.2.1.1 Communications Lost

MS will monitor the communications link quality and link statistics of its capabilities onboard. If MS determines that a specific capability is degraded, MS will perform a failover activity to switch communications to an operational, redundant communications link if it's available. This degradation of a capability will be reported with *CommTerminalCapabilityStatusMT*, where *CapabilityStatus.AvailabilityInfo* [*Availability* and *AvailabilityReason*] fields provide details regarding the degraded capability. MS will also populate the updated *CapabilityPrecedence* as a result of the degraded capability. If a redundant link is not available and MS determines that each of its communications links are unavailable, MS will send a *SystemStatusMT* to MA with the *Communications.MissionCommunicationsState* of its *Communications* property set to *LOST* in order to trigger the "Communications Lost" contingency.

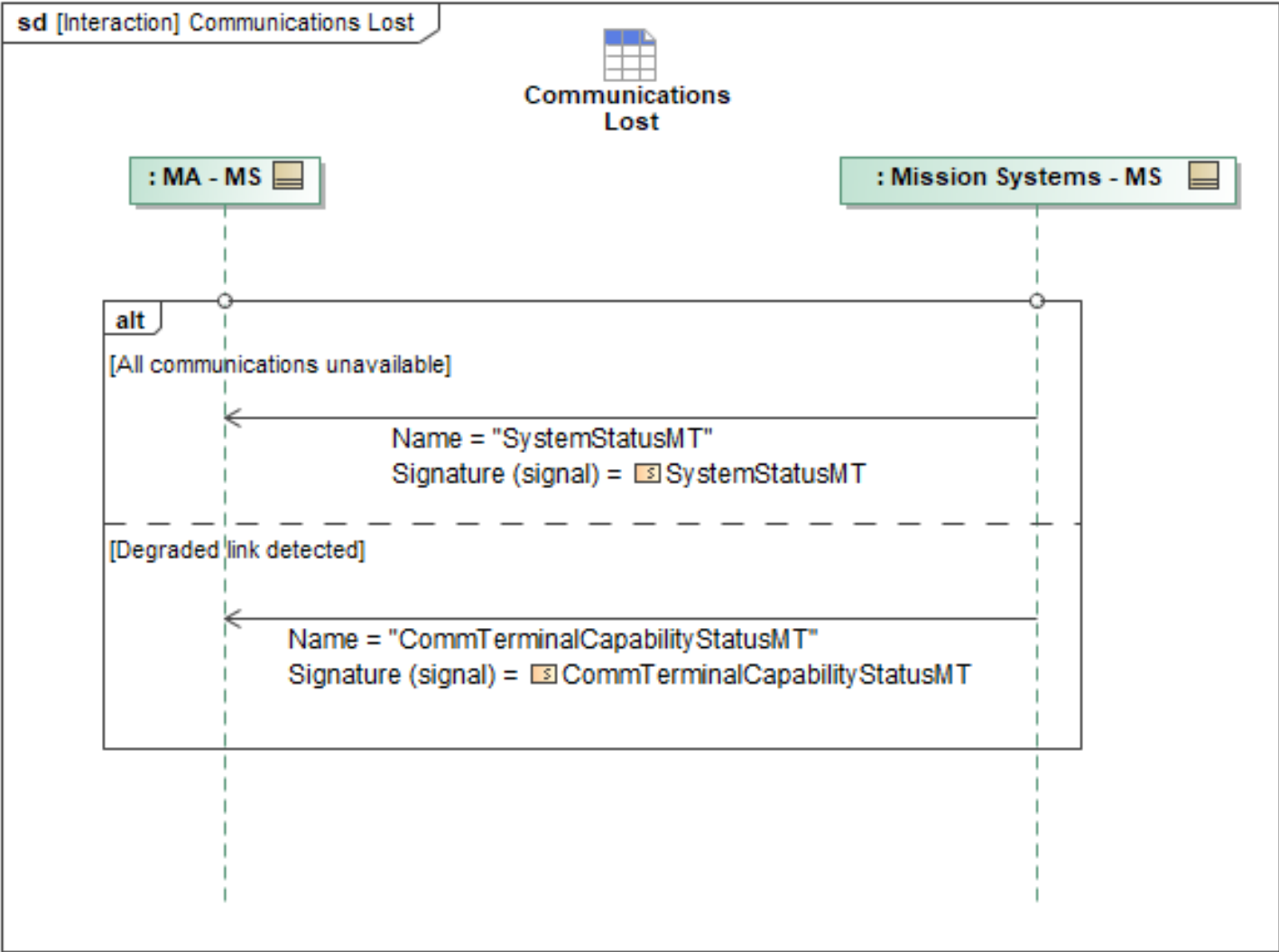


Figure 1-2: Communications Lost

Table 1-1: Communications Lost

Message Name (Name:Type)	Required Fields (Name:Type)
CommTerminalCapabilityStatusMT: CommTerminalCapabilityStatusMT	CapabilityPrecedence: unsignedInt AvailabilityReason: CannotComplyType
SystemStatusMT: SystemStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.2 Configure Comm Terminal

Mission Use Case: Additional Comms Capabilities

Typically MS would handle the configuration of a Comm Terminal Capability using the Comm Terminal Plan, but if MA needs to make a direct modification MA can send a *CommTerminalCommandMT* directly to MS. If the command is accepted, MS responds with

CommTerminalCommandStatusMT_1 with the *Activity* that satisfies the associated *CommTerminalCommandMT*. MS will then provide status of the resulting *Activity* to MA through *CommTerminalActivityMT*. If the command is not accepted, MS responds with *CommTerminalCommandStatusMT_2* with *CommandProcessingStateReason* indicating why the command was not in an acceptable state.

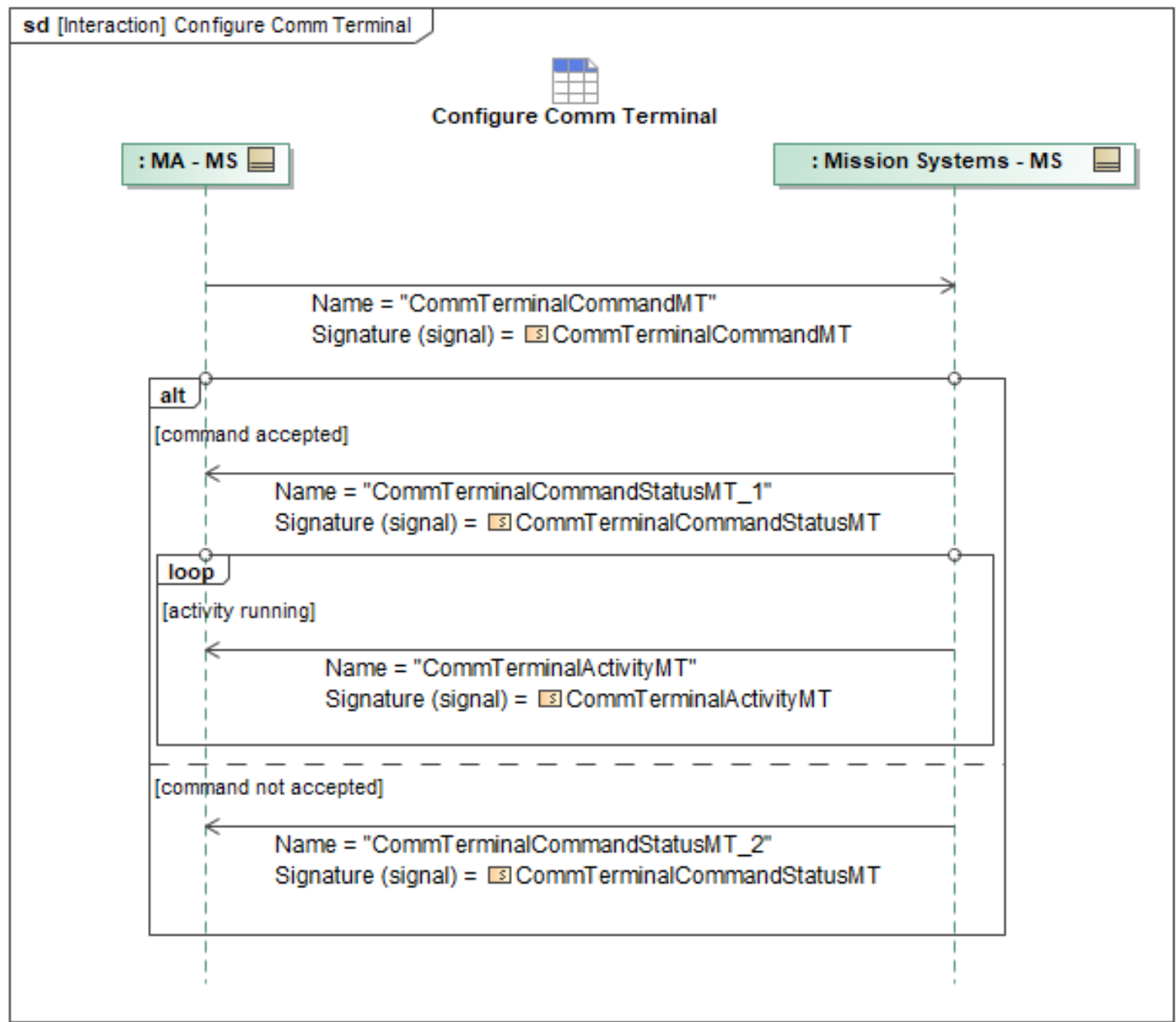


Figure 1-3: Configure Comm Terminal

Table 1-2: Configure Comm Terminal

Message Name (Name:Type)	Required Fields (Name:Type)
CommTerminalActivityMT: CommTerminalActivityMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
CommTerminalCommandMT: CommTerminalCommandMT	None
CommTerminalCommandStatusMT_1: CommTerminalCommandStatusMT	Activity: CommTerminalActivityCommandType
CommTerminalCommandStatusMT_2: CommTerminalCommandStatusMT	CommandProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.3 Configure Gateway

Mission Use Case: Additional Comms Capabilities

Typically MS would handle the configuration of a Gateway Capability using the Data Dissemination Plan, but if MA needs to make a direct modification MA can send a *MA_GatewayCommandMT* message to MS directly. Then, if the command is accepted, MS sends the *MA_GatewayCommandStatusMT_1* to MA. If a Gateway Activity is running, MS also sends the *MA_GatewayActivityMT* message to MA. If the command is NOT accepted, MS sends the *MA_GatewayCommandStatusMT_2* message to MA with *CommandProcessingStateEnum* set to "REJECTED" and the *CommandProcessingStateReason* filled out as needed.

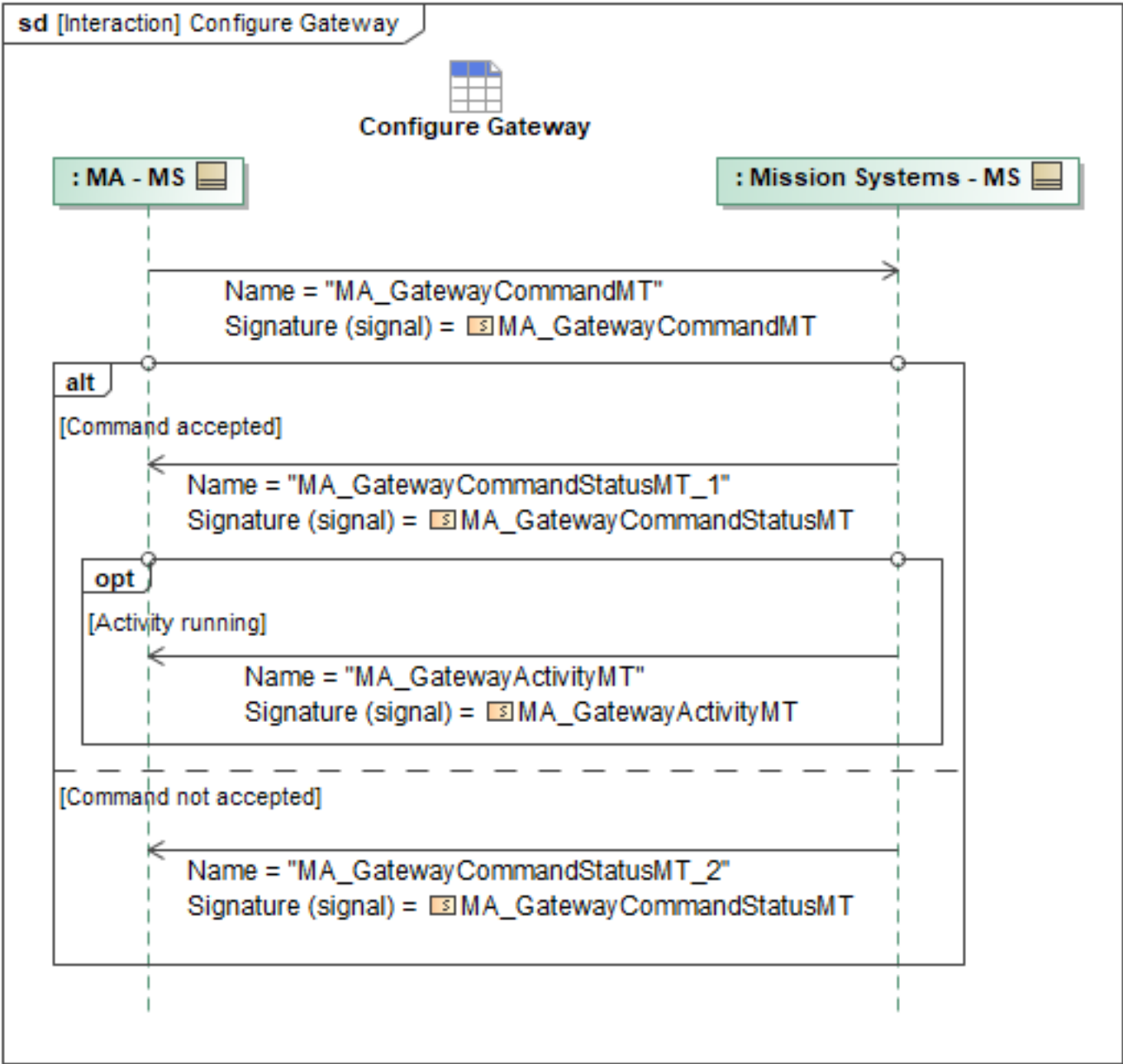


Figure 1-4: Configure Gateway

Table 1-3: Configure Gateway

Message Name (Name:Type)	Required Fields (Name:Type)
MA_GatewayActivityMT: MA_GatewayActivityMT	None
MA_GatewayCommandMT: MA_GatewayCommandMT	None
MA_GatewayCommandStatusMT_1: MA_GatewayCommandStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_GatewayCommandStatusMT_2: MA_GatewayCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.4 Establish Comm Terminal

The Establish Comm Terminal interaction provides MA with a periodic description of the MS communications system, communications assets, and communication capabilities that may be utilized. The communications system and its assets are defined through *CommSystemMT* and *CommAssetMT*. A *CommAssetMT* is provided for each *CommAssetID* that is defined in *CommSystemMT*. *CommTerminalCapabilityMT* is provided by Mission Systems to establish the capabilities of the communication terminal along with each capability's configurable characteristics and settings. Lastly, the message sequence provides MA the current schedule for communication activities that are scheduled for execution with *CommScheduleAllocationMT*.

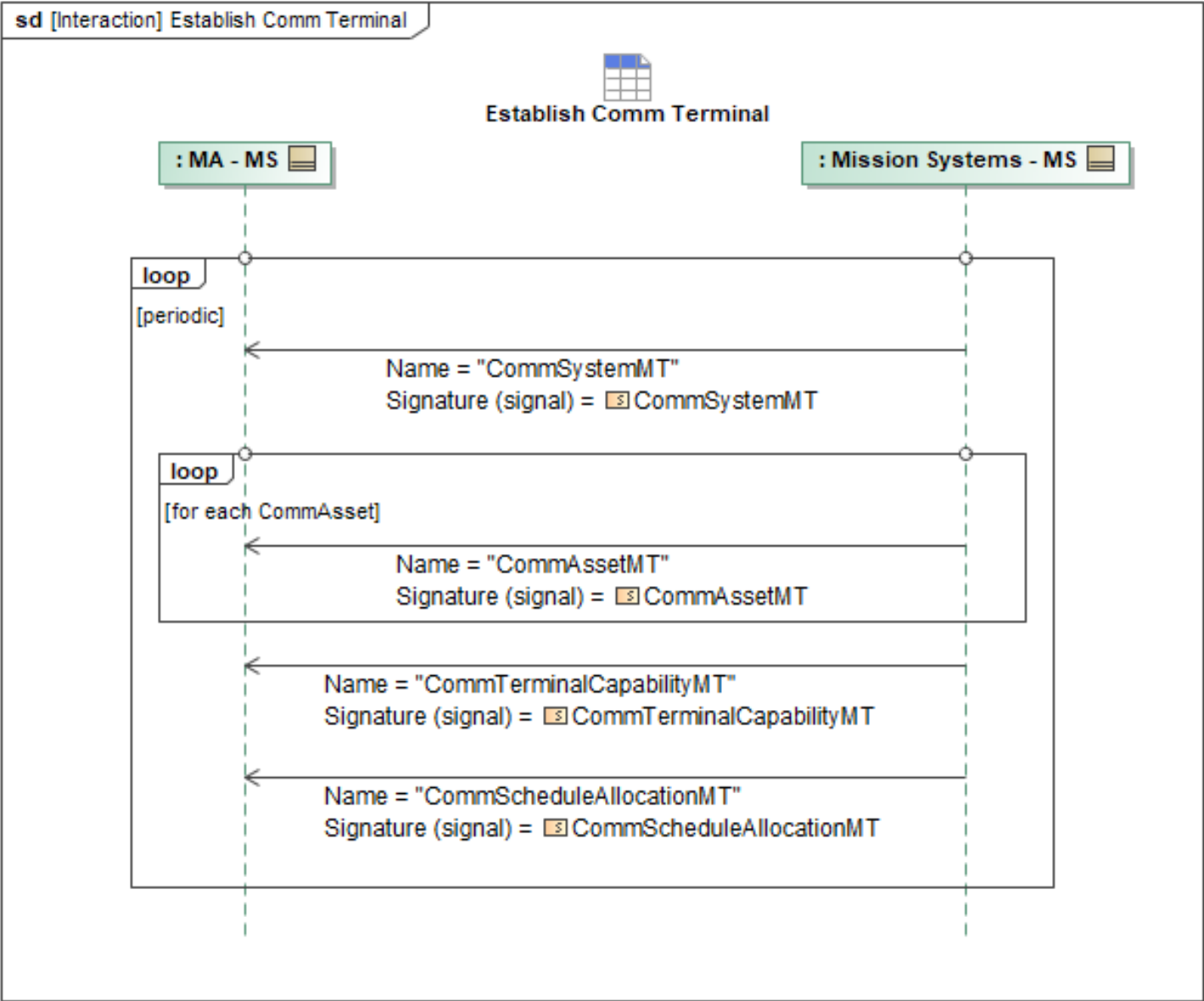


Figure 1-5: Establish Comm Terminal

Table 1-4: Establish Comm Terminal

Message Name (Name:Type)	Required Fields (Name:Type)
CommAssetMT: CommAssetMT	None
CommScheduleAllocationMT: CommScheduleAllocationMT	None
CommSystemMT: CommSystemMT	None
CommTerminalCapabilityMT: CommTerminalCapabilityMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.5 Maintain Comm Terminal

MS will periodically send *CommTerminalCapabilityStatusMT* to MA to report the current states of waveforms and the capabilities those waveforms are being used for within the comm terminal and system. If a communications activity is running, MS will send *CommTerminalActivityMT*, which provides detailed information on what exactly a waveform capability is doing at any point in time through the *SubsystemID* and *Activity* information fields. MS specifically statuses the data links settings and statistics through *ConnectionStatus*, *CommonSettingsStatus*, and *LinkStatistics* information fields. MS reports specific waveform settings status (such as Link 16 waveform settings status) through the *WaveformSpecificSettingsStatus* field. This series of message exchanges is intended to be exchanged while comm terminals are active within an operational state.

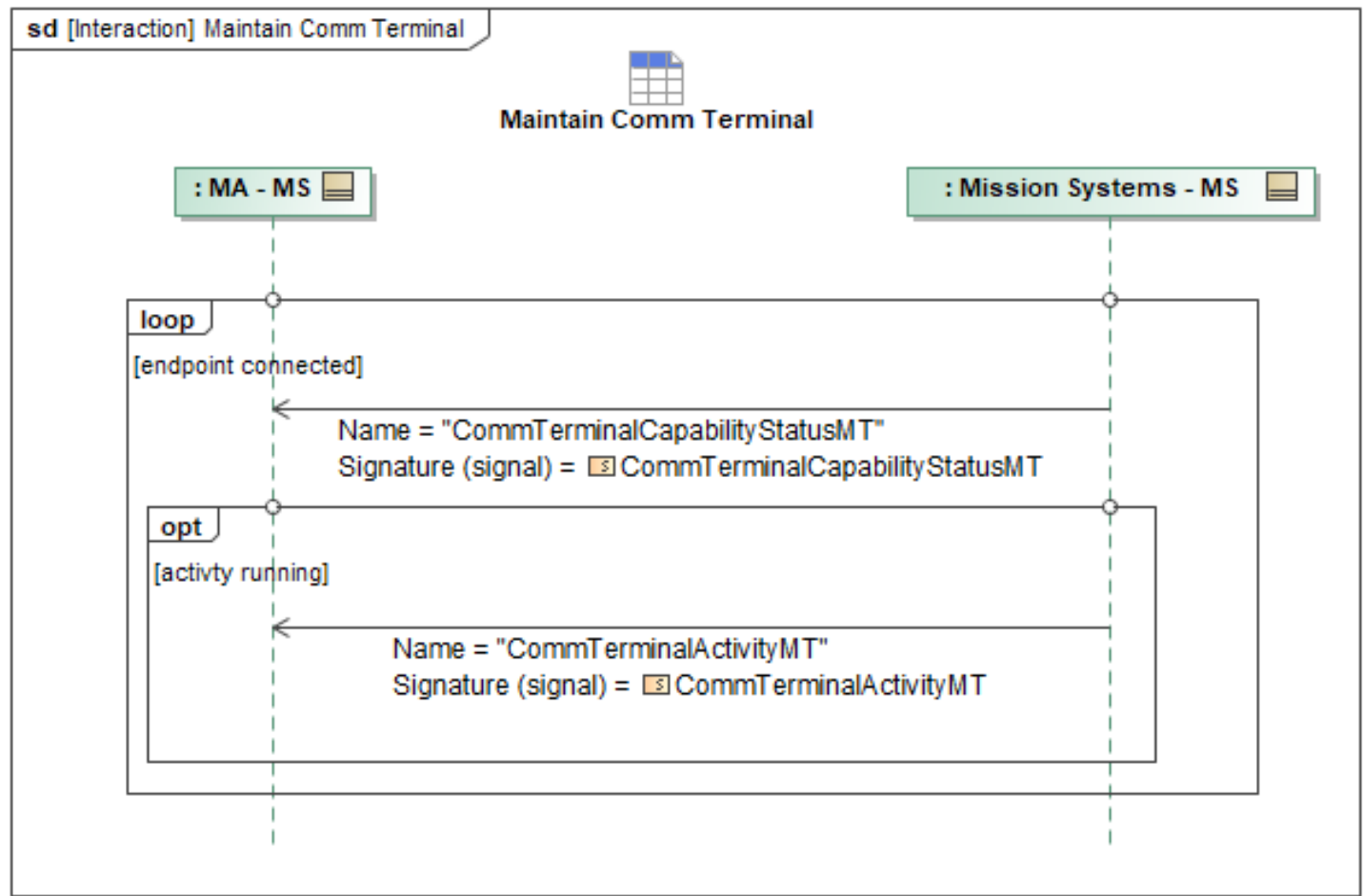


Figure 1-6: Maintain Comm Terminal

Table 1-5: Maintain Comm Terminal

Message Name (Name:Type)	Required Fields (Name:Type)
CommTerminalActivityMT: CommTerminalActivityMT	CommonSettingsStatus: CommControlType LinkStatistics: CommLinkStatisticsType WaveformSpecificSettingsStatus: CommWaveformActivityPET ConnectionStatus: CommConnectionStateEnum

Message Name (Name:Type)	Required Fields (Name:Type)
CommTerminalCapabilityStatusMT: CommTerminalCapabilityStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.6 Override Data Dissemination Plan

Mission Use Case: Additional Comms Capabilities

In some situations a modification to the Data Dissemination Plan is required that are short term in nature and a complete update to the *MA_DataDisseminationPlanMT* message is not required. The *MA_DataDisseminationPlanOverrideRequestMT* can be used for these sorts of modifications. MA sends the *MA_DataDisseminationPlanOverrideRequestMT* message to MS. MS can then accept or reject the request(s). If the request(s) are accepted, MS sends the *MA_DataDisseminationPlanOverrideRequestStatusMT_1* message to MA with the Request(s) having the *Request.RequestProcessingState* set to "QUEUED", "PROCESSING", "CANCELED", or "COMPLETED" as needed. Depending on the original request, the *EffectivityActivationStatus* or *OperatorModificationStatus* fields will be filled out as needed. The interaction "Update Data Dissemination Plan Activation Status" is initiated which will provide an update to the active status and include the *OverrideModificationState* associated with the override request status. For each applicable Gateway Activity, MS sends the *MA_GatewayActivityMT* message(s) to MA that will include any modifications that were made. If the request is NOT accepted, MS sends the *MA_DataDisseminationPlanOverrideRequestStatusMT_2* message to MA with the *RequestProcessingState* set to "REJECTED" or "FAILED", and the *RequestProcessingStateReason* filled out as needed. The *EffectivityActivationStatus* or *OperatorModificationStatus* fields will be filled out outlining the reason for the rejections/failures.

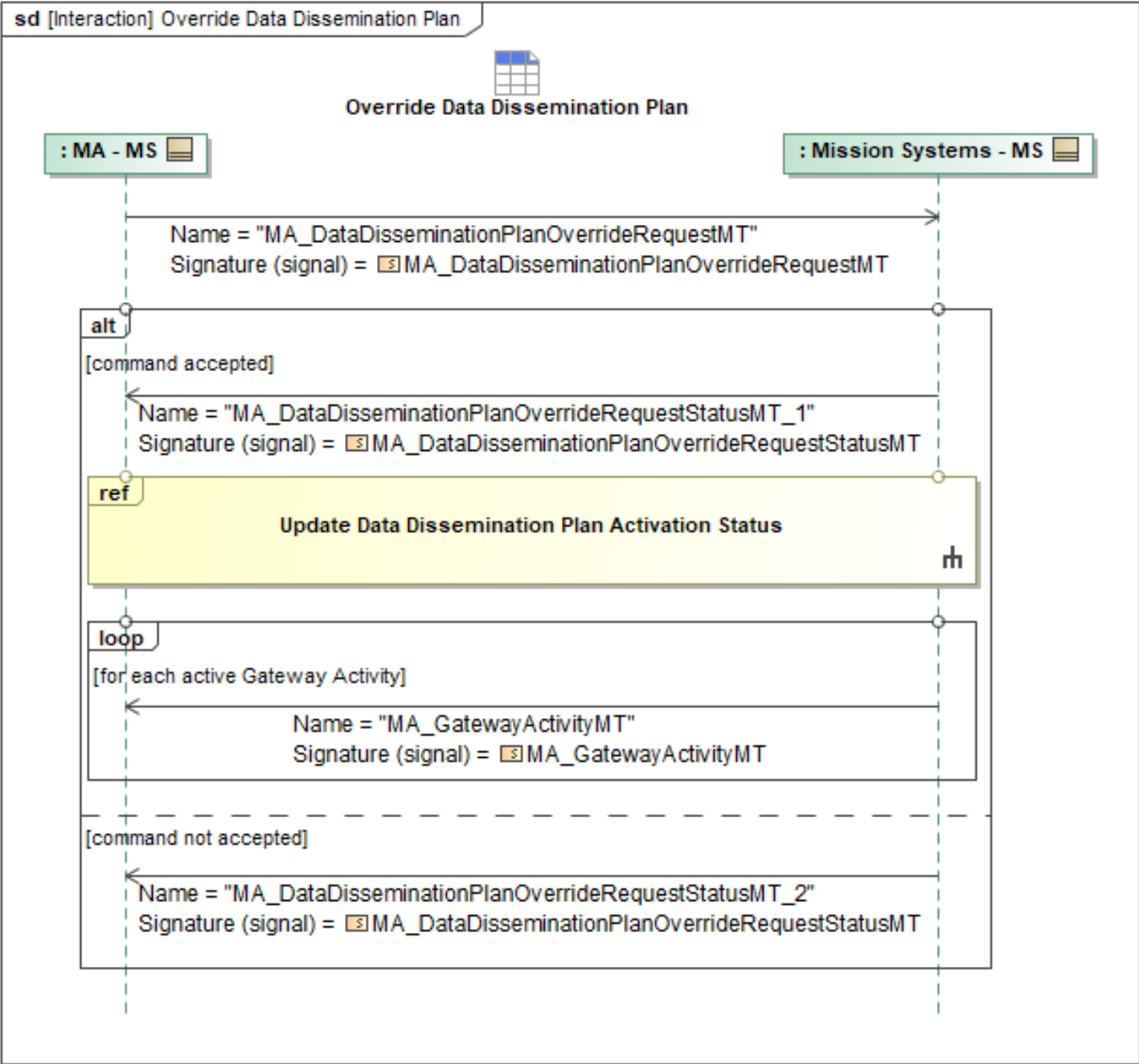


Figure 1-7: Override Data Dissemination Plan

Table 1-6: Override Data Dissemination Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_DataDisseminationPlanOverrideRequestMT : MA_DataDisseminationPlanOverrideRequestMT	AllocatedToSystemIDs: MA_PackageSystemType

Message Name (Name:Type)	Required Fields (Name:Type)
MA_DataDisseminationPlanOverrideRequestStatusMT_1: MA_DataDisseminationPlanOverrideRequestStatusMT	None
MA_DataDisseminationPlanOverrideRequestStatusMT_2: MA_DataDisseminationPlanOverrideRequestStatusMT	RequestProcessingStateReason: CannotComplyType
MA_GatewayActivityMT: MA_GatewayActivityMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.7 Publish Comm Team Report

MS publishes the Communication Team Report for MA's situational awareness when a new node/endpoint is discovered or leaves the team. If a team exists, the *MA_CommTeamReportMT_1* message reports a team's composition to MA by using the *SystemID* and *EndpointLabel* for each endpoint. If there are no node/endpoints to report, MS reports *MA_CommTeamReportMT_2*.

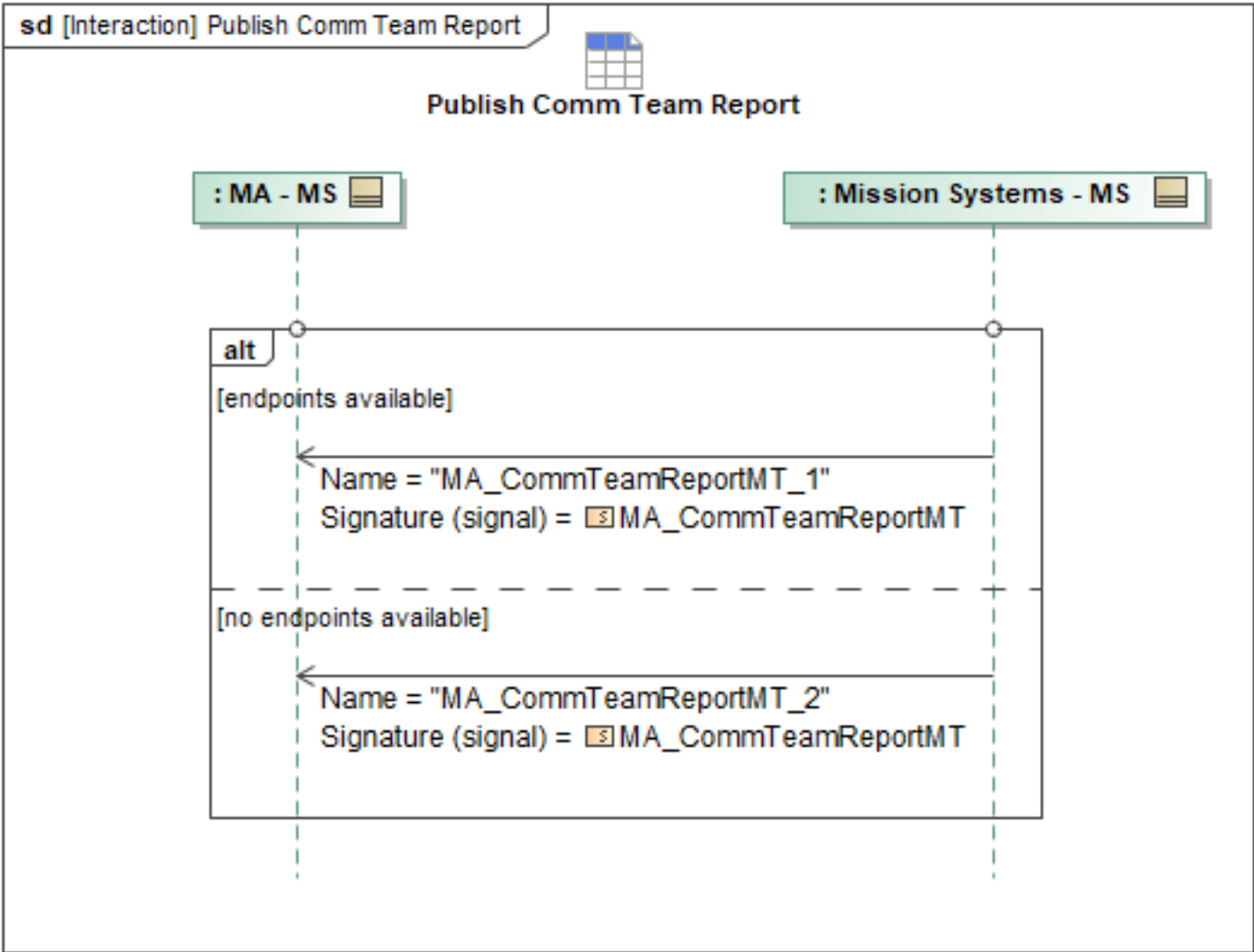


Figure 1-8: Publish Comm Team Report

Table 1-7: Publish Comm Team Report

Message Name (Name:Type)	Required Fields (Name:Type)
MA_CommTeamReportMT_1: MA_CommTeamReportMT	EndpointLabel: VisibleString256Type
MA_CommTeamReportMT_2: MA_CommTeamReportMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.8 Publish Network Endpoint Availability

The MS information service will publish the *MA_CommAvailableEndpointsMT* message when new endpoints become available in the network, or when the network membership is updated. This message conveys to MA the available network endpoints by its *SystemID* and *EndpointLabel* in the *AvailableNetworkEndpoints* field, if endpoints are available to report.

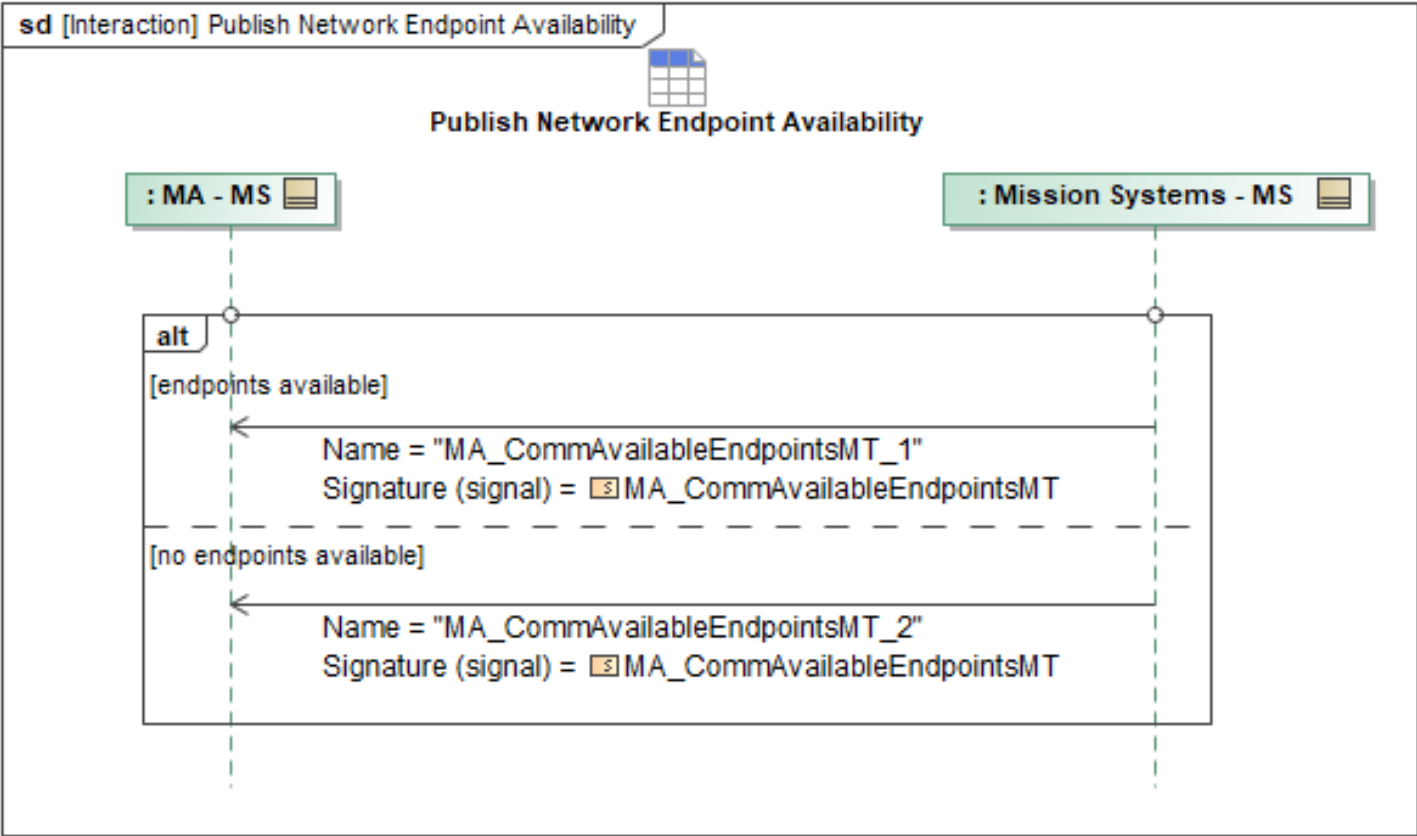


Figure 1-9: Publish Network Endpoint Availability

Table 1-8: Publish Network Endpoint Availability

Message Name (Name:Type)	Required Fields (Name:Type)
MA_CommAvailableEndpointsMT_1: MA_CommAvailableEndpointsMT	EndpointLabel: VisibleString256Type AvailableNetworkEndpoints: MA_AvailableNetworkEndPointsType
MA_CommAvailableEndpointsMT_2: MA_CommAvailableEndpointsMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.9 Receive Communication Error Report

MS can report communication faults to MA using the *MA_CommErrorReportMT* message. This message includes a *CommErrorReportID* field to identify the communication error report and an *ErrorDetail* field to specify details, including the source of the fault for a specified *SystemID*.

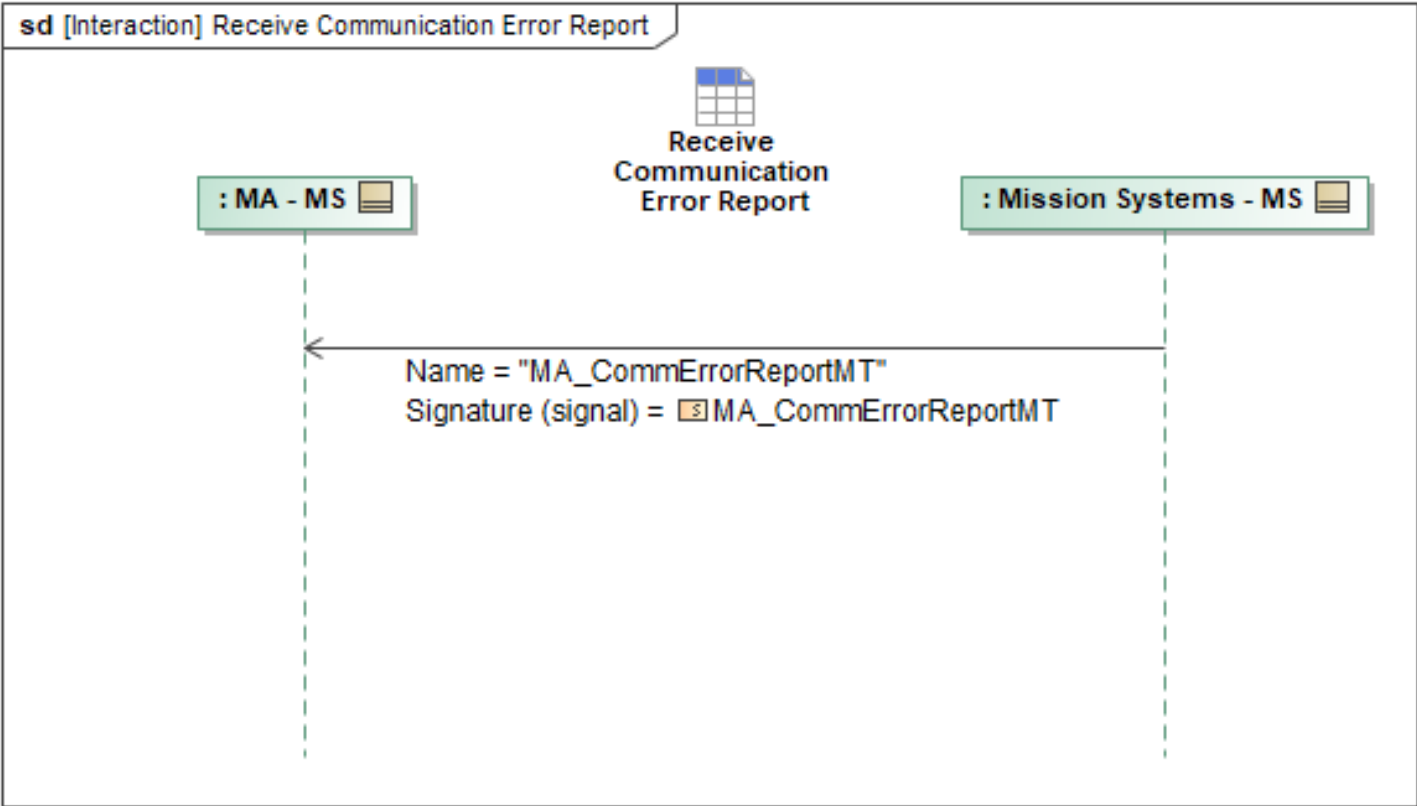


Figure 1-10: Receive Communication Error Report

Table 1-9: Receive Communication Error Report

Message Name (Name:Type)	Required Fields (Name:Type)
MA_CommErrorReportMT: MA_CommErrorReportMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.10 Request Communication Asset Status

Mission Use Case: Additional Comms Capabilities

The Request Communication Asset Status sequence is used by MA to request the estimated position and status of a MS communication asset. MA would request information on the communication asset through *CommAssetEstimationRequestMT* with the *CommAssetID* of the asset of interest and *ResultsInNativeMessage* set to FALSE. MS would respond with the *EstimationResult* of the communication asset that was requested in the *CommAssetEstimationRequestStatusMT*.

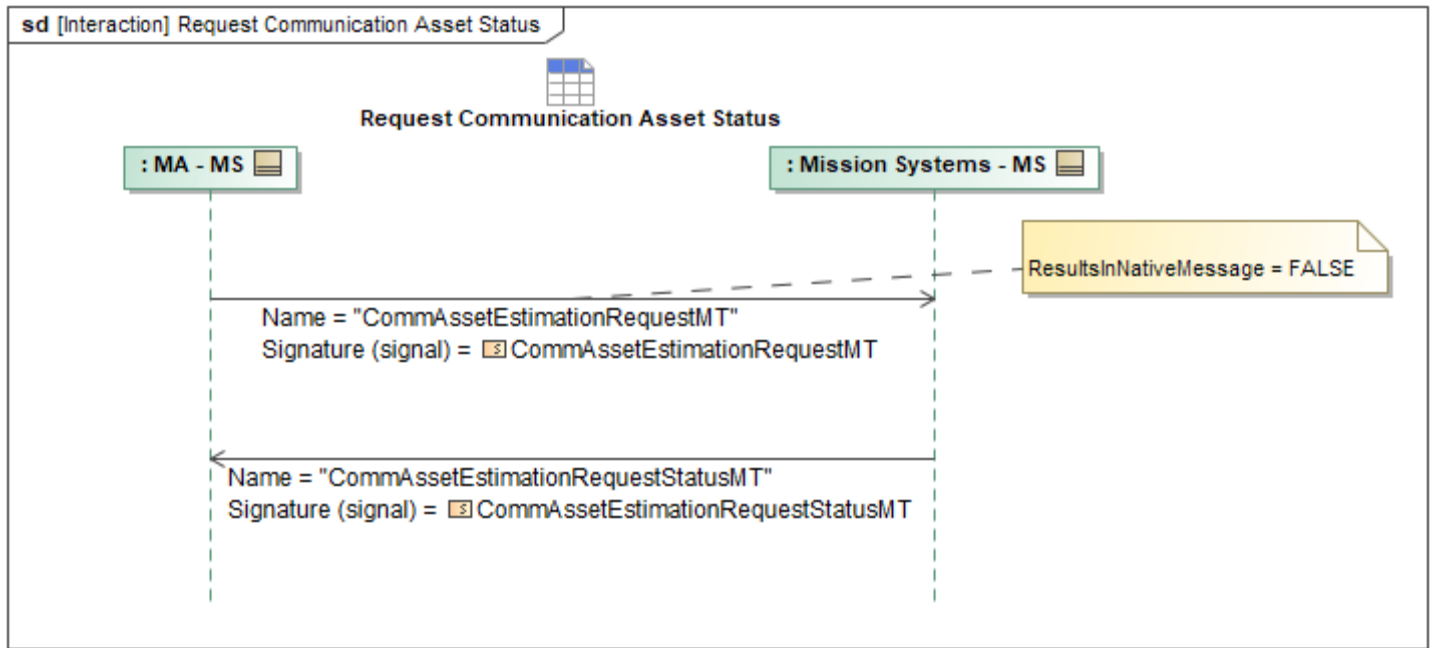


Figure 1-11: Request Communication Asset Status

Table 1-10: Request Communication Asset Status

Message Name (Name:Type)	Required Fields (Name:Type)
CommAssetEstimationRequestMT: CommAssetEstimationRequestMT	None
CommAssetEstimationRequestStatusMT: CommAssetEstimationRequestStatusMT	EstimationResult: CommAssetEstimationResultType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.11 Update Comm Terminal Plan

Mission Use Case: Additional Comms Capabilities

The *CommTerminalPlanMT* is used to configure how individual Comm Terminal Capabilities should be configured for an individual system. The plan contains a set of *Effectivities* that can be activated directly or setup with *Triggers* for different Mission Plans, sub-plans, etc. The *Effectivity* can specify the configuration of the entire Comm Terminal subsystem using a Mission Data File (MDF) or using Capability specific settings. These configurations of the Comm Terminal Capabilities allows a system to connect to other systems over various communication networks. First, MA shall post a *CommTerminalPlanMT* message that the communication service, in MS, will use when a command is issued. *CommTerminalPlanMT* will specify a set of desired states of the network communication capability and how they can be activated with the configuration setup in individual *Effectivities* and associated *Triggers*. Second, MA shall send a *CommTerminalPlanActivationCommandMT* to MS to command the activation of the previously posted *CommTerminalPlanMT* referencing the associated *CommTerminalPlanID*, and optionally populating the *EffectivityActivation* with an *EffectivityID* and *ConfigurationID*. Finally, MS replies to MA with a *CommTerminalPlanActivationCommandStatusMT_1* message if the command is accepted. While the plan activation is in-progress, until the activation

status is in a success or fail state, MS will report *CommTerminalPlanActivationCommandStatusMT_2*. If the command is not accepted, MS will report *CommTerminalPlanActivationCommandStatusMT_3*.

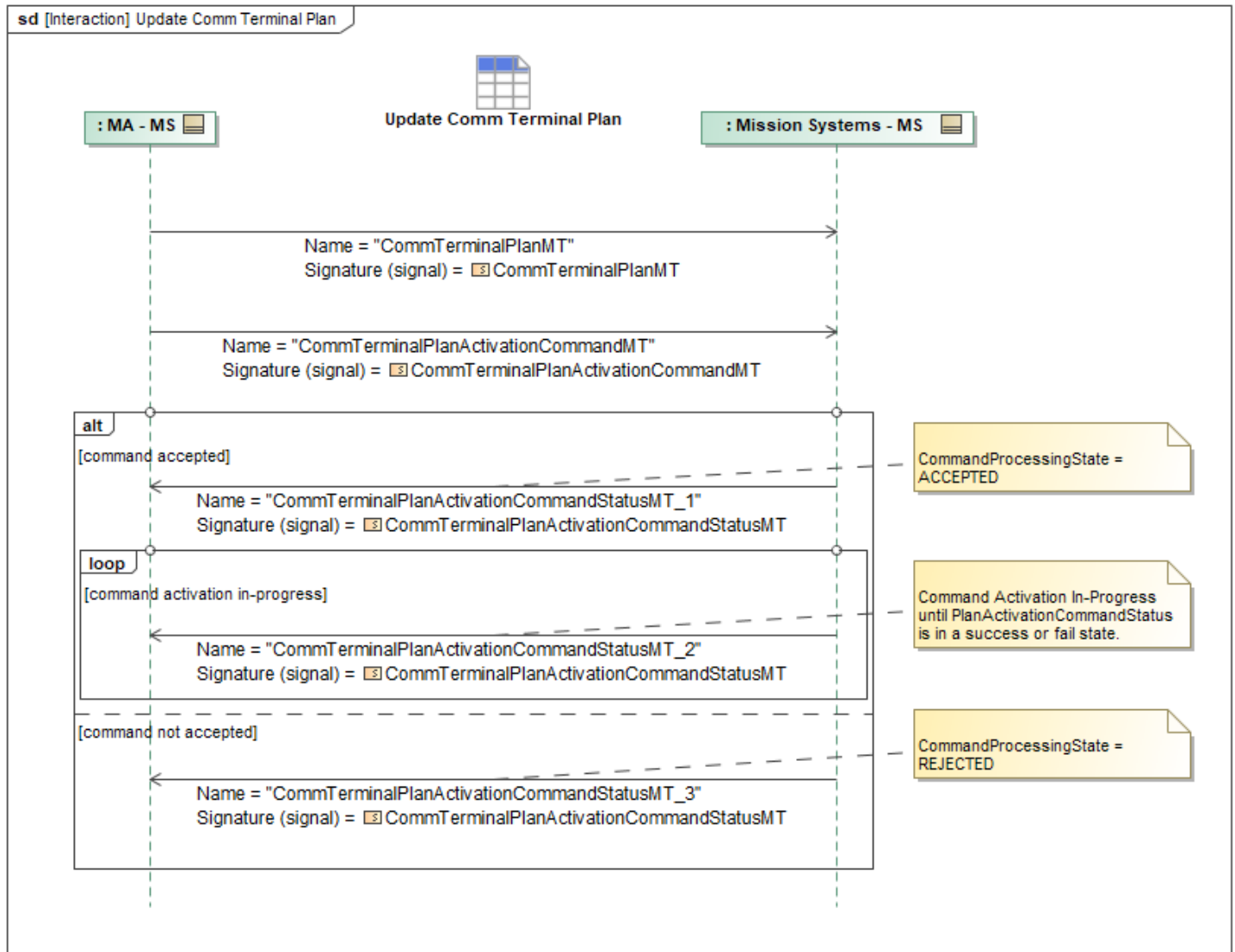


Figure 1-12: Update Comm Terminal Plan

Table 1-11: Update Comm Terminal Plan

Message Name (Name:Type)	Required Fields (Name:Type)
CommTerminalPlanActivationCommandMT: CommTerminalPlanActivationCommandMT	EffectivityActivation: CommTerminalPlanEffectivityActivationType
CommTerminalPlanActivationCommandStatus MT_1: CommTerminalPlanActivationCommandStatus MT	None
CommTerminalPlanActivationCommandStatus MT_2: CommTerminalPlanActivationCommandStatus MT	None

Message Name (Name:Type)	Required Fields (Name:Type)
CommTerminalPlanActivationCommandStatus MT_3: CommTerminalPlanActivationCommandStatus MT	CommandProcessingStateReason: CannotComplyType
CommTerminalPlanMT: CommTerminalPlanMT	PlannedCommCapability: PlannedCommCapabilityType ConnectionRequest: CommConnectionCommandEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.12 Update Comms Gateway Status

Mission Use Case: Additional Comms Capabilities

The Gateway Capability is how messages egress or ingress over the air. To provide information about each of the active gateways available for use, MS sends the *GatewayCapabilityMT* message and the *GatewayCapabilityStatusMT* message to MA. Then, for each active Gateway Activity, MS send the *MA_GatewayActivityMT* message(s) to MA. These messages provide the information needed to understand the state of the over the air messaging, they will provide context for how things are currently configured, and provide information for how to configure over the air messaging. MS typically configures the gateways, but the information can be useful when creating the plans that the MS uses.

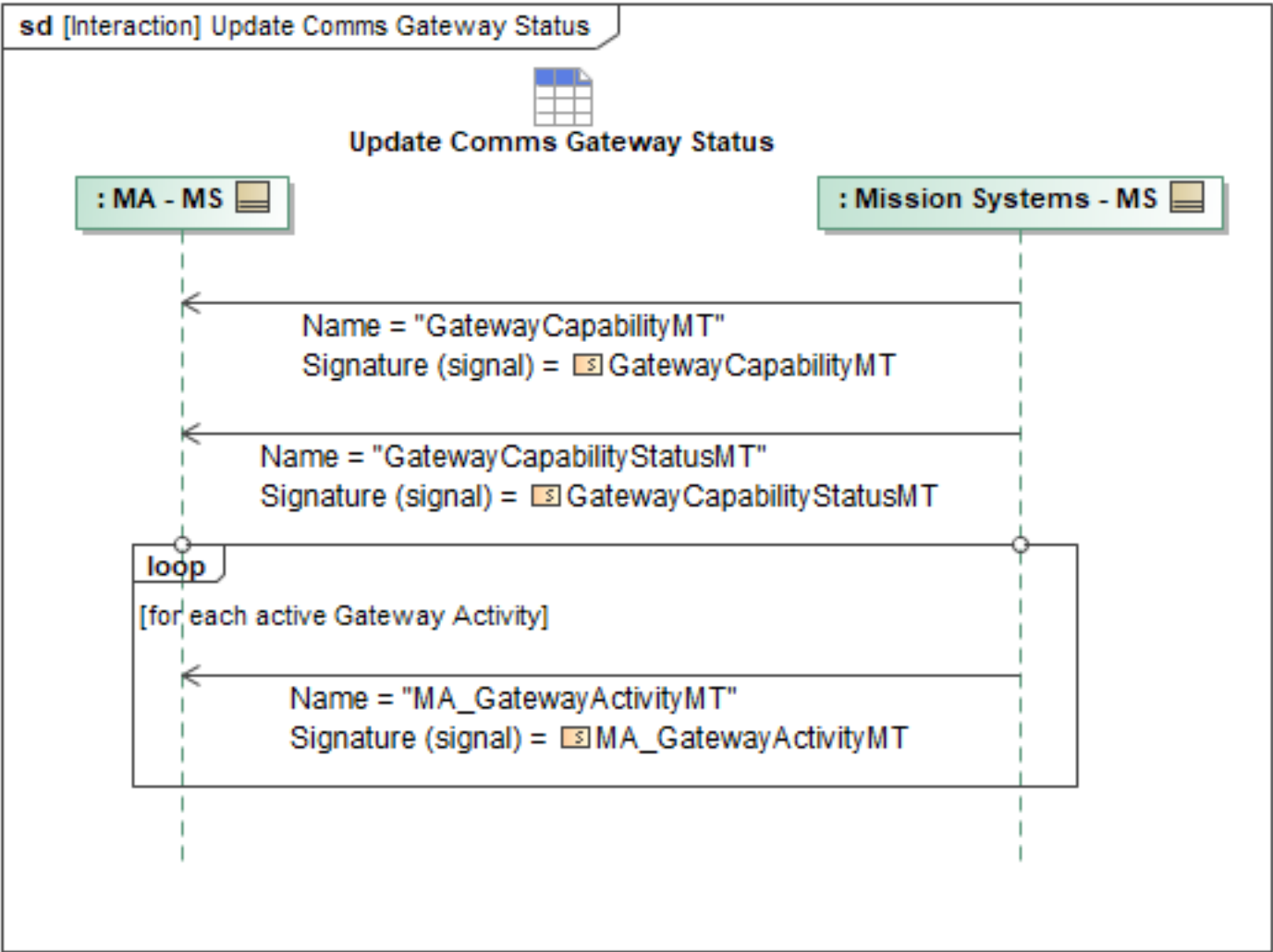


Figure 1-13: Update Comms Gateway Status

Table 1-12: Update Comms Gateway Status

Message Name (Name:Type)	Required Fields (Name:Type)
GatewayCapabilityMT: GatewayCapabilityMT	None
GatewayCapabilityStatusMT: GatewayCapabilityStatusMT	None
MA_GatewayActivityMT: MA_GatewayActivityMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.13 Update Data Dissemination Plan

Mission Use Case: Additional Comms Capabilities

The *MA_DataDisseminationPlanMT* is used to configure how messages ingress and egress over the air. The message contains a set of Effectivities that can be activated (using the

MA_DataDisseminationPlanActivationCommandMT message) or overridden (using the *MA_DataDisseminationPlanOverrideRequestMT* message) as needed. This implementation focuses using DDS as the over the air transport. The *Effectivity* field is used to match Information Exchange Requirement(s) (IERs) to DistributionGroup(s) and RateGroup(s) if the messages are periodic. The Applicability field in the *MA_DataDisseminationPlanMT* is used to set what systems or packages this messages applies to. The *AssociatedPlanID* field can be used to link together multiple *MA_DataDisseminationPlanMT* as needed.

The *IER* field is used to define the set of messages that will ingress or egress from the local system. The *IER.ContentType* field specifies the message filters. Periodic messages must not be combined into a single *IER.ContentType*, but non-periodic messages that have other matching items like Direction or Query can be combined. The *IER.ContentType.Type.TacticalDataLink* can be used to define messages that are meant to be sent over Tactical Data Links specifically. Alternatively, *IER.ContentType.Type.UCI_Message* can be used for DDS type connections when matched with a Distribution Group configured for DDS. When DDS is configured and the *IER.ContentType.Direction* is set to "EGRESS", "INGRESS", or "BOTH" they would match to "PUBLISH", "SUBSCRIBE", "PUB_AND_SUB", if the *DistributionGroup.NetworkNativeEndpoint.TopicPublishSubscribe* field is not set. If the *IER.ContentType.AllowWrappedEgressMessages* field is set to "TRUE", this means that the implementing Gateway will accept a wrapped *MA_TxDataPayloadCommandMT* message along with the non-wrapped message before using the Query filter for that IER. It is up to the implementation to not send duplicate messages if the same message is received both wrapped and unwrapped. The *IER.Priority* field will define the priority of that IER compared to other IERs and will be used as the DDS topic priority, if the *DistributionGroup.NetworkNativeEndpoint.TopicQosPriority* field is not set.

The DistributionGroup field is used to configure the DDS specific items using the *DistributionGroup.NetworkNativeEndpoint* with the *DDS_DataLinkNativeInfoType*. The settings in this field will override settings from other fields. The Domain field is used to define the DDS Domain ID the Gateway will connect to. The TopicName field is used to define an individual topic name specifically, or if left blank will default to the message type name defined in the connected IER. The *TopicPublishSubscribe* field can be used to specifically set the intended direction of the message(s) from the IER, or if left blank will default to the connected IER settings for Direction. The TopicQosReliability field can be use to specifically set "RELIABLE" or "BEST_EFFORT", but if set to "NONE" or left unset the A-GRA method for UCI_PRIMITIVE_TYPE is used (defined in the DMS section). Setting the TopicQosReliability field will also override the DeliveryProtocol. If the DirectMessagingAllowed field is set to "TRUE" in any of the active Distribution Groups, Specific Destination messages to/from SystemIDs via a System UUID topic name is enabled. Alternatively the *MA_TxDataPayloadCommandMT* wrapper can be used with message query filters applied where the entire *MA_TxDataPayloadCommandMT* message would be sent and receiving systems would have to filter for their System UUID from the *SpecificDestinationID* field.

The *RateGroup* field is used to define rate values that can be matched to multiple IERs. The *RateType* field should be set to "PERIODIC". The *TransmitFrequency* is applied to all messages inside of the connected IER(s) individually. Therefore it is recommended that any connected IER only contain a single message so that the rate is applied per message. The *TransmitFrequency* field is the target delta time between messages sent (e.g. 1 msg per second would mean a value of "PT1.0S" or 1 second).

The *Effectivity.IER_Assignment* field is used to match an *IER_ID* to *Endpoint_Assignment.DistributionGroupID*. If the *IER* is considered periodic, a *RateGroupID* can also

be set. The *Endpoint_Assignment.VectorID* can be used if the Plan is meant for specific point to point type configuration. The *Effectivity.ActivationTriggerID* field can also be used to make an Effectivity that is activated automatically given the specific Trigger.

If the Data Dissemination Plan needs update or is a new plan, MA sends the *MA_DataDisseminationPlanMT* message and the *MA_DataDisseminationPlanActivationCommandMT* message to MS. If the command is accepted by MS, MS sends the *MA_DataDisseminationPlanActivationCommandStatusMT_1* message to MA with the *CommandProcessingStateEnum* set to "ACCEPTED", the interaction "Update Data Dissemination Plan Activation Status" is initiated, and for each applicable Gateway Activity MS sends the *MA_GatewayActivityMT* message(s) to MA. If the command is NOT accepted, MS send the *MA_DataDisseminationPlanActivationCommandStatusMT_2* message to MA with the *CommandProcessingStateEnum* set to "REJECTED" and the *PlanActivationCommandStatus* is set according to the reason it was rejected.

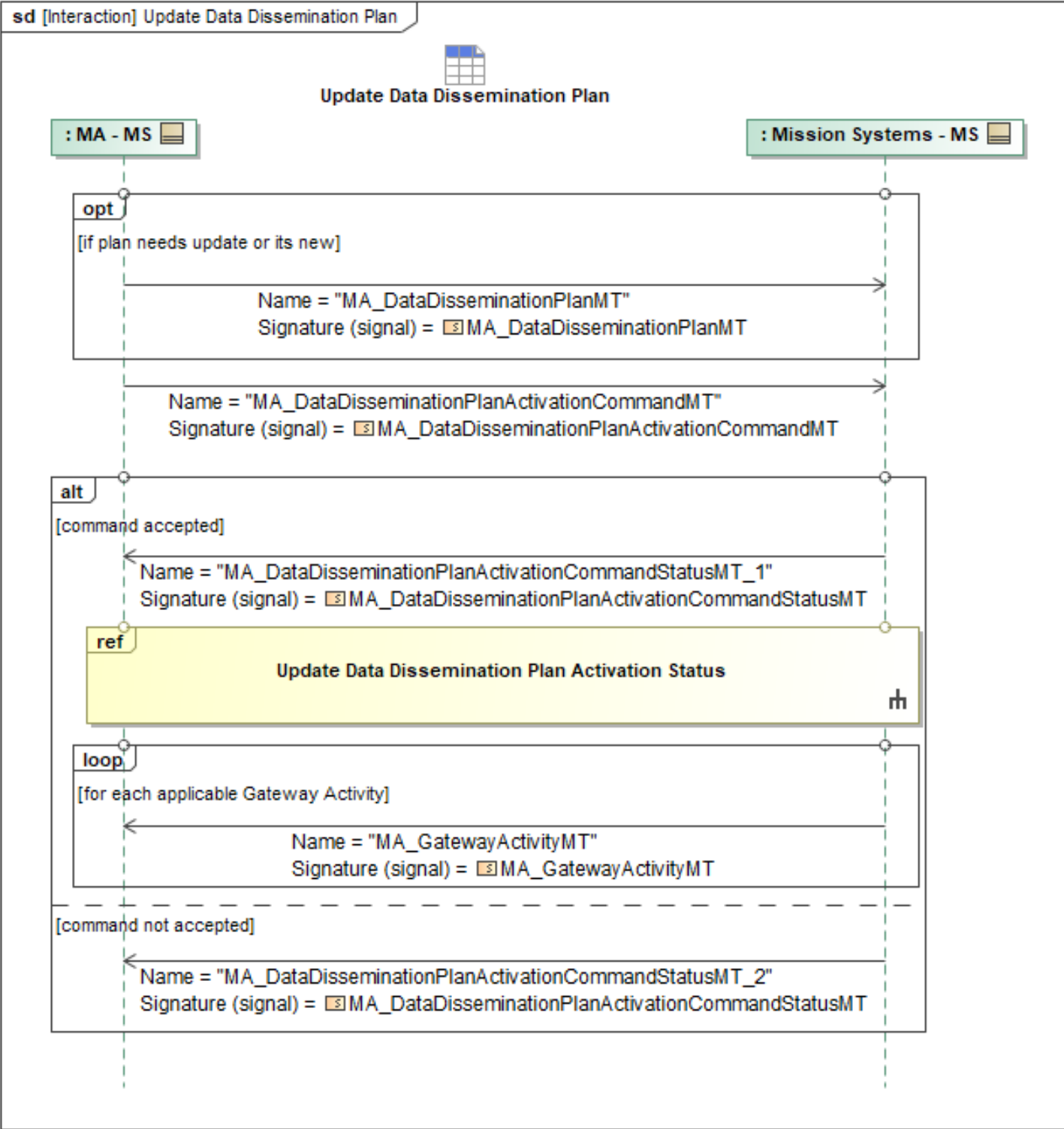


Figure 1-14: Update Data Dissemination Plan

Table 1-13: Update Data Dissemination Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_DataDisseminationPlanActivationCommandMT: MA_DataDisseminationPlanActivationCommandMT	Applicability: MA_PackageSystemType
MA_DataDisseminationPlanActivationCommandStatusMT_1: MA_DataDisseminationPlanActivationCommandStatusMT	None
MA_DataDisseminationPlanActivationCommandStatusMT_2: MA_DataDisseminationPlanActivationCommandStatusMT	RequestProcessingStateReason: CannotComplyType
MA_DataDisseminationPlanMT: MA_DataDisseminationPlanMT	None
MA_GatewayActivityMT: MA_GatewayActivityMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.14 Update Data Dissemination Plan Activation Status

Mission Use Case: Additional Comms Capabilities

The MS will send the *MA_DataDisseminationPlanActivationStatusMT* message to MA that will reflect the active state of Data Dissemination Plan. This can be sent periodically, but will always be sent upon change to the active state. The message will reflect the active Effectivities and Override Modifications of the Data Dissemination Plan.

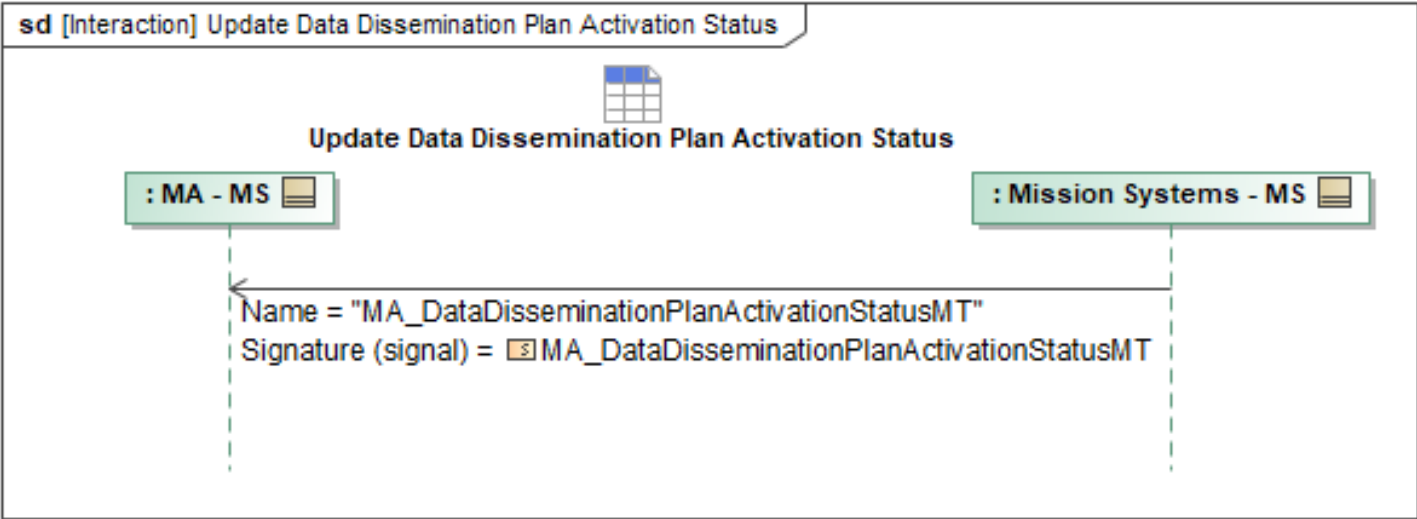


Figure 1-15: Update Data Dissemination Plan Activation Status

Table 1-14: Update Data Dissemination Plan Activation Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_DataDisseminationPlanActivationStatusMT : MA_DataDisseminationPlanActivationStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.2 Contingencies Package

1.2.2.1 Intra-Vehicle Comms Failure_MS

Mission Systems (MS) will provide regular status to Mission Autonomy (MA) via *ServiceStatusMT*. In the event MA doesn't receive a status from MS within the designated time frame, MA will request status from MS via the *ServiceStatusDataRequestMT*. MS will continue to attempt to report status via *ServiceStatusMT*. If communication between MA and MS is not restored, MA will refer to and follow the contingency plan associated with a loss of communication with MS.

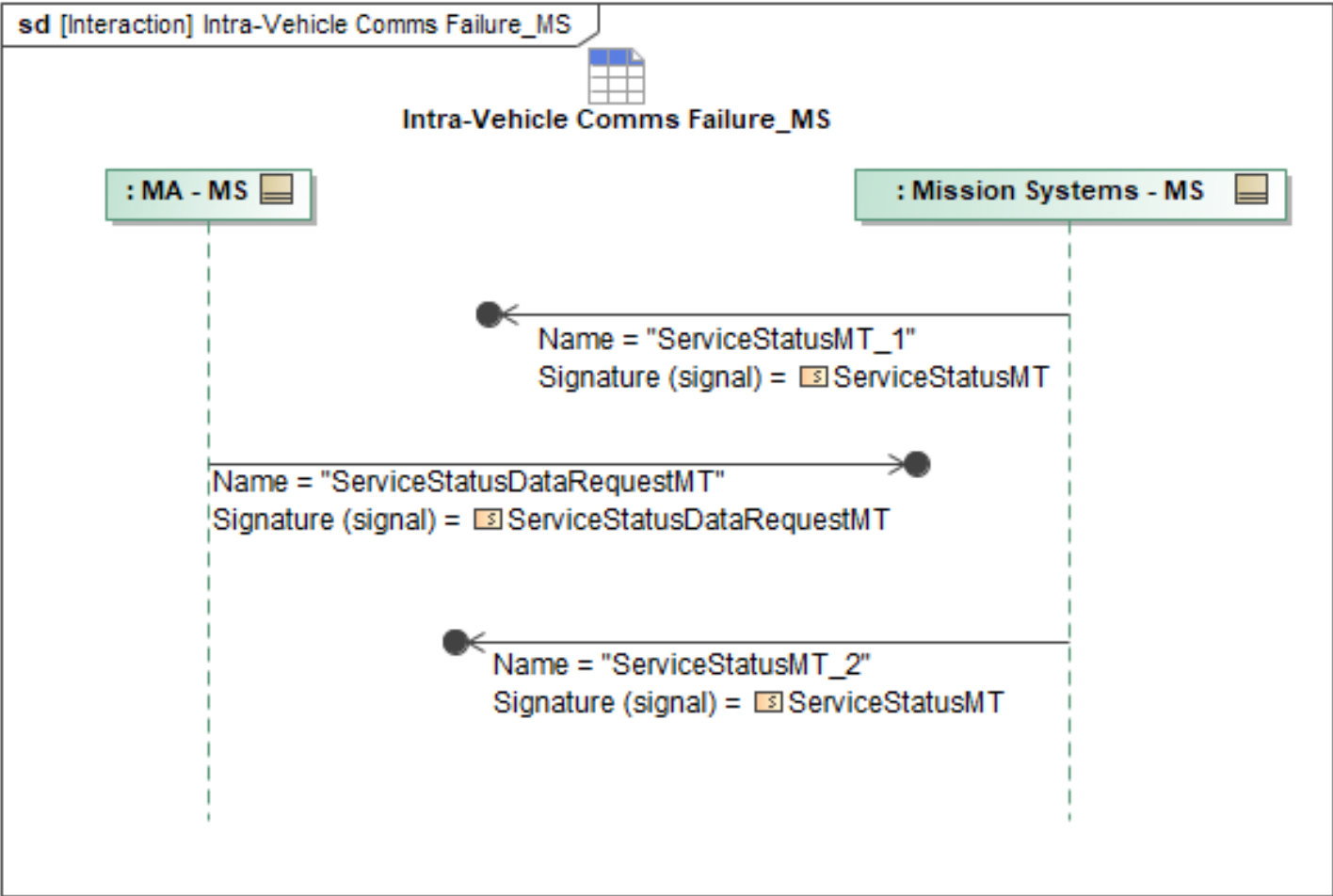


Figure 1-16: Intra-Vehicle Comms Failure_MS

Table 1-15: Intra-Vehicle Comms Failure_MS

Message Name (Name:Type)	Required Fields (Name:Type)
ServiceStatusDataRequestMT: ServiceStatusDataRequestMT	None
ServiceStatusMT_1: ServiceStatusMT	None
ServiceStatusMT_2: ServiceStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.2.2 Logging Failure

Mission Systems (MS) will report the status of the services it provides to Mission Autonomy (MA) via the *ServiceStatusMT* UCI message. In the event of a service failure, MS will report what service has failed after running a self initiated BIT. Status of the failed service will be reported via the *SubsystemBIT_StatusMT* UCI message.

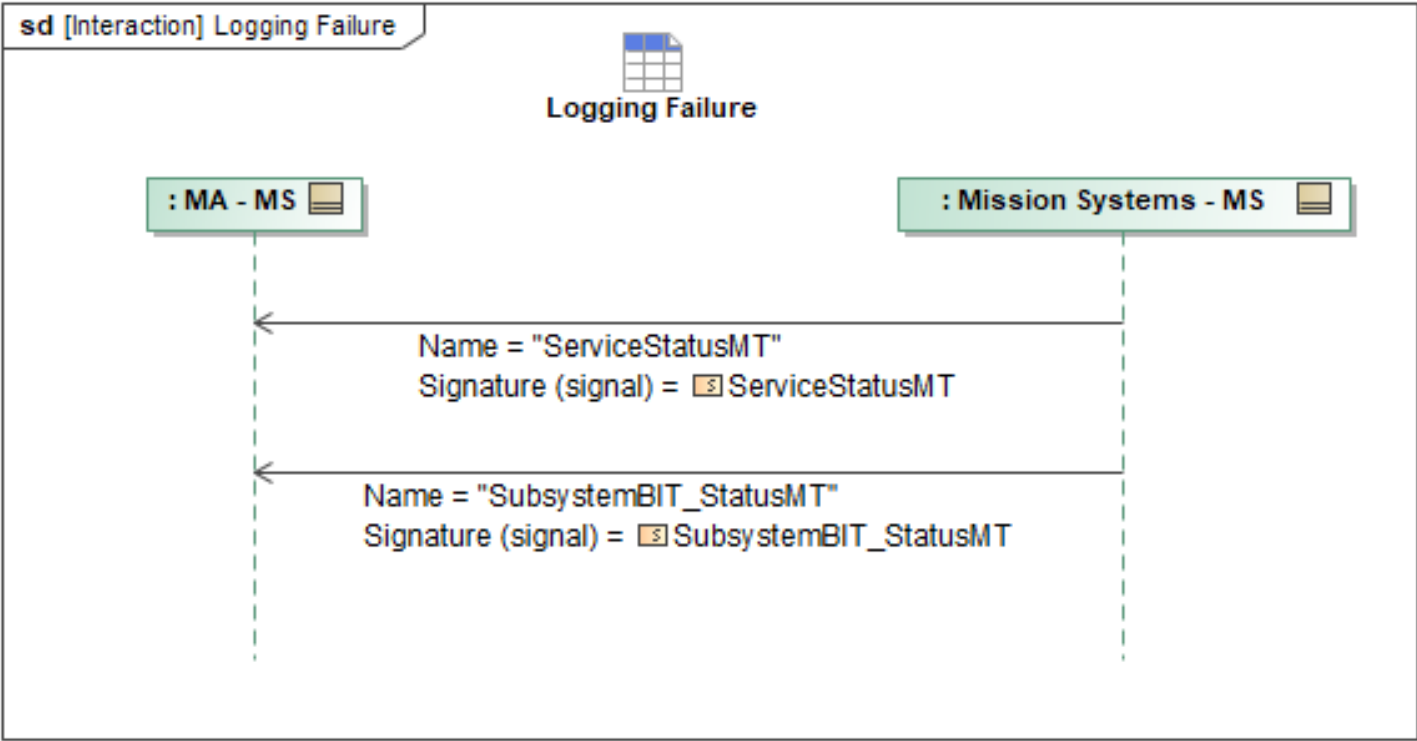


Figure 1-17: Logging Failure

Table 1-16: Logging Failure

Message Name (Name:Type)	Required Fields (Name:Type)
ServiceStatusMT: ServiceStatusMT	None
SubsystemBIT_StatusMT: SubsystemBIT_StatusMT	CompletedBIT: SubsystemCompletedBIT_Type FailReason: VisibleString256Type

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.2.3 MS Malformed Message Response

Upon receipt of a malformed message, MA will send *MA_SystemNotificationMT* to MS, indicating that the message could not be successfully processed. Within the message, MA sets *NotificationState* to ACTIVE and populates *NotificationDetail.Reason* with *INVALID_INPUT_PARAMETER*. *NotificationDetail.Description* will contain a human-readable string stating which fields are malformed. *AssociatedMessage* will reference the malformed message. The *MessageType* will be set to the same type as the malformed message. *AssociatedID* will be populated with a reference ID from the malformed message if applicable, the *UUID* of that message. If additional data is required to resolve the condition and the fault remains active, MA will issue *QueryDataRequestMT* with the *Query* field containing the ID of the specific message. MS will respond with *QueryDataRequestStatusMT*, specifically the *Message* field within the *Result* field will contain the corrected message containing the originally missing data. If the fault has been addressed, then MA will only send a new *MA_SystemNotificationMT* with the same fields as above and *NotificationState* set to CLOSED.

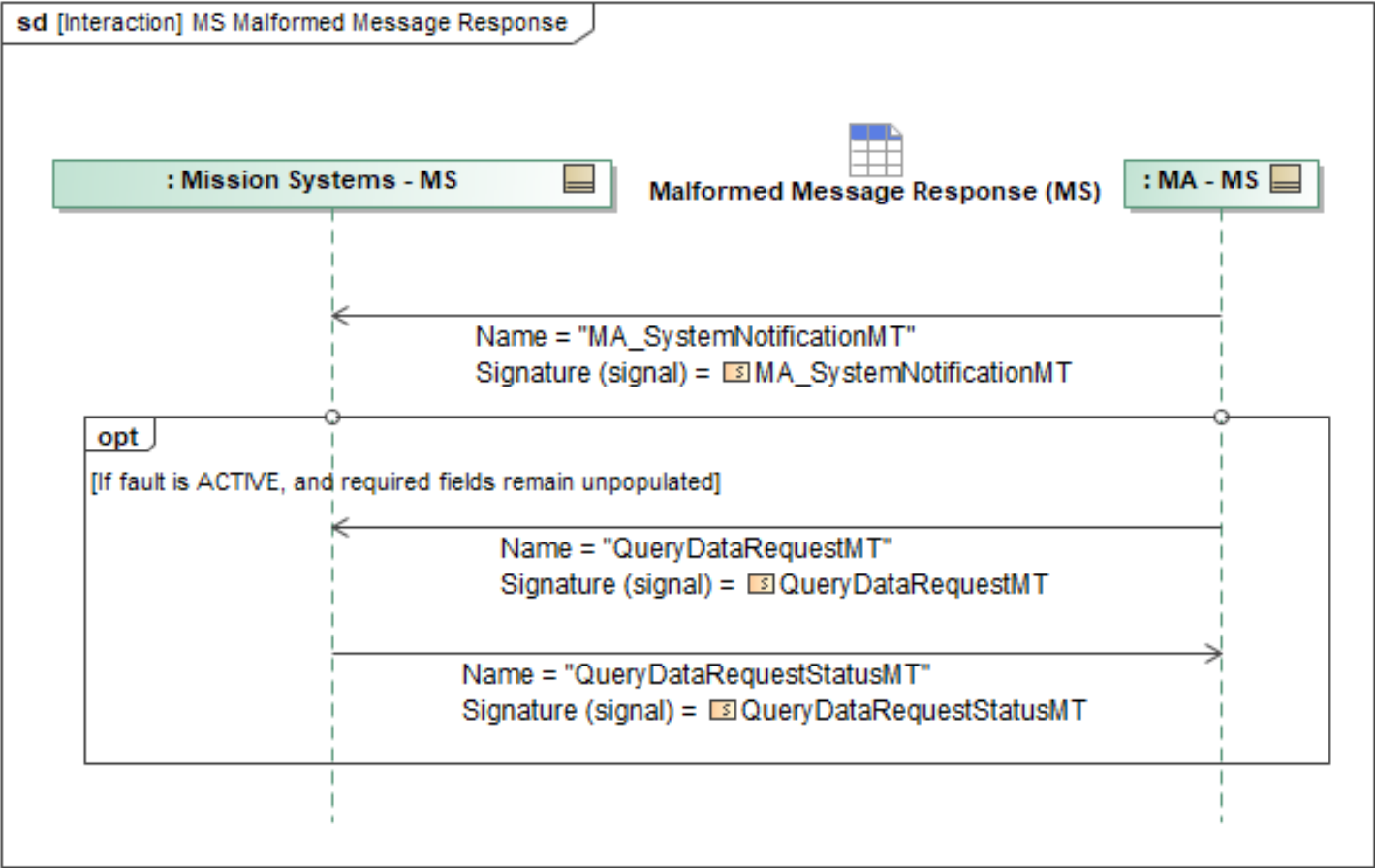


Figure 1-18: MS Malformed Message Response

Table 1-17: MS Malformed Message Response

Message Name (Name:Type)	Required Fields (Name:Type)
MA_SystemNotificationMT: MA_SystemNotificationMT	Description: VisibleString1024Type AssociatedID: ID_Type
QueryDataRequestMT: QueryDataRequestMT	Query: QueryPET
QueryDataRequestStatusMT: QueryDataRequestStatusMT	Message: MessageType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.2.4 Payload Management Failure

MS will report status of the onboard payloads MS is responsible for via *SubsystemStatusMT*. In the event that a fault is detected, or there is a payload deployment failure, MS may optionally report the fault and necessary details to MA via the *MA_FaultMT* message, depending on whether MS supports this functionality. Faults are expected to be periodically reported by MS in the Receive Periodic Health, Status, & BIT Information sequence.

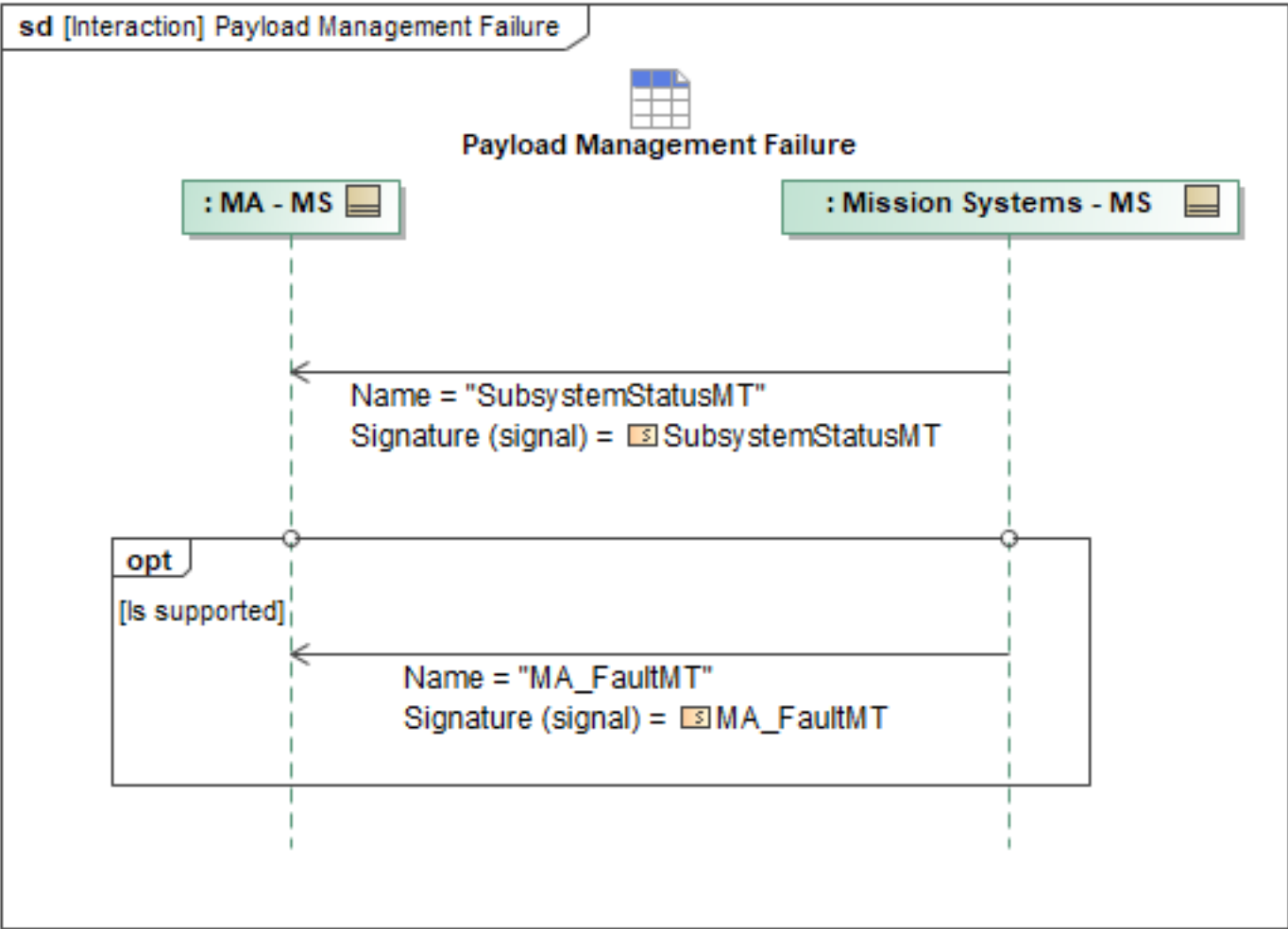


Figure 1-19: Payload Management Failure

Table 1-18: Payload Management Failure

Message Name (Name:Type)	Required Fields (Name:Type)
MA_FaultMT: MA_FaultMT	Severity: MA_FaultSeverityEnum FaultState: FaultStateEnum FaultDescription: VisibleString256Type
SubsystemStatusMT: SubsystemStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.2.5 Sensor Failure

ServiceStatusMT is the UCI message that covers the status of all services for which MS is responsible. In the event a sensor subsystem fault is detected, MS may optionally report the fault and necessary fault detail to MA via the MA_FaultMT message, depending on whether MS supports this functionality. This reporting sequence is specific to sensor subsystem failures only. Faults are expected to be periodically reported in the Receive Periodic Health, Status, & BIT Information

sequence.

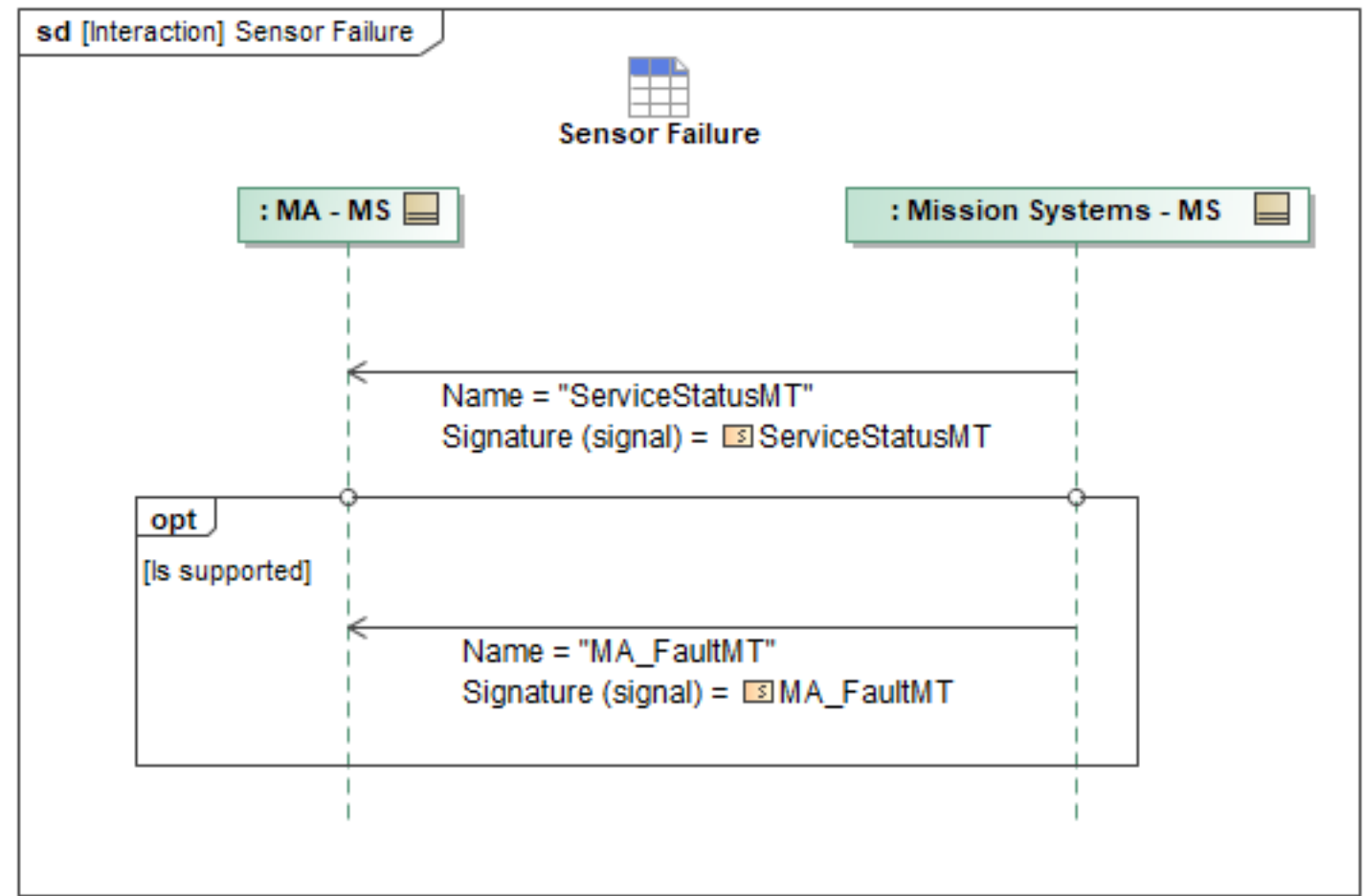


Figure 1-20: Sensor Failure

Table 1-19: Sensor Failure

Message Name (Name:Type)	Required Fields (Name:Type)
MA_FaultMT: MA_FaultMT	CapabilityID: CapabilityID_Type ServiceID: ServiceID_Type AmbiguityGroup: SubsystemFaultAmbiguityGroupType SubsystemID: SubsystemID_Type SupportCapabilityID: SupportCapabilityID_Type ComponentID: ComponentID_Type
ServiceStatusMT: ServiceStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.3 COP Package

1.2.3.1 MS Comms Status Updates to COP

The MS Comms Status Updates to COP sequence diagram specifies the interaction of the MS sending updates of the comms state to MA.

At a predefined period, the *SubsystemStatusMT* is sent from MS to MA. The *SubsystemStatusMT* must have the *SubsystemID* of the comms subsystem and *SubsystemState* set to the state of the comms subsystem and its links.

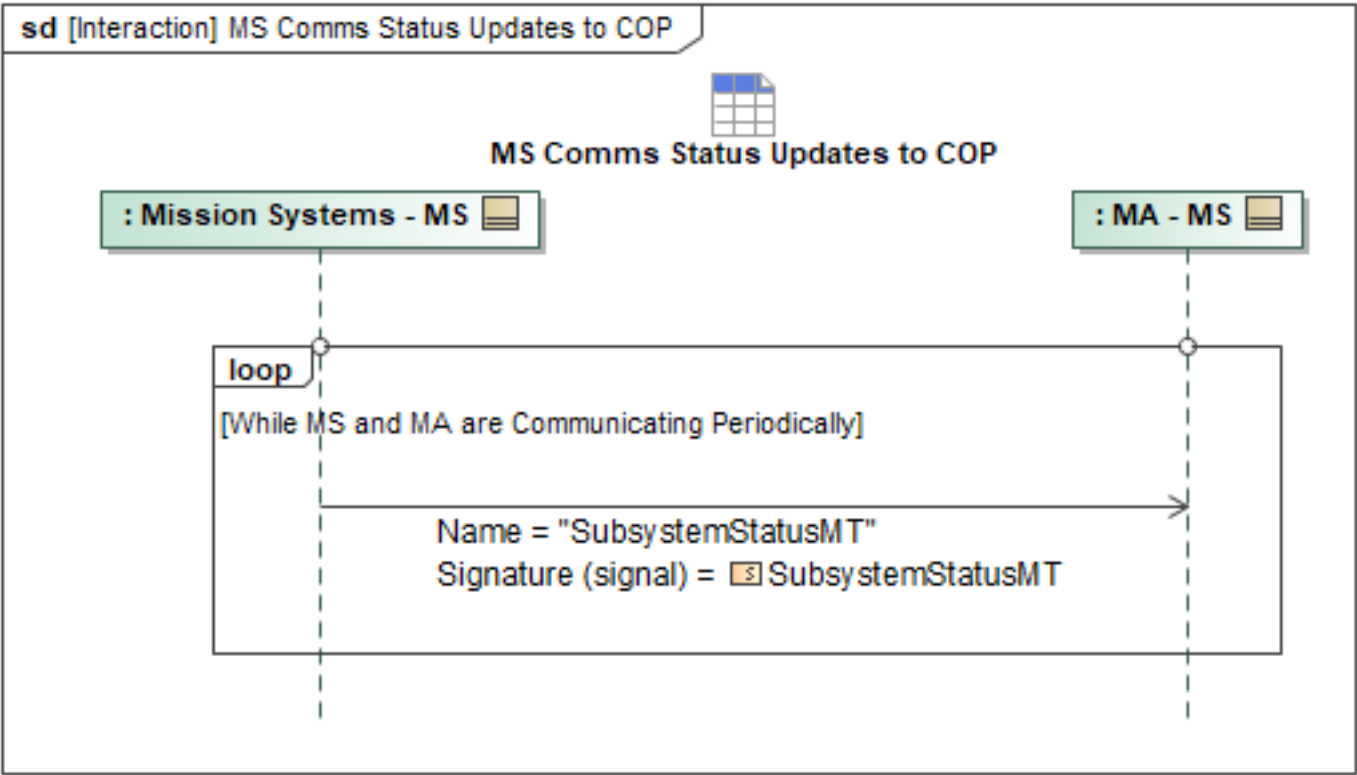


Figure 1-21: MS Comms Status Updates to COP

Table 1-20: MS Comms Status Updates to COP

Message Name (Name:Type)	Required Fields (Name:Type)
SubsystemStatusMT: SubsystemStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.3.2 MS Sensor AMTI Status Updates to COP

Mission Use Case: AMTI

The MS AMTI Sensor Status Updates to COP sequence diagram specifies the interaction of MS communicating the status of the sensor Airborne Moving Target Indicator (AMTI) sensor capability to MA for common operating picture (COP) reporting purposes. This sequence is only present if AMTI sensing capabilities exist in MS. MS publishes *MA_AMTI_CapabilityMT* with at least one *Capability*

specified. MS periodically publishes *AMTI_CapabilityStatusMT* and *AMTI_ActivityMT* to MA to report status and usage of the AMTI capability. COP uses *MA_AMTI_CapabilityMT* and *AMTI_CapabilityStatusMT* to know if the AMTI sensor capability is available to command. COP uses *AMTI_ActivityMT* to know if the AMTI capability can effectively be used (e.g. not fully scheduled). Requirements put upon *MA_AMTI_CapabilityMT*, *AMTI_CapabilityStatusMT*, and *AMTI_ActivityMT* are set by other sequences requiring use-case specific functionality (e.g. “Advertise Sensor AMTI Capability for Search and Track”, “Report Sensor AMTI Capability State”), for COP purposes these messages are received and aggregated into general status.

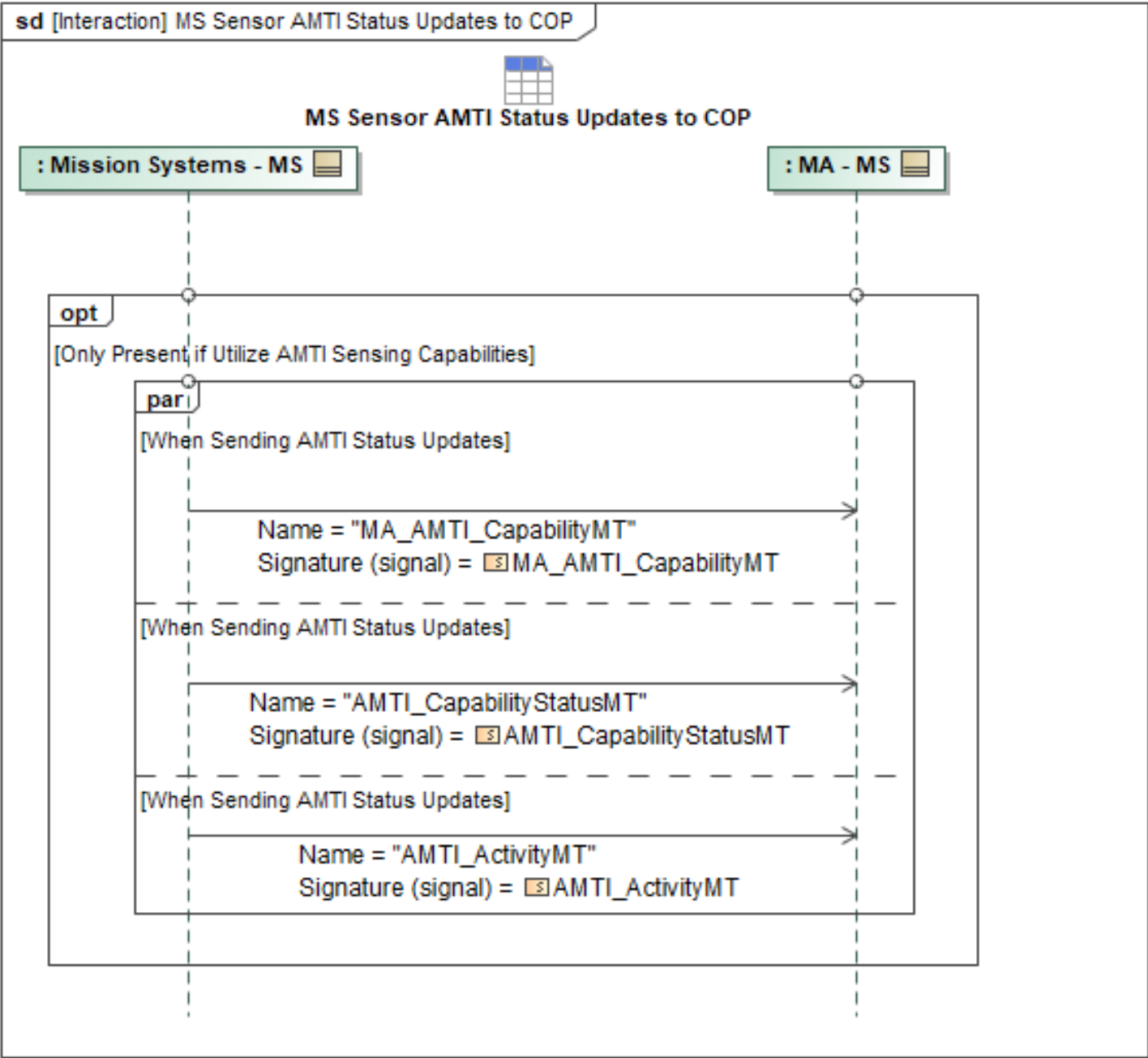


Figure 1-22: MS Sensor AMTI Status Updates to COP

Table 1-21: MS Sensor AMTI Status Updates to COP

Message Name (Name:Type)	Required Fields (Name:Type)
AMTI_ActivityMT: AMTI_ActivityMT	None
AMTI_CapabilityStatusMT: AMTI_CapabilityStatusMT	None
MA_AMTI_CapabilityMT: MA_AMTI_CapabilityMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy

Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.3.3 MS Sensor PO Status Updates to COP

Mission Use Case: PO

The MS PO Sensor Status Updates to COP sequence diagram specifies the interaction of MS communicating the status of the sensor Passive Optical (PO) sensor capability to MA for common operating picture (COP) reporting purposes. This sequence is only present if PO sensing capabilities exist in MS. MS publishes *PO_CapabilityMT* with at least one *Capability* specified. MS periodically publishes *PO_CapabilityStatusMT* and *PO_ActivityMT* to MA to report status and usage of the PO capability. COP uses *PO_CapabilityMT* and *PO_CapabilityStatusMT* to know if the PO sensor capability is available to command. COP uses *PO_ActivityMT* to know if the PO capability can effectively be used (e.g. not fully scheduled). Requirements put upon *PO_CapabilityMT*, *PO_CapabilityStatusMT*, and *PO_ActivityMT* are set by other sequences requiring use-case specific functionality (e.g. “Advertise Sensor PO Capability for Search and Track”, “Report Sensor PO Capability State”), for COP purposes these messages are received and aggregated into general status.

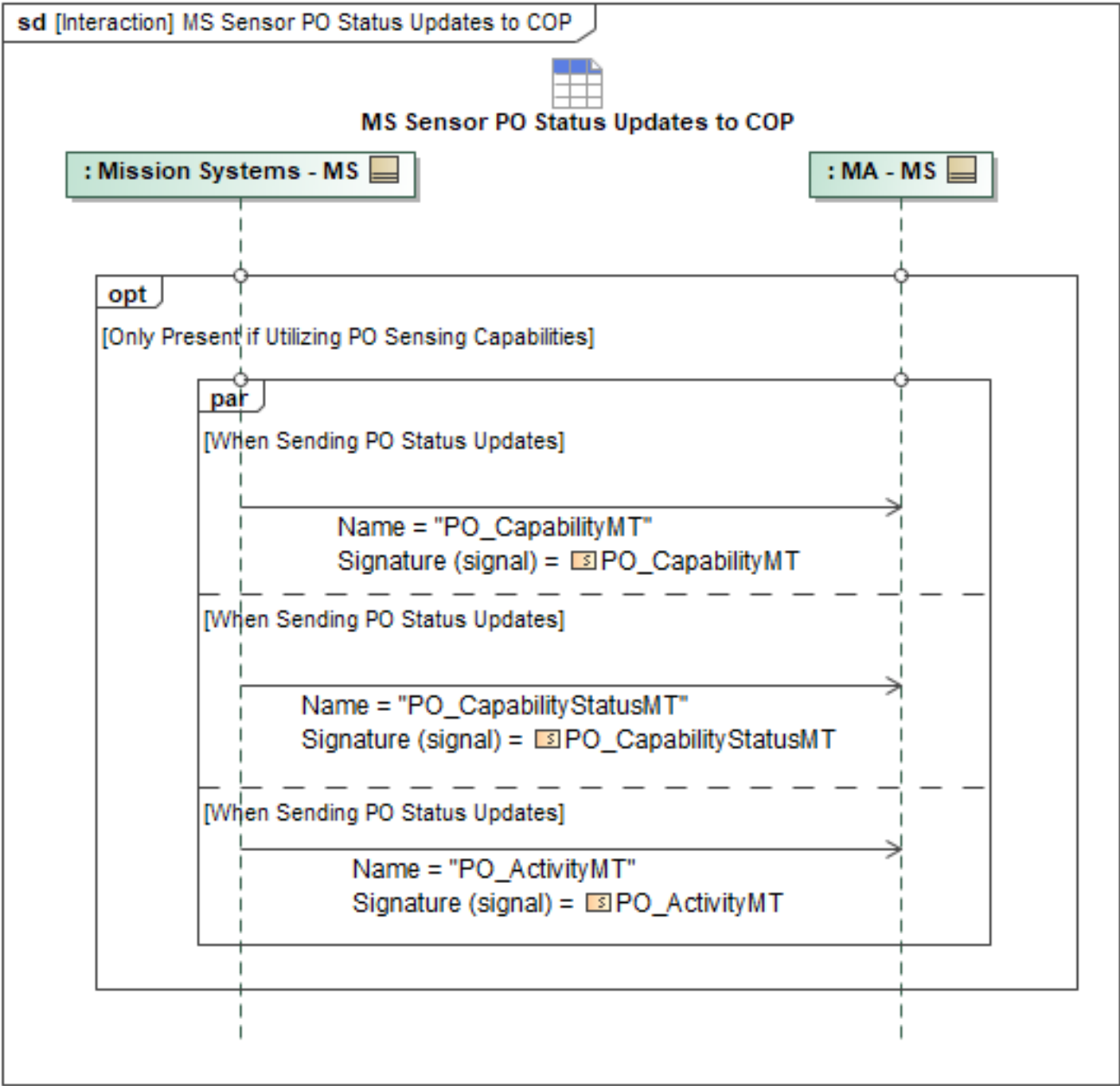


Figure 1-23: MS Sensor PO Status Updates to COP

Table 1-22: MS Sensor PO Status Updates to COP

Message Name (Name:Type)	Required Fields (Name:Type)
PO_ActivityMT: PO_ActivityMT	None
PO_CapabilityMT: PO_CapabilityMT	None
PO_CapabilityStatusMT: PO_CapabilityStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.3.4 MS Weapon Status Updates to COP

Mission Use Case: Store Management

The MS Weapon Status Updates to COP sequence diagram specifies the interaction of Mission Systems (MS) communicating the status of the weapons subsystem to Mission Autonomy (MA) for common operating picture (COP) reporting purposes.

MS publishes *StoreManagementStatusMT* messages to report the status of individual weapons and allow MA to know weapon quantities and individual statuses. This is complementary to general usage status provided by *MA_StrikeCapabilityMT* and *StrikeCapabilityStatusMT*. The *SupportCapabilityID* field specifies a unique identifier used by other messaging (not in scope) to manipulate store settings. Each *StoreStation* has a *StationIndex* to identify station using MS implementation specific schemes. The *CompatibleStore* field specifies the type of weapons that station can hold. The *InstalledStore* field is optional but required when the station currently holds a weapon. When specified, the *InstalledStore* field identifies the weapon type. If the *InstalledStore* field and sub-fields are omitted, it is interpreted that this station does not hold a weapon. Each station is associated with a *Capability* defined by *MA_StrikeCapabilityMT* by specifying the *CapabilityID* field, which is required.

MS publishes *MA_StrikeCapabilityMT* messages with *Capability* required. Each *Capability* has its own unique identifier set in the *CapabilityID*. The *AcceptedInterface* field must include at least *CAPABILITY_COMMAND*. The *StoreType* and *StoreCategory* fields are used to identify the type of weapons the *Capability* is used to status and command. MS implementation determines how the *InterruptOtherActivities* field is set. Other schema-optional fields are not required but may be specified for more information.

MS publishes *StrikeCapabilityStatusMT* messages to report status of the weapon capabilities identified by the *Capability* fields in the *MA_StrikeCapabilityMT* message. Each *CapabilityStatus* has the *CapabilityID* set to match the *CapabilityID* field of the corresponding *Capability*. The *Availability* field is used to report general usability of the weapons *Capability* by MA. The *StoreStatus* field is required with *StoreState* and *StoreStatusDetail* communicating the specific state information of the weapon(s) associated with the *Capability*. Other schema-optional fields are not required but may be specified for more information.

MS publishes *MA_StrikeActivityMT* to communicate more information about the current employment of weapons to MA. Each *Activity* specifies a unique identifier in the *ActivityID* field, corresponding to a scheduled weapon employment. The *CapabilityID* field associates the employment with a specific *Capability* defined in the *MA_StrikeCapabilityMT* message. *Interactive* is set according to MS implementation. The *ActivityState* field is used to specify the state of the weapon employment activity. The *SelectedForKeyLoad*, *WeaponArmed*, *ReadyForRelease*, *SelectedForRelease*, and *SelectedForJettison* fields provide information about the current state of the weapon while it prepares for the weapon employment activity. The *AssignedTarget* field is required when the weapon is scheduled to be launched at a target and not jettisoned. The *TimeToRelease* field is required and specifies the timing of the weapon release. Optionally, MS can provide MA with the *MunitionActive* field that pertains to the specific weapon state data and orientation during weapon flyout. Active weapon states include *TARGETED*, *REQUESTED*, *SEEKER_INACTIVE*, *SEEKER_ACTIVE*, *FUZE_INACTIVE*, *FUZE_ACTIVE*, *NOT_FULLY_AUTONOMOUS*, *FULLY_AUTONOMOUS*,

DEFEATED, TIMED_OUT, SUCCESS.

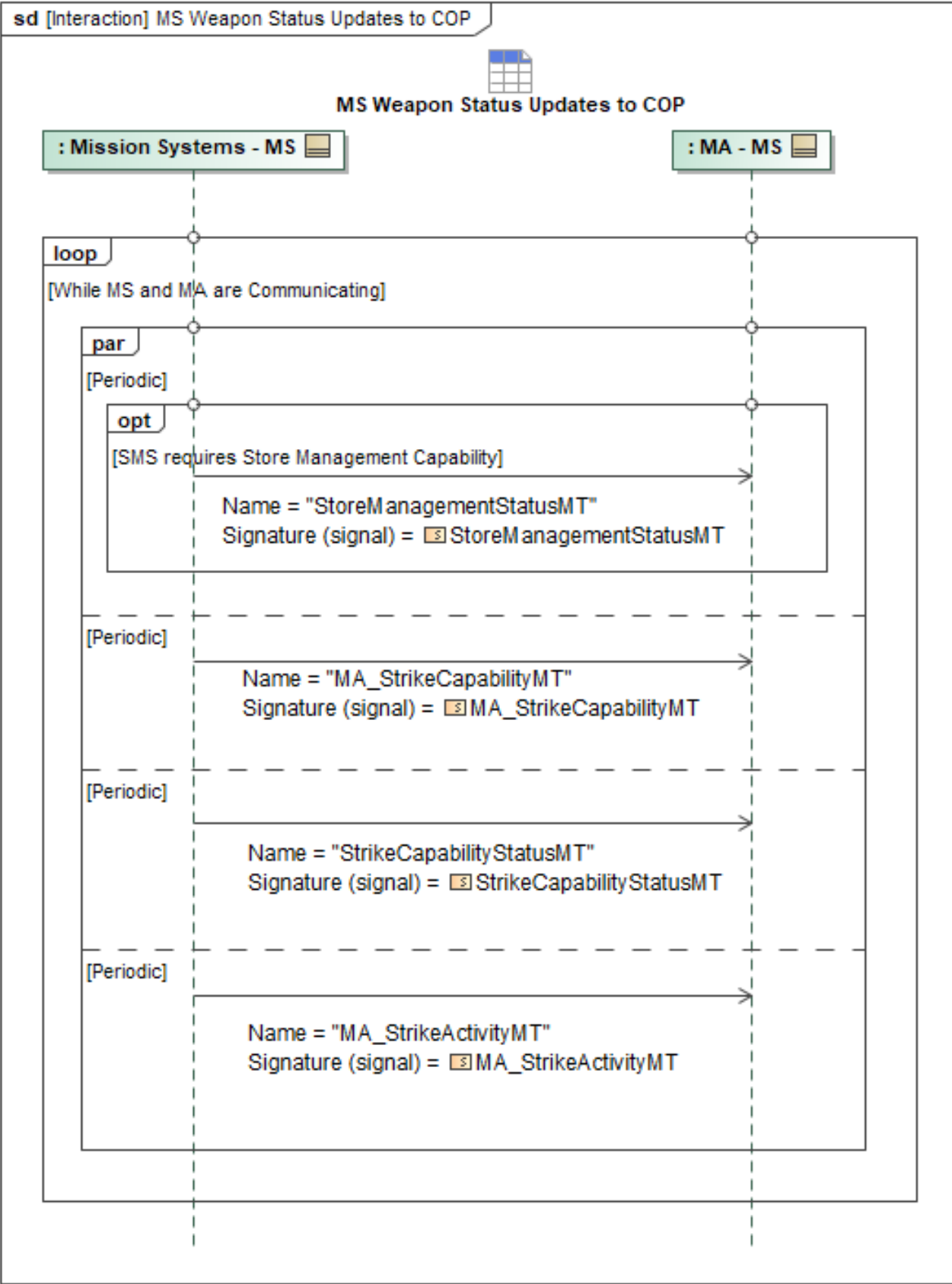


Figure 1-24: MS Weapon Status Updates to COP

Table 1-23: MS Weapon Status Updates to COP

Message Name (Name:Type)	Required Fields (Name:Type)
MA_StrikeActivityMT: MA_StrikeActivityMT	TimeToRelease: DurationType
MA_StrikeCapabilityMT: MA_StrikeCapabilityMT	None
StoreManagementStatusMT: StoreManagementStatusMT	CapabilityID: CapabilityID_Type InstalledStore: StoreType
StrikeCapabilityStatusMT: StrikeCapabilityStatusMT	StoreStatus: StoreStatusType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4 Data Fusion Package

1.2.4.1 Distribute Single Sensor Track Data to Perform Local Fusion

This interaction describes how MS provides tracks from onboard sensors to MA to enable local fusion. This is accomplished via the *EntityMT* message, which contains the kinematic data of the track. Additionally, the *ObservationMeasurementReportMT* message contains the sensor data that is used to generate the track or entity. Track data sources include both onboard cooperative sensors (e.g., Mode 5 IFF, ADS-B) and onboard non-cooperative sensors. It is important to note that only a subset of the total Observation Measurement Reports (OMRs) recorded are associated with a given track, and can therefore be referred to as Associated Measurement Reports (AMRs). These associated OMRs are identified in the *EntityMT* by the *MeasurementID* field. Once fusion has taken place, the resultant fused tracks can then be utilized by and integrated into the Common Operating Picture (COP). Alternately, if a platform is incapable of performing local fusion, the pre-fused tracks may still be used by COP.

Additionally, a kinematic and Combat ID (CID) fusion implementation will likely want the capability that OMRs were generated from, and which system/subsystem had the capability generating the OMR.

Within the *ObservationMeasurementReportMT* message, there are two optional paths to different sets of *SubsystemIDs* and *SystemIDs*:

1) *ObservationMeasurementReportMDT* → *ObservationMeasurement* → *MeasurementSource* → *ElementDetails* → *SourceElementIdentifier* → *SystemID* + *SubsystemID*

- These messages provide the system/subsystem that's the source of the OMR's kinematics.

2) *ObservationMeasurementReportMDT* → *ObservationMeasurement* → *MeasurementSource* → *SourceIdentity* → *CapabilityID* + *SystemID* + *SubsystemID*

- These messages provide the system/subsystem and capability for the source of identity in the OMR (it's expected this will usually be the same system/subsystem as the kinematics). A client would want to know the capability type of the provided entity ID, and may have to subscribe to additional messages to understand the type (e.g. [Capability]StatusMT messages).

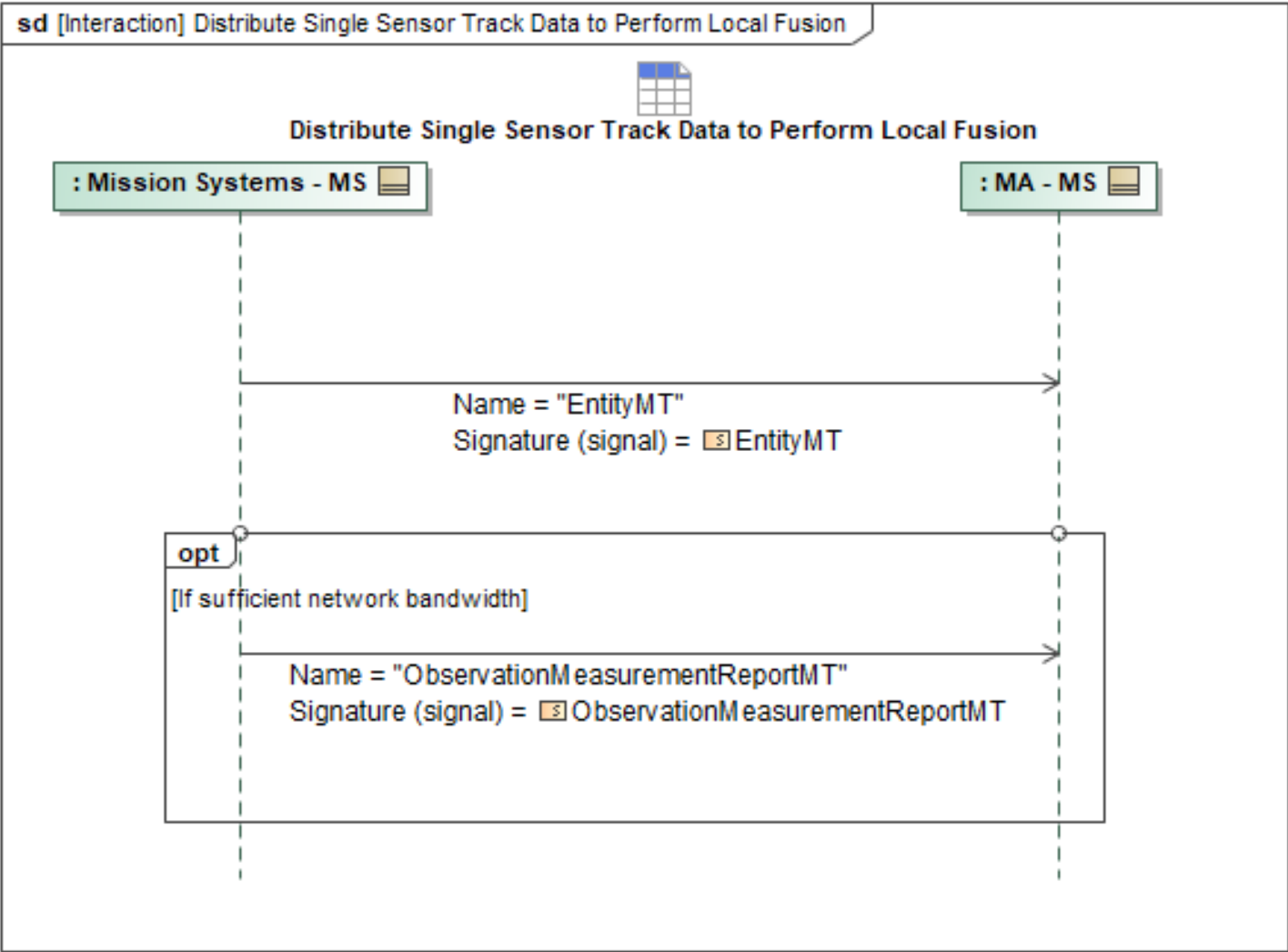


Figure 1-25: Distribute Single Sensor Track Data to Perform Local Fusion

Table 1-24: Distribute Single Sensor Track Data to Perform Local Fusion

Message Name (Name:Type)	Required Fields (Name:Type)
EntityMT: EntityMT	None
ObservationMeasurementReportMT: ObservationMeasurementReportMT	MeasurementSource: ObservationMeasurementSourceType ElementDetails: ElementDetailsType SourceElementIdentifier: SourceKinematicsElementIdentityType SystemID: SystemID_Type SubsystemID: SubsystemID_Type SourceIdentity: MeasurementSourceIdentityType CapabilityID: CapabilityID_Type SystemID: SystemID_Type SubsystemID: SubsystemID_Type MeasurementKinematics: MeasurementKinematicsChoiceType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.2 Receive Processed Track

The Receive Processed Track sequence diagram describes how MA receives track information from an onboard data fusion service. The basic functionality of a data fusion service is to process and merge target reports from various internal and external sources to provide a resulting track, called *Entity*. Fields that assist MA on decision making include *Source*, *EntityStatus*, *Identity*, *Kinematics*, and *EstimatedKinematics*. This information, processed from 2D unfused tracks or 3D tracks (cues) via *EntityMT*, not only provides MA a picture of the battlespace, but also includes information about an entity's kinematic characteristics to support a variety of mission objectives, such as following an Entity. This sequence shows a loop of *EntityMT* messages as long as there is more than one *Entity* to convey to MA. This message is also triggered when new tracks are determined, or there are changes to already established tracks.

Losing a track is a special optional case that happens when a sensor can no longer detect and track an entity; when this occurs, the *EntityLostMT* message will notify Mission Autonomy of this event.

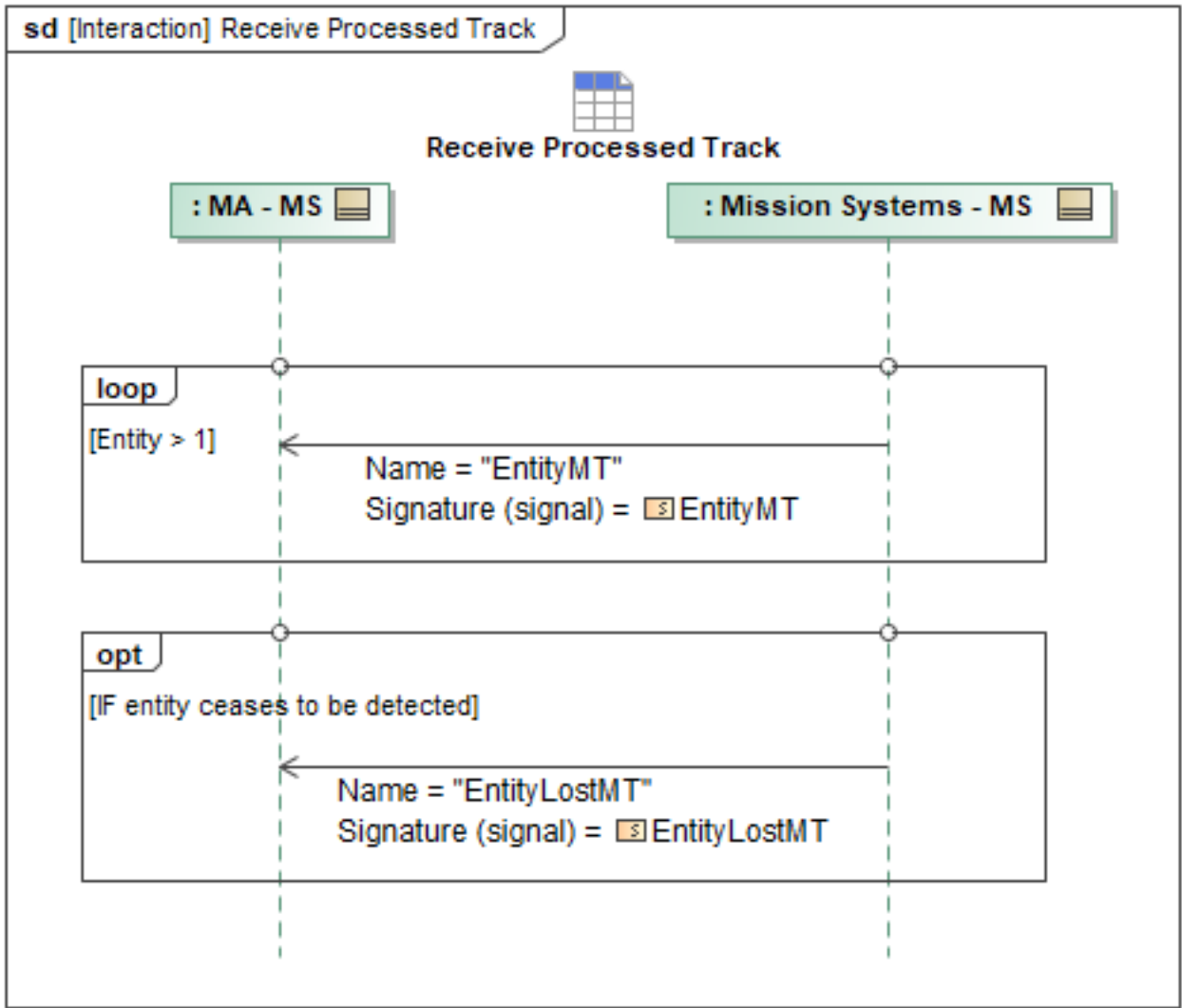


Figure 1-26: Receive Processed Track

Table 1-25: Receive Processed Track

Message Name (Name:Type)	Required Fields (Name:Type)
EntityLostMT: EntityLostMT	None
EntityMT: EntityMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.5 Decentralized Messaging Service Package

The Decentralized Messaging Service (DMS) Interface is a communications Gateway Capability implementation that provides an interface for MA to ingress/egress UCI messages offbaord. This interface facilitates the passing of UCI messages to/from C2 and between ACP's over an IP network using the Data Distribution Service (DDS) standard. In the case when a non-IP or link-specific encoding must be used, such as Link 16, that encoding will be specified separately from the general offbaord message format. The core Minimum Message Set (MMS) accounted for in the DMS consists

of the messages required to transmit data payloads off-board (*MA_TxDataPayloadCommandMT*, *MA_TxDataPayloadCommandStatusMT*), a message to pass a received message from MS to MA (*MA_RxDataPayload*), a set of messages for MA to indicate desired subscriptions to MS (*MA_MessagingSubscriptionSettingCommandMT*, *MA_MessagingSubscriptionSettingCommandStatusMT*) and a message for MS to send existing subscription information to MA (*MA_MessagingSubscriptionStatusMT*). MA can populate a *MA_TxDataPayloadCommandMT* with a UCI message payload, and optionally a list of network endpoints by *SystemID* to send to MS. In order to send the message off-board, MS will use the DMS to send the message OTA to another system's DMS. The receiving system's DMS is responsible for receiving the message, formatting it as a *MA_RxDataPayloadMT* message, and forwarding the message to the appropriate interface for decoding and processing. The *MA_TxDataPayloadCommandMT* and *MA_RxDataPayloadMT* UCI wrapper messages require an *EncodedPayload* field which will have a datatype while crossing the L1 boundary between MA and MS is UTF-8 XML represented as hexBinary. The OTA wrapper message sent between the DMSs, *MA_DataPayloadWrapper* IDL, also requires an *EncodedPayload* field. The *MA_DataPayloadWrapper's* *EncodedPayload* field datatype while crossing between DMSs will be in EXI representation. The "Serialization and Data Encoding" section describes the EXI representation in detail.

With the "Additional Comms Capability" MUC, a Gateway can be configured to convert any message on the ASB to be transmitted offboard by either subscribing to it directly or by using a UCI wrapper (*MA_TxDataPayloadCommandMT* with the corresponding status message *MA_TxDataPayloadCommandStatusMT*). A Gateway can be configured, by the *MA_DataDisseminationPlanMT* or by a direct *MA_GatewayCommandMT*, to receive any message from the offboard network and either publish it out directly or use a UCI wrapper (*MA_RxDataPayloadMT*). In either case the UCI message header will contain the source information from the sender of the message for authentication and other processing. The configuration of ingress/receive is typically done through the *MA_DataDisseminationPlanMT* with associated activated *Effectivities*, *IERs*, and *DistributionGroups*, but can also be done by using the *MA_MessagingSubscriptionSettingCommandMT* and associated status messages *MA_MessagingSubscriptionSettingCommandStatusMT* and *MA_MessagingSubscriptionStatusMT*. These and the *MA_GatewayCommandMT* provides a way to directly configure the DMS, but the *MA_DataDisseminationPlanMT* allows MS to manage and update all the Gateways as needed based on the plan. The *MA_DataDisseminationPlanMT* has an optional *IER_DesiredService.CompressionType* field that can be set to "EXI" to defines the encoding/compression format for the *EncodedPayload* field in the OTA *MA_DataPayloadWrapper* IDL that is used for communication between DMSs.

In order to successfully send data between instances of DMS via the OTA network, the Data Distribution Service (DDS) standard shall be used as the middleware. DDS provides a scalable, real-time, and decentralized publish-subscribe communication model well-suited to dynamic and distributed environments. It allows systems to exchange structured data via shared topics in a flexible deployment, resilient to node failure, and provides easier integration.

All implementations of the DMS must conform to the DDS specification (see <https://www.omg.org/spec/DDS/1.4/>) and be compatible with the Real-Time Publish-Subscribe(RTPS) wire protocol, minimum version 2.3. While a specific DDS implementation is not mandated, all system components must comply with the interface and behavioral expectations described herein to ensure interoperability. This section covers discovery mechanisms, topic definitions, quality of service

requirements, serialization and data encoding, and security features.

To enable interoperability, it is necessary to standardize several aspects of the DDS configuration. First, two types of DDS topics will be used:

1. Pub/sub topics, by default, are named according to the UCI message type that will be sent via those topics. Examples include "SystemStatus" or "ActionCommand". Subscriptions to these topics can be managed via the "Subscribe to Message Topic" interaction. The "Additional Comms Capability" MUC uses the "Update Data Dissemination Plan" interaction to manage subscripsts and publications.
2. Specific destination topics shall be used to send messages according to *MA_TxDataPayloadCommandMT.SpecificDestinationID* fields. The DMS shall subscribe to a DDS topic that is equal to the UUID of its own SystemID. Since DDS uses pub/sub rather than point-to-point semantics, these topics will be used to send OTA messages to systems listed in *SpecificDestination* fields of the *MA_TxDataPayloadCommand*. These topics will use RELIABLE QoS settings. Topic names that correspond to SystemIDs will be prefixed with "id_", use all uppercase letters, and use "_" instead of "-" for separators to comply with DDS Topic naming rules. A SystemID of "3b39Bb83ee-d994-5f64-ae73-e62df8ec3c72" becomes a topic name of "id_B39B83EE_D994_5F64_AE73_E62DF8EC3C72". When using the "Additional Comms Capability MUC, UCI wrapped messaging and Direct Messaging can be enabled through a configurations option.

DMS implementations shall perform content filtering to ensure that duplicate messages are not publish to MA when a platform receives the same message via a topic subscription and via UCI wrapped *SpecificDestination* routing. This will be accomplished by requiring DDS Readers for normal topics (those other than the *SpecificDestination* topics described in Item 2 above) to create a content filter which excludes any message that is also sent to that system via *SpecificDestination*. Since the message will be delivered via *SpecificDestination* topic it will use RELIABLE QoS rather than whatever QoS are used by the normal topic. This may be implemented via DDS content filters which can accomplish content filtering at the DataWriter to save bandwidth or implementations may implement this feature via other means within the DMS.

To facilitate use of the *MA_TxDataPayloadCommandStatusMT.TransmissionStatus* field, every DDS DomainParticipant, DataWriter, and DataReader shall set its UserQoS data to be a vector of bytes equal to its host system's SystemID.UUID.

Disclaimer: If the "Additional Comms Capability" MUC is not implemented, the DMS can function as a stand alone service without the need of the Data Dissemination Plan sequences. Any references to the *MA_DataDisseminationPlanMT* in this section can be ignored. This means the Minimum Message Set (MMS) accounted for in the DMS consists of the messages required to transmit data payloads off-board (*MA_TxDataPayloadCommandMT*, *MA_TxDataPayloadCommandStatusMT*), a message to pass a received message from MS to MA (*MA_RxDataPayloadMT*), a set of messages for MA to indicate desired subscriptions to MS (*MA_MessagingSubscriptionSettingCommandMT*, *MA_MessagingSubscriptionSettingCommandStatusMT*) and a message for MS to send existing subscription information to MA (*MA_MessagingSubscriptionStatusMT*).

1.2.5.1 Receive Message Subscription Status

If the DMS was subscribing to topics using the MA configured by the "Subscribe to Message Topic" sequence or by a configuration file, MS will send MA information about the existing manually configured subscriptions using the *MA_MessagingSubscriptionStatusMT* message. Subscriptions

may be loaded at startup using a DMS config file, therefore this interaction enables MA to receive information about subscriptions that were not directly requested by MA. This interaction also enables MA to manage any topics which were manually configured by the "Subscribe to Message Topic" sequence. This message may be sent periodically by the DMS or upon any change to existing subscriptions. When using the "Additional Comms Capability" MUC, subscription information on other topics implemented from the active *Effectivities* from a *MA_DataDisseminationPlanMT* will be provided to MA through the *MA_GatewayActivityMT* status message.

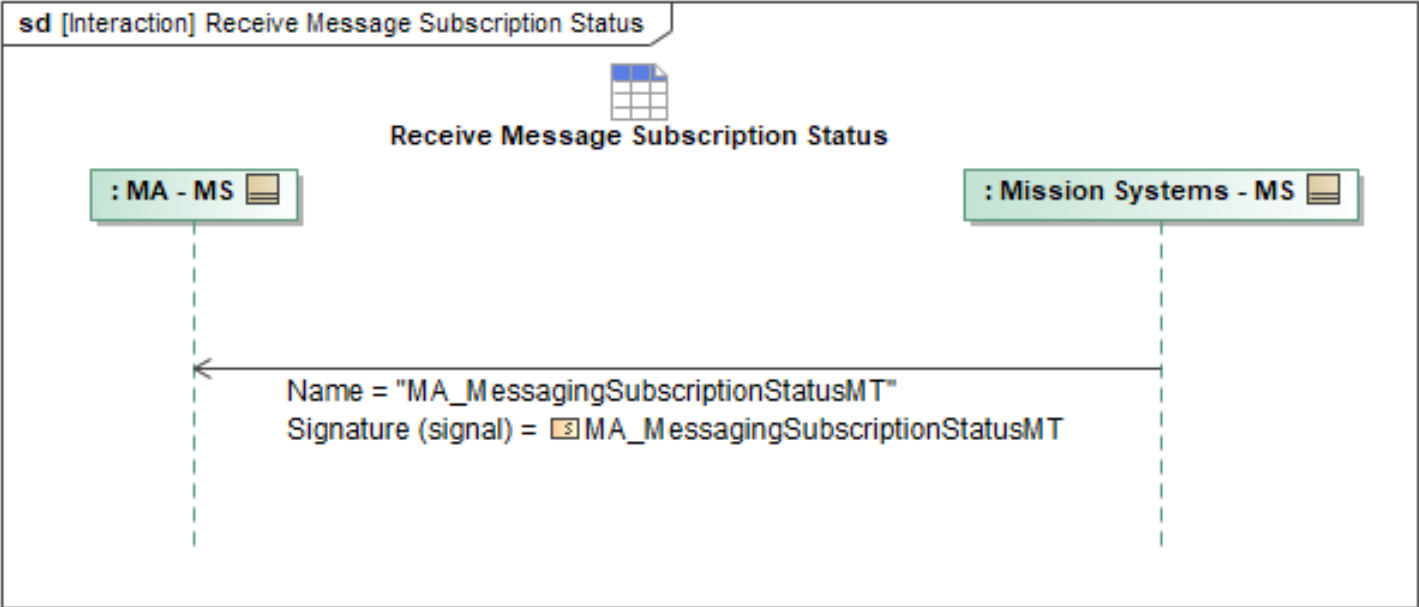


Figure 1-27: Receive Message Subscription Status

Table 1-26: Receive Message Subscription Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MessagingSubscriptionStatusMT: MA_MessagingSubscriptionStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.5.2 Receive UCI Wrapped Message

When MS receives an offboard message from C2 or a peer ACP it can send the message to MA using the UCI wrapper *MA_RxDataPayloadMT*. When using the "Additional Comms Capability" MUC, the active *Effectivity* specified by the activated *MA_DataDisseminationPlanMT* must be configured for *WrapIngressMessages*="TRUE" for the DMS to UCI wrap the receiving messages in a *MA_RxDataPayloadMT* before sending to MA. The *MA_RxDataPayloadMT* contains a *hexBinary* field called *EncodedPayload* that contains a UTF-8 XML UCI message represented as hexBinary. This message also has message fields *RxDataPayloadID*, *DataPayloadOriginator*, *Priority*, *MessageType*, *DestinationRouting*, *SpecificDestinationID*, *Timestamp* and *MessageExpiry*. *RxDataPayloadID* is a unique ID for this received data payload message. The *DataPayloadOriginator* field in the *MA_RxDataPayloadMT* specifies the ID of the ACP that originally sent the message. The *Priority* field indicates the priority of a received message using a number as follows: 0-10 indicates high priority,

10-20 indicates medium priority, and 20-30 indicates low priority. *MessageType* specifies the UCI type of the message in the *EncodedPayload* field. *DestinationRouting* documents whether the message was routed only to specific destinations or also to the topic corresponding to the *MessageType*. *SpecificDestinationID* is optional, but one or more *SpecificDestinationIDs* will be included if the message was send to those systems directly. The *Timestamp* field indicates the time at which the message was received and the *MessageExpiry* field indicates when the payload message expires. The fields contained within the *MA_RxDataPayloadMT* enable implementers to receive UCI messages for facilitating communication with other ACPs.

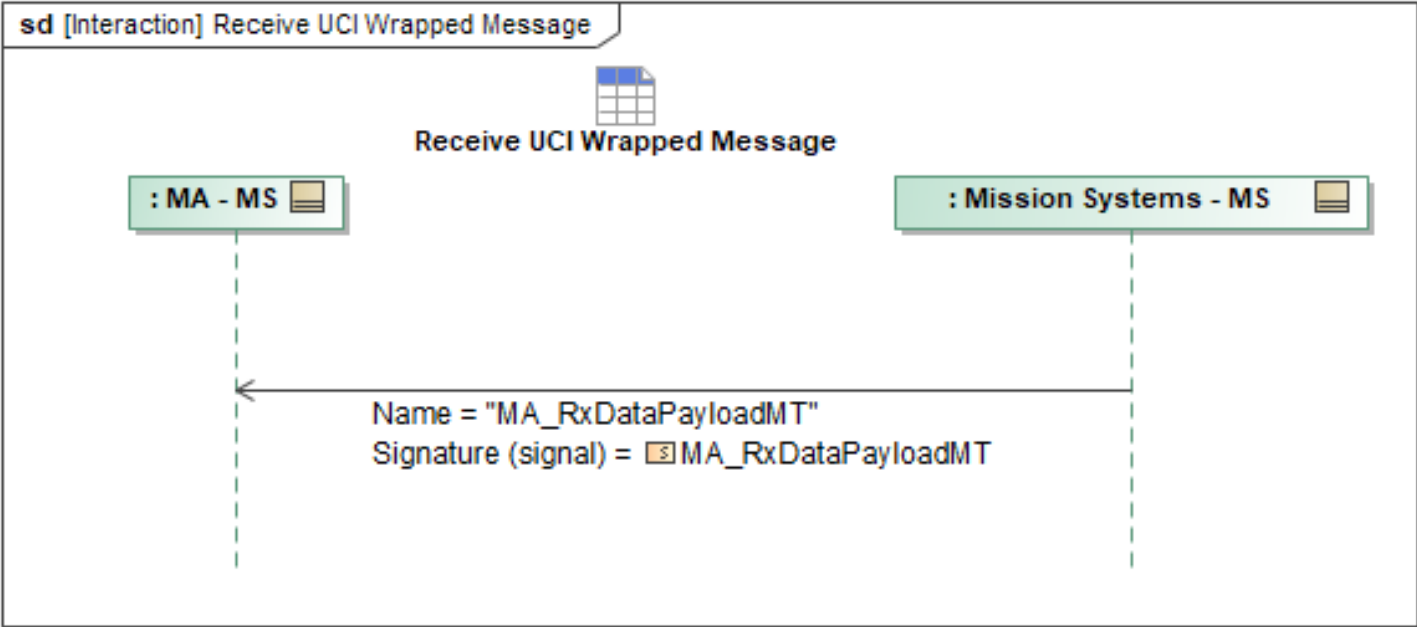


Figure 1-28: Receive UCI Wrapped Message

Table 1-27: Receive UCI Wrapped Message

Message Name (Name:Type)	Required Fields (Name:Type)
MA_RxDataPayloadMT: MA_RxDataPayloadMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.5.3 Send UCI Wrapped Message

When the MA needs to send or publish messages to an external system, such as a peer or group of peer ACPs, it can send the UCI message to MS using a *MA_TxDataPayloadCommandMT* message. When Using the "Additional Comms Capability" MUC, the *IER.AllowWrappedEgressMessages* needs to be set to "TRUE" in the active *Effectivities* in the related *MA_DataDisseminationPlanMT* to enable UCI wrapped egress messages. The *MA_DataDisseminationPlanMT* can also be configured to subscribe to UCI messages on the local ASB directly for sending message offboard. The *MA_TxDataPayloadCommandMT* contains a hexBinary field called *EncodedPayload* that contains a UTF-8 XML UCI message represented as hexBinary. The *MA_TxDataPayloadCommandMT* message has fields *MessageType*, *Priority*, *Timestamp*, *MessageExpiry*, *SpecificDestinationID*, and *DestinationRouting* fields. The *MessageType* indicates the UCI type of the message in the

EncodedPayload field. The *Priority* field indicates the priority of a message being sent using a number as follows: 0-10 indicates high priority, 10-20 indicates medium priority, and 20-30 indicates low priority. The *Timestamp* field indicates the time at which the message was created and the *MessageExpiry* field indicates when the payload message expires. The *SpecificDestinationID* field is either a list of the ACP endpoints the data is to be sent to or is an empty list. When using the "Additional Comms Capability" MUC, both sender and receiver must have the *DistributaionGroups.DirectedMessagingAllowed* field set to "TRUE" in the related *MA_DataDisseminationPlanMTs* for Specific Destination messaging to be sent/received. The *DestinationRouting* field specifies whether the message should be sent only to *SpecificDestinationID* destinations or to both the appropriate topic (based on *MessageType*) as well as the *SpecificDestinationIDs*. If *DestinationRouting*=SPECIFIC_DESTINATION_ONLY then at least one of *SpecificDestinationID* must be provided. Otherwise, a failure status message will be sent to MA. The fields contained within the *MA_TxDataPayloadCommandMT* enable implementers to send UCI messages for facilitating communication with other ACPs. One or more accompanying *MA_TxDataPayloadCommandStatusMT* messages are returned from MS to MA to indicate the status of the process.

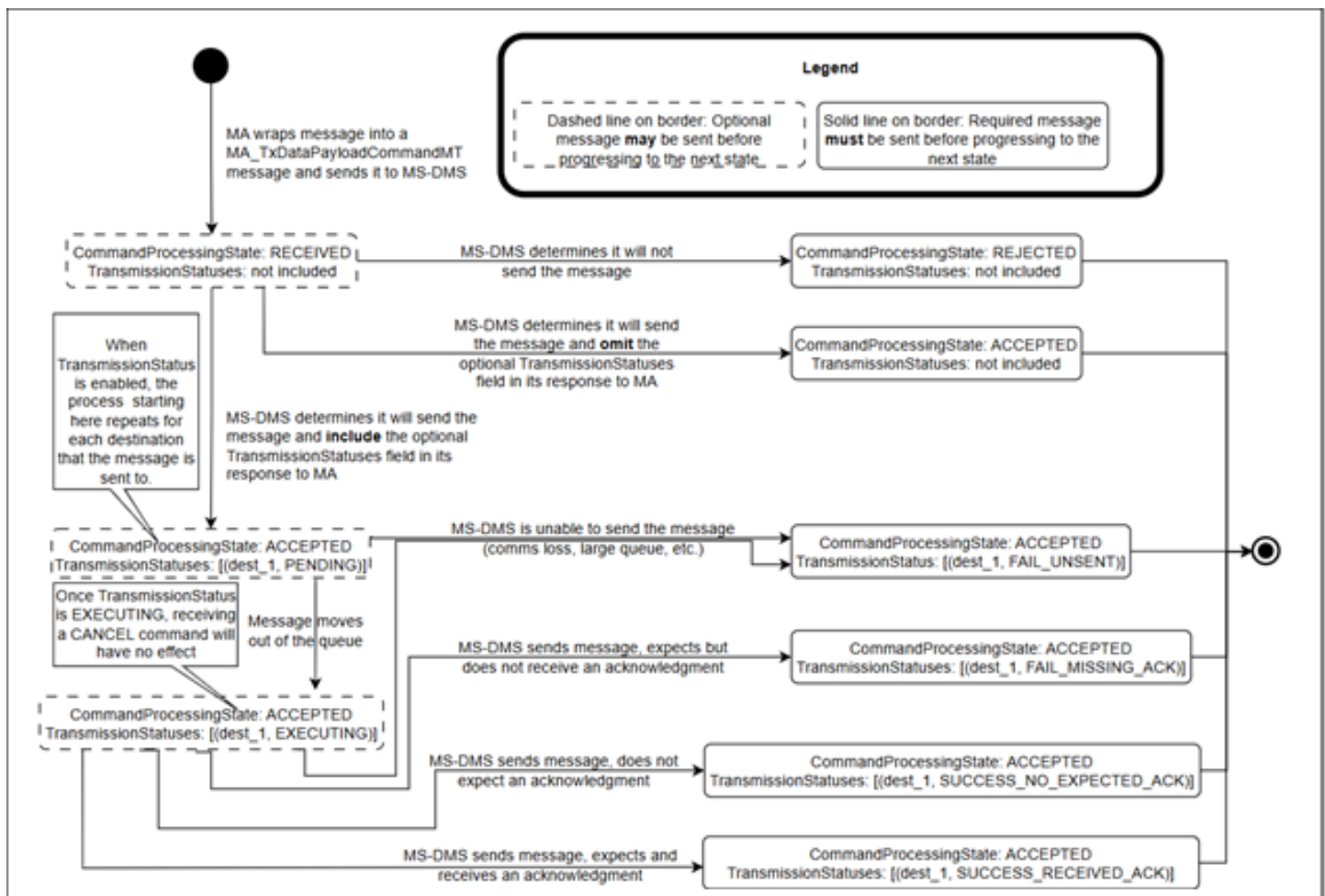


Figure 1-29: State Machine for Send UCI Wrapped Message UCI Diagram

State machine diagram for the "Send UCI Wrapped Message" interaction with transmission to one destination with *SpecificDestinationID*=dest_1.

The steps of sending a message to a single destination are detailed in the state machine diagram shown above, which is an alternative representation of the interaction below. First, MA packs a UTF-8 XML UCI message into the payload of *MA_TxDataPayloadCommandMT*. MA sends this message to MS-DMS. Next, MS-DMS may send an optional confirmation message, *MA_TxDataPayloadCommandStatusMT_1* (with *CommandProcessingState*=RECEIVED), to confirm that it received the message and will attempt to send it offboard. If the message is determined to be invalid (e.g. *DestinationRouting*= SPECIFIC_DESTINATION_ONLY and no *SpecificDestinationID* fields are specified), MS-DMS sends a failure message back to MA with *CommandProcessingState*=REJECTED. This message, represented by *MA_TxDataPayloadCommandStatusMT_2* in the interaction, indicates that the message will not be processed. This case terminates the interaction and no further processing occurs. Otherwise, the message is valid, and MS-DMS will attempt to send it offboard to all specified destinations, as shown in the table below:

Logic for determining set of all destinations to send message to:

Table 1-28: Message Send Destination Logic

		Number of <i>SpecificDestinationID</i> fields	
		None	One or more
<i>DestinationRouting</i>	SPECIFIC_DESTINATION_ONLY	Error: A failure status message with <i>CommandProcessingState</i> =REJECTED will be sent to MA.	The message will be sent to all <i>SpecificDestinationID</i> systems.
	TOPIC_AND_SPECIFIC_DESTINATION	The message will be published to the relevant topic.	The message will be published to the relevant topic and sent to all <i>SpecificDestinationID</i> systems.

Note that after a message is determined to be valid, all subsequent messages in the interaction have a value of ACCEPTED for the *CommandProcessingState* field.

MS-DMS sends one or more status messages back to MA to indicate the result of sending the message offboard. The decision of which optional messages to send may be influenced by configuration and internal logic, the details of which are considered an implementation detail that is out of the scope of this interaction. If the optional (but recommended) *DestinationTransmissionStatus* field is omitted, MS-DMS simply sends *MA_TxDataPayloadCommandStatusMT_3* to MA, indicating that it processed the message, and the interaction terminates. In this case, it cannot provide any additional information such as whether transmission was successful. Without this information, MA cannot perform potentially useful behaviors like application-level resend and smart deletion of sent messages from MD logs when storage is low. Otherwise, MS-DMS provides more detailed information regarding the process of sending the message offboard to each destination. To facilitate use of the *MA_TxDataPayloadCommandStatusMT.TransmissionStatus* field, every DDS DomainParticipant, DataWriter, and DataReader shall set its UserQoS data to be a vector of bytes equal to its host system's *SystemID.UUID*. MS-DMS obtains the set of all destinations specified by *MA_TxDataPayloadCommandMT*. This is either read directly in the *SpecificDestinationID* field or can be determined by retrieving the UserData for all the subscribers to the topic corresponding to the *MessageType* (if publishing to a topic). Then, MS-DMS attempts to send the message to each destination. Transmission to multiple destinations may happen in parallel. For each destination, one or more *MA_TxDataPayloadCommandStatusMT_4* messages are sent from MS-DMS to MA to provide information on the status or result of transmission. The *DestinationTransmissionStatus* field in each of these messages has one or more elements. Each element consists of a

MA_TransmissionStatusType, which contains the *SpecificDestinationID* indicating where the message is transmitted to and the associated *TransmissionStatus* containing the status. For transmission to a single destination “dest_1”, the process is as follows:

- First, MS-DMS may optionally send the message with *DestinationTransmissionStatus* =[(dest_1, PENDING)] to MA. This confirms that the message has been added into the queue to be sent offboard.
- Next, if a problem occurs that results in the message being removed from the queue, such as a loss of comms, MS-DMS send it with *DestinationTransmissionStatus* =[(dest_1, FAIL_UNSENT to MA)].
- Otherwise, MS-DMS may optionally send the message with *DestinationTransmissionStatus* =[(dest_1, EXECUTING)] to MA. This indicates that MS-DMS is actively attempting to send the message offboard and the sending can no longer be canceled or updated. While canceling and updating are not explicitly part of this interaction, if MS-DMS supports canceling or updating (via a *MA_TxDataPayloadCommandMT* with *CommandState*=CANCEL or UPDATE), it must happen before this point.
- Finally, MS-DMS sends the message to MA to indicate the final status of transmission. This message must have one of the following values for the entry in the *DestinationTransmissionStatus* field for the associated destination:
 - **FAIL_UNSENT:** Indicates that some problem (such as a loss of comms) prevented the message from being sent.
 - **FAIL_MISSING_ACK:** Indicates that the message was sent via a protocol that supports confirmation that the message was received (e.g., TCP), but the protocol was unable to confirm that the message was received.
 - **SUCCESS_NO_ACK_EXPECTED:** Indicates that the message was sent using a protocol that does not support confirmation that the message was received (e.g., UDP.)
 - **SUCCESS_RECEIVED_ACK:** Indicates that the message was sent via a protocol that supports confirmation that the message was received (e.g., TCP), and the protocol was able to confirm that the message was received.

This process repeats until a final transmission status has been sent for every destination that MS-DMS attempts to send the UCI message to. Note that DDS does not provide a standard way to access message receipt acknowledgments so SUCCESS_NO_ACK_EXPECTED may be the most common status.

Examples

To clarify the details of the interaction described above, the following examples show two different cases for this interaction.

Example 1. Simplest possible success case for the "Send UCI Wrapped Message" interaction.

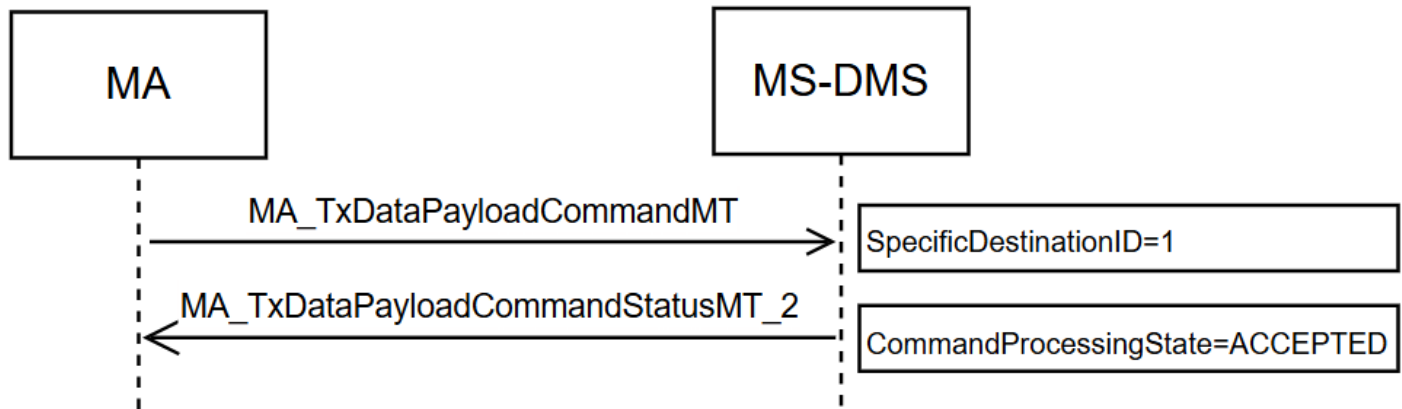


Figure 1-30: Example 1

Example 1 above shows the simplest success case for the interaction where all optional messages are omitted:

- MA sends *MA_TxDataPayloadCommandMT* to MS-DMS
- MS-DMS replies with *MA_TxDataPayloadCommandStatusMT_2* message indicating that the command was accepted.

Example 2. General case for the "Send UCI Wrapped Message" interaction.

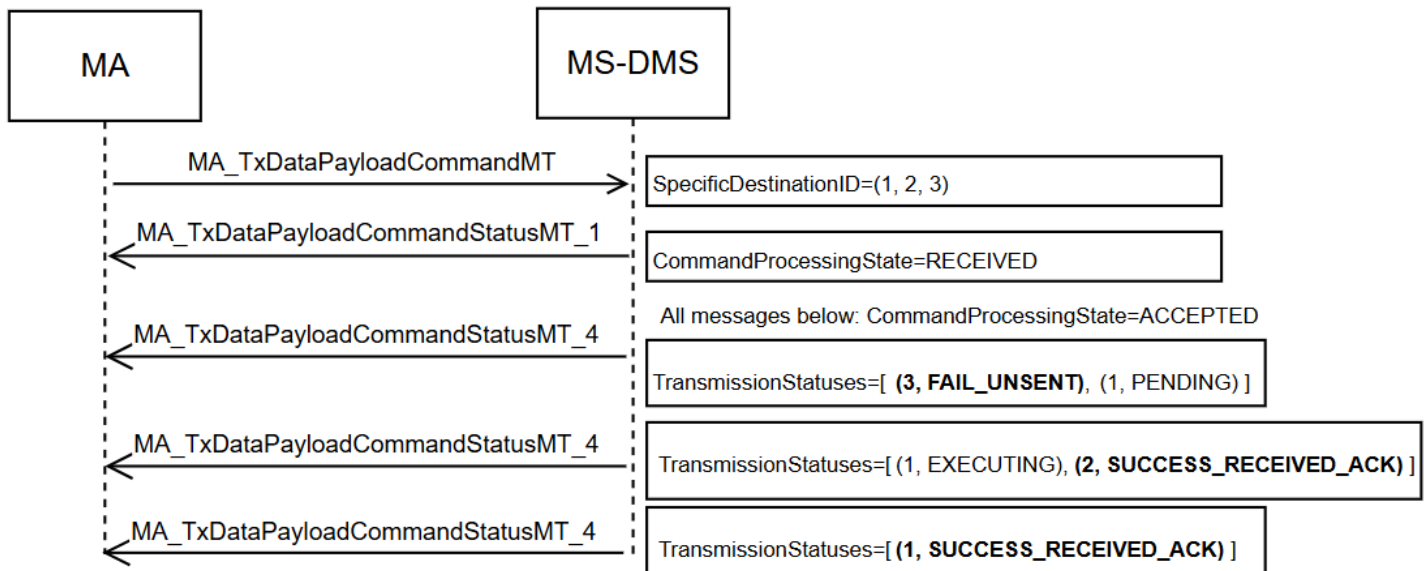


Figure 1-31: Example 2

Example 2 shows an example of the interaction when some optional messages are included:

- MA sends *MA_TxDataPayloadCommandMT* to MS-DMS indicating that the message should be sent to 3 destinations.
- MS-DMS replies with *MA_TxDataPayloadCommandStatusMT_1* message indicating that the command was received.
- MS-DMS sends back a series of *MA_TxDataPayloadCommandStatusMT_4* messages.

- Each message contains *TransmissionStatus* entries for one or more destinations. The destinations are not ordered and different destinations in the same message may have different values for *TransmissionStatus*.
- The interaction terminates when the final transmission status (shown in bold for each destination) is sent for all destinations.

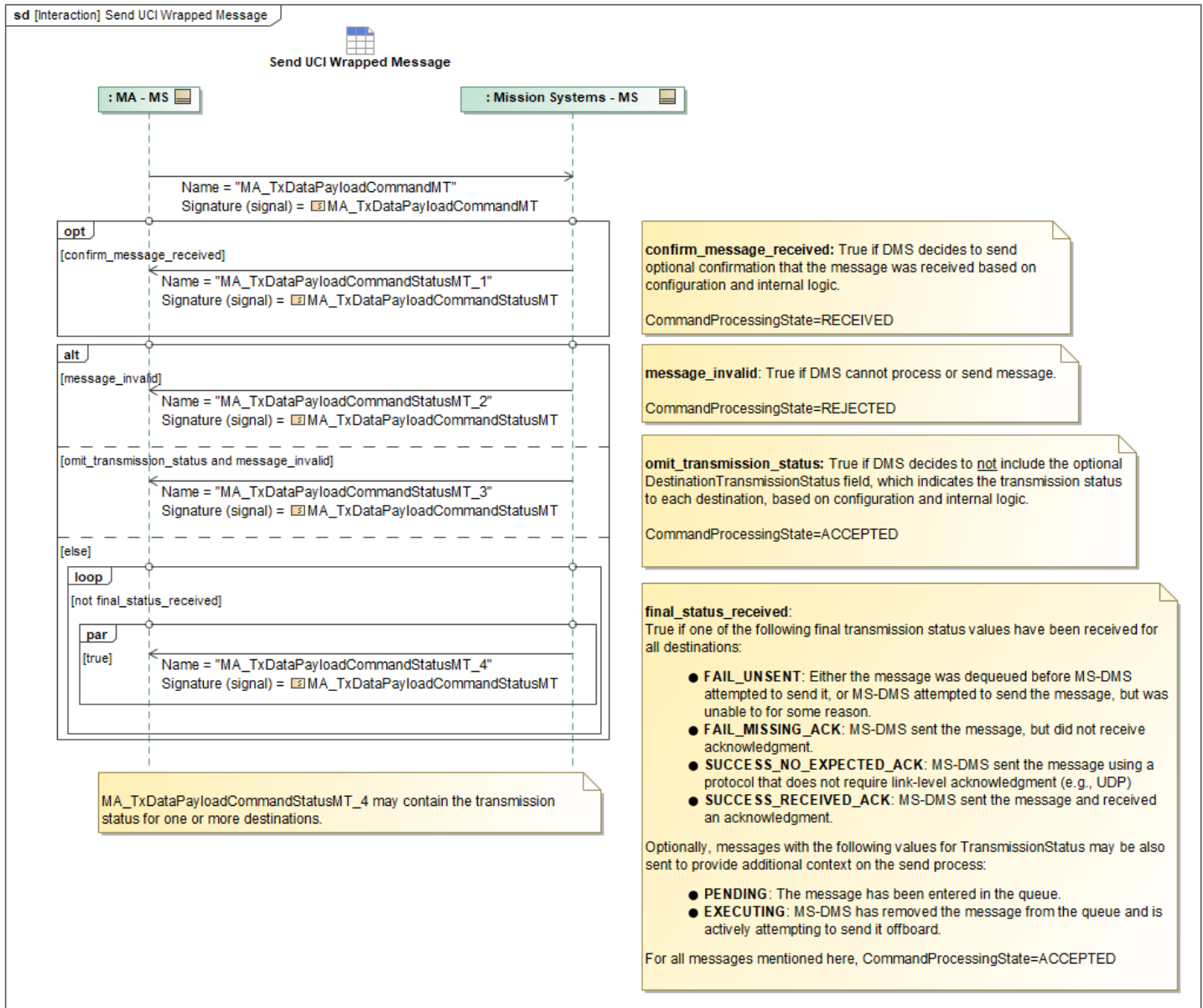


Figure 1-32: Send UCI Wrapped Message

Table 1-29: Send UCI Wrapped Message

Message Name (Name:Type)	Required Fields (Name:Type)
MA_TxDataPayloadCommandMT: MA_TxDataPayloadCommandMT	SpecificDestinationID: SystemID_Type

Message Name (Name:Type)	Required Fields (Name:Type)
MA_TxDataPayloadCommandStatusMT_1: MA_TxDataPayloadCommandStatusMT	None
MA_TxDataPayloadCommandStatusMT_2: MA_TxDataPayloadCommandStatusMT	None
MA_TxDataPayloadCommandStatusMT_3: MA_TxDataPayloadCommandStatusMT	None
MA_TxDataPayloadCommandStatusMT_4: MA_TxDataPayloadCommandStatusMT	DestinationTransmissionStatus: MA_TransmissionStatusType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.5.4 Subscribe to Message Topic

For inter-ACP communication utilizing the DMS, the DMS will subscribe to message topics to let other DMSs know it wants to receive those messages on that topic. When using the "Additional Comms Capability" MUC, MS would handle the configuration of the DMS Gateway Capability using the *MA_DataDisseminationPlanMT*, but if MA needs to make a direct modification of the subscriptions this sequence can be used. The active *Effectivity* in the related *MA_DataDisseminationPlanMT* specifies the set of ingress messages and associated topics the DMS should subscribe to. If subscriptions are needed outside of the active *MA_DataDisseminationPlanMT*, MA can send MS a *MA_MessagingSubscriptionSettingCommandMT_1* with the *AddTopicSubscription* field populated with the *Topic(s)* to subscribe to and the *SubscriptionID*. Optionally, MA can specify the *Priority* of the subscription as well as additional filtering of the subscription using the *SubscriptionFilterType* field. A *Topic* must be provided for a valid subscription. MA can also optionally specify the *DataPayloadOriginator* for the subscription. If there is no *DataPayloadOriginator* in the message, then MA will be subscribed to all publishers of the *Topic*. If the *DataPayloadOriginator* is populated, then the subscription will only be for data payloads of that specified *Topic* that were generated by the specified originator. To modify an existing subscription created by this sequence in the past or configuration file, MA will send MS a *MA_MessagingSubscriptionSettingCommandMT_2* with the *ChangeTopicSubscription* field populated with the *SubscriptionChangeType* and the *SubscriptionID*. To unsubscribe from a topic created by this sequence in the past or configuration file, the same *MA_MessagingSubscriptionSettingCommandMT* is passed from MA to MS, but the *ChangeTopicSubscription* field is populated with the *SubscriptionID(s)* that are to be removed and the *SubscriptionChangeType* set to *DELETE*. In response to the *MA_MessagingSubscriptionSettingCommandMT* message, MS will return a *MA_MessagingSubscriptionSettingCommandStatusMT* message to MA to report the status of the subscription settings command.

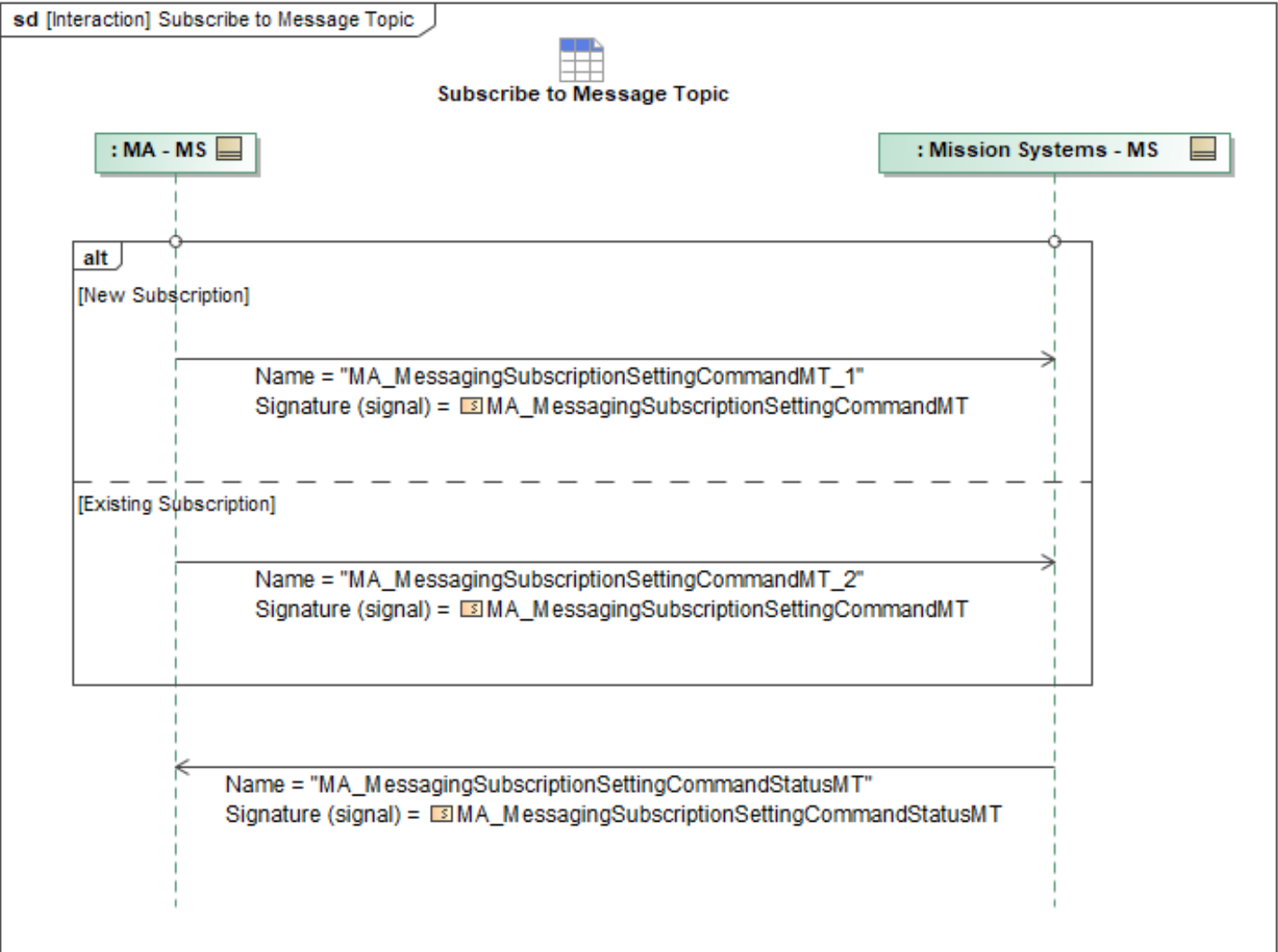


Figure 1-33: Subscribe to Message Topic

Table 1-30: Subscribe to Message Topic

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MessagingSubscriptionSettingCommandMT_1: MA_MessagingSubscriptionSettingCommandMT	AddTopicSubscription: MA_TopicSubscriptionType
MA_MessagingSubscriptionSettingCommandMT_2: MA_MessagingSubscriptionSettingCommandMT	ChangeTopicSubscription: MA_ChangeTopicSubscriptionType
MA_MessagingSubscriptionSettingCommandStatusMT: MA_MessagingSubscriptionSettingCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set

Tables section)

1.2.5.5 Quality of Service

To meet system requirements for reliability, latency, and data freshness, each topic, on the DDS network, is governed by a set of Quality of Service (QoS) policies. These policies control how data is transmitted, stored, and delivered between systems. At a minimum, all topics must declare their QoS settings explicitly.

The default DDS Reliability QoS policies include:

- **RELIABLE** delivery for critical data. This includes messages that require a response, such as acknowledgments or other types of responses, ensuring that messages are delivered and a link level acknowledgment is received, providing guaranteed delivery. Specifically, this applies to UCI primitive types: *Command-2*, *ActionRequest-2*, *DataRequest-2*, *DataRecord-1*, and the *MA_SystemNotificationMT*, *OperatorNotificationMT*, and *OperatorNotificationAckMT* messages.
- **BEST_EFFORT** delivery where data loss is acceptable. This includes messages that do not require an acknowledgment or response. Specifically, this applies to UCI primitive types: *Data-1* and *Status-1*.

When using the "Additional Comms Capability" MUC, the DDS QoS configuration can be configured or changed from the default by using the *DistributionGroup.NetworkNativeEndpoint* field in the *MA_DataDisseminationPlanMT* message and linking it to one or multiple *IER*s. The "Update Data Dissemination Plan" sequence discusses the configuration in more detail. The *DistributionGroup.NetworkNativeEndpoint* field allows for setting the following QoS specific items:

- *TopicQosReliability*: Setting this to "BEST_EFFORT" or "RELIABLE" will directly set this value for the topic(s). If set to "NONE" or not set, either the associated *DeliveryProtocol* is used or the DDS Reliability values above is used to determine the QoS Reliability Setting.
- *TopicQosKeepLast*: If "TRUE", upon reconnect the last message published on the ASB will be sent out to the subscriber. This will configure the DDS publisher with the Keep Last QoS setting. This should only be done for status and some data type messages and it should not be used for commands, requests, or requirement messages that could cause issues with the receiving system. This provides a nice way for status messages to be delivered specifically for messages that are non-periodic.
- *TopicQosPriority*: If Empty, the *IER.Priority* is used. In DDS, the higher the value, the higher the priority. Therefore the *IER.Priority* would need to be inverted or mapped.

QoS compatibility is essential for interoperability and successful communication, ensuring that a subscriber's QoS expectations are compatible with those of the publisher by ensuring that the *MA_DataDisseminationPlanMT*s are aligned and activated appropriately.

1.2.5.6 Security

System communication requires authentication, encryption, and access control to meet security requirements. The DDS Security specification (see <https://www.omg.org/spec/DDS-SECURITY/1.2/>) shall be leveraged to enforce these measures. Systems must be authenticated using X.509 certificates or equivalent credentials before joining the network. Implementers are responsible for managing keys, certificates, and revocation policies.

All data transmitted over the network must be encrypted to protect it from eavesdropping and tampering. Implementers must use the DDS Security plugins to enforce Transport Layer Security (TLS) with mutual TLS (mTLS). This ensures data confidentiality, integrity, and authenticity during transmission.

1.2.5.7 Serialization and Data Encoding

Serialization and data encoding are critical for ensuring interoperability between different systems in the DMS. The DDS middleware supports various serialization formats to encode data structures defined in the Interface Definition Language (IDL). The Common Data Representation (CDR) format shall be used for serialization due to its interoperability, performance, and compatibility with DDS. All data structures shared between systems must be defined in the IDL and serialized according to the defined schema.

All topics transmit data using the *MA_DataPayloadWrapper* IDL type. The actual UCI message content will be EXI encoded into the *EncodedPayload* field. The IDL of the *MA_DataPayloadWrapper* is included in the Autonomy Starter Kit release with the Decentralized Messaging Service component of the A-GRI. When using the "Additional Comms Capability" MUC the *MA_DataDisseminationPlanMT* has an optional *IER_DesiredService.CompressionType* field that can be set to "EXI" to defines the encoding/compression format for the *EncodedPayload* field in the OTA *MA_DataPayloadWrapper* IDL that is used for communication between DMSs.

The table below specifies in detail the EXI specific Settings that should be used:

Table 1-31: EXI Settings

EXI Settings	Details
Schema Informed	True
EXI Strict	True
EXI Compression	False
EXI Cookie	Omitted
EXI Options Field	Include except for Link16
UUID DTR	UUIDs shall be represented as an EXI string table partition
ScaledDecimal DTR	Decimal precision will be as follows: 7 - Latitude, Longitude 6 - VnVn, VnVe, VeVe 4 - NorthAcceleration, EastAcceleration, DownAcceleration, YawRate, PitchRate, RollRate, PnPn, PnPp, PePe, PnVn, PnVe, PnVn, PeVe 2 - Yaw, Pitch, Roll, Bearing, BearingError, MagneticHeading, Azimuth, AzimuthError, Heading/Value, Kollsman 1 - Altitude, NorthSpeed, EastSpeed, DownSpeed, Confidence, TrueAirspeed, Fuel, ProbabilityOfCorrectAssociation, IndicatedBaroAltitude, PowerLevel, Percent, EstimatedPercentComplete, TargetRange
DateTimeTable DTR	The DateTimeTable DTR uses an EXI string table partition for all xsd:DateTimes to represent those that fully or partially match previous DateTime stamps in a message more efficiently. It starts with an EXI Boolean indicating whether this is a repeated DateTime (i.e., whether it exists in the DateTime string table partition). If the DateTime is repeated,

the Boolean is set to true and is followed by the index of the DateTime in the table represented an EXI n-bit Unsigned Integer, where n is $\lceil \log_2 m \rceil$ and m is the number of entries in DateTime string table partition. If the DateTime only partially matches one of the entries in the DateTime string table partition,

the Boolean is set to false and is followed by a special year value indicating how much of the DateTime matched, followed by the table index of the partially matching DateTime, represented as an EXI n-bit Unsigned Integer (as described above), followed by the non-matching parts of the DateTime. Table below shows the special year values used to indicate which parts of the DateTime matched.

Matching Parts	Year Value	Non-matching parts
Year,Month,Day,Hour,Minute,Second	-1	presence,[FractionalSeconds]
Year,Month,Day,Hour,Minute	-2	Seconds,presence,[FractionalSeconds]
Year,Month,Day,Hour	-3	Minutes*64+Seconds,presence,[FractionalSeconds]
Year,Month,Day	-4	Time,presence,[FractionalSeconds]
Year,Month	-5	Day,Time,presence,[FractionalSeconds]

The non-matching parts are represented according to section 7.1.87 of the EXI specification. If the DateTime does not fully or partially match any of the entries in the DateTime string table partition, the Boolean is set to false and is followed by the entire DateTime. All parts of the DateTime are represented according to section 7.1.8 of the EXI specification, except the Year. The DateTimeTable DTR represents years after 1999 as an EXI Unsigned Integer with the value $(y - 2000) * 2 + s$, where y is the Year and s is 0. The DateTimeTable DTR represents years before 2000 as an EXI Unsigned Integer with the value $(y - 2000 - 6) * 2 + s$, where y is the Year

and s is 1. This representation eliminates the need for a separate sign bit and reserves the values -1 through -5 to signal partial matches.

1.2.5.8 System Discovery

DDS enables distributed components to automatically discover each other on the network and exchange metadata about the topics they publish or subscribe to. The DDS middleware relies on the RTPS (Real-Time Publish-Subscribe) protocol, specifically the Simple Participant Discovery Protocol (SPDP) and Simple Endpoint Discovery Protocol (SEDP), to dynamically form a mesh of communicators without requiring a centralized broker. Upon startup, each node registers itself and advertises its capabilities using these discovery protocols.

Participants must be uniquely identifiable by their *SystemID*, defined via configuration. Multicast-based discovery is the default mechanism, but support for static discovery and peer list configuration is required for network-constrained environments.

1.2.5.9 Topic Publication and Subscription

Communication between system components occurs over well-defined topics. Each topic represents a stream of typed data, with any number of publishers and subscribers. Topics are uniquely named and associated with a specific data structure defined in an Interface Definition Language (IDL). All systems must agree on this schema and the semantics of each topic they interact with

As described previously, by default each message type is assigned a dedicated topic, with the topic name exactly matching the message name. This enforces a consistent mapping between data and transport, simplifies subscription management, and allows fine-grained QoS control. Publishers and subscribers may join or leave at any time, and DDS guarantees delivery according to the configured QoS policies. When using the "Additional Comms Capability" MUC, the

MA_DataDisseminationPlanMT does allow for configuring unique topic names, but this should be

limited as it makes interoperability harder to manage. Additionally, each DMS subscribes to a topic that is equal to its host system's *SystemID.UUID* (with a "id_" prefix and "-"→"_") to facilitate *SpecificDestinationID* messaging.

All topics transmit data using the *MA_DataPayloadWrapper* IDL type. The actual UCI message content will be EXI Encoded into the *EncodedPayload* field. The "Serialization and Data Encoding" section describes the EXI representation in detail. The IDL of the *MA_DataPayloadWrapper* is included in the Autonomy Starter Kit release with the Decentralized Messaging Service component of the A-GRI.

The DMS will advertise topics on the DDS interface based on the *MA_TxDataPayloadCommandMT*. When using the "Additional Comms Capability" MUC, topics are advertised using the *IERs* and *DistributionGroups* in the active *Effectivities* in the related *MA_DataDisseminationPlanMT*. The message format on all topics will be the *MA_DataPayloadWrapper* IDL that is filled with either the contents from a *MA_TxDataPayloadCommandMT* message or, when using the "Additional Comms Capability" MUC, the directly ASB-subscribed UCI message. The DMS will publish messages on the appropriate topics to go off-board based on the message type or, when using the "Additional Comms Capability" MUC, the mapping of messages to topics in the *MA_DataDisseminationPlanMT*. For subscriptions when using the "Additional Comms Capability" MUC, the DMS will use the *IERs* and *DistributionGroups* in the active *Effectivities* in the related *MA_DataDisseminationPlanMT* to determine what topics it should subscribe to on the DDS interface. Along with the *SystemID* topic specified above. When a *MA_DataPayloadWrapper* IDL message is received, it will be processed through the filter in the *MA_DataDisseminationPlanMT* and will either be published to the local ASB directly as the UCI message (preserving the senders message header information) or will be wrapped in a *MA_RxDataPayloadMT* message. MA will then receive and process these messages as needed.

1.2.5.10 Transport Protocols

To ensure robust and flexible communication, systems must support both UDP and TCP transport protocols. TCP is recommended for scenarios requiring reliable, ordered delivery of messages. It ensures data integrity and is suitable for messages that require acknowledgments or responses. UDP is recommended for scenarios where low latency and minimal transmission overhead are critical. It is suitable for non-critical data where occasional message loss is acceptable and where speed is prioritized over reliability.

1.2.6 Mission Plan Behaviors Package

1.2.6.1 Receive Plans

The diagram contains the collection of plan updates specifically for MS, performed following the update logic described below for each individual plan represented in the diagram.

Plan Data is updated the via the specific *XYZ_PlanMT* message for the plan that is being updated. If the message being sent represents the first instance of the data/object, the *ObjectState* field is set to "NEW". When sending data to revise an existing set or object, *ObjectState* is set to "UPDATED".

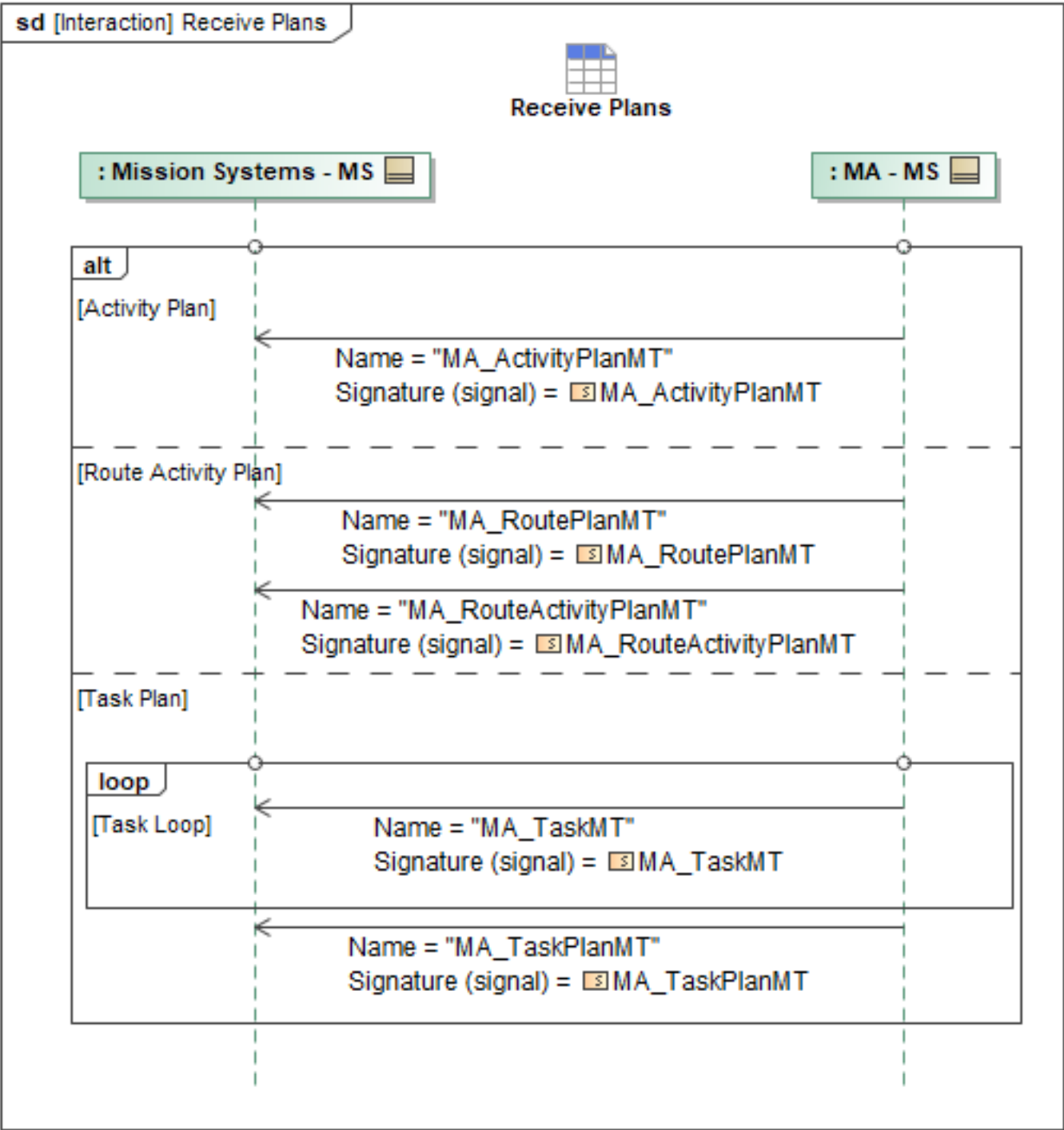


Figure 1-34: Receive Plans

Table 1-32: Receive Plans

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActivityPlanMT: MA_ActivityPlanMT	ObjectState: ObjectStateEnum
MA_RouteActivityPlanMT: MA_RouteActivityPlanMT	ObjectState: ObjectStateEnum

Message Name (Name:Type)	Required Fields (Name:Type)
MA_RoutePlanMT: MA_RoutePlanMT	ObjectState: ObjectStateEnum
MA_TaskMT: MA_TaskMT	ObjectState: ObjectStateEnum
MA_TaskPlanMT: MA_TaskPlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.6.2 Verify Plan Patch Is Possible - MS

Depending on the type of *Plan* being patched, one of *MA_MissionPlanValidationCommandMT*, *TaskPlanValidationCommandMT*, *RouteActivityPlanValidationCommandMT*, *ActivityPlanValidationCommandMT*, or *ActionPlanValidationCommandMT* is sent from MA to MS. MS then responds with the corresponding *CommandStatusMT*, i.e., *MA_MissionPlanValidationCommandStatusMT*, *TaskPlanValidationCommandStatusMT*, *RouteActivityPlanValidationCommandStatusMT*, *ActivityPlanValidationCommandStatusMT*, or *ActionPlanValidationCommandStatusMT*. Within the **CommandMT* message, the *PlanPart* contains the plan patch that needs to be validated.

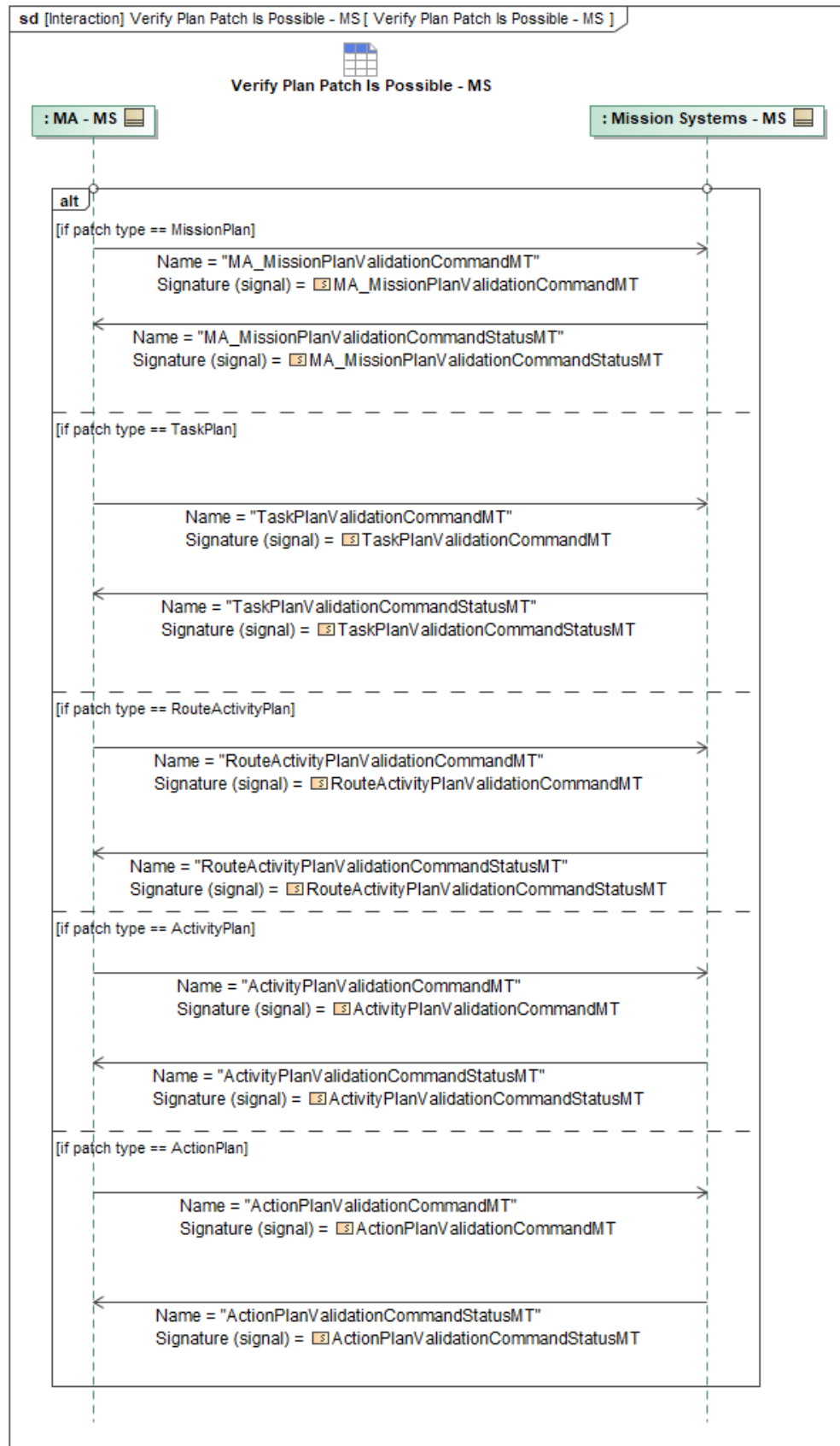


Figure 1-35: Verify Plan Patch Is Possible - MS

Table 1-33: Verify Plan Patch Is Possible - MS

Message Name (Name:Type)	Required Fields (Name:Type)
ActionPlanValidationCommandMT: ActionPlanValidationCommandMT	PlanPart: ActionTypeEnum
ActionPlanValidationCommandStatusMT: ActionPlanValidationCommandStatusMT	None
ActivityPlanValidationCommandMT: ActivityPlanValidationCommandMT	PlanPart: ActivityPlanPartsEnum
ActivityPlanValidationCommandStatusMT: ActivityPlanValidationCommandStatusMT	None
MA_MissionPlanValidationCommandMT: MA_MissionPlanValidationCommandMT	PlanPart: PlanPartsType
MA_MissionPlanValidationCommandStatusMT: MA_MissionPlanValidationCommandStatusMT	None
RouteActivityPlanValidationCommandMT: RouteActivityPlanValidationCommandMT	PlanPart: ActivityPlanPartsEnum
RouteActivityPlanValidationCommandStatusMT: RouteActivityPlanValidationCommandStatusMT	None
TaskPlanValidationCommandMT: TaskPlanValidationCommandMT	PlanPart: TaskTypeEnum
TaskPlanValidationCommandStatusMT: TaskPlanValidationCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7 PNT Service Package

1.2.7.1 Request PNT Navigation Data

Mission Use Case: PNT

This interaction sequence enables MA to request precision navigation and time from the MS in order to perform its functions that are position- and time-dependent. First MA requests the status of the MS to verify it can provide position information using the *SubsystemStatusDataRequestMT* message. The MS will respond to the request with a *SubsystemStatusDataRequestStatusMT* message that provides a *SubsystemStatusData* field that reports the health and status of the MS. MA will then command the PNT service to adjust the frequency it posts PNT data to other services using the *SubsystemSettingsCommandMT*. The PNT service will reply, providing MA with a *MA_PositionReportDetailedMT* with detailed position information of the ACP. This interaction provides MA integrators a way to retrieve precision time and navigation information from the MS.

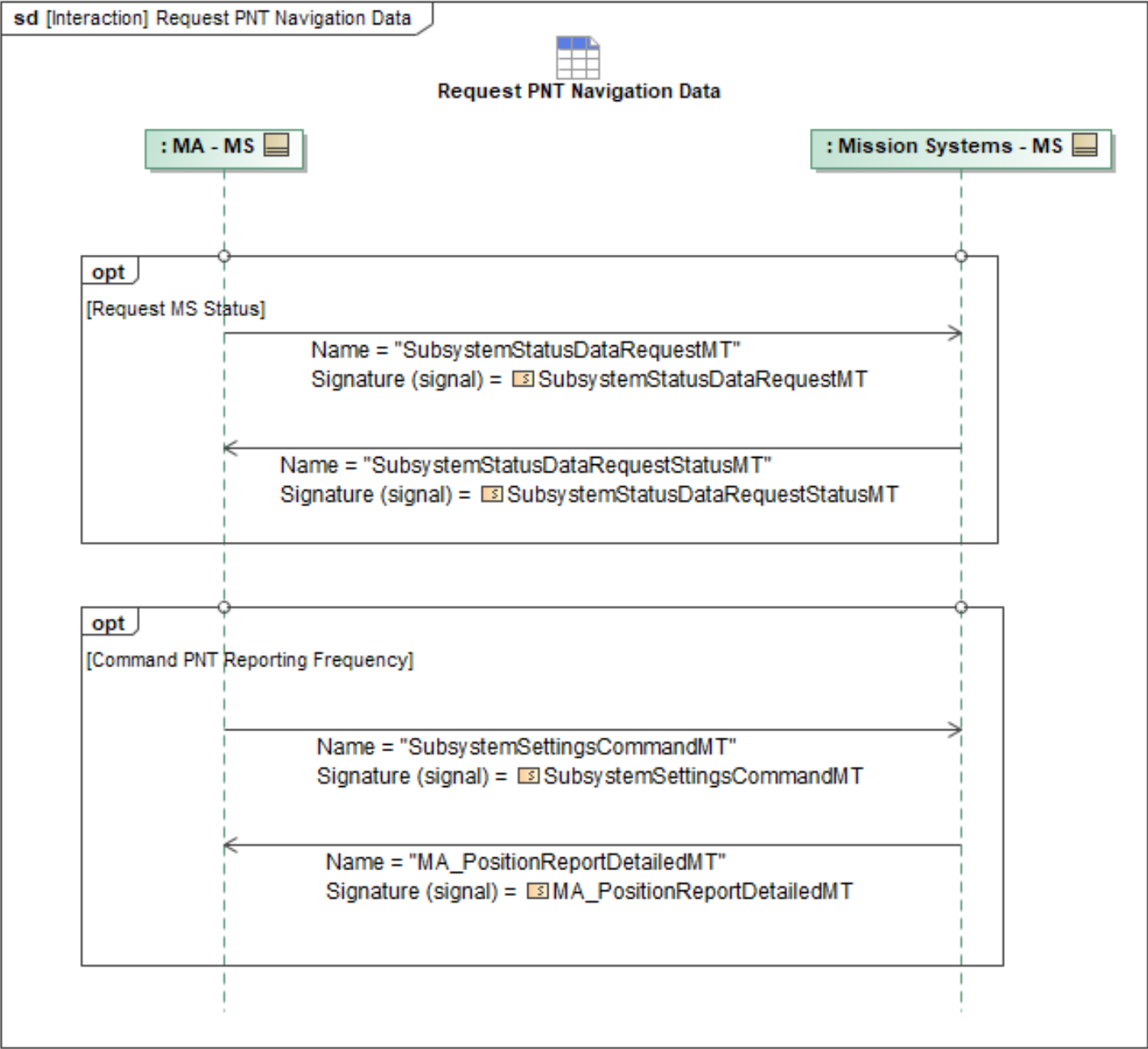


Figure 1-36: Request PNT Navigation Data

Table 1-34: Request PNT Navigation Data

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PositionReportDetailedMT: MA_PositionReportDetailedMT	Orientation: OrientationType MagneticHeading: AngleType WanderAngle: AngleType OrientationRate: OrientationRateType OrientationAcceleration: OrientationAccelerationType

Message Name (Name:Type)	Required Fields (Name:Type)
	Acceleration: Acceleration3D_Type AirData: MA_AirDataType
SubsystemSettingsCommandMT: SubsystemSettingsCommandMT	None
SubsystemStatusDataRequestMT: SubsystemStatusDataRequestMT	None
SubsystemStatusDataRequestStatusMT: SubsystemStatusDataRequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8 Sensor/Vehicle Tasking Package

1.2.8.1 Advertise Sensor AMTI Capability

Mission Use Case: AMTI

The Advertise Sensor AMTI Capability sequence diagram below specifies the interaction of MS communicating the Airborne Moving Target Indicator (AMTI) sensor capability to MA so that it may be used by MA to detect aircraft through active radar. This sequence is only present if AMTI sensing capabilities exist in MS. MS publishes *MA_AMTI_CapabilityMT* with at least one *Capability* specified. Multiple *Capability* fields are specified to advertise different command use cases. Each capability must have a unique identifier set in *CapabilityID*. The *AcceptedInterface* field must specify at least *CAPABILITY_COMMAND*. The *CapabilityType* should be set to the types of commands that the capability supports. The *SubCapabilityType* field can be set to any of the four options, as MS is implemented. The *CapabilitiesOptions* field and sub-fields will be set by MS according to the ability of MS (e.g. *InterruptOtherActivities* and *ConcurrentOperationSupported* are descriptive of particular MS implementation). Similarly, The *SupportedFrequencies* and *ComponentFieldofRegard* fields and sub-fields are set by MS according to the ability of MS as well.

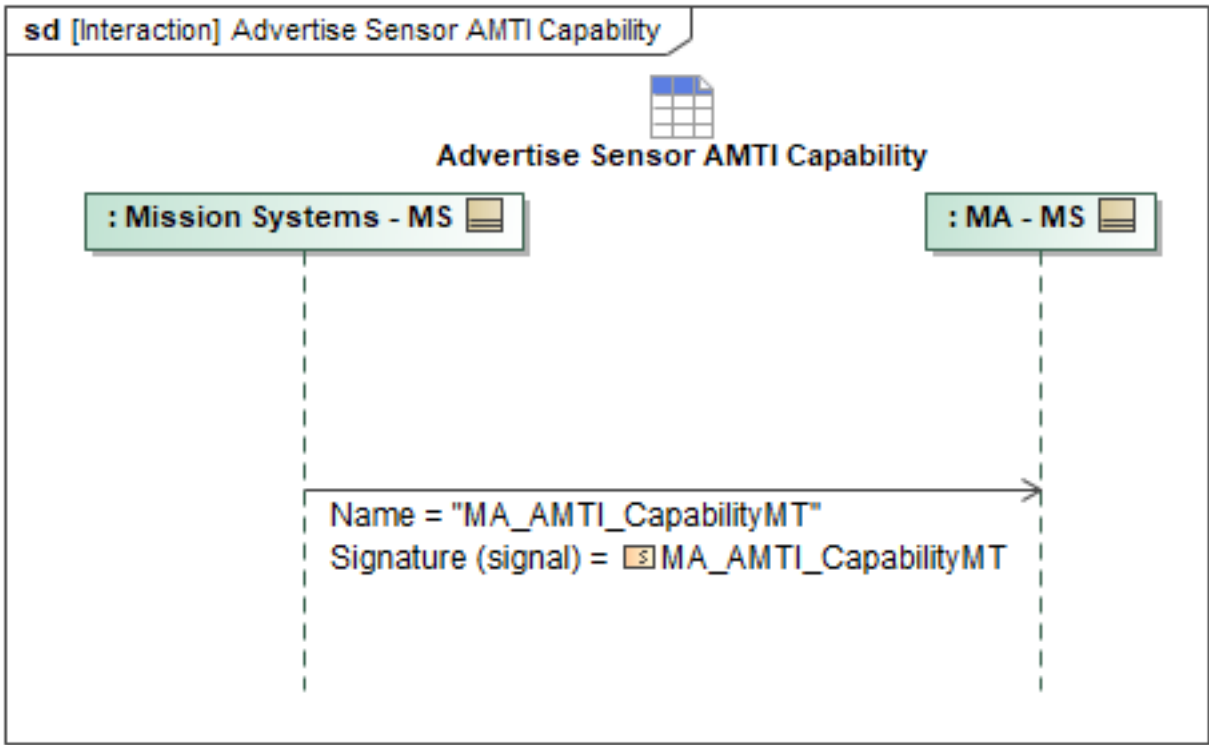


Figure 1-37: Advertise Sensor AMTI Capability

Table 1-35: Advertise Sensor AMTI Capability

Message Name (Name:Type)	Required Fields (Name:Type)
MA_AMTI_CapabilityMT: MA_AMTI_CapabilityMT	ComponentFieldOfRegard: ComponentFieldOfRegardType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.2 Advertise Sensor ESM Capability

Mission Use Case: ESM

The Advertise Sensor ESM Capability sequence diagram below specifies the interaction of MS communicating the Electronic Support Measures (ESM) sensor capability to MA so that it may be used by MA to detect aircraft through electronic support measures. This sequence is only present if ESM sensing capabilities exist in MS. MS publishes *ESM_CapabilityMT* with at least one *Capability* specified. Multiple *Capability* fields are specified to advertise multiple command use cases. Each *Capability* must have a unique identifier set in *CapabilityID*. The *AcceptedInterface* field must specify at least *CAPABILITY_COMMAND*. The *CapabilityType* should either be *SEARCH_VOLUME* (for searching a volume of space to generate tracks) or *TRACK* (for updating a specified track). The *CapabilitiesOptions* field and sub-fields will be set by MS according to the ability of MS (e.g. *InterruptOtherActivities* and *CalibrateOnActivity* are descriptive of particular MS implementation). If the capability supports providing a Field of Regard, the *CapabilityFOR* field must specify the field of regard of the sensor ESM capability. No other schema-optional fields are required but may be specified for more information.

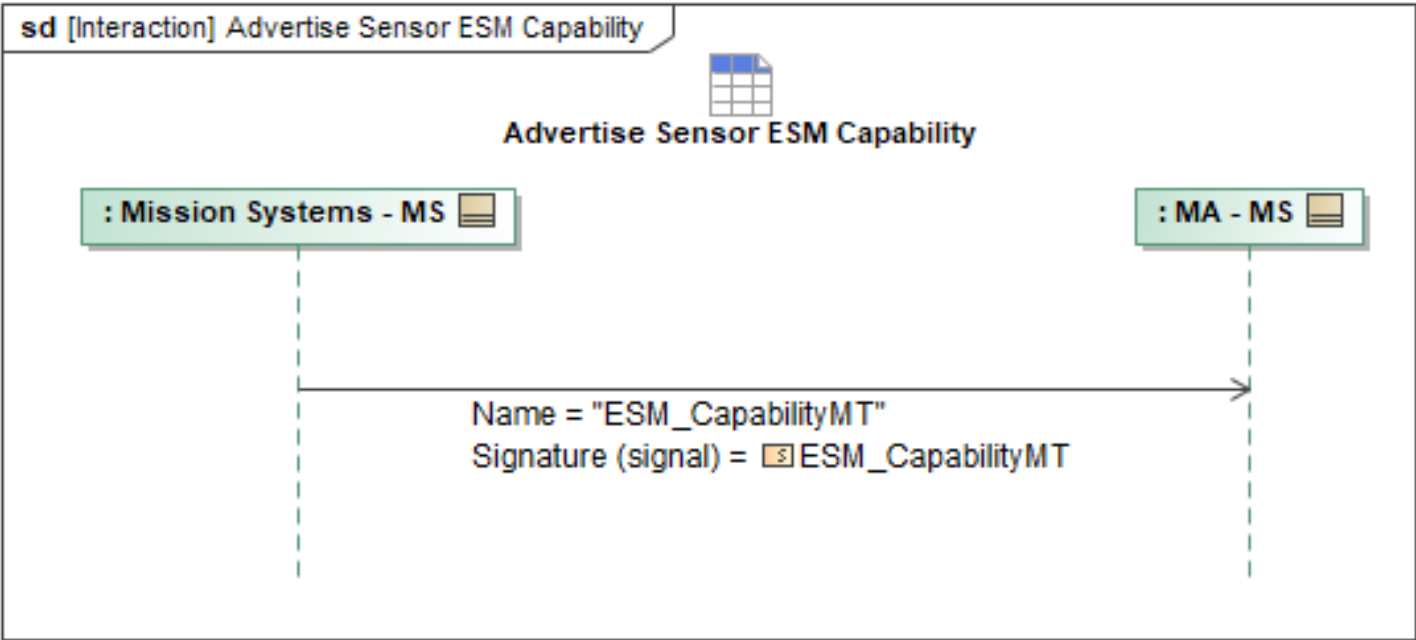


Figure 1-38: Advertise Sensor ESM Capability

Table 1-36: Advertise Sensor ESM Capability

Message Name (Name:Type)	Required Fields (Name:Type)
ESM_CapabilityMT: ESM_CapabilityMT	CapabilityFOR: FOR_Type

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.3 Advertise Sensor PO Capability

Mission Use Case: PO

The Advertise Sensor PO Capability sequence diagram specifies the interaction of MS communicating the Passive Optical (PO) sensor capability to MA so that it may be used by MA to detect aircraft through passive optical search and track. This sequence is only present if PO sensing capabilities exist in MS. MS publishes *PO_CapabilityMT* with at least one Capability specified. Multiple *Capability* fields are specified to advertise multiple command use cases. Each *Capability* must have a unique identifier set in *CapabilityID*. The *AcceptedInterface* field must specify at least *CAPABILITY_COMMAND*. The *CapabilityType* should either be *SEARCH_VOLUME* (for searching a volume of space to generate tracks) or *TRACK* (for updating a specified track). The *CapabilitiesOptions* field and sub-fields will be set by MS according to the ability of MS (e.g. *InterruptOtherActivities* and *CalibrateOnActivity* are descriptive of particular MS implementation). The *CapabilityFOR* field is required to specify the field of regard of the sensor PO capability.

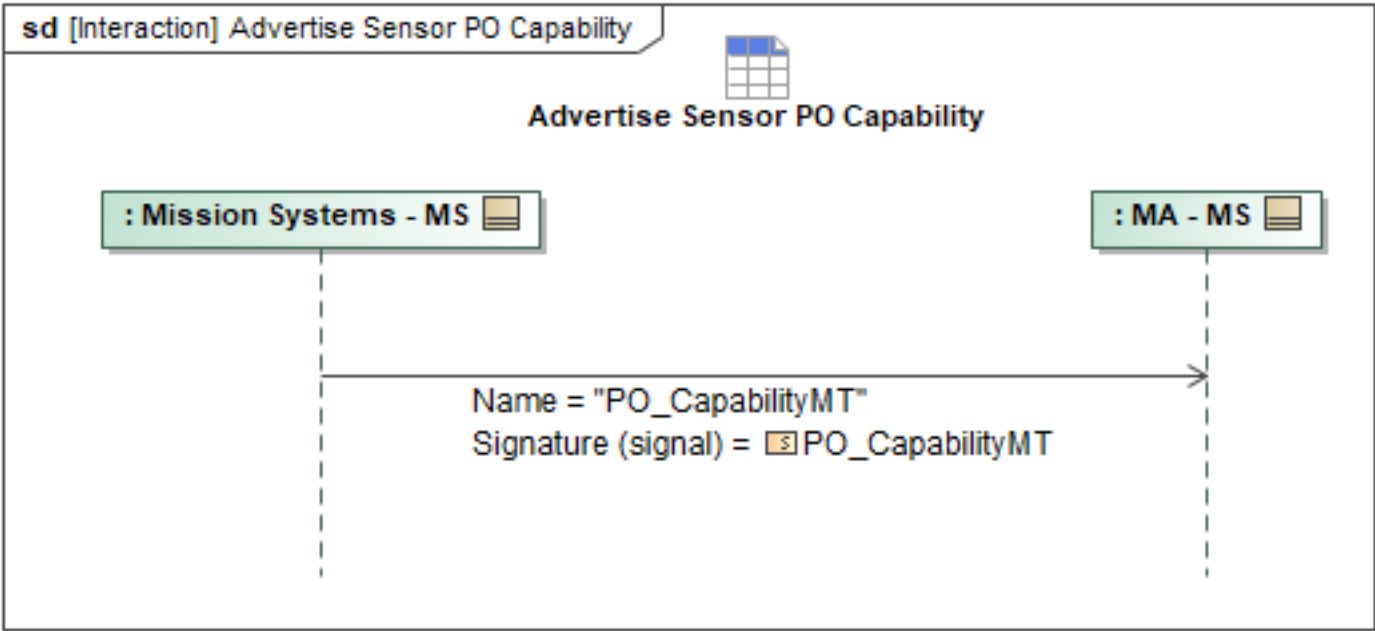


Figure 1-39: Advertise Sensor PO Capability

Table 1-37: Advertise Sensor PO Capability

Message Name (Name:Type)	Required Fields (Name:Type)
PO_CapabilityMT: PO_CapabilityMT	CapabilityFOR: FOR_Type

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.4 Command Sensor AMTI Capability

Mission Use Case: AMTI

The Command Sensor AMTI Capability State sequence diagram specifies the interaction of MA commanding MS to schedule activity of the Airborne Moving Target Indicator (AMTI) sensor capability and then MS responding to the command with status. MA publishes *AMTI_CommandMT* with the Command choice set to *Capability* and a unique *CommandID*. The *CommandState* is set to *NEW* to schedule new AMTI capability execution, *UPDATE* to change a previous command (scheduled or not), and *CANCEL* to stop or remove a scheduled AMTI capability execution. The *CapabilityID* specified corresponds to one of the *CapabilityID* fields published by MS in *AMTI_CapabilityMT*. The *Ranking* field and sub-fields are set according to MS implementation and desired effect on MS scheduling priorities. The schema-optional field *TemporalConstraints* must be specified with select sub-fields. *StartTimeWindow* containing both *Begin* and *End* values must be specified to provide the MS scheduler a time window to start the AMTI capability execution. *EndTimeWindow* containing both *Begin* and *End* must be specified to provide the MS scheduler a time window to the end the AMTI capability execution.

The *TargetVolume* choice field is set based upon the *CapabilityType* of the *Capability* from *MA_AMTI_CapabilityMT* that the *AMTI_CommandMT* *CapabilityID* specifies. There should only be one *TargetVolume* specified. If the *CapabilityType* is *VOLUME* (for searching a volume of space to generate tracks), then *TargetVolume* must be set to *AirVolume* with either of the required

VolumeLocation choices used to specify a volume of space by sensor reference values or 3D LLA geometry. If the *CapabilityType* is TRACK (for updating a specified track), then *TargetVolume* must be set to *EntityID* with a unique identifier that both MA and MS are aware of corresponding to the track that should be updated by MS. No remaining schema-optional fields are required but may be specified for more information.

In response to the *AMTI_CommandMT* message sent from MA, MS sends back *AMTI_CommandStatusMT* first acknowledging that the command was received. The *CommandID* must match the *CommandID* set in the *AMTI_CommandMT* being responded to. *CommandProcessingState* must be set to *RECEIVED*. Following this, MS assesses the command and sends another *AMTI_CommandStatusMT* with the same *CommandID* but with *CommandProcessingState* set to *ACCEPTED* if MS has scheduled the AMTI command, *REJECTED* if MS rejected the command, or *CANCELED* if the *CommandState* in the *AMTI_CommandMT* is set to *CANCEL* and the command was approved (removing an AMTI capability execution from the schedule).

In addition, *AMTI_ActivityMT* is published periodically with the *SubsystemID* set to a unique identifier for MS or the MS subsystem responsible for AMTI command scheduling. The Activity sequence specified must be ordered temporally according to the AMTI command schedule, with earlier commands coming first. Each Activity has a unique *ActivityID* and the *CapabilityID* corresponds to one of the *Capability* fields specified in the *MA_AMTI_CapabilityMT* message published by MS. Interactive is set according to MS implementation (MS is only required to support *MA_AMTI_CapabilityMT* AcceptedInterface of *CAPABILITY_COMMAND*, which is not modifying activities, only scheduling or cancelling them). Each Activity *ActivityState* is populated appropriately as required by the schema. *EstimatedStartTime* and *EstimatedCompletionTime* are required to be populated, showing that capability commands are scheduled by MS as commanded.

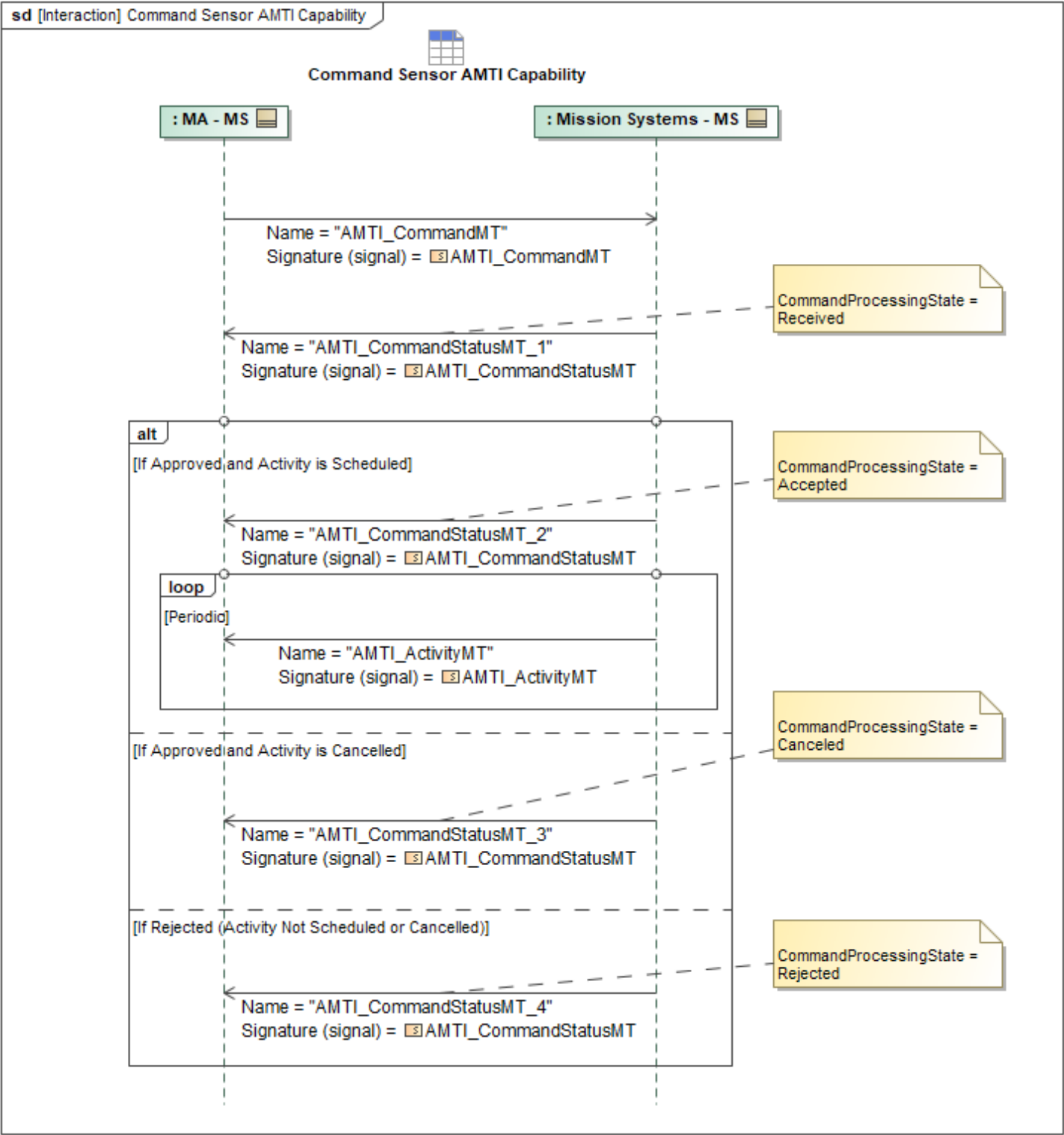


Figure 1-40: Command Sensor AMTI Capability

Table 1-38: Command Sensor AMTI Capability

Message Name (Name:Type)	Required Fields (Name:Type)
AMTI_ActivityMT: AMTI_ActivityMT	EstimatedStartTime: DateTimeType

Message Name (Name:Type)	Required Fields (Name:Type)
	EstimatedCompletionTime: DateTimeType
AMTI_CommandMT: AMTI_CommandMT	StartTimeWindow: DateTimeRangeType Begin: DateTimeType End: DateTimeType EndTimeWindow: DateTimeRangeType Capability: AMTI_CapabilityCommandType
AMTI_CommandStatusMT_1: AMTI_CommandStatusMT	None
AMTI_CommandStatusMT_2: AMTI_CommandStatusMT	None
AMTI_CommandStatusMT_3: AMTI_CommandStatusMT	None
AMTI_CommandStatusMT_4: AMTI_CommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.5 Command Sensor ESM Capability

Mission Use Case: ESM

The Command Sensor ESM Capability State sequence diagram below specifies the interaction of MA commanding MS to schedule activity of the Electronic Support Measures (ESM) sensor capability and then MS responding to the command with status. MA publishes *ESM_CommandMT* with the *Command* choice set to *Capability* and a unique *CommandID* (see “Advertise Sensor ESM Capability”). The *CommandState* is set to *NEW* to schedule new ESM capability execution, *UPDATE* to change a previous command (scheduled or not), and *CANCEL* to stop or remove a scheduled ESM capability execution. The *CapabilityID* specified corresponds to one of the *CapabilityID* fields published by MS in *ESM_CapabilityMT*. The *Ranking* field and sub-fields are set according to MS implementation and desired effect on MS scheduling priorities. The schema-optional field *TemporalConstraints* must be specified with select sub-fields. *StartTimeWindow* containing both *Begin* and *End* values must be specified to provide the MS scheduler a time window to start the ESM capability execution. *EndTimeWindow* containing both *Begin* and *End* must be specified to provide the MS scheduler a time window to the end the ESM capability execution.

The schema-optional *ESM_Location* choice field is required. The choice field is set based upon the *CapabilityType* of the *Capability* from *ESM_CapabilityMT* that the *ESM_CommandMT CapabilityID* specifies. If the *SubCapabilityType* is *ACQUISITION* (for searching a volume of space to generate tracks), then *ESM_Location* must be set to *ESM_AirVolume* with the required parameters to specify the volume. If the *CapabilityType* is *SINGLE_SUBSYSTEM_GEOLOCATION* or *MULTI_SUBSYSTEM_GEOLOCATION* (for locating a specific emitter), then *TargetLocationData* must be set to *EntityID* with a unique identifier that both MA and MS are aware of corresponding to the track that should be updated by MS. No remaining schema-optional fields are required but may be specified for more information.

In response to the *ESM_CommandMT* message sent from MA, MS sends back *ESM_CommandStatusMT* first acknowledging that the command was received. The *CommandID* must match the *CommandID* set in the *ESM_CommandMT* being responded to. *CommandProcessingState* must be set to *RECEIVED*. Following this, MS assesses the command and sends another *ESM_CommandStatusMT* with the same *CommandID* but with *CommandProcessingState* set to *ACCEPTED* if MS has scheduled the ESM command, *REJECTED* if MS rejected the command, or *CANCELED* if the *CommandState* in the *ESM_CommandMT* is set to *CANCEL* and the command was approved (removing an ESM capability execution from the schedule).

In addition, *ESM_ActivityMT* is published periodically with the *SubsystemID* set to a unique identifier for MS or the MS subsystem responsible for ESM command scheduling. The *Activity* sequence specified must be ordered temporally according to the ESM command schedule, with earlier commands coming first. Each *Activity* has a unique *ActivityID* and the *CapabilityID* corresponds to one of the *Capability* fields specified in the *ESM_CapabilityMT* message published by MS. *Interactive* is set according to MS implementation (MS is only required to support *ESM_CapabilityMT AcceptedInterface* of *CAPABILITY_COMMAND*, which is not modifying activities, only scheduling or canceling them). Each *Activity* *ActivityState* is populated appropriately as required by the schema. *EstimatedStartTime* and *EstimatedCompletionTime* are required to be populated, showing that capability commands are scheduled by MS as commanded. The rest of the schema-optional fields are not required but may be specified for more information.

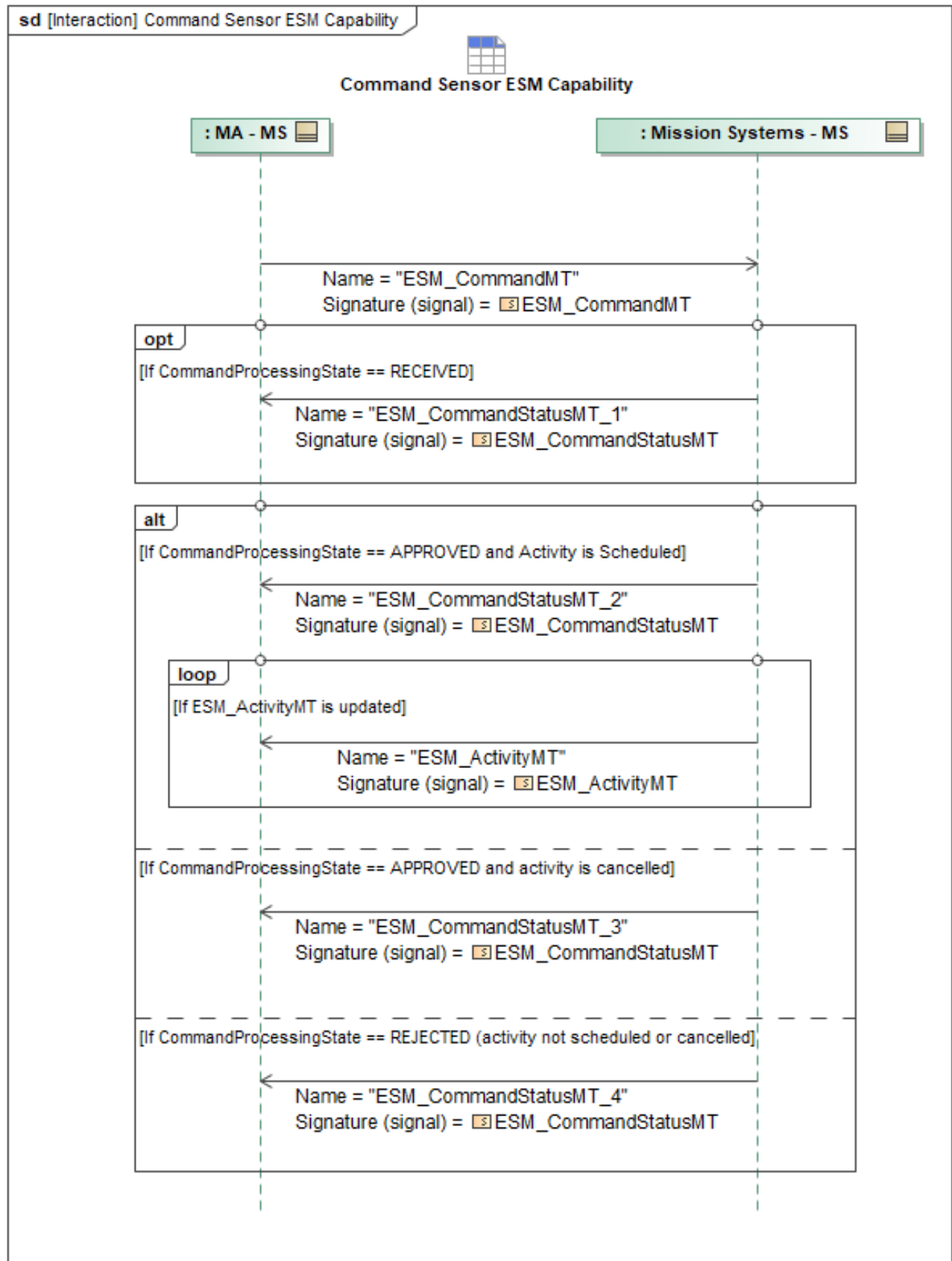


Figure 1-41: Command Sensor ESM Capability

Table 1-39: Command Sensor ESM Capability

Message Name (Name:Type)	Required Fields (Name:Type)
ESM_ActivityMT: ESM_ActivityMT	EstimatedStartTime: DateTimeType EstimatedCompletionTime: DateTimeType
ESM_CommandMT: ESM_CommandMT	ESM_Location: ESM_LocationType EndTimeWindow: DateTimeRangeType StartTimeWindow: DateTimeRangeType Capability: PO_CapabilityCommandType Begin: DateTimeType End: DateTimeType
ESM_CommandStatusMT_1: ESM_CommandStatusMT	None
ESM_CommandStatusMT_2: ESM_CommandStatusMT	None
ESM_CommandStatusMT_3: ESM_CommandStatusMT	None
ESM_CommandStatusMT_4: ESM_CommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.6 Command Sensor PO Capability

Mission Use Case: PO

The Command Sensor PO Capability State sequence diagram specifies the interaction of MA commanding MS to schedule activity of the Passive Optical (PO) sensor capability and then MS responding to the command with status. MA publishes *PO_CommandMT* with the Command choice set to *Capability* and a unique *CommandID* (see “Advertise Sensor PO Capability”). The *CommandState* is set to *NEW* to schedule new PO capability execution, *UPDATE* to change a previous command (scheduled or not), and *CANCEL* to stop or remove a scheduled PO capability execution. The *CapabilityID* specified corresponds to one of the *CapabilityID* fields published by MS in *PO_CapabilityMT*. The *Ranking* field and sub-fields are set according to MS implementation and desired effect on MS scheduling priorities. The schema-optional field *TemporalConstraints* must be specified with select sub-fields. *StartTimeWindow* containing both *Begin* and *End* values must be specified to provide the MS scheduler a time window to start the PO capability execution. *EndTimeWindow* containing both *Begin* and *End* must be specified to provide the MS scheduler a time window to the end the PO capability execution.

The schema-optional *Pointing* choice field is required and set to the *Volume* choice type. There should be only one *Volume* specified, and the *Volume* choice field is set based upon the *CapabilityType* of the Capability from *PO_CapabilityMT* that the *PO_CommandMT* *CapabilityID* specifies. If the *CapabilityType* is *SEARCH_VOLUME* (for searching a volume of space to generate tracks), then *Volume* must be set to *AirVolume* with either of the required choices used to specify a volume of space by sensor reference values or 3D LLA geometry. If the *CapabilityType* is *TRACK* (for

updating a specified track), then *Volume* must be set to *EntityID* with a unique identifier that both MA and MS are aware of corresponding to the track that should be updated by MS. No remaining schema-optional fields are required but may be specified for more information.

In response to the *PO_CommandMT* message sent from MA, MS sends back *PO_CommandStatusMT* first acknowledging that the command was received. The *CommandID* must match the *CommandID* set in the *PO_CommandMT* being responded to. *CommandProcessingState* must be set to RECEIVED. Following this, MS assesses the command and sends another *PO_CommandStatusMT* with the same *CommandID* but with *CommandProcessingState* set to ACCEPTED if MS has scheduled the PO command, REJECTED if MS rejected the command, or CANCELED if the *CommandState* in the *PO_CommandMT* is set to CANCEL and the command was approved (removing an PO capability execution from the schedule).

In addition, *PO_ActivityMT* is published periodically with the *SubsystemID* set to a unique identifier for MS or the MS subsystem responsible for PO command scheduling. The Activity sequence specified must be ordered temporally according to the PO command schedule, with earlier commands coming first. Each Activity has a unique *ActivityID* and the *CapabilityID* corresponds to one of the *Capability* fields specified in the *PO_CapabilityMT* message published by MS. *Interactive* is set according to MS implementation (MS is only required to support *PO_CapabilityMT AcceptedInterface* of *CAPABILITY_COMMAND*, which is not modifying activities, only scheduling or cancelling them). Each Activity *ActivityState* is populated appropriately as required by the schema. *EstimatedStartTime* and *EstimatedCompletionTime* are required to be populated, showing that capability commands are scheduled by MS as commanded.

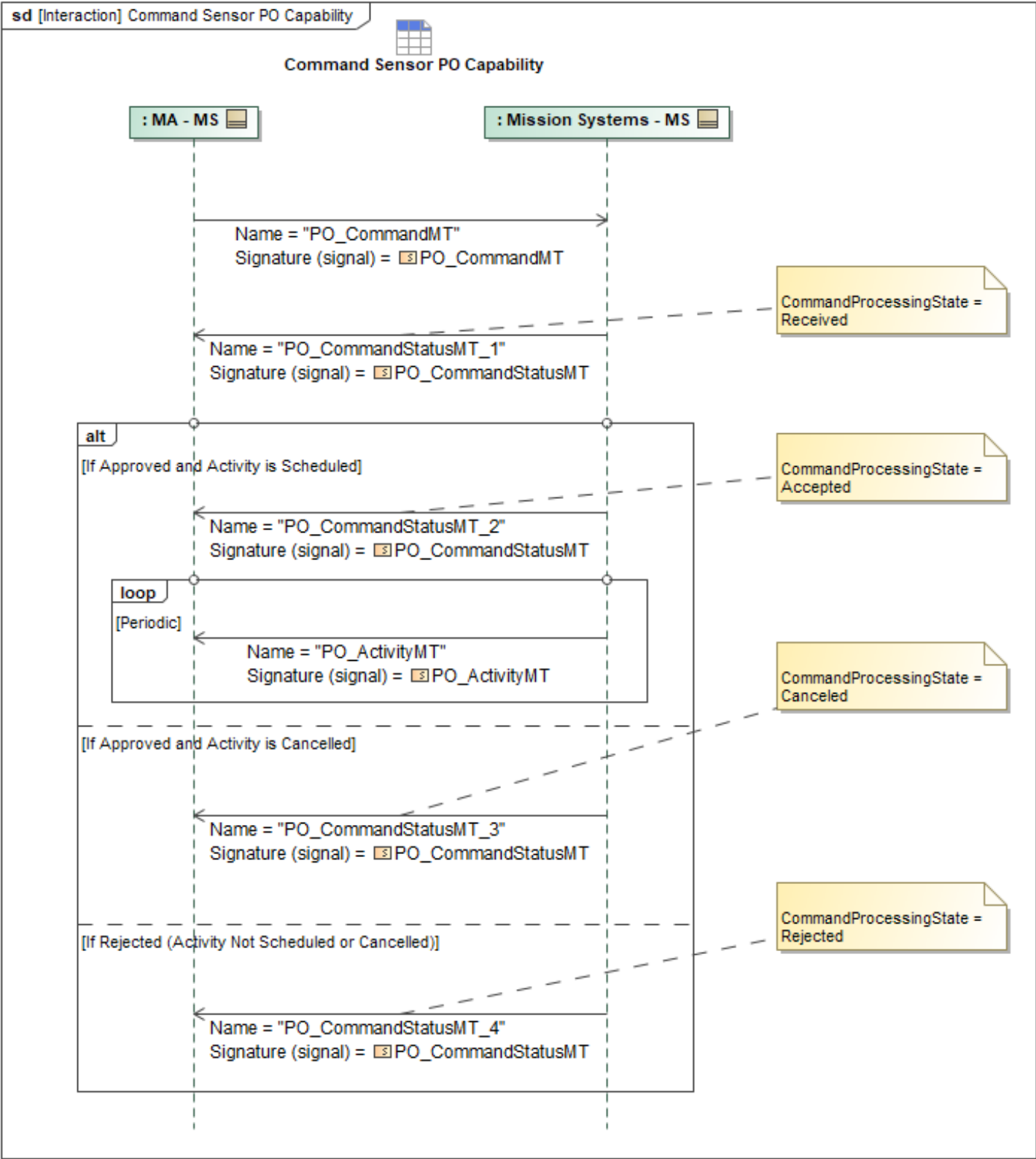


Figure 1-42: Command Sensor PO Capability

Table 1-40: Command Sensor PO Capability

Message Name (Name:Type)	Required Fields (Name:Type)
PO_ActivityMT: PO_ActivityMT	None
PO_CommandMT: PO_CommandMT	Begin: DateTimeType Pointing: PO_TargetType Capability: PO_CapabilityCommandType End: DateTimeType
PO_CommandStatusMT_1: PO_CommandStatusMT	None
PO_CommandStatusMT_2: PO_CommandStatusMT	None
PO_CommandStatusMT_3: PO_CommandStatusMT	None
PO_CommandStatusMT_4: PO_CommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.7 Publish External Weather Reports

MS can request weather data from off-board, periodically-updated weather services through comms channels (ex: XM weather from satellite). MA can initiate a request of that data by sending MS *DataUpdateRequestMT* with the *RequestType* set to *Query*, *Query.MessageType* set to *WEATHER*, and *Query.Query* included. The query will contain details for specifics of the off-board weather data being requested.

MS will respond with *DataUpdateRequestStatusMTs* to confirm that it has received the *DataUpdateRequestMT* from MA and to subsequently provide updates on the state of the request using the *RequestProcessingState* field. Finally, once MS has received the updated weather data, it will send it to MA as a *WeatherMT* message, and send a final *DataUpdateRequestStatusMT* with *RequestProcessingState* set to *COMPLETED*. If MS was unable to obtain the updated weather data, it will not send *WeatherMT*, and the final *DataUpdateRequestStatusMT* will contain *RequestProcessingState* set to *FAILED*.

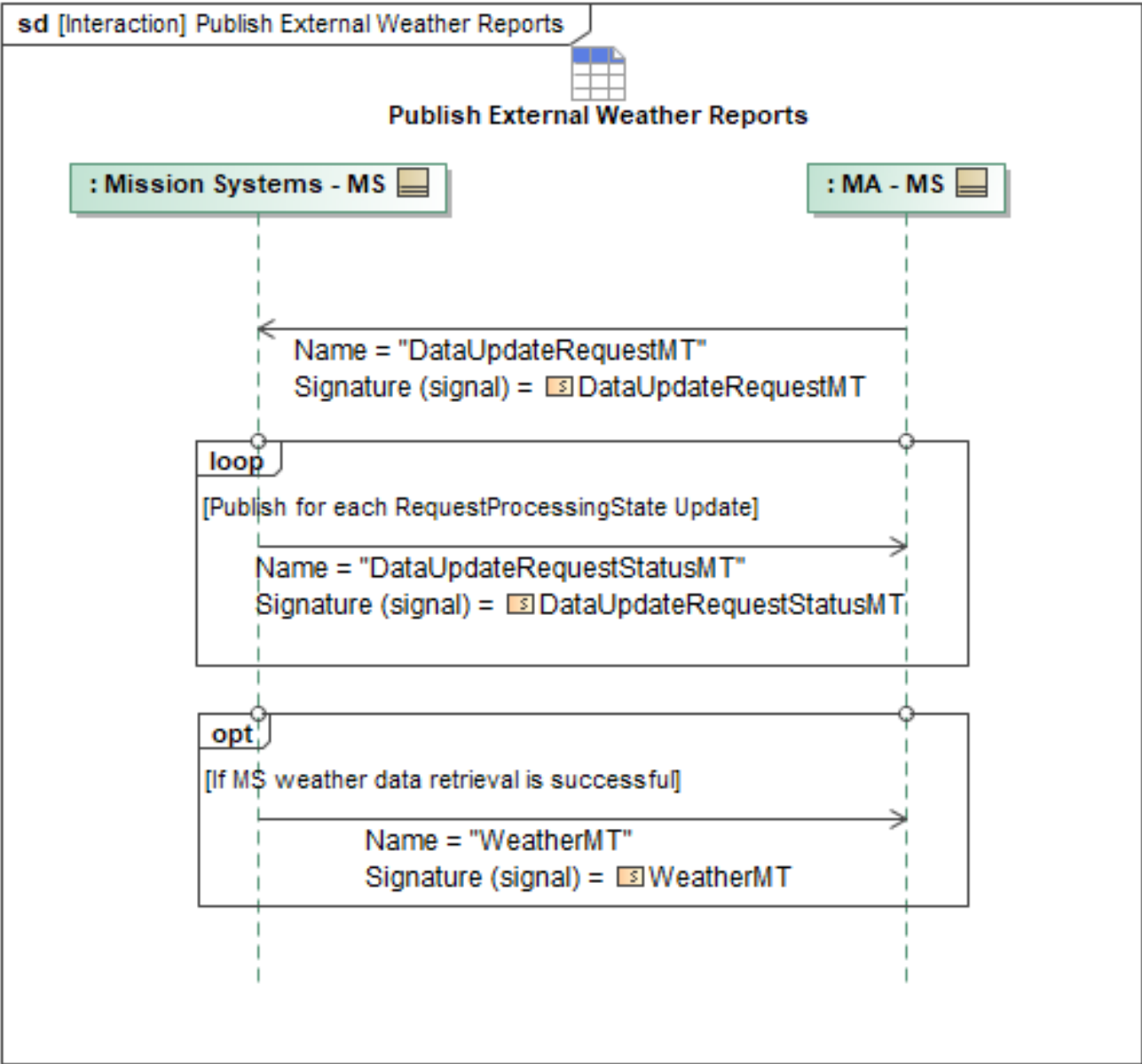


Figure 1-43: Publish External Weather Reports

Table 1-41: Publish External Weather Reports

Message Name (Name:Type)	Required Fields (Name:Type)
DataUpdateRequestMT: DataUpdateRequestMT	MessageType: MessageTypeEnum Query: QueryPET
DataUpdateRequestStatusMT: DataUpdateRequestStatusMT	None
WeatherMT: WeatherMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.8 Report Sensor AMTI Capability State

Mission Use Case: AMTI

The Report Sensor AMTI Capability State sequence diagram specifies the interaction of MS communicating the Airborne Moving Target Indicator (AMTI) sensor capability availability and scheduled activity to MA. This sequence is only present if AMTI sensing capabilities exist in MS (see “Advertise Sensor AMTI Capability for Search and Track”). MS periodically publishes *AMTI_CapabilityStatusMT* and *AMTI_ActivityMT* to MA. MS also publishes *AMTI_CapabilityStatusMT* asynchronously in response to changes in the AMTI capability availability to MA (e.g. sensors are in an unusable state). MS also publishes *AMTI_ActivityMT* asynchronously in response to changes in the scheduled AMTI capability commands MS will execute (e.g. a new command was accepted).

AMTI_CapabilityStatusMT is published with at least one *CapabilityStatus* specified. Each *CapabilityStatus* must have a unique identifier set in *CapabilityID* that corresponds to a *Capability* included in *MA_AMTI_CapabilityMT* previously published by MS. The *Availability* enumeration is used to specify if MA can command this capability. *Availability* of *AVAILABLE* signifies that the capability is functioning, and MA can issue commands. All other *Availability* enumerations signify that MA cannot issue commands. No schema-optional fields are required but may be specified for more information.

AMTI_ActivityMT is published with the *SubsystemID* set to a unique identifier for MS or the MS subsystem responsible for AMTI command scheduling. The *Activity* sequence specified must be ordered temporally according to the AMTI command schedule, with earlier commands coming first. Each *Activity* has a unique *ActivityID* and the *CapabilityID* corresponds to one of the *Capability* fields specified in the *MA_AMTI_CapabilityMT* message published by MS. *Interactive* is set according to MS implementation (MS is only required to support *MA_AMTI_CapabilityMT AcceptedInterface* of *CAPABILITY_COMMAND*, which is not modifying activities, only scheduling or cancelling them). Each *Activity* *ActivityState* is populated appropriately as required by the schema. *EstimatedStartTime* and *EstimatedCompletionTime* are required to be populated, showing that capability commands are scheduled by MS as commanded.

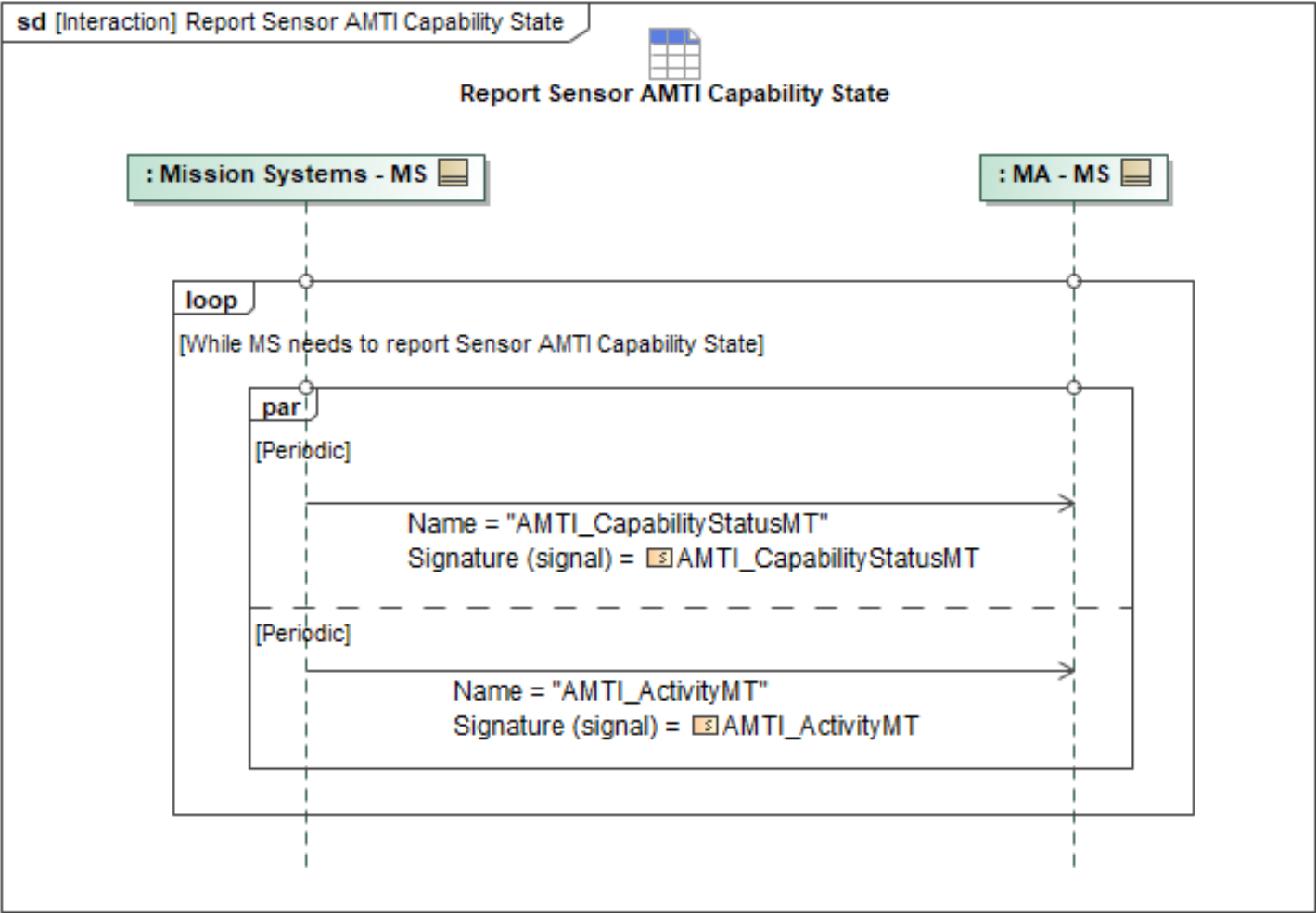


Figure 1-44: Report Sensor AMTI Capability State

Table 1-42: Report Sensor AMTI Capability State

Message Name (Name:Type)	Required Fields (Name:Type)
AMTI_ActivityMT: AMTI_ActivityMT	EstimatedCompletionTime: DateTimeType EstimatedStartTime: DateTimeType
AMTI_CapabilityStatusMT: AMTI_CapabilityStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.9 Report Sensor ESM Capability State

Mission Use Case: ESM

The Report Sensor ESM Capability State sequence diagram below specifies the interaction of MS communicating the Electronic Support Measures (ESM) sensor capability availability and scheduled activity to MA. This sequence is only present if ESM sensing capabilities exist in MS (see “Advertise Sensor ESM Capability for Search and Track”). MS publishes *ESM_ActivityMT* asynchronously in

response to changes in the scheduled ESM capability commands MS will execute (e.g. a new command was accepted).

ESM_ActivityMT is published with the *SubsystemID* set to a unique identifier for MS or the MS subsystem responsible for ESM command scheduling. The *Activity* sequence specified must be ordered temporally according to the ESM command schedule, with earlier commands coming first. Each *Activity* has a unique *ActivityID*. The *CapabilityID* corresponds to one of the *Capability* fields specified in the *ESM_CapabilityMT* message published by MS. *Interactive* is set according to MS implementation (MS is only required to support *ESM_CapabilityMT AcceptedInterface* of *CAPABILITY_COMMAND*, which is not modifying activities, only scheduling or cancelling them). Each *Activity* *ActivityState* is populated appropriately as required by the schema. *EstimatedStartTime* and *EstimatedCompletionTime* are required to be populated, showing that capability commands are scheduled by MS as commanded. The rest of the schema-optional fields are not required but may be specified for more information.

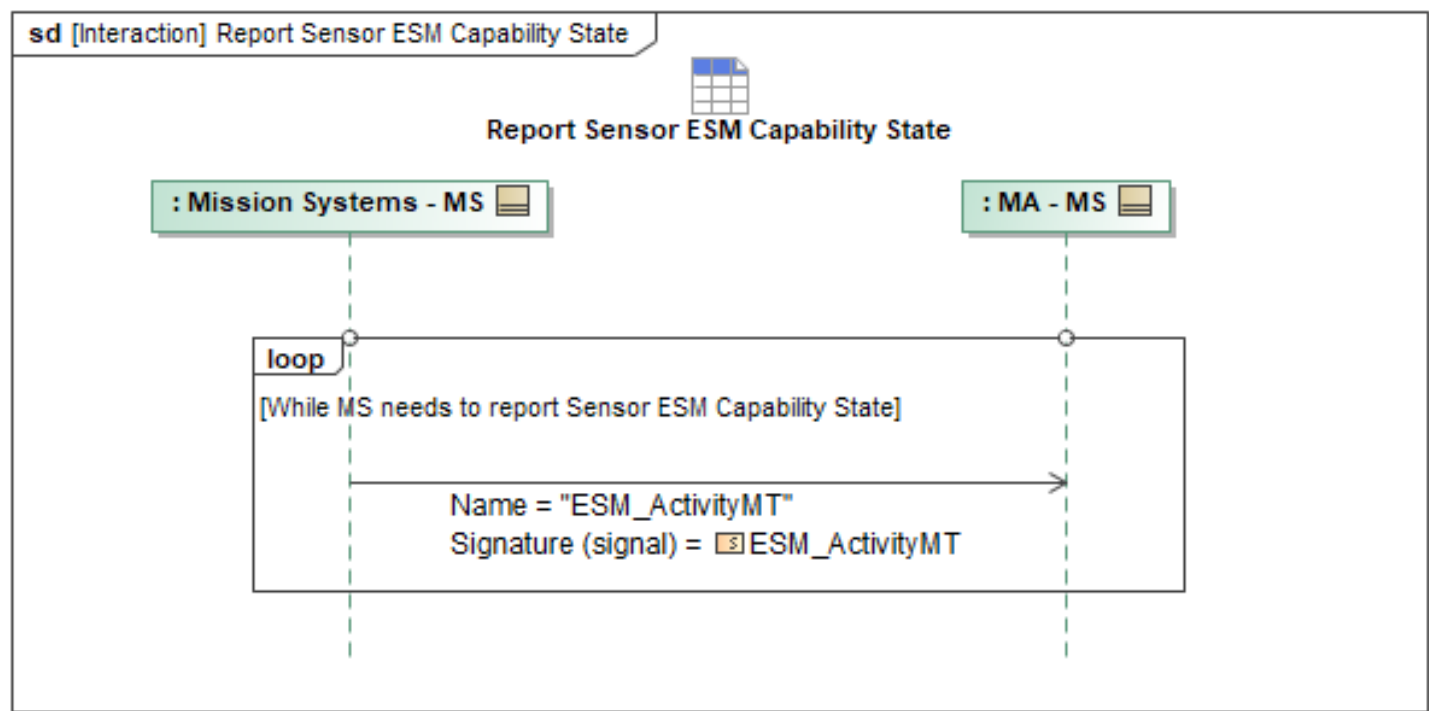


Figure 1-45: Report Sensor ESM Capability State

Table 1-43: Report Sensor ESM Capability State

Message Name (Name:Type)	Required Fields (Name:Type)
ESM_ActivityMT: ESM_ActivityMT	EstimatedStartTime: DateTimeType EstimatedCompletionTime: DateTimeType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.10 Report Sensor PO Capability State

Mission Use Case: PO

The Report Sensor PO Capability State sequence diagram specifies the interaction of MS communicating the Passive Optical (PO) sensor capability availability and scheduled activity to MA. This sequence is only present if PO sensing capabilities exist in MS (see “Advertise Sensor PO Capability for Search and Track”). MS periodically publishes *PO_CapabilityStatusMT* and *PO_ActivityMT* to MA. MS also publishes *PO_CapabilityStatusMT* asynchronously in response to changes in the PO capability availability to MA (e.g. sensors are in an unusable state). MS also publishes *PO_ActivityMT* asynchronously in response to changes in the scheduled PO capability commands MS will execute (e.g. a new command was accepted).

PO_CapabilityStatusMT is published with at least one *CapabilityStatus* specified. Each *CapabilityStatus* must have a unique identifier set in *CapabilityID* that corresponds to a *Capability* included in *PO_CapabilityMT* previously published by MS. The *Availability* enumeration is used to specify if MA can command this capability. *Availability* of *AVAILABLE* signifies that the capability is functioning, and MA can issue commands. All other *Availability* enumerations signify that MA cannot issue commands. No schema-optional fields are required but may be specified for more information.

PO_ActivityMT is published with the *SubsystemID* set to a unique identifier for MS or the MS subsystem responsible for PO command scheduling. The *Activity* sequence specified must be ordered temporally according to the PO command schedule, with earlier commands coming first. Each *Activity* has a unique *ActivityID*. The *CapabilityID* corresponds to one of the *Capability* fields specified in the *PO_CapabilityMT* message published by MS. *Interactive* is set according to MS implementation (MS is only required to support *PO_CapabilityMT AcceptedInterface* of *CAPABILITY_COMMAND*, which is not modifying activities, only scheduling or cancelling them). Each *Activity* *ActivityState* is populated appropriately as required by the schema. *EstimatedStartTime* and *EstimatedCompletionTime* are required to be populated, showing that capability commands are scheduled by MS as commanded.

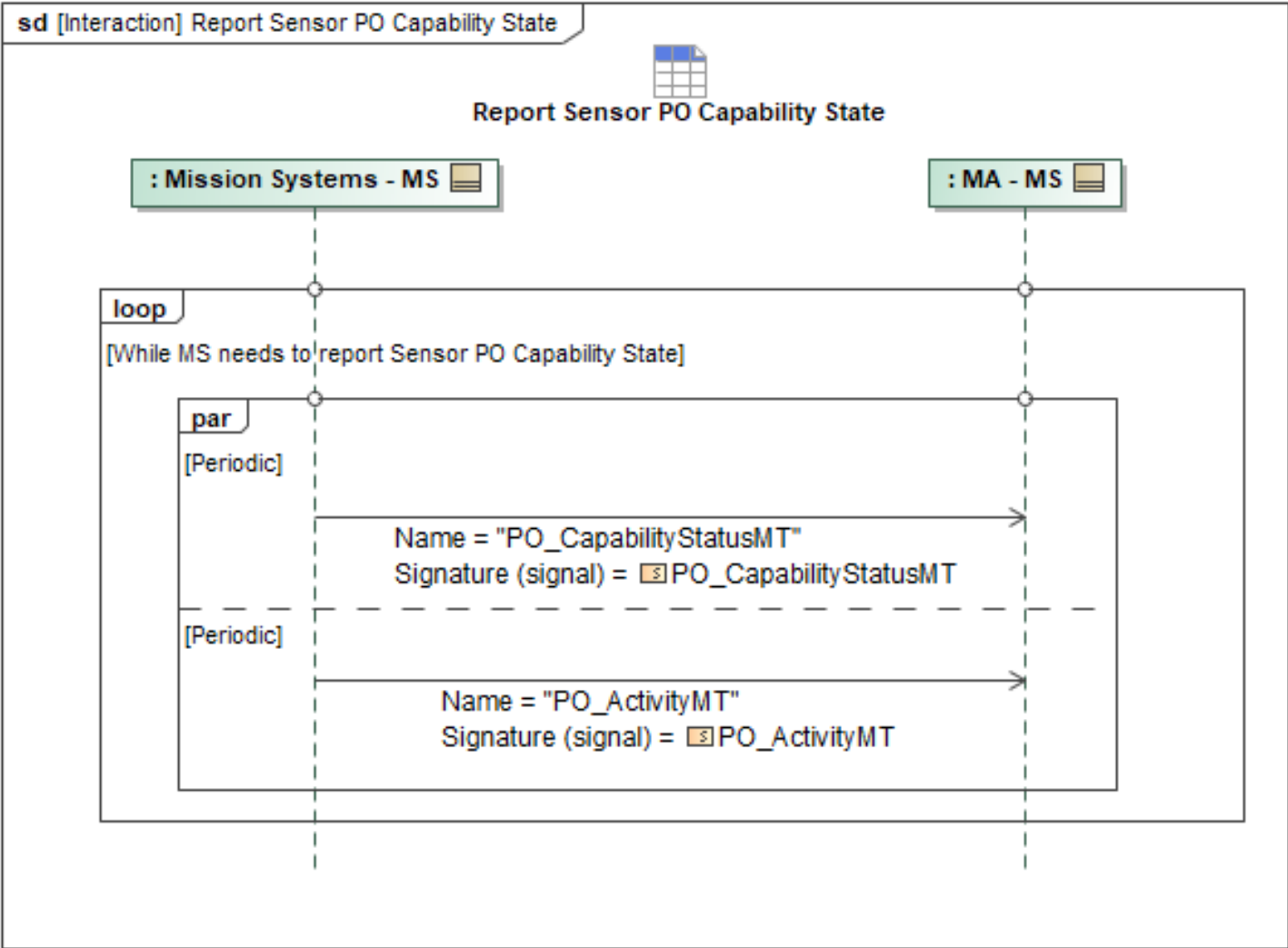


Figure 1-46: Report Sensor PO Capability State

Table 1-44: Report Sensor PO Capability State

Message Name (Name:Type)	Required Fields (Name:Type)
PO_ActivityMT: PO_ActivityMT	None
PO_CapabilityStatusMT: PO_CapabilityStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.11 Report Weather Radar Capability Observations

Mission Use Case: Weather Radar

If a platform has a weather radar capability, *WeatherRadarObservationMT* will be used to to send weather data collected by the radar from MS to MA.

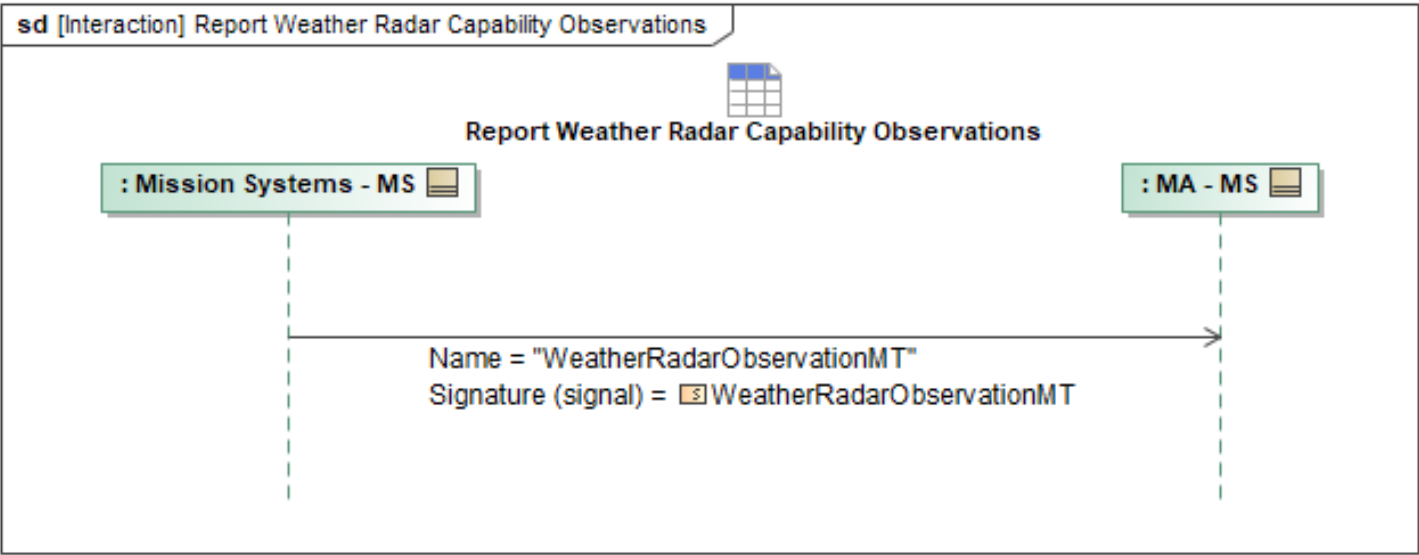


Figure 1-47: Report Weather Radar Capability Observations

Table 1-45: Report Weather Radar Capability Observations

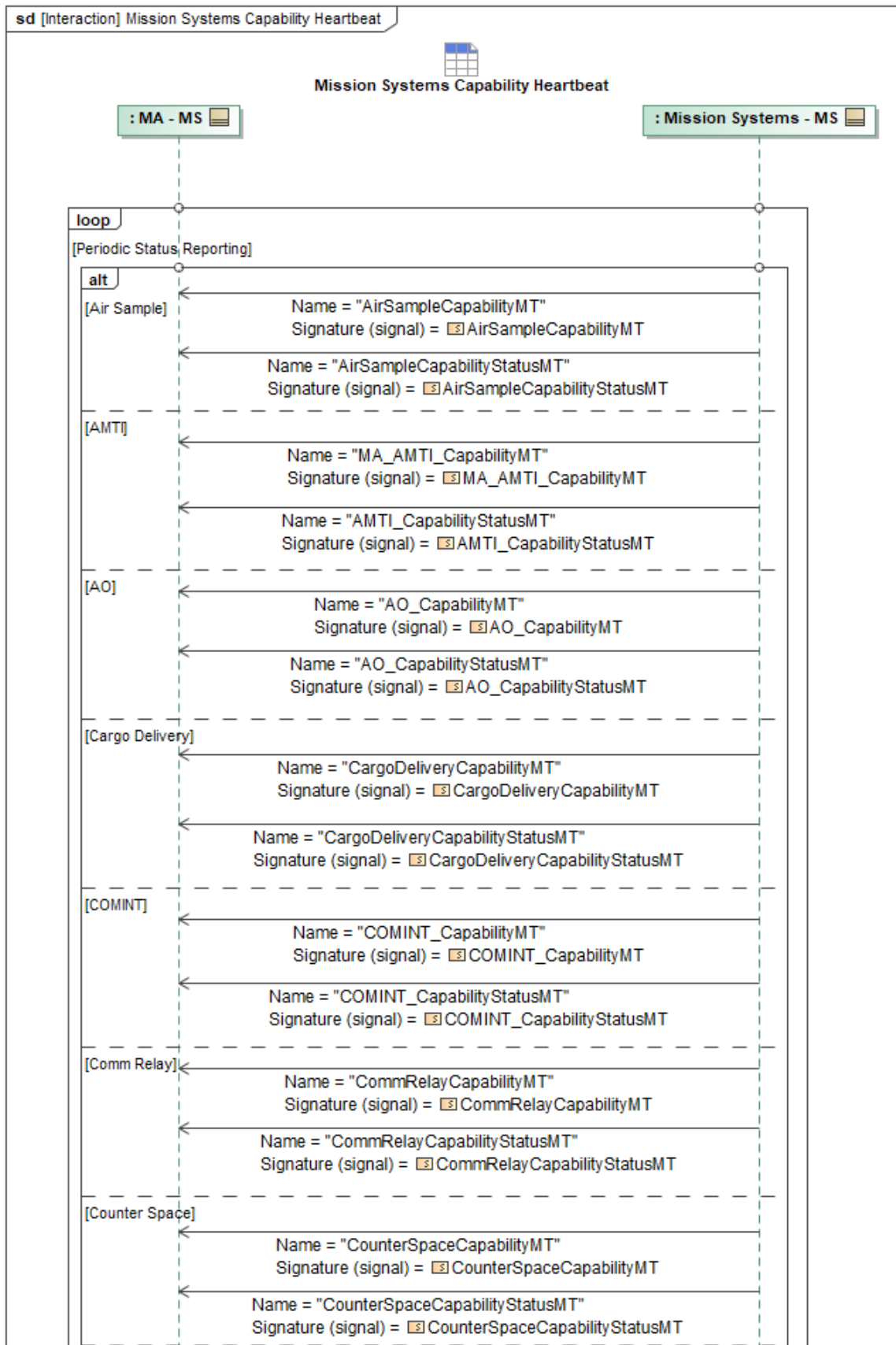
Message Name (Name:Type)	Required Fields (Name:Type)
WeatherRadarObservationMT: WeatherRadarObservationMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.9 Status Service Package

1.2.9.1 Mission Systems Capability Heartbeat

MS periodically sends **CapabilityMT* and **CapabilityStatusMT* messages to MA to describe the capabilities that MS is generally able to complete and those capabilities that are currently available for MS to execute, respectively. This periodic "heartbeat" continues throughout the duration of the mission. As these messages are published periodically, MA checks the provided status messages against the Minimum Essential Capabilities List (MECL) to identify potential contingencies.



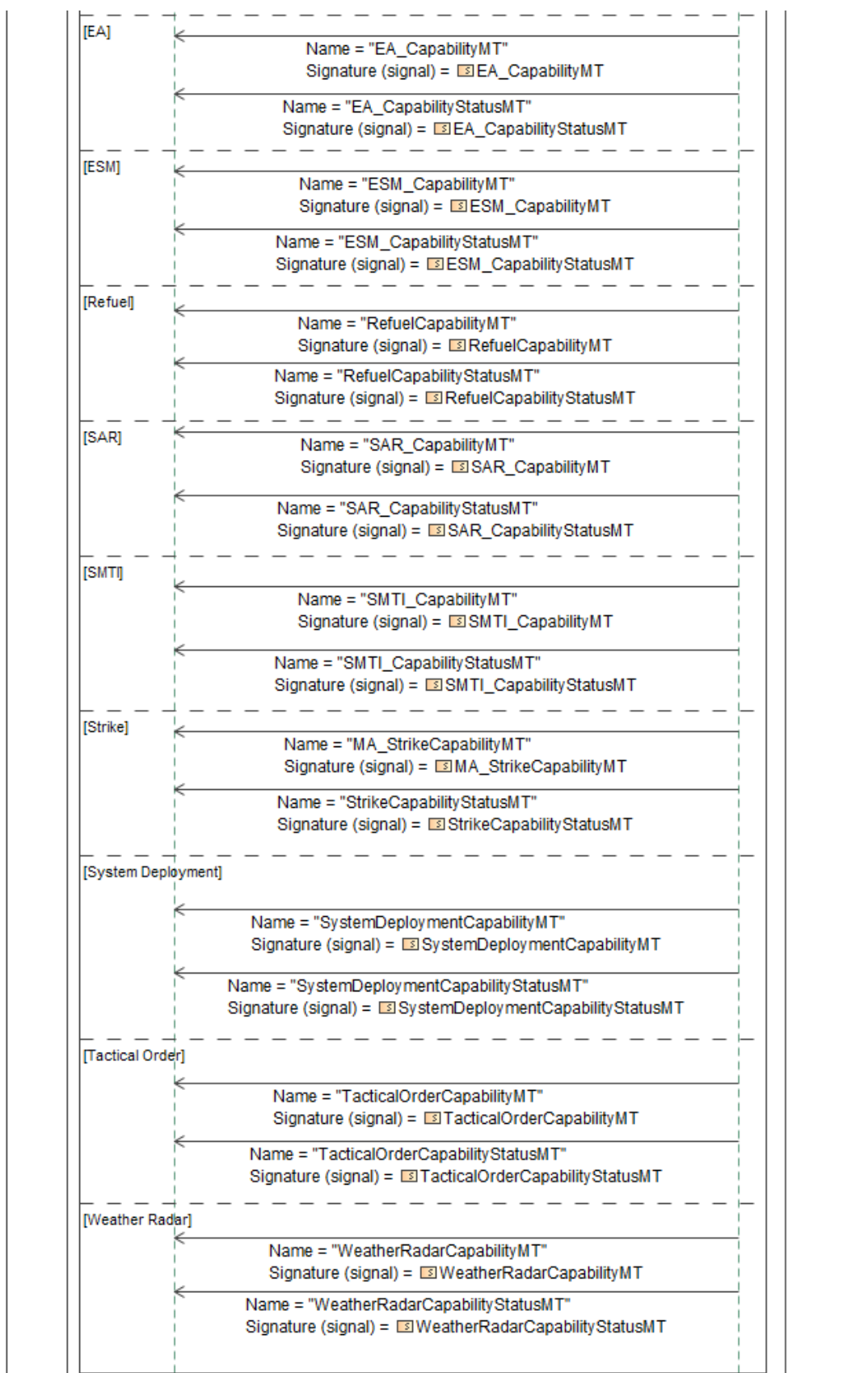


Figure 1-48: Mission Systems Capability Heartbeat

Table 1-46: Mission Systems Capability Heartbeat

Message Name (Name:Type)	Required Fields (Name:Type)
AirSampleCapabilityMT: AirSampleCapabilityMT	None
AirSampleCapabilityStatusMT: AirSampleCapabilityStatusMT	None
AMTI_CapabilityStatusMT: AMTI_CapabilityStatusMT	None
AO_CapabilityMT: AO_CapabilityMT	None
AO_CapabilityStatusMT: AO_CapabilityStatusMT	None
CargoDeliveryCapabilityMT: CargoDeliveryCapabilityMT	None
CargoDeliveryCapabilityStatusMT: CargoDeliveryCapabilityStatusMT	None
COMINT_CapabilityMT: COMINT_CapabilityMT	None
COMINT_CapabilityStatusMT: COMINT_CapabilityStatusMT	None
CommRelayCapabilityMT: CommRelayCapabilityMT	None
CommRelayCapabilityStatusMT: CommRelayCapabilityStatusMT	None
CounterSpaceCapabilityMT: CounterSpaceCapabilityMT	None
CounterSpaceCapabilityStatusMT: CounterSpaceCapabilityStatusMT	None
EA_CapabilityMT: EA_CapabilityMT	None
EA_CapabilityStatusMT: EA_CapabilityStatusMT	None
ESM_CapabilityMT: ESM_CapabilityMT	None
ESM_CapabilityStatusMT: ESM_CapabilityStatusMT	None
MA_AMTI_CapabilityMT: MA_AMTI_CapabilityMT	None
MA_StrikeCapabilityMT: MA_StrikeCapabilityMT	None
RefuelCapabilityMT: RefuelCapabilityMT	None
RefuelCapabilityStatusMT: RefuelCapabilityStatusMT	None
SAR_CapabilityMT: SAR_CapabilityMT	None
SAR_CapabilityStatusMT: SAR_CapabilityStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
SMTI_CapabilityMT: SMTI_CapabilityMT	None
SMTI_CapabilityStatusMT: SMTI_CapabilityStatusMT	None
StrikeCapabilityStatusMT: StrikeCapabilityStatusMT	None
SystemDeploymentCapabilityMT: SystemDeploymentCapabilityMT	None
SystemDeploymentCapabilityStatusMT: SystemDeploymentCapabilityStatusMT	None
TacticalOrderCapabilityMT: TacticalOrderCapabilityMT	None
TacticalOrderCapabilityStatusMT: TacticalOrderCapabilityStatusMT	None
WeatherRadarCapabilityMT: WeatherRadarCapabilityMT	None
WeatherRadarCapabilityStatusMT: WeatherRadarCapabilityStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.9.2 MS Responds to Query for Capabilities

Mission Systems Capability Messages are delivered from MS to MA in the Mission Systems Capability Heartbeat. This data is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may also respond with intermediate *QueryDataRequestStatusMT* messages with *RequestProcessingStateEnum* set to indicate additional in-process states, such as "QUEUED" and "PROCESSING".

When complete, the *QueryDataRequestStatusMT* message's *RequestProcessingStateEnum* is set to "COMPLETED" and contains the requested data.

When the original *QueryDataRequestMT*'s optional field *ResultsInNative* is populated with the matching message type, the underlying data management services instead publishes the requested data in its original or raw format as a separate message and indicates *QueryDataRequestStatusMT*.

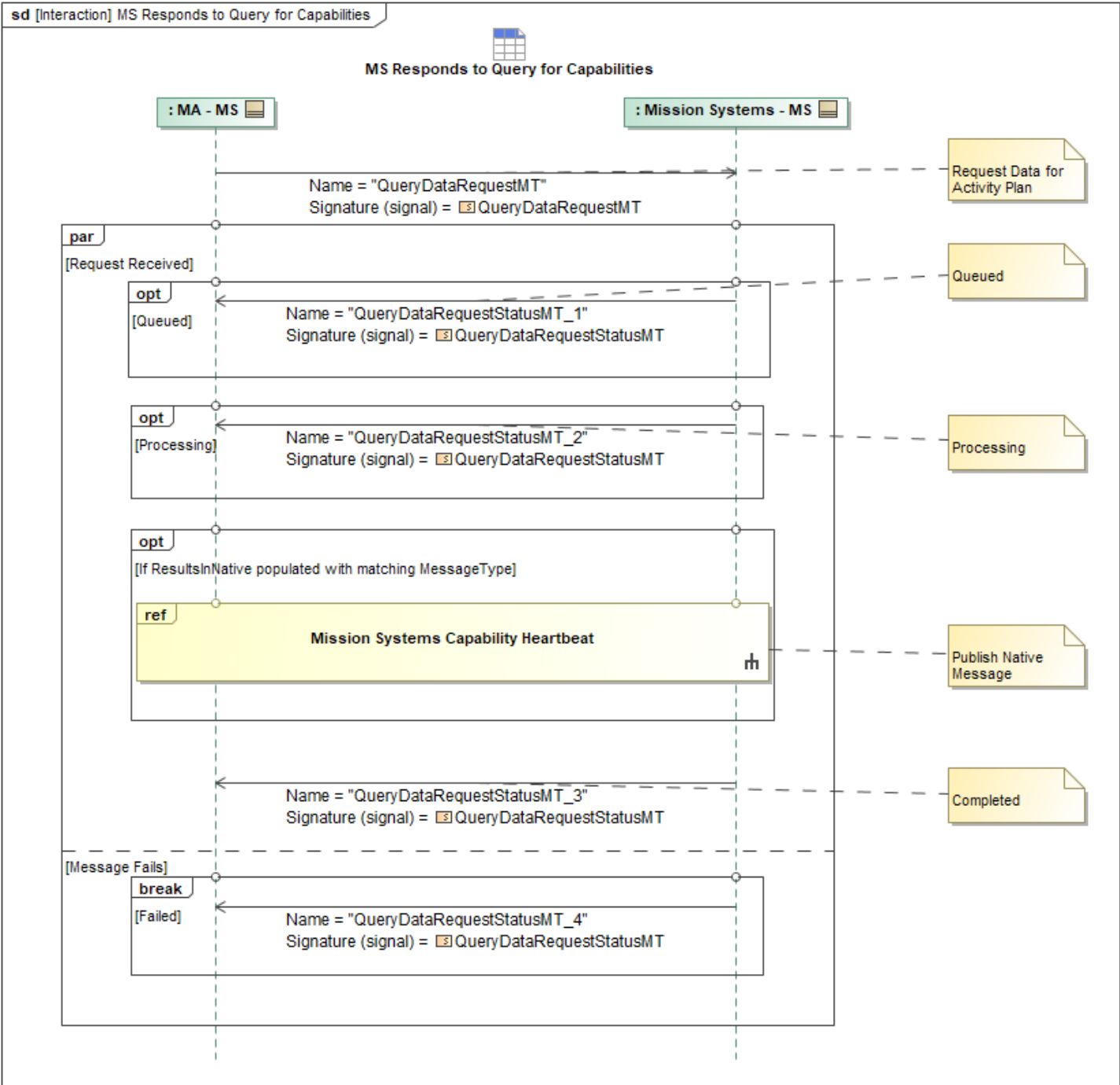


Figure 1-49: MS Responds to Query for Capabilities

Table 1-47: MS Responds to Query for Capabilities

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.9.3 Publish Periodic Passive Optical Sensor Configuration

Mission Use Case: PO

The following sequence shows the interface between MS and MA to support Sensor Configurations. To enable MA to be platform agnostic, MA does not need to have a priori knowledge of the vehicle and its subsystems. This is achieved via the *PO_CapabilityMT* message that "advertises the mostly static characteristics of a mission-ready Passive Optical Capability (of a Subsystem of System) so that it can be "discovered" and used. It is assumed that the underlying sensor subsystem routinely publishes the *PO_CapabilityMT* and *PO_CapabilityStatusMT* messages, which proceed to MA through the MS L1 interface. The Passive Optical sensing capability is exemplified in this sequence but exchanges following the same pattern are expected for all MS-based Capabilities.

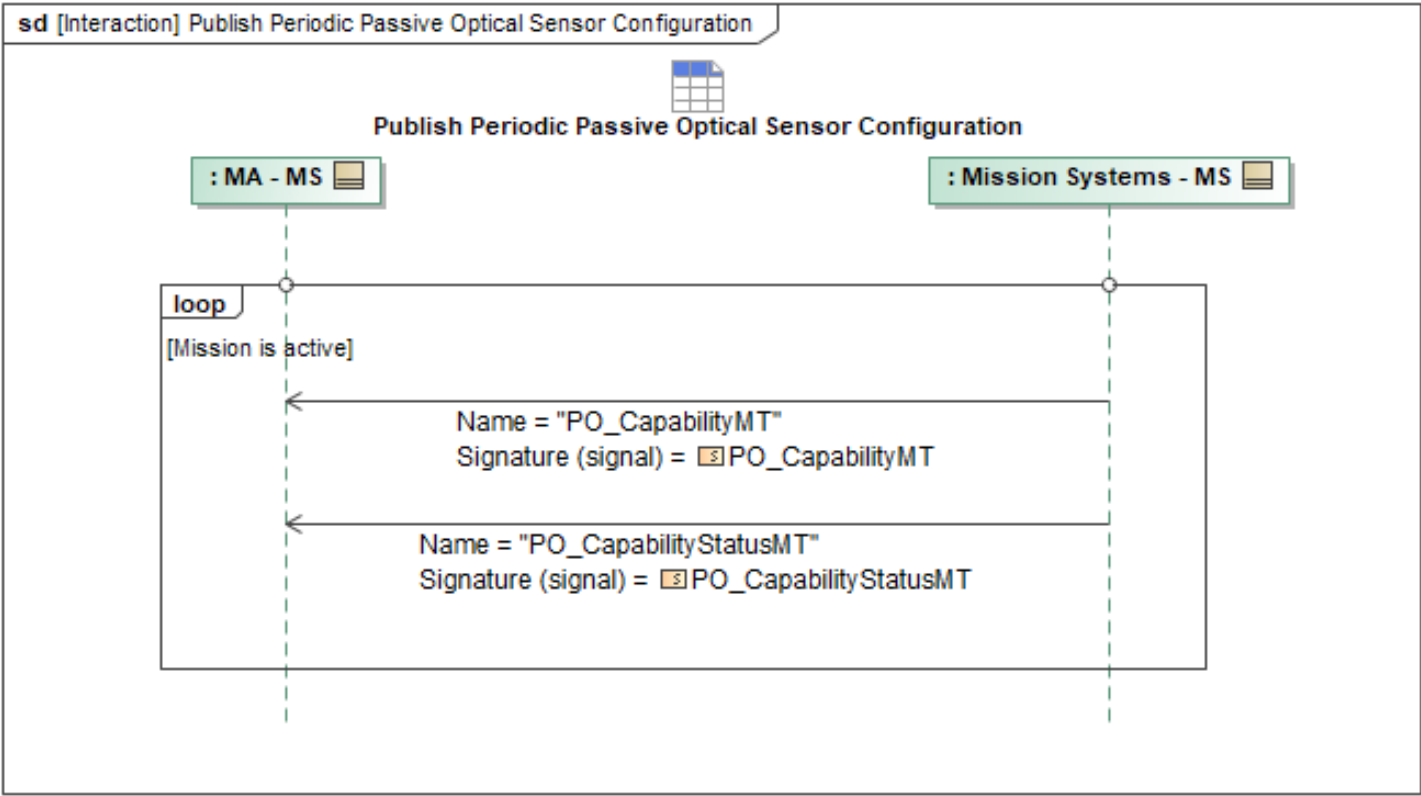


Figure 1-50: Publish Periodic Passive Optical Sensor Configuration

Table 1-48: Publish Periodic Passive Optical Sensor Configuration

Message Name (Name:Type)	Required Fields (Name:Type)
PO_CapabilityMT: PO_CapabilityMT	DomainPairing: EnvironmentPairingEnum PointingCapability: PointingTypeEnum PackageOperation: PackageOperationEnum Lens: PO_ComponentLensAssemblyType AvailableMDF: MDF_Type MessageOutput: PO_MessageOutputsEnum CapabilityFOR: FOR_Type TrackControlOptions: PO_TrackCapabilityTypeEnum PO_Component: PO_ComponentType TrackingRange: DoubleMinMaxType ATR_Filter: IdentityBasicType
PO_CapabilityStatusMT: PO_CapabilityStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.9.4 Receive Periodic Health, Status, & BIT Information

The Mission System reports subsystem status, service status, and subsystem BIT status continuously, at a configurable periodic rate, after initializing. Mission Autonomy uses these messages to verify the Mission System is operational. Subsystems contained within the Mission System will relay status information to Mission Autonomy using the *SubsystemStatusMT* message. The *SubsystemStatusMT* contains the fields *SubsystemID*, *SubsystemState* and other optional fields in order to indicate the subsystem's current state. Subsystems can also report periodic BIT information to Mission Autonomy using the *SubsystemBIT_StatusMT* message. BIT or Built In Test is a function of hardware and software to test its functionality to verify it is operating as intended and the *SubsystemBIT_StatusMT* message facilitates the reporting of this information to Mission Autonomy. The *SubsystemBIT_StatusMT* contains the fields *ActiveBIT*, *CompletedBIT* and *Fault* to allow the implementer to report the progress of an active BIT and information about completed BITs. Finally, the Mission System can report the status of services inside the subsystem using the *ServiceStatusMT* message. The *ServiceStatusMT* message contains the fields *TimeUp*, *ServiceStatus* and other optional fields that provide context on the health of the service. This periodic interaction allows Mission Autonomy to understand the current state of the Missions System subsystems and services.

The optional *ServiceConfigurationParamsMT* message informs Mission Autonomy which service specific configuration values can be manipulated. Mission Autonomy may use the *ServiceConfigurationChangeRequestMT* message to update any of the configurable parameters identified in the *ServiceConfigurationParamsMT*. If a service does not provide a *ServiceConfigurationParamsMT* it is assumed that the service will not respond to *ServiceConfigurationChangeRequestMT* messages. A configurable Service will respond to the Mission Autonomy with a *ServiceConfigurationChangeRequestStatusMT* which should indicate if the configuration change succeed (*RequestProcessingState* = "COMPLETED") or failed (*RequestProcessingState* = "FAILED").

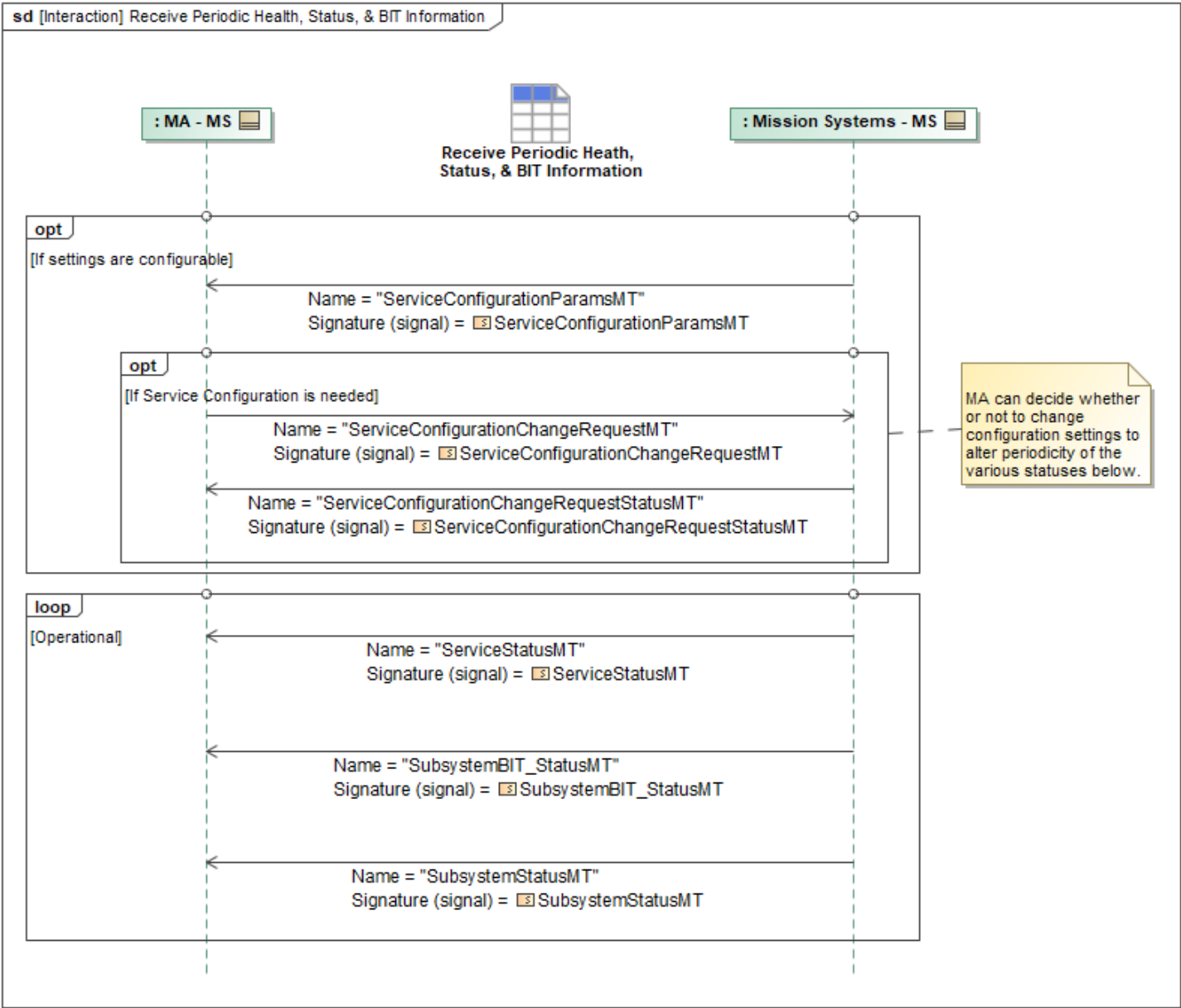


Figure 1-51: Receive Periodic Health, Status, & BIT Information

Table 1-49: Receive Periodic Health, Status, & BIT Information

Message Name (Name:Type)	Required Fields (Name:Type)
ServiceConfigurationChangeRequestMT: ServiceConfigurationChangeRequestMT	None
ServiceConfigurationChangeRequestStatusMT: ServiceConfigurationChangeRequestStatusMT	None
ServiceConfigurationParamsMT: ServiceConfigurationParamsMT	None
ServiceStatusMT: ServiceStatusMT	None
SubsystemBIT_StatusMT: SubsystemBIT_StatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
SubsystemStatusMT: SubsystemStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.9.5 Request Mission System Health, Status & BIT Information

MA can request health and status information from MS in order to verify MS is operational. Subsystem status can be requested through the *SubsystemStatusDataRequestMT* message. The *SubsystemStatusDataRequestMT* message specifies the *SubsystemID* it wants a status update from and the subsystem inside MS will respond with a *SubsystemStatusDataRequestStatusMT*. The Subsystem contained within MS will report status information to MA using the *SubsystemStatusMT* message. The *SubsystemStatusMT* contains the fields *SubsystemID*, *SubsystemState* and other optional fields in order to indicate the subsystem's current state. Similarly, MA can also request subsystem BIT and service status information by sending a *SubsystemBIT_CommandMT* or *ServiceStatusDataRequestMT* respectively. It is required that the *SubsystemID* or *ServiceID* for the subsystem or service the information was requested from is provided in the request message. Once MS receives either a *SubsystemBIT_CommandMT* or *ServiceStatusDataRequestMT* it will respond with a *SubsystemBIT_CommandStatusMT* or *ServiceStatusDataRequestStatusMT* detailing the processing state of the command or request. Subsystems report BIT information to MA using the *SubsystemBIT_StatusMT* message. Built In Test (BIT) is a function of hardware and software to test its functionality to verify it is operating as intended and the *SubsystemBIT_StatusMT* message facilitates the reporting of this information to MA. The *SubsystemBIT_StatusMT* contains the fields *ActiveBIT*, *CompletedBIT* and *Fault* to allow the implementer to report the progress of an active BIT and information about completed BITs. Finally, MS reports the status of services inside the subsystem using the *ServiceStatusMT* message. The *ServiceStatusMT* message contains the fields *TimeUp*, *ServiceStatus* and other optional fields that provide context on the health of the service. This interaction allows MA to understand the current state of MS subsystems and services on an on-demand basis.

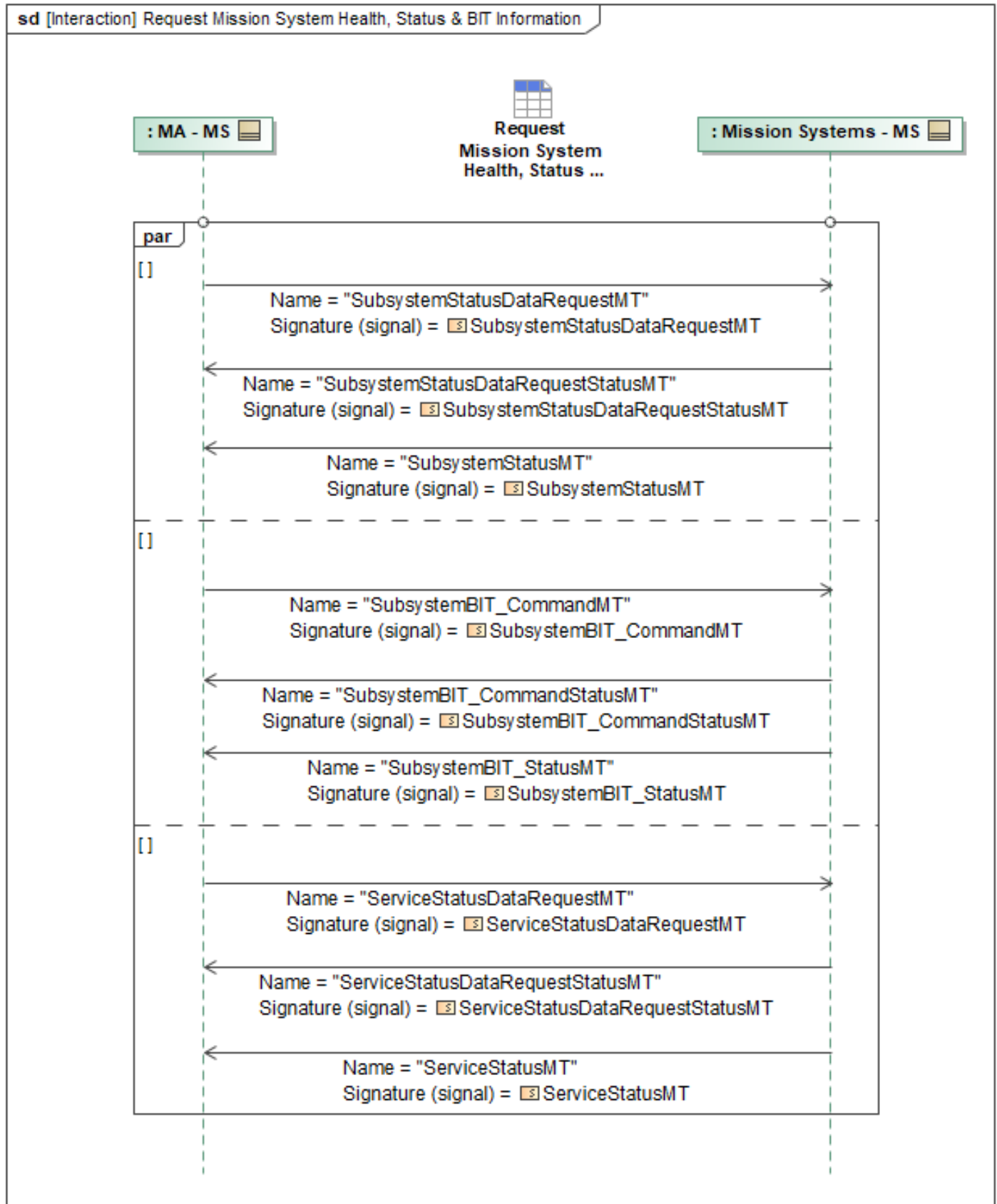


Figure 1-52: Request Mission System Health, Status & BIT Information

Table 1-50: Request Mission System Health, Status & BIT Information

Message Name (Name:Type)	Required Fields (Name:Type)
ServiceStatusDataRequestMT: ServiceStatusDataRequestMT	None
ServiceStatusDataRequestStatusMT: ServiceStatusDataRequestStatusMT	None
ServiceStatusMT: ServiceStatusMT	None
SubsystemBIT_CommandMT: SubsystemBIT_CommandMT	None
SubsystemBIT_CommandStatusMT: SubsystemBIT_CommandStatusMT	None
SubsystemBIT_StatusMT: SubsystemBIT_StatusMT	CompletedBIT: SubsystemCompletedBIT_Type Fault: SubsystemFaultType ActiveBIT: SubsystemActiveBIT_Type
SubsystemStatusDataRequestMT: SubsystemStatusDataRequestMT	None
SubsystemStatusDataRequestStatusMT: SubsystemStatusDataRequestStatusMT	None
SubsystemStatusMT: SubsystemStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10 Weapon Employment Package

1.2.10.1 Abort Release Sequence

Once a task type of strike has been commanded to be executed, MA may need to abort the weapon release sequence. To abort the strike task, MA will send a *MA_TaskCommandMT* to MS with the corresponding *TaskID* and *CommandState* of *CANCEL*. MS will respond with *MA_TaskStatusMT* with *ExecutionState* set to *CANCELLED* and Mission Store Status messages from MS to verify the abort was successful.

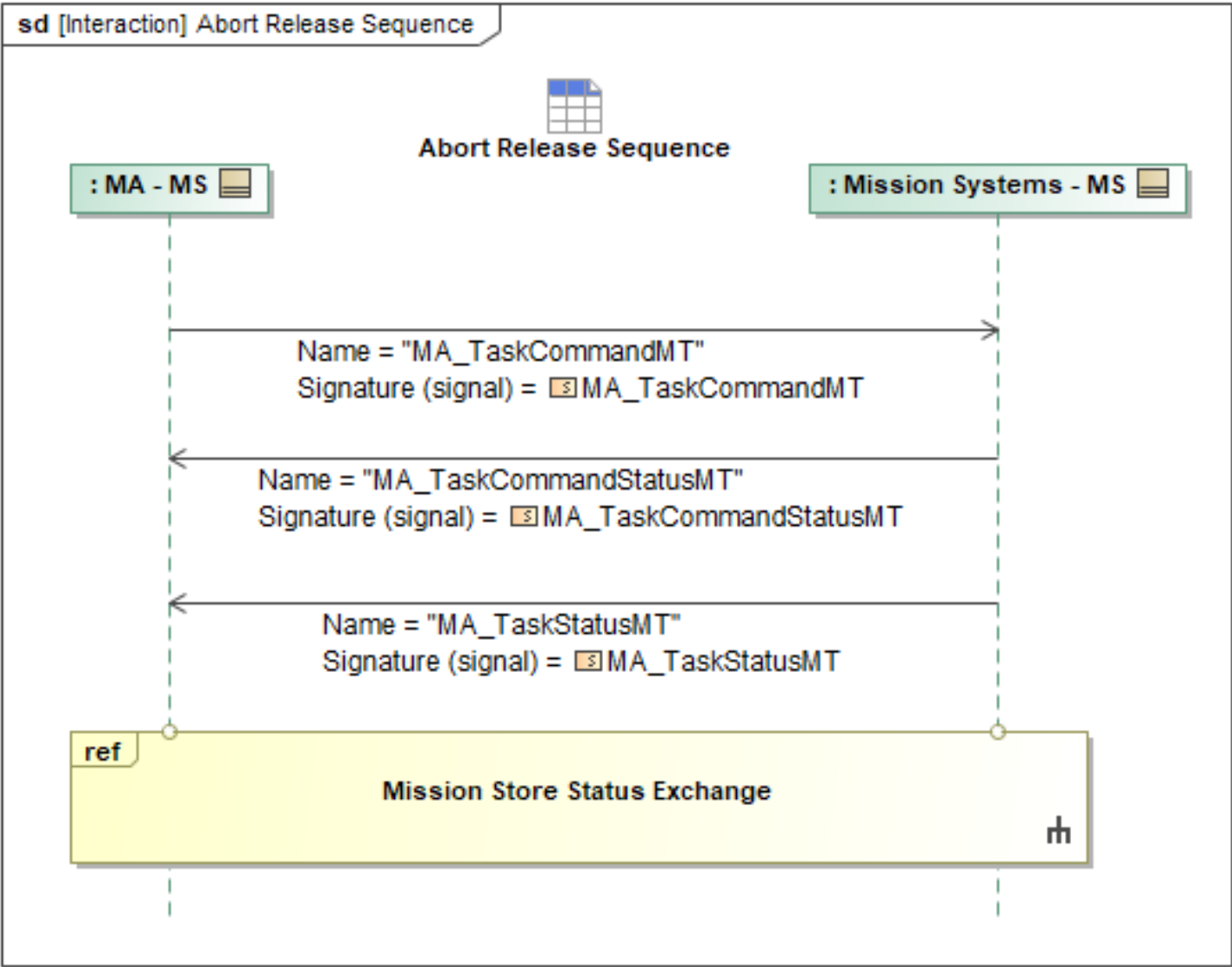


Figure 1-53: Abort Release Sequence

Table 1-51: Abort Release Sequence

Message Name (Name:Type)	Required Fields (Name:Type)
MA_TaskCommandMT: MA_TaskCommandMT	CANCEL: CommandState: CommandStateEnum TaskID: TaskID_Type
MA_TaskCommandStatusMT: MA_TaskCommandStatusMT	None
MA_TaskStatusMT: MA_TaskStatusMT	CapabilityID: CapabilityID_Type CANCELED:

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.2 Confirm Threat ID

The following sequence is meant to standardize how MA assesses MS data to confirm identification criteria for each target in the queue or matrix. The *EntityMT* represents a track in the battle space and is ingested by MA through the MS interface. MS will classify the detected enemy as friendly, hostile, or ambiguous. The *StandardIdentityType* contains the enumeration, *StandardIdentityEnum* that can classify the target as either *PENDING*, *UNKNOWN*, *ASSUMED_FRIEND*, *FRIEND*, *NEUTRAL*, *SUSPECT*, *HOSTILE*, or *UNDEFINED*. The result of target classification is included in the corresponding *EntityMT* and can be used in combination with active engagement criteria to ensure entity ID criteria is met during or in the lead up to an engagement. MA may also require updates to an *EntityMT* to maintain positive custody during weapons employment activities.

The *ThreatType* field will also be passed to validate the ID of the track. The *ThreatType* categorizes the threat platform to validate it is a hostile entity.

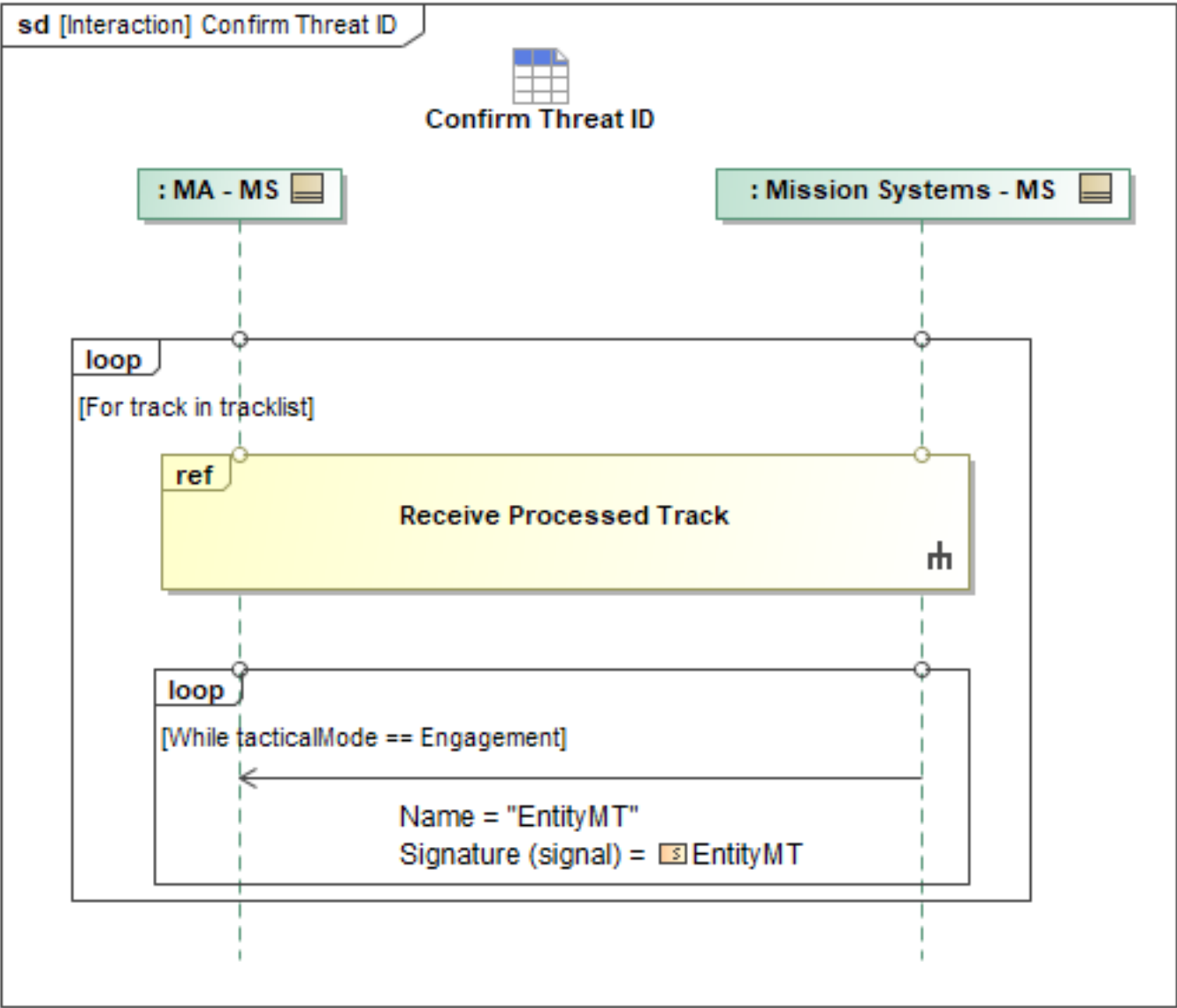


Figure 1-54: Confirm Threat ID

Table 1-52: Confirm Threat ID

Message Name (Name:Type)	Required Fields (Name:Type)
EntityMT: EntityMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.3 Exchange DLZ

MA sends out a *DLZ_RequestMT* with required field *ResultsInNativeMessage* set to *TRUE* to MS and receives a *DLZ_RequestStatusMT* response back to provide processing state. When processing is complete MS sends a *DLZ_MT* back to MA and utilizes *DLZ_RequestID* to reference the originating request.

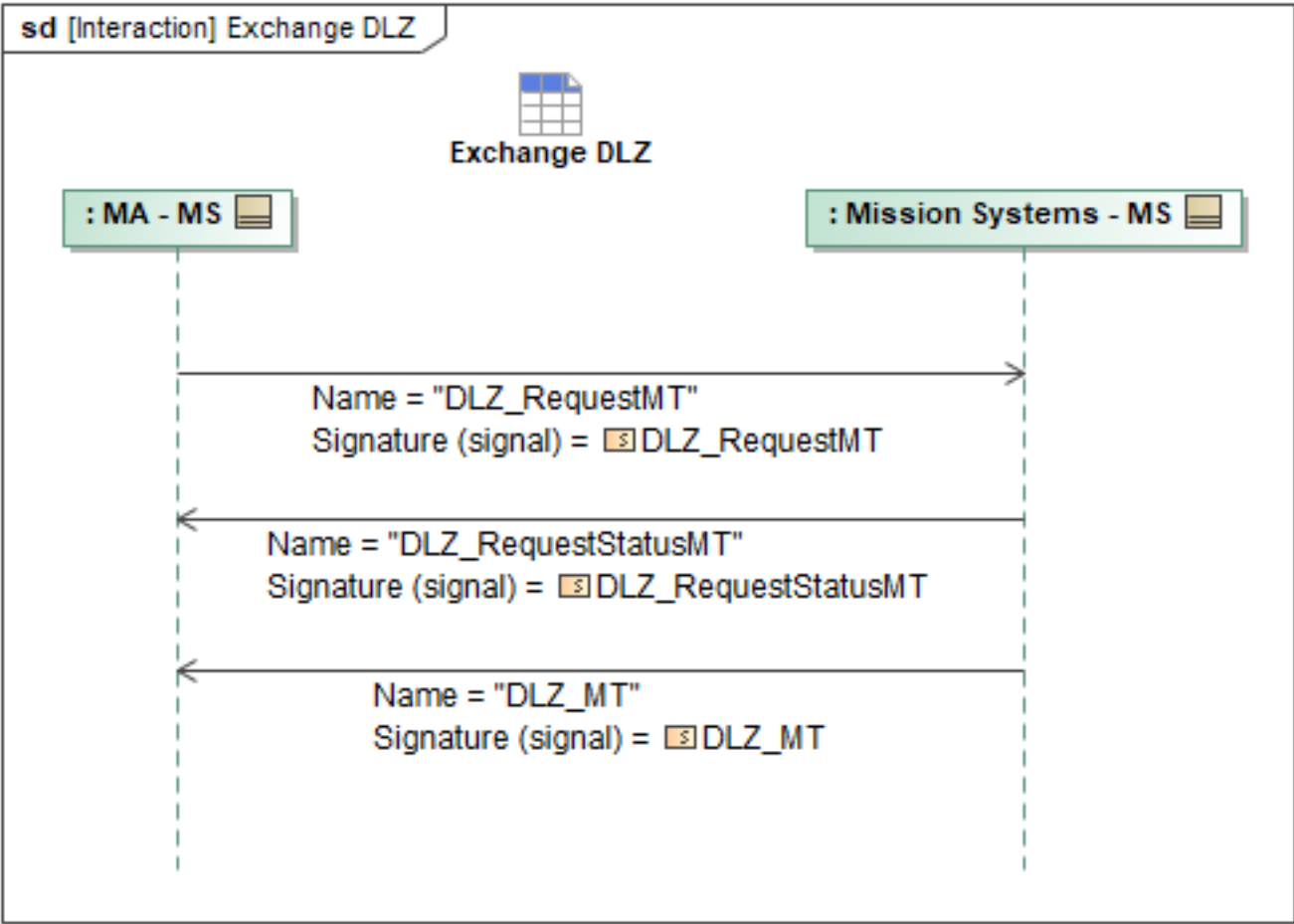


Figure 1-55: Exchange DLZ

Table 1-53: Exchange DLZ

Message Name (Name:Type)	Required Fields (Name:Type)
DLZ_MT: DLZ_MT	DLZ_RequestID: RequestID_Type
DLZ_RequestMT: DLZ_RequestMT	ResultsInNativeMessage: boolean
DLZ_RequestStatusMT: DLZ_RequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.4 Exchange WEZ

MA will receive the Weapon Engagement Zone (WEZ) through the MS interface. The WEZ represents a region around the ownship in which an adversary is vulnerable to a shot and is defined by the range of the weapon system. The WEZ can be utilized in addition to the Dynamic Launch Zone (DLZ) in the case that there is no target state data, targets have not been assigned, or target state data lacks fidelity. The WEZ is made up of a Weapon's various ranges. Those ranges are a function of the weapon type, the azimuth relative to the ownship which the weapon would need to turn to, the ownship state/position data, and a target's heading, velocity and altitude. The *MA_WEZ_RequestMT* message uses the *StoreInformation* field to identify the specific capability the WEZ is being requested

for. The schema then uses the *WEZ_FieldOfView* field to identify the region for the WEZ by indicating the desired azimuth range around the ownship for which the capability ranges are being requested for. For example, the *FRONT_WEZ* enumeration represents an array of azimuths from 0 to 180 deg off the nose of the ownship with a default 5 deg step defining the array. This array of ranges can be pre-calculated by the service prior to flight or calculated upon the request while in flight. To account for ownship and potential target state/position data, the implementer has a choice of options that are outlined by the *WEZ_ChoiceType*.

The *StaticWEZ* choice uses the ownship's position data as reported by the vehicle interface and then the service assumes that the target heading is pointed at the ownship per azimuth outlined in the *WEZ_FieldOfView*. The service also will assume the target as the same altitude and velocity as the ownship. This is the most static, least dynamic representation of the WEZ.

The *PlannedStaticWEZ* choice allows for the user to specify a planned ownship state/position using the *PlannedReleaseConditions* field. The service then also assumes the target's heading is pointed at the ownship per azimuth but assumes that its velocity and altitude is the same as what is identified in the *PlannedReleaseConditions* field.

The *PredictedWEZ* choice allows for a more dynamic WEZ and requires the user to input the target state/position data using the *PredictedTargetStateData* field. This allows the user to input a specific velocity, altitude, and target heading aspect off the ownship per azimuth. The service would then subscribe to the ownship state/position data as reported by the vehicle interface.

Lastly, the *PlannedPredictedWEZ* choice, allows the implementer to specify the ownship and predicted target state/position data using both the *PlannedReleaseConditions* and *PredictedTargetStateData* fields.

This WEZ request is then acknowledged by MS using *MA_WEZ_RequestStatusMT*. With the store information and required inputs for a capability range provided by MA, MS can now send MA the WEZ specific to that store. MS does so by providing the *MA_WEZ* message in which provides a various set of ranges per azimuth requested, which builds out the requested 2D region. The required ranges that define the WEZ are *RangeMinimum*, *RangeTurnAndRun*, *RangeProbabilityOfIntercept*, *RangeMaxGuidance*, *RangeOptimal*, and *RangeMaxAero*. The WEZ UCI message structures are very similar to that of the DLZ messages with the difference being WEZ does not require target state data

Potential WEZ Use Cases:

- Course Planning - Plan a specific course based on weapon capability
- Set up Formation - Specific ACP spacing based on weapon Capability
- Projection of Power consideration - radius overlapping the correct amount
- Swarm impacts
- CAP Geometry - CAP radius based off weapon capability
- Task Allocation - Allocate tasks based on weapon capability - Platform A is in the range of detected entity (not a target yet) - DLZ would follow on as entity becomes target set

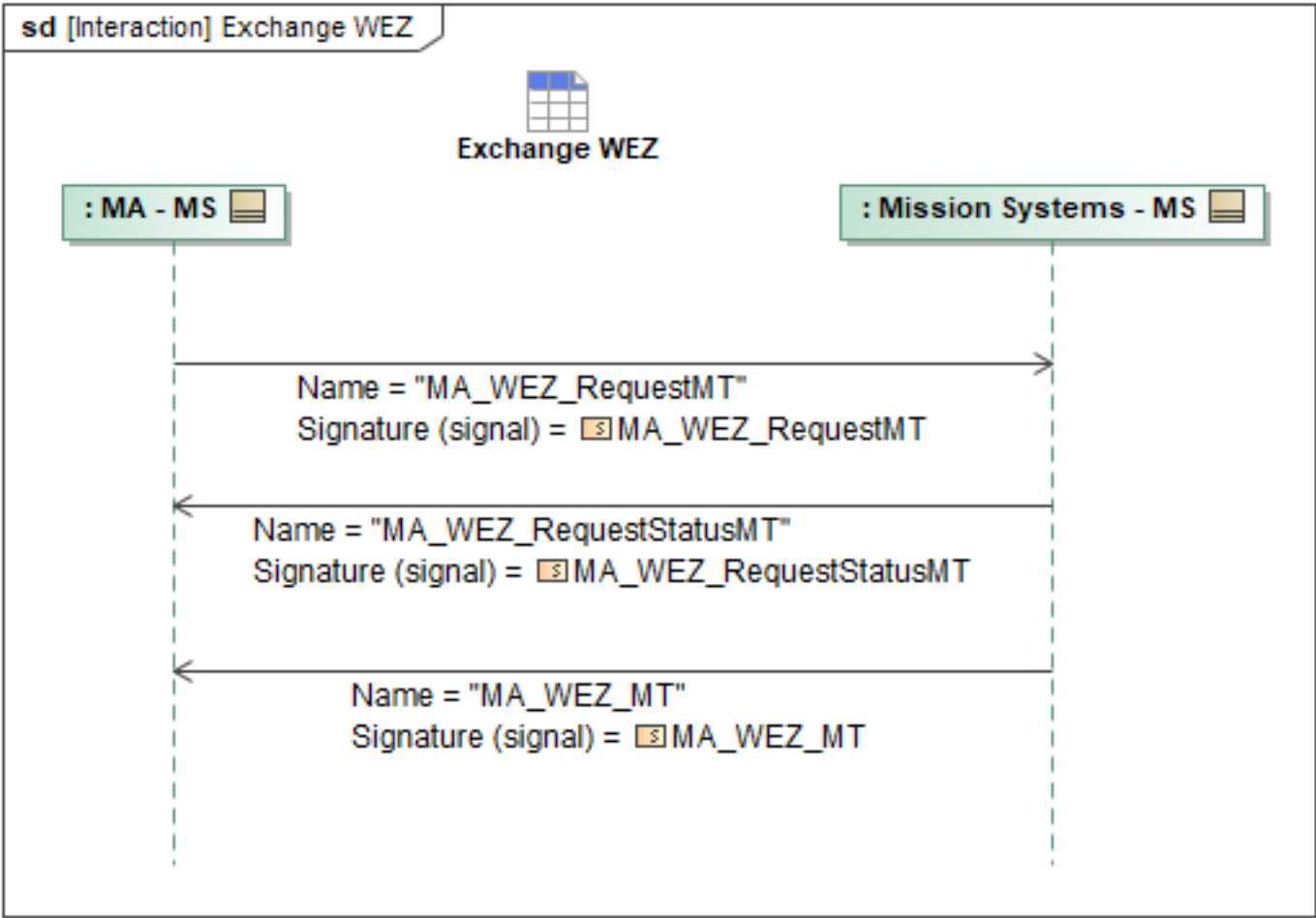


Figure 1-56: Exchange WEZ

Table 1-54: Exchange WEZ

Message Name (Name:Type)	Required Fields (Name:Type)
MA_WEZ_MT: MA_WEZ_MT	None
MA_WEZ_RequestMT: MA_WEZ_RequestMT	None
MA_WEZ_RequestStatusMT: MA_WEZ_RequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.5 Fire Command Exchange

As a prerequisite to initiating a weapons release, MA must assess and determine the threat prioritization, maintain custody of hostile threat(s), and obtain release consent authorization in accordance with the engagement criteria. MA will update *MA_Task.MessageData.TaskType.Strike* contents as required to support weapons employment. MS will respond with *MA_TaskStatus* with the selected *CapabilityID* and a *TaskID*. MA will then send a *TaskCommand* message to MS to command the execution of the previously referenced *TaskID*. MS will respond to *TaskCommand* with *TaskCommandStatus* indicating the command was accepted or rejected. MS will periodically update

MA with *MA_TaskStatus* on the progress and state changes of the task being executed. At task completion, MA will also receive *StrikeCapability* and *StrikeCapabilityStatus* to confirm the weapon (*CapabilityID*) was successfully released.

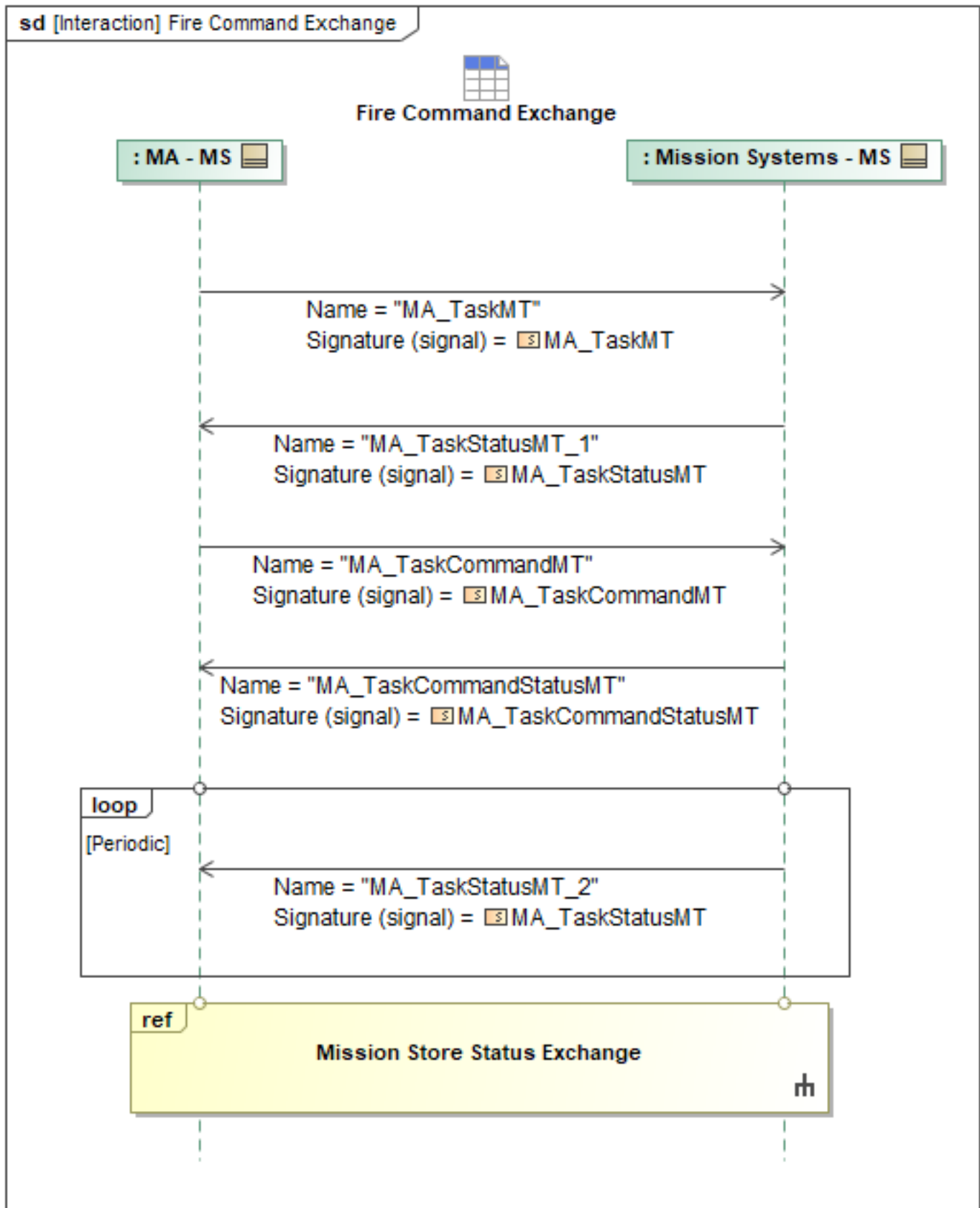


Figure 1-57: Fire Command Exchange

Table 1-55: Fire Command Exchange

Message Name (Name:Type)	Required Fields (Name:Type)
MA_TaskCommandMT: MA_TaskCommandMT	None
MA_TaskCommandStatusMT: MA_TaskCommandStatusMT	None
MA_TaskMT: MA_TaskMT	None
MA_TaskStatusMT_1: MA_TaskStatusMT	None
MA_TaskStatusMT_2: MA_TaskStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.6 Jettison Release Sequence

The jettison function enables MA to send a jettison task to MS to perform an off nominal release sequence of the weapon regardless of the weapon's powered state. MA will send a task, *MA_Task.MessageData.TaskType.Jettison*, to MS to initiate the Jettison sequence. The *MA_JettisonTaskType* requires both *MA_JettisonStoreSelection* and *MA_StoreJettisonType* to be provided. *MA_JettisonStoreSelection* specifies either *JettisonAllCapableWeapons (EmptyType)* to jettison all weapons capable of being jettisoned, or the *CapabilityID(s)* of the stores to be jettisoned. *MA_StoreJettisonType* has two enumerations of Standard or Emergency. Standard jettison will inform MS that if the weapon is powered on, it is to zeroize and power down before jettison. Emergency jettison will inform MS that the weapon is to be released immediately regardless of powered state. MS will respond to *MA_Task* with *MA_TaskStatusMT* with each of the *CapabilityID(s)* for the selected stores for jettison. MA will execute the task with *MA_TaskCommandMT*. MS will respond with *MA_TaskCommandStatusMT*. MA will monitor *MA_TaskStatusMT* and Mission Store Status sequence for the completion of the task.

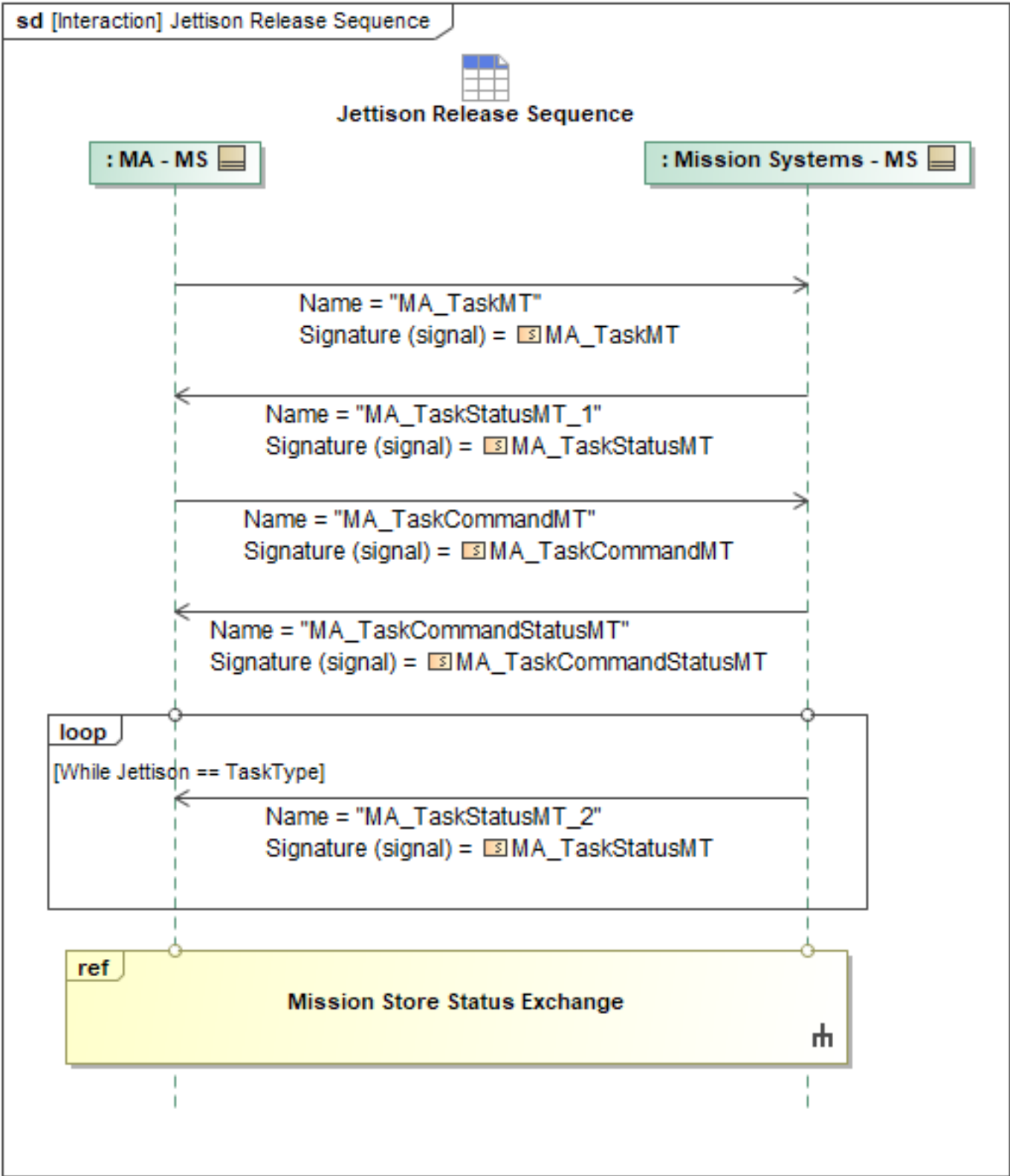


Figure 1-58: Jettison Release Sequence

Table 1-56: Jettison Release Sequence

Message Name (Name:Type)	Required Fields (Name:Type)
MA_TaskCommandMT: MA_TaskCommandMT	None
MA_TaskCommandStatusMT: MA_TaskCommandStatusMT	None
MA_TaskMT: MA_TaskMT	Jettison: MA_JettisonTaskType
MA_TaskStatusMT_1: MA_TaskStatusMT	Traceability: RequirementStatusTraceabilityType
MA_TaskStatusMT_2: MA_TaskStatusMT	Traceability: RequirementStatusTraceabilityType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.7 Mission Store Release Initialization Sequence

This sequence is the start of the weapons employment release sequence and initiates mission store initialization. MA will send the *TaskID* with *MA_Task.MessageData.TaskType.Strike*. The *Strike* task shall include the following: *TaskType.Strike.Target (Entity or SpecificType)*, *TaskType.Strike.WeaponList.Weapon.RequiredWeapon*, *TaskType.Strike.WeaponList.Weapon.StoreInformation.StoreType* and *TaskType.Strike.WeaponList.Weapon.StoreInformation.StoreQuantity*. MS will reply with *MA_TaskStatusMT* with the assigned CapabilityID(s) via *Traceability.CapabilityID*. MS will periodically send Mission Store Status Exchange to provide details of each of the installed capabilities.

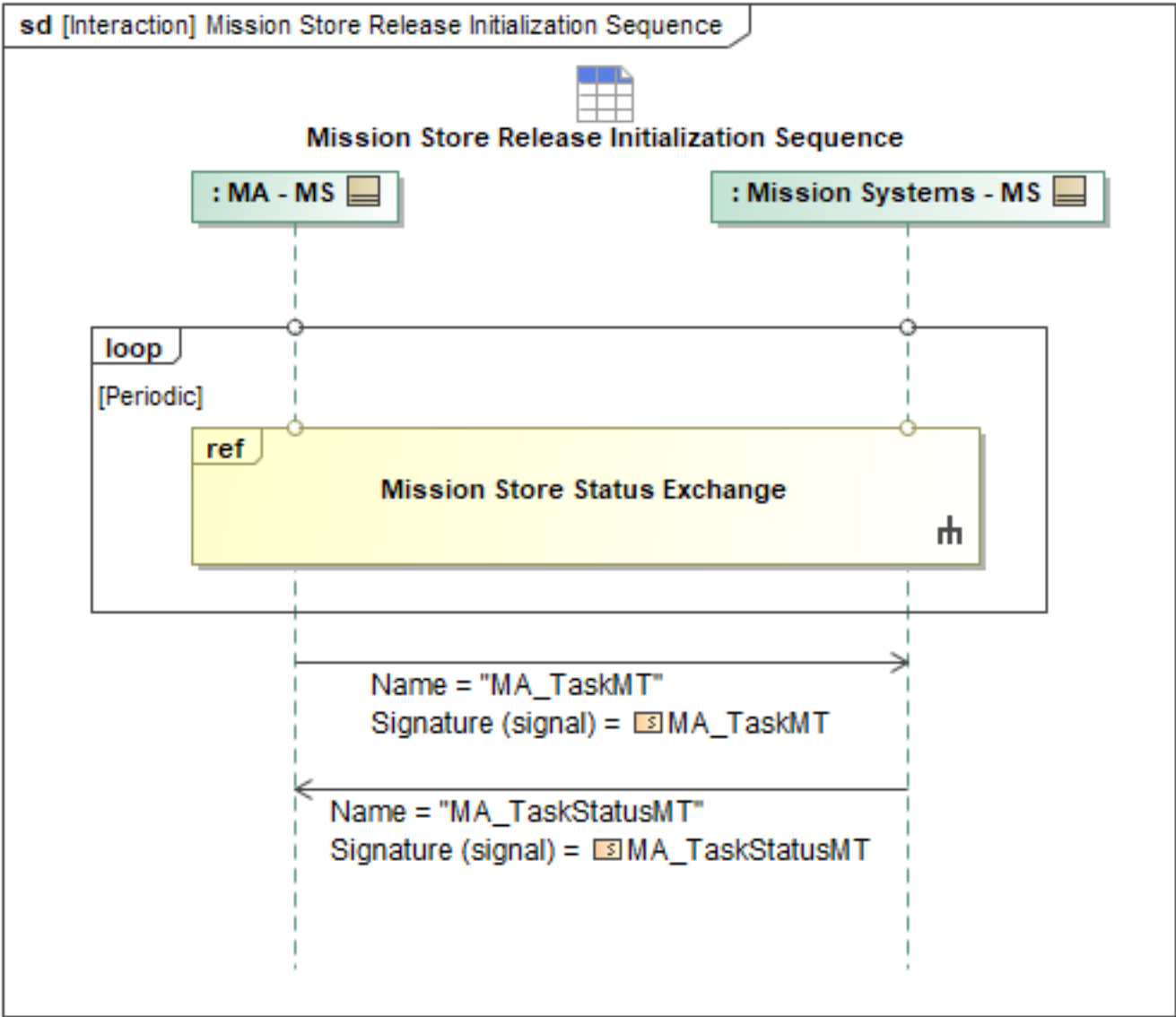


Figure 1-59: Mission Store Release Initialization Sequence

Table 1-57: Mission Store Release Initialization Sequence

Message Name (Name:Type)	Required Fields (Name:Type)
MA_TaskMT: MA_TaskMT	Weapon: StrikeTaskWeaponType StoreInformation: WeaponeeringStoreType Strike: StrikeTaskType
MA_TaskStatusMT: MA_TaskStatusMT	CapabilityID: CapabilityID_Type

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.8 Mission Store Status Exchange

The store management system within the MS will periodically publish *StrikeCapabilityStatus* for each

store capability ID regardless of the store presence. MA will utilize *StrikeCapabilityStatusMT* and *MA_StrikeCapabilityMT* to maintain persistent awareness of the store state, loaded mission data, and mission store parameters in accordance with the selected Mission Data File (MDF). In the event that MS has an unexpected or undesirable state reported, MS is then required to update *StrikeCapabilityStatus.MessageData.CapabilityStatus.AvailabilityInfo.AvailabilityReason*.

Upon strike tasking from MA to MS and the strike command being sent internal to MS, the Strike Activity will become true. In the case the Strike Activity becomes true, MS must pass the *MA_StrikeActivityMT_1* to MA to include the *ActivityState* of the *MA_StrikeActivity*. The *ActivityState* pertains to the state of the activity and includes enumerations for *DISABLED*, *ENABLED*, *ACTIVE_UNCONSTRAINED*, *ACTIVE_PARTIALLY_CONSTRAINED*, *ACTIVE_FULLY_CONSTRAINED*, *COMPLETED*, *FAILED*, *DELETED*. MS should publish the *MA_StrikeActivityMT* once to MA regardless of the *ActivityState*. If the *ActivityState* is not equal to *DISABLED*, *FAILED* or *DELETED* MS will publish the *MA_StrikeActivityMT_2* to MA periodically throughout the strike activity. Optionally, MS can provide MA with the *MunitionActive* field that pertains to the specific weapon state data and orientation during weapon flyout. Active weapon states include *TARGETED*, *REQUESTED*, *SEEKER_INACTIVE*, *SEEKER_ACTIVE*, *FUZE_INACTIVE*, *FUZE_ACTIVE*, *NOT_FULLY_AUTONOMOUS*, *FULLY_AUTONOMOUS*, *DEFEATED*, *TIMED_OUT*, *SUCCESS*.

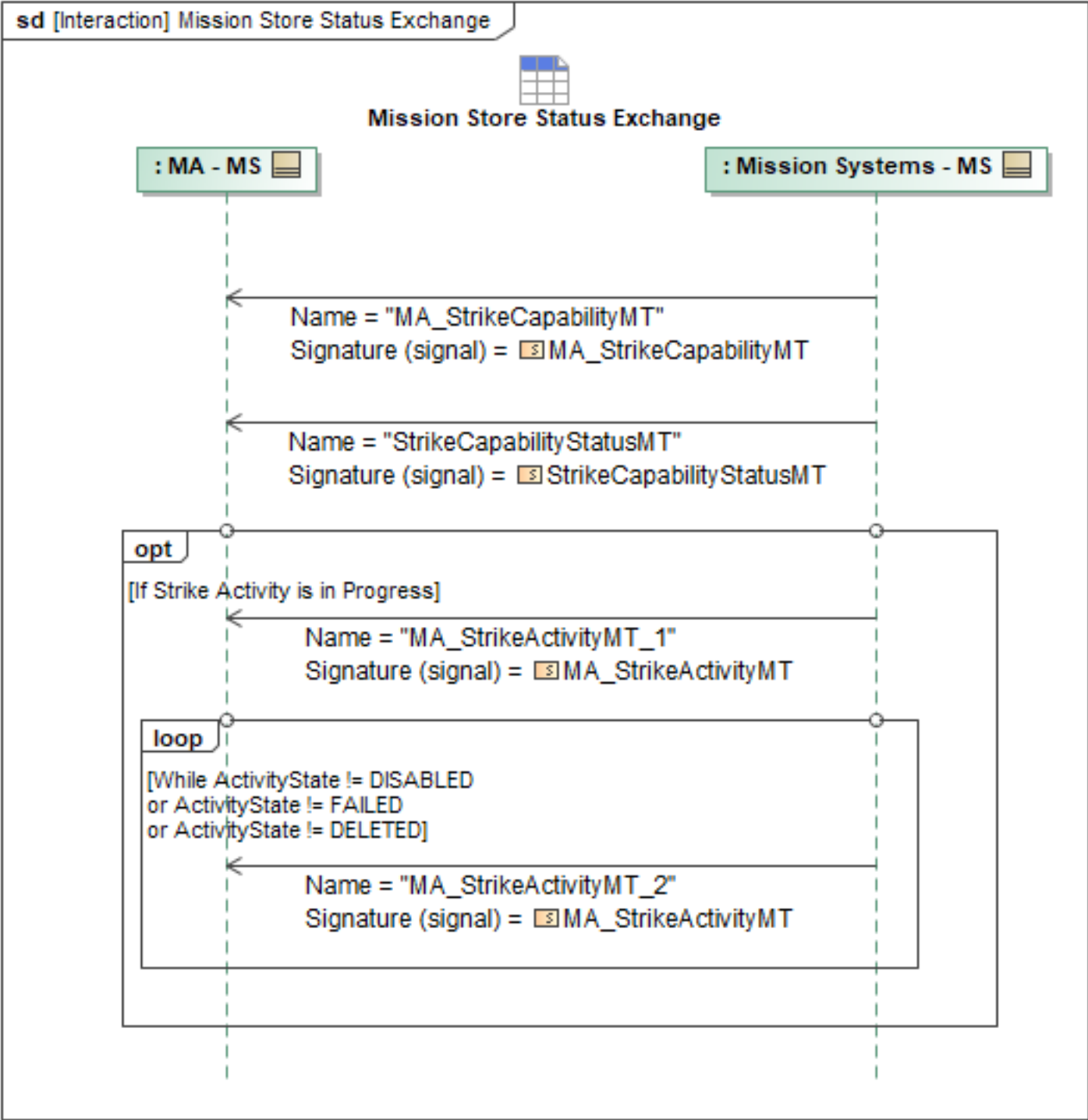


Figure 1-60: Mission Store Status Exchange

Table 1-58: Mission Store Status Exchange

Message Name (Name:Type)	Required Fields (Name:Type)
MA_StrikeActivityMT_1: MA_StrikeActivityMT	None
MA_StrikeActivityMT_2: MA_StrikeActivityMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_StrikeCapabilityMT: MA_StrikeCapabilityMT	None
StrikeCapabilityStatusMT: StrikeCapabilityStatusMT	StoreStatus: StoreStatusType AvailabilityReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.9 Receive Changes to Stores Primary Mission Plan Data

Prior to the communication exchange between MS and MA, MS sends a *ServiceStatus* message to MA to indicate that it is ready to receive *DataRequest* messages. After receipt of *ServiceStatus* message, there are two different sets of interactions that can occur - updating existing parameters on the current MDF or loading a new MDF.

To update existing MDF parameters, MA sends a *ServiceConfigurationParamsDataRequest* message in order to retrieve the latest configuration parameters for the mission data set. MS will reply with the *ServiceConfigurationParamsDataRequestStatus* and afterwards send the *ServiceConfigurationParams* message to MA in order to fulfill the receipt of primary mission data set updates. The *ServiceConfigurationChangeRequest* will then be submitted including all parameters to be modified along with their respective updated values. The updated parameter should be among the list of configuration parameters provided in the *ServiceConfigurationParams* message.

To load a new Mission Data File, MA sends *StrikeSettingsCommandMDT.ActivateMDF* command to MS and MS responds accordingly in the *StrikeSettingsCommandStatus* message. The updated *MA_StrikeCapabilityMT* message will be relayed from MS to MA, as well as the *StrikeCapabilityStatus* message.

This is an optional capability, and not every service may have it. The "Receive Changes to Stores Primary Mission Plan Data" sequence should be implemented by services that have this capability if MA deems that it needs updated or new MDF parameter(s).

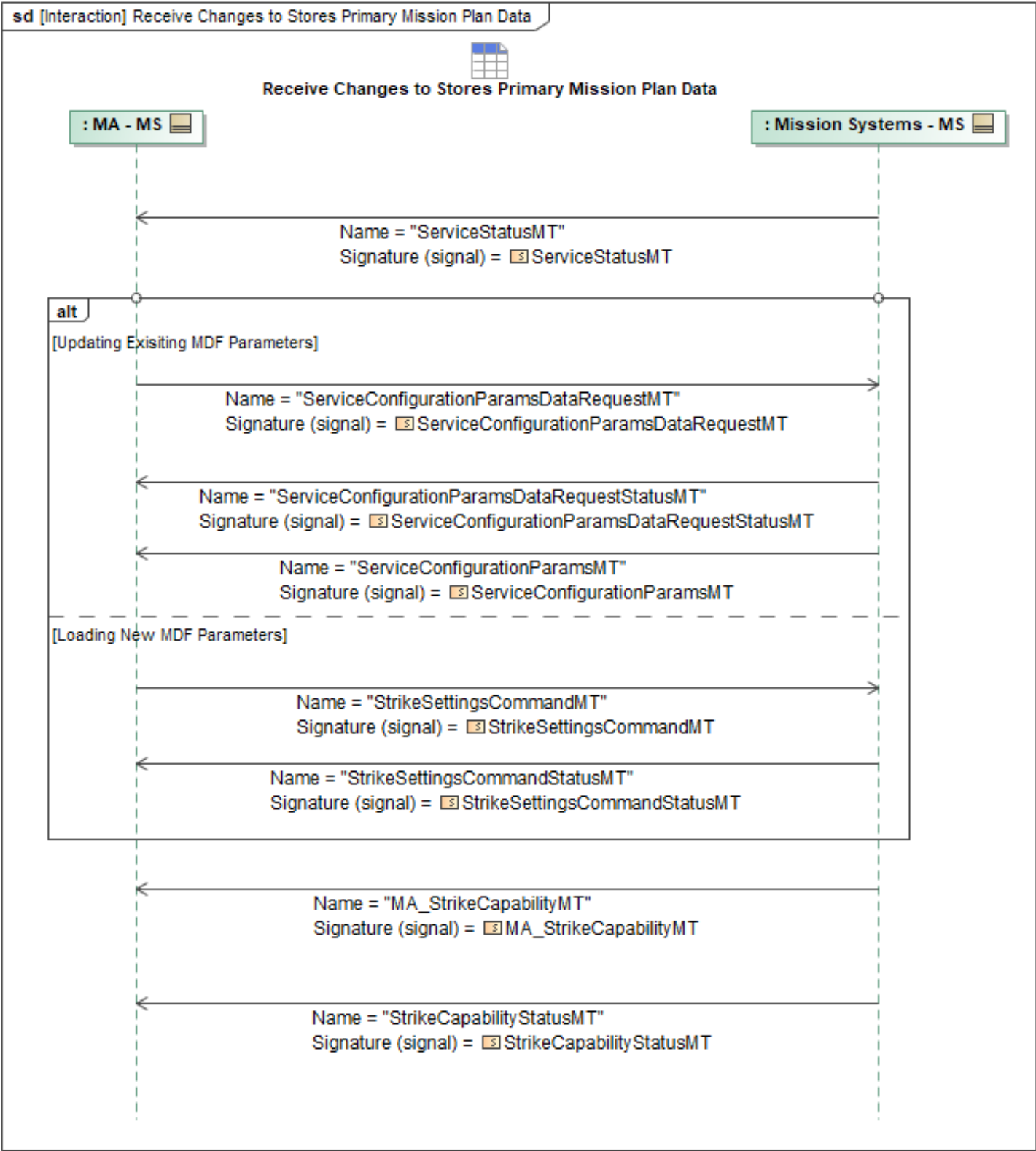


Figure 1-61: Receive Changes to Stores Primary Mission Plan Data

Table 1-59: Receive Changes to Stores Primary Mission Plan Data

Message Name (Name:Type)	Required Fields (Name:Type)
MA_StrikeCapabilityMT: MA_StrikeCapabilityMT	AvailableMDF: MDF_Type
ServiceConfigurationParamsDataRequestMT: ServiceConfigurationParamsDataRequestMT	None
ServiceConfigurationParamsDataRequestStatusMT: ServiceConfigurationParamsDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType
ServiceConfigurationParamsMT: ServiceConfigurationParamsMT	None
ServiceStatusMT: ServiceStatusMT	None
StrikeCapabilityStatusMT: StrikeCapabilityStatusMT	StoreStatus: StoreStatusType
StrikeSettingsCommandMT: StrikeSettingsCommandMT	ActivateMDF: MDF_CommandType
StrikeSettingsCommandStatusMT: StrikeSettingsCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.10 Receive Weapon Flyout State

Mission Use Case: Store Management

MA sends an *AssessmentRequestMT*. MS will reply with an *AssessmentRequestStatusMT_1* of *accepted* or *AssessmentRequestStatusMT_2* of *rejected*. If request is *accepted*, MS sends a periodic *MA_EntityMT* until it returns the *EngagementState* of *LOST*, *SOFT_LOST*, *END_OF_FLIGHT*.

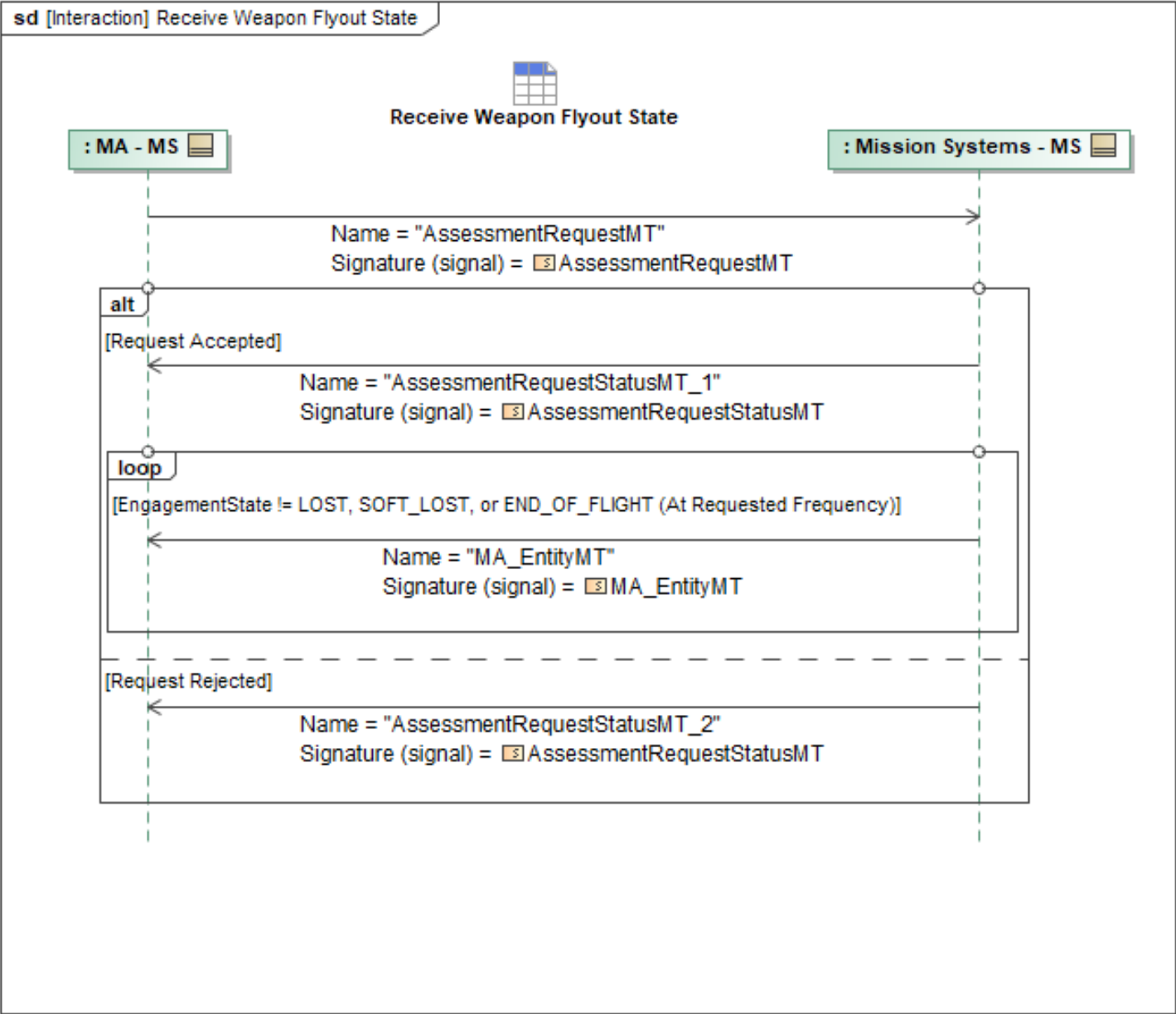


Figure 1-62: Receive Weapon Flyout State

Table 1-60: Receive Weapon Flyout State

Message Name (Name:Type)	Required Fields (Name:Type)
AssessmentRequestMT: AssessmentRequestMT	None
AssessmentRequestStatusMT_1: AssessmentRequestStatusMT	Assessment: MA_AssessmentType
AssessmentRequestStatusMT_2: AssessmentRequestStatusMT	Assessment: MA_AssessmentType
MA_EntityMT: MA_EntityMT	EngagementState: MA_EngagementStateEnum TimeToActive: DurationType

Message Name (Name:Type)	Required Fields (Name:Type)
	TimeToIntercept: DurationType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11 Release Consent Sequence

The following sequence is meant to standardize the interface between MS and MA for Weapon Release Consent. Prior to this exchange, the MA has received information that specifies system "authorizers" for final fire consent, i.e. which system(s) can authorize the weapons release action. The MA will use that list of authorizers when enforcing the Weapons Release Authority to decide where to send the *StrikeConsentRequestMT*. The Weapons Release Authority Criteria allows (if configured) for MA to automatically respond to the *StrikeConsentRequestMT*. The 3 options, configured independently, are to automatically respond if C2 has sent a *MA_DesignationMT*, C2 has approved a *ResponseMT* that starts the Requirement against the target, or if C2 has sent an arming command. After MA executes the Fire Command exchange, MS will respond with a request for final authorization to fire. This involves the following messaging:

- MS will send a *MA_StrikeActivityMT* message to MA with the *ActivityID*. The *ActivityID* identifies the fire activity. For Activity/Task association in the MA, MS must send a *MA_StrikeActivityMT* message in response to a strike command within MS. The Fire Command sequence provides the MA with the *TaskID*; however, MA needs the *ActivityID* to be able to enforce a Weapons Release Authority Criteria from the Engagement Criteria message. Otherwise, MA will not have sufficient information to send a *StrikeConsentRequestMT* to C2.
- MS will send a *StrikeConsentRequestMT* to MA, requesting authorization to fire.
- Once the appropriate "authorizers" have been determined, MA either sends a *StrikeConsentRequestMT* to C2 to obtain final authorization to fire or automatically confirms final authorization (modeled in the C2 profile's Release Consent section).
- C2 responds with *StrikeConsentRequestStatusMT* if this was required (modeled in the C2 profile's Release Consent section).
- MA sends a *StrikeConsentRequestStatusMT* to MS.

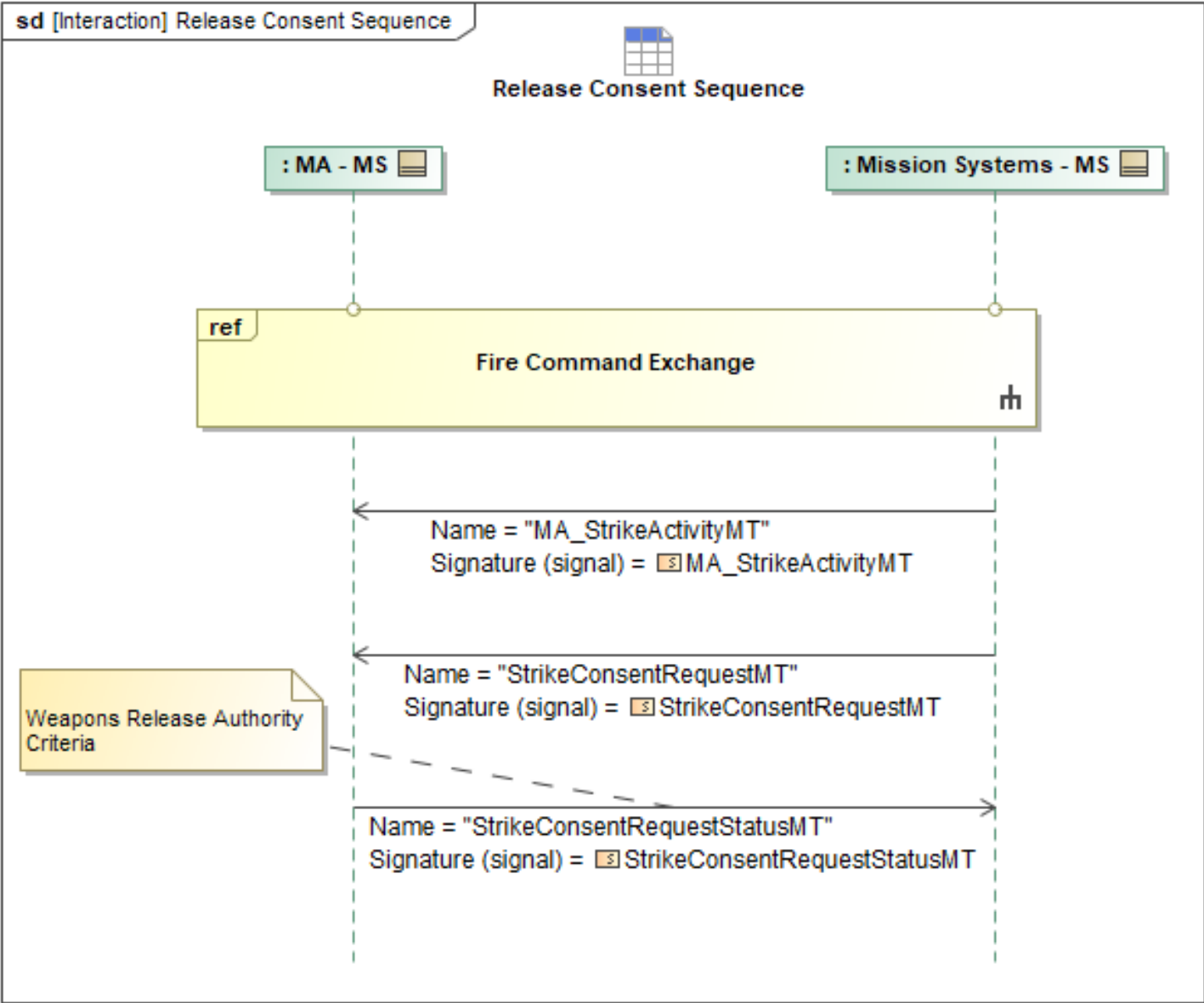


Figure 1-63: Release Consent Sequence

Table 1-61: Release Consent Sequence

Message Name (Name:Type)	Required Fields (Name:Type)
MA_StrikeActivityMT: MA_StrikeActivityMT	None
StrikeConsentRequestMT: StrikeConsentRequestMT	None
StrikeConsentRequestStatusMT: StrikeConsentRequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.12 Send Weapon Entity Data

The Send Weapon Entity Data sequence diagram describes how MA distributes fused tracks and global COP data to the Mission System to inform weapon systems of assigned Entity details. The basic functionality of a data fusion service is to process and merge target reports from various internal and external sources to provide a resulting track, called *Entity*. MA sends the updated Global COP *EntityMT* to MS to provide the latest and best information for both internal and external sources on a given target. Fields that are necessary for MS to inform weapons systems include *EntityID*, *Source*, *EntityStatus*, *Identity*, and *Kinematics*. The *Kinematics* field is used to provide current kinematic information for the given Entity, including *Position* of the target. This position information provides data to MS and the corresponding weapon system for navigation to the assigned target. The "current" nature of the data is determined by the given system, and the necessary *KinematicTypes* used are determined by the given implementation. This message is also triggered when new tracks are determined, or there are changes to already established tracks.

Losing a track is a special optional case that happens when a sensor can no longer detect and track an entity; when this occurs, the *EntityLostMT* message will notify the Mission System of this event. MS may already have this information, but in the instance that MS is not the detecting sensor, the Global COP updates from MA are necessary to inform MS of updates for itself and the weapons systems.

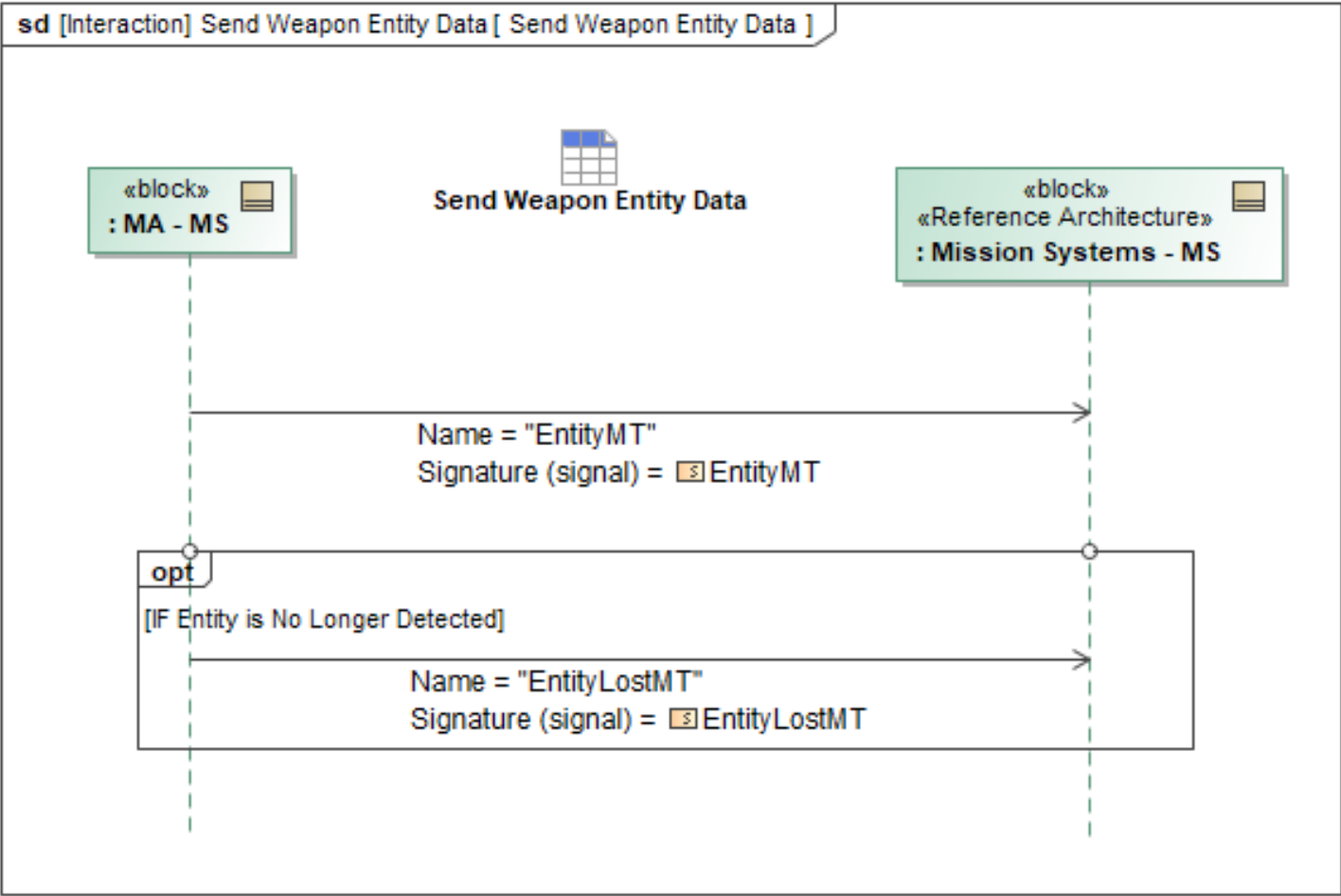


Figure 1-64: Send Weapon Entity Data

Table 1-62: Send Weapon Entity Data

Message Name (Name:Type)	Required Fields (Name:Type)
EntityLostMT: EntityLostMT	None
EntityMT: EntityMT	Kinematics: KinematicsType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.13 Store Inventory and Descriptions Exchange

The supporting weapon subsystems (i.e. Stores Management System, Weapons Resource Manager) within the Mission Systems (MS) are required to publish store inventory data (i.e. installed mission stores and carriages) using *MA_StrikeCapabilityMT*. Mission Autonomy (MA) will continuously monitor updates to *MA_StrikeCapabilityMT* to determine the inventory of the installed store types, station locations and installed capabilities. Mission Systems will assign a capability ID for each installed mission store. The supporting capability is related to the carriage or rack that the store is attached. MA can request the weapon subsystem to verify the reported inventory is compatible with the platform using *StoreManagementCommand* and receive a response with *StoreManagementCommandStatus*. Store compatibility infers the platform has completed weapon

system integration testing to include safe separation testing.

This message is used for inventory and mission capability. This message will be updated as required when a tactic is selected and a Mission Data File (MDF) is loaded. The table below is only focused on the required fields for MA to determine initial store inventory and store descriptions.

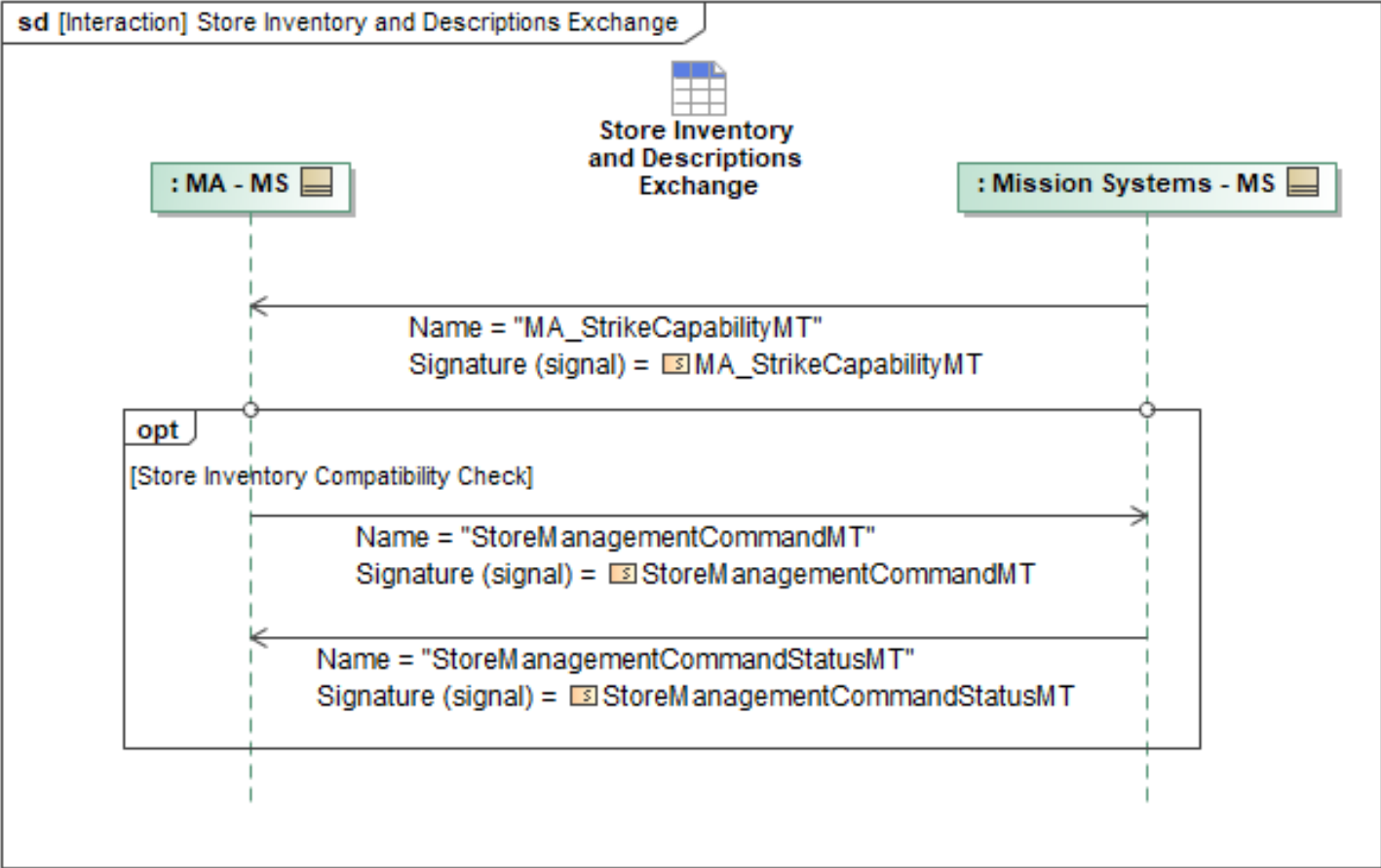


Figure 1-65: Store Inventory and Descriptions Exchange

Table 1-63: Store Inventory and Descriptions Exchange

Message Name (Name:Type)	Required Fields (Name:Type)
MA_StrikeCapabilityMT: MA_StrikeCapabilityMT	StoreTypeVariant: ForeignKeyType Mnemonic: VisibleString256Type Powerable: boolean LoadoutMnemonic: VisibleString256Type Carriage: StoreCarriageCapabilityType
StoreManagementCommandMT: StoreManagementCommandMT	None
StoreManagementCommandStatusMT: StoreManagementCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set

Tables section)

1.3 Interface Compliance

The Interface Compliance table contains the compliance requirements for the L1 MS interface.

Table 1-64: Mission Systems Interface Compliance Requirements

ID	Requirement
MA-L1-008	For a Level 1 Mission Systems interface, MA shall only communicate with Mission Systems via the platform hosted ASB.
MA-L1-007	A Level 1 Mission Systems interface shall implement all required messaging sequences in the MS feature profile (A-GRA Location: 1 Problem Domain > 2 White Box > 3 Functional Analysis > L1 Functional Analysis > L1 Interface Behaviors > MS Interface Behavior), marked Core and all non-Core Mission Use Case sequences that apply to the platform specific implementation.
MA-L1-006	A Level 1 Mission Systems interface shall support the Mission Systems Interface Minimum Message Set defined in the MS MMS Table in the Mission Systems feature profile (A-GRA location: 1 Problem Domain > 3 Exchange Items > Minimum Message Set (MMS)) for all messages in Core Mission Use Case sequences and all non-Core Mission Use Case sequences required for platform specific implementations.
MA-L1-010	The DMS's DDS middleware shall use the MA_DataPayloadWrapper IDL (Interface Definition Language) to send messages OTA across all topics.
MA-L1-009	The DMS's DDS middleware shall use the RTPS (Real-Time Publish-Subscribe) protocol, specifically the Simple Participant Discovery Protocol (SPDP) and Simple Endpoint Discovery Protocol (SEDP) for system discovery.
MA-L1-011	The DMS's DDS middleware shall use topic names that exactly match the message name of the EncodedPayload message. The only messages sent on that topic shall have an EncodedPayload that matches the corresponding topic name.
MA-L1-012	The DMS shall use a DDS middleware to handle OTA communication with other DMS instances.
MA-L1-019	Upon receipt of a Malformed Message (a message missing expected fields (i.e., required by the behavior but not the UCI schema) or with invalid context-dependent information within the scope of that individual message) from the MS interface, MA shall respond to the error by notifying MS of the malformed message and requesting the corrected message according to the Malformed Message Response (MS) sequence.
MA-L1-025	The DMS shall use the EXI standard for the MA_DataPayloadWrapper IDL EncodedPayload field using the settings table found in the "Serialization and Data Encoding" section of the L1 MA ICD. The contents to decode/encode will come from the EncodedPayload field within the MA_TxDataPayloadCommandMT. The datatype of the MA_TxDataPayloadCommandMT/MA_RxDataPayloadMT EncodedPayload fields is UTF-8 XML represented as hexBinary.

1.3.1 MS MMS

The Minimum Message Set referenced in the Interface Compliance Requirements is shown in this section. For an implementor to be compliant, all messages shown in the table shall be supported. The "Direction" column specifies the message direction relative to the MS interface. The "MUC" column specifies the message's Mission Use Case. The message and its respective interaction(s) are only required for the particular Mission Use Case listed.

Mission Use Cases (MUCs) identify specific implementation needs for different ACP solutions. These include a Core set of behaviors that all A-GRA-compliant systems must meet, as well as select use cases such as Carrier-based or Electronic Support Measure-capable. MUCs are considered to be L0 as they describe the use of an ACP, not necessarily a single L1 system. For L1 interface compliance, all implementations must meet the interactions delegated to the Core MUC and all interactions delegated to other MUCs relevant to their implementation. Compliance to interactions not delegated to MUCs do not count towards A-GRA compliance.

Table 1-65: MS MMS

Message Type	Message Name	Interface	Direction	Applicable MUCs
UCI 2.3	ActionPlanValidationCommandMT	MS	in	Core
UCI 2.3	ActionPlanValidationCommandStatusMT	MS	out	Core
UCI 2.3	ActivityPlanValidationCommandMT	MS	in	Core
UCI 2.3	ActivityPlanValidationCommandStatusMT	MS	out	Core
UCI 2.3	AirSampleCapabilityMT	MS	out	Core
UCI 2.3	AirSampleCapabilityStatusMT	MS	out	Core
UCI 2.3	AMTI_ActivityMT	MS	out	AMTI
UCI 2.3	AMTI_CapabilityStatusMT	MS	out	Core, AMTI
UCI 2.3	AMTI_CommandMT	MS	in	AMTI
UCI 2.3	AMTI_CommandStatusMT	MS	out	AMTI
UCI 2.3	AO_CapabilityMT	MS	out	Core
UCI 2.3	AO_CapabilityStatusMT	MS	out	Core
UCI 2.3	AssessmentRequestMT	MS	in	Store Management
UCI 2.3	AssessmentRequestStatusMT	MS	out	Store Management
UCI 2.3	CargoDeliveryCapabilityMT	MS	out	Core
UCI 2.3	CargoDeliveryCapabilityStatusMT	MS	out	Core
UCI 2.3	COMINT_CapabilityMT	MS	out	Core
UCI 2.3	COMINT_CapabilityStatusMT	MS	out	Core
UCI 2.3	CommAssetEstimationRequestMT	MS	in	Additional Comms Capabilities
UCI 2.3	CommAssetEstimationRequestStatusMT	MS	out	Additional Comms Capabilities
UCI 2.3	CommAssetMT	MS	out	Core
UCI 2.3	CommRelayCapabilityMT	MS	out	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
UCI 2.3	CommRelayCapabilityStatusMT	MS	out	Core
UCI 2.3	CommScheduleAllocationMT	MS	out	Core
UCI 2.3	CommSystemMT	MS	out	Core
UCI 2.3	CommTerminalActivityMT	MS	out	Additional Comms Capabilities, Core
UCI 2.3	CommTerminalCapabilityMT	MS	out	Core
UCI 2.3	CommTerminalCapabilityStatusMT	MS	out	Core
UCI 2.3	CommTerminalCommandMT	MS	in	Additional Comms Capabilities
UCI 2.3	CommTerminalCommandStatusMT	MS	out	Additional Comms Capabilities
UCI 2.3	CommTerminalPlanActivationCommandMT	MS	in	Additional Comms Capabilities
UCI 2.3	CommTerminalPlanActivationCommandStatusMT	MS	out	Additional Comms Capabilities
UCI 2.3	CommTerminalPlanMT	MS	in	Additional Comms Capabilities
UCI 2.3	CounterSpaceCapabilityMT	MS	out	Core
UCI 2.3	CounterSpaceCapabilityStatusMT	MS	out	Core
UCI 2.3	DataUpdateRequestMT	MS	in	Core
UCI 2.3	DataUpdateRequestStatusMT	MS	out	Core
UCI 2.3	DLZ_MT	MS	out	Core
UCI 2.3	DLZ_RequestMT	MS	in	Core
UCI 2.3	DLZ_RequestStatusMT	MS	out	Core
UCI 2.3	EA_CapabilityMT	MS	out	Core
UCI 2.3	EA_CapabilityStatusMT	MS	out	Core
UCI 2.3	EntityLostMT	MS	inout	Core
UCI 2.3	EntityMT	MS	inout	Core
UCI 2.3	ESM_ActivityMT	MS	out	ESM
UCI 2.3	ESM_CapabilityMT	MS	out	Core, ESM
UCI 2.3	ESM_CapabilityStatusMT	MS	out	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
UCI 2.3	ESM_CommandMT	MS	in	ESM
UCI 2.3	ESM_CommandStatusMT	MS	out	ESM
UCI 2.3	GatewayCapabilityMT	MS	out	Additional Comms Capabilities
UCI 2.3	GatewayCapabilityStatusMT	MS	out	Additional Comms Capabilities
A-GRA Custom Extension	MA_ActivityPlanMT	MS	in	Core
A-GRA Custom Extension	MA_AMTI_CapabilityMT	MS	out	Core, AMTI
A-GRA Custom Extension	MA_CommAvailableEndpointsMT	MS	out	Core
A-GRA Custom Extension	MA_CommErrorReportMT	MS	out	Core
A-GRA Custom Extension	MA_CommTeamReportMT	MS	out	Core
A-GRA Custom Extension	MA_DataDisseminationPlanActivationComm andMT	MS	in	Additional Comms Capabilities
A-GRA Custom Extension	MA_DataDisseminationPlanActivationComm andStatusMT	MS	out	Additional Comms Capabilities
A-GRA Custom Extension	MA_DataDisseminationPlanActivationStatus MT	MS	out	Additional Comms Capabilities
A-GRA Custom Extension	MA_DataDisseminationPlanMT	MS	in	Additional Comms Capabilities
A-GRA Custom Extension	MA_DataDisseminationPlanOverrideRequest MT	MS	in	Additional Comms Capabilities
A-GRA Custom Extension	MA_DataDisseminationPlanOverrideRequest StatusMT	MS	out	Additional Comms Capabilities
A-GRA Custom Extension	MA_EntityMT	MS	out	Store Management
A-GRA Custom Extension	MA_FaultMT	MS	out	Core
A-GRA Custom Extension	MA_GatewayActivityMT	MS	out	Additional Comms Capabilities

Message Type	Message Name	Interface	Direction	Applicable MUCs
A-GRA Custom Extension	MA_GatewayCommandMT	MS	in	Additional Comms Capabilities
A-GRA Custom Extension	MA_GatewayCommandStatusMT	MS	out	Additional Comms Capabilities
A-GRA Custom Extension	MA_MessagingSubscriptionSettingCommandMT	MS	in	Core
A-GRA Custom Extension	MA_MessagingSubscriptionSettingCommandStatusMT	MS	out	Core
A-GRA Custom Extension	MA_MessagingSubscriptionStatusMT	MS	out	Core
A-GRA Custom Extension	MA_MissionPlanValidationCommandMT	MS	in	Core
A-GRA Custom Extension	MA_MissionPlanValidationCommandStatusMT	MS	out	Core
A-GRA Custom Extension	MA_PositionReportDetailedMT	MS	out	PNT
A-GRA Custom Extension	MA_RouteActivityPlanMT	MS	in	Core
A-GRA Custom Extension	MA_RoutePlanMT	MS	in	Core
A-GRA Custom Extension	MA_RxDataPayloadMT	MS	out	Core
A-GRA Custom Extension	MA_StrikeActivityMT	MS	out	Core, Store Management
A-GRA Custom Extension	MA_StrikeCapabilityMT	MS	out	Core, Store Management
A-GRA Custom Extension	MA_SystemNotificationMT	MS	in	Core
A-GRA Custom Extension	MA_TaskCommandMT	MS	in	Core
A-GRA Custom Extension	MA_TaskCommandStatusMT	MS	out	Core
A-GRA Custom Extension	MA_TaskMT	MS	in	Core
A-GRA Custom Extension	MA_TaskPlanMT	MS	in	Core
A-GRA Custom Extension	MA_TaskStatusMT	MS	inout	Core
A-GRA Custom Extension	MA_TxDataPayloadCommandMT	MS	in	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
A-GRA Custom Extension	MA_TxDataPayloadCommandStatusMT	MS	out	Core
A-GRA Custom Extension	MA_WEZ_MT	MS	out	Core
A-GRA Custom Extension	MA_WEZ_RequestMT	MS	in	Core
A-GRA Custom Extension	MA_WEZ_RequestStatusMT	MS	out	Core
UCI 2.3	ObservationMeasurementReportMT	MS	out	Core
UCI 2.3	PO_ActivityMT	MS	out	PO
UCI 2.3	PO_CapabilityMT	MS	out	PO
UCI 2.3	PO_CapabilityStatusMT	MS	out	PO
UCI 2.3	PO_CommandMT	MS	in	PO
UCI 2.3	PO_CommandStatusMT	MS	out	PO
UCI 2.3	QueryDataRequestMT	MS	in	Core
UCI 2.3	QueryDataRequestStatusMT	MS	out	Core
UCI 2.3	RefuelCapabilityMT	MS	out	Core
UCI 2.3	RefuelCapabilityStatusMT	MS	out	Core
UCI 2.3	RouteActivityPlanValidationCommandMT	MS	in	Core
UCI 2.3	RouteActivityPlanValidationCommandStatus MT	MS	out	Core
UCI 2.3	SAR_CapabilityMT	MS	out	Core
UCI 2.3	SAR_CapabilityStatusMT	MS	out	Core
UCI 2.3	ServiceConfigurationChangeRequestMT	MS	in	Core
UCI 2.3	ServiceConfigurationChangeRequestStatus MT	MS	out	Core
UCI 2.3	ServiceConfigurationParamsDataRequestMT	MS	in	Core
UCI 2.3	ServiceConfigurationParamsDataRequestSta tusMT	MS	out	Core
UCI 2.3	ServiceConfigurationParamsMT	MS	out	Core
UCI 2.3	ServiceStatusDataRequestMT	MS	in	Core
UCI 2.3	ServiceStatusDataRequestStatusMT	MS	out	Core
UCI 2.3	ServiceStatusMT	MS	out	Core
UCI 2.3	SMTI_CapabilityMT	MS	out	Core
UCI 2.3	SMTI_CapabilityStatusMT	MS	out	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
UCI 2.3	StoreManagementCommandMT	MS	in	Core
UCI 2.3	StoreManagementCommandStatusMT	MS	out	Core
UCI 2.3	StoreManagementStatusMT	MS	out	Store Management
UCI 2.3	StrikeCapabilityStatusMT	MS	out	Core, Store Management
UCI 2.3	StrikeConsentRequestMT	MS	out	Core
UCI 2.3	StrikeConsentRequestStatusMT	MS	in	Core
UCI 2.3	StrikeSettingsCommandMT	MS	in	Core
UCI 2.3	StrikeSettingsCommandStatusMT	MS	out	Core
UCI 2.3	SubsystemBIT_CommandMT	MS	in	Core
UCI 2.3	SubsystemBIT_CommandStatusMT	MS	out	Core
UCI 2.3	SubsystemBIT_StatusMT	MS	out	Core
UCI 2.3	SubsystemSettingsCommandMT	MS	in	PNT
UCI 2.3	SubsystemStatusDataRequestMT	MS	in	PNT, Core
UCI 2.3	SubsystemStatusDataRequestStatusMT	MS	out	PNT, Core
UCI 2.3	SubsystemStatusMT	MS	out	Carrier, Core
UCI 2.3	SystemDeploymentCapabilityMT	MS	out	Core
UCI 2.3	SystemDeploymentCapabilityStatusMT	MS	out	Core
UCI 2.3	SystemStatusMT	MS	inout	Core, Carrier
UCI 2.3	TacticalOrderCapabilityMT	MS	out	Core
UCI 2.3	TacticalOrderCapabilityStatusMT	MS	out	Core
UCI 2.3	TaskPlanValidationCommandMT	MS	in	Core
UCI 2.3	TaskPlanValidationCommandStatusMT	MS	out	Core
UCI 2.3	WeatherMT	MS	out	Core
UCI 2.3	WeatherRadarCapabilityMT	MS	out	Core
UCI 2.3	WeatherRadarCapabilityStatusMT	MS	out	Core
UCI 2.3	WeatherRadarObservationMT	MS	out	Weather Radar

1.3.2 MS UCI Extensions

The tables in the following sections provide the specifications for extended UCI 2.3 messages utilized by the A-GRA to communicate Autonomy L1 interface interactions. The tables include both extended messages and message types. Each table includes the message name, the functional description, and fields contained within. Top-level messages include images of the nested message structure. For a better view of the diagram and the message structure, it is strongly recommended to look at the

schema in a XML viewer tool.

1.3.2.1 MA_StrikeCapabilityMT UCI Extension

MA_StrikeCapabilityMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_StrikeCapabilityMDT

Table A-1-66: MS UCI Extensions MA_StrikeCapabilityMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_StrikeCapabilityMDT	
Message Structure		
Field Name	Field Type	Field Usage
Capability	MA_StrikeCapabilityType	Repeated
Description	Indicates a Strike Capability provided by the Subsystem.	
Carriage	StoreCarriageCapabilityType	Repeated
Description	Optional list of carriage related to the StrikeCapability.	
Verification	StoreVerificationStatusType	Optional
Description	This is a rollup of all verification of each capability in this message.	
LoadoutMnemonic	VisibleString256Type	Optional
Description	Mnemonic representation of the store loadout that the reported strike capabilities and carriage support capabilities are derived from.	
AvailableMDF	MDF_Type	Repeated
Description	Indicates a Mission Data File (MDF) that is available for use. All MDFs currently available for use should be listed. An MDF is a collection of data, organized into files, that is used to tailor, configure or otherwise affect the behavior of a Capability. MDFs can vary from mission to mission or even by phases of a single mission. They are generally developed in pre-mission planning and are unique for each Subsystem model. Their content is not standardized by UCI, but the mechanism for advertising their availability and activating them is. Their content is abstracted by the UCI concept of Capability.	

Table A-1-67: MS UCI Extensions MA_StrikeCapabilityMT - MessageData UCI Extension Field

1.3.2.2 MA_CommErrorReportMT UCI Extension

MA_CommErrorReportMT	
Description	This message represents a general report of an communication error that is of significance to other systems, services, etc. in the network. This message is not intended for remote, centralized error logging; it is meant for "error awareness".

	See the XSD File for more info
Ext. Fields	MessageData : MA_CommErrorReportMDT CommErrorReportID : MA_CommErrorReportID_Type

Table A-1-68: MS UCI Extensions MA_CommErrorReportMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_CommErrorReportMDT	
Message Structure		
Field Name	Field Type	Field Usage
CommErrorReportID	MA_CommErrorReportID_Type	Required
Description	The unique identifier for the Comm error report.	
SystemID	SystemID_Type	Required
Description	This element represents the SystemID of the service that experienced the error.	
SubsystemID	SubsystemID_Type	Optional
Description	This element represents the SubsystemID of the service that experienced the error.	
ServiceID	ServiceID_Type	Optional
Description	This element represents the SystemID of the service that experienced the error.	
ErrorDetail	VisibleString1024Type	Required
Description	Free text field to provide detail about the error in the communication.	
SoftwareExceptionDetail	VisibleString1024Type	Optional
Description	Free text field to provide detail about a software exception related to this Comm error.	
AssociatedMessage	AssociatedMessageType	Repeated
Description	Indicates an input message that is associated with or the cause of the communication error. This element should be omitted if an input message was not the cause of the error.	

Table A-1-69: MS UCI Extensions MA_CommErrorReportMT - MessageData UCI Extension Field

CommErrorReportID		
Description	The unique identifier for the Comm error report.	
Base Type	MA_CommErrorReportID_Type	
Message Structure		
Field Name	Field Type	Field Usage
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-70: MS UCI Extensions MA_CommErrorReportMT - CommErrorReportID UCI Extension Field

1.3.2.3 MA_PositionReportDetailedMT UCI Extension

MA_PositionReportDetailedMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	PositionReportData : MA_PositionReportDataType MessageData : MA_PositionReportDetailedMDT

Table A-1-71: MS UCI Extensions MA_PositionReportDetailedMT UCI Extension

PositionReportData		
Description	If multiple instances are given, each should be of a different navigation solution type as indicated by the child element.	
Base Type	MA_PositionReportDataType	
Message Structure		
Field Name	Field Type	Field Usage
PositionSource	PositionSourceID_ChoiceType	Required
Description	The source of the positional data.	
ComponentID	ComponentID_Type	Optional
Description	The ID of the specific component within the subsystem that is the source of this data. This can be used if multiple redundant INS/GPS units are represented as a single subsystem.	
NavigationSolutionState	NavigationSolutionStateEnum	Required
Description	The navigation solution state indicates what sensor inputs the navigation unit is using to generate its kinematic solution.	
FigureOfMerit	BytePositiveType	Optional
Description	A FOM is typically calculated as the root-sum-square of the estimated errors contributing to the solution accuracy. Example criteria include: a. GPS receiver state (e.g., carrier tracking, code tracking, acquisition) b. Carrier to noise ratio c. Satellite geometry (DOP value) d. Satellite range accuracy (URA value) e. Ionospheric measurement or modeling error f. Receiver aiding used g. Kalman filter error estimates.	

	The resultant FOM is presented as a numerical value from 1 to 9, where 1 indicates the best navigation performance.	
Kinematics	MA_DetailedKinematicsType	Required
Description	The kinematic state of the vehicle.	
KinematicsError	DetailedKinematicsErrorType	Required
Description	Indicates the covariance matrix of the vehicle, containing detailed kinematic errors of the position, velocity, and orientation. See the DetailedKinematicsErrorType element annotations for details.	
SolutionCorrections	NavigationSolutionCorrectionsType	Optional
Description	This element contains the corrections applied by the measurement unit to the reported values.	

Table A-1-72: MS UCI Extensions MA_PositionReportDetailedMT - PositionReportData UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_PositionReportDetailedMDT	
Message Structure		
Field Name	Field Type	Field Usage
PositionReportData	MA_PositionReportDataType	Repeated
Description	If multiple instances are given, each should be of a different navigation solution type as indicated by the child element.	

Table A-1-73: MS UCI Extensions MA_PositionReportDetailedMT - MessageData UCI Extension Field

1.3.2.4 MA_CommAvailableEndpointsMT UCI Extension

MA_CommAvailableEndpointsMT	
Description	Publishes available network endpoints as provided by network overhead messages. See the XSD File for more info
Ext. Fields	MessageData : MA_CommAvailableEndpointsMDT AvailableNetworkEndpoints : MA_AvailableNetworkEndPointsType MessageNumbering : MA_MessageNumberingType SystemID : SystemID_Type

Table A-1-74: MS UCI Extensions MA_CommAvailableEndpointsMT UCI Extension

MessageData	
Description	See annotations in child elements and messages/elements that use this type for details.

Base Type	MA_CommAvailableEndpointsMDT	
Message Structure		
Field Name	Field Type	Field Usage
AvailableNetworkEndpoints	MA_AvailableNetworkEndPointsType	Repeated
Description	Publishes available network endpoints as discovered on network in accordance with the published comm plan.	
MessageNumbering	MA_MessageNumberingType	Optional
Description	Optional field for identification of total messages required to pass all network endpoints.	

Table A-1-75: MS UCI Extensions MA_CommAvailableEndpointsMT - MessageData UCI Extension Field

AvailableNetworkEndpoints		
Description	Publishes available network endpoints as discovered on network in accordance with the published comm plan.	
Base Type	MA_AvailableNetworkEndPointsType	
Message Structure		
Field Name	Field Type	Field Usage
SystemID	SystemID_Type	Required
Description	Individual network endpoint's unique endpoint ID.	
EndpointLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-76: MS UCI Extensions MA_CommAvailableEndpointsMT - AvailableNetworkEndpoints UCI Extension Field

MessageNumbering		
Description	Optional field for identification of total messages required to pass all network endpoints.	
Base Type	MA_MessageNumberingType	
Message Structure		
Field Name	Field Type	Field Usage
CurrentMessage	double	Required
Description	Current message number.	
TotalMessageCount	double	Required
Description	Total number of messages to complete endpoint availability list.	

Table A-1-77: MS UCI Extensions MA_CommAvailableEndpointsMT - MessageNumbering UCI Extension Field

SystemID	
Description	Individual network endpoint's unique endpoint ID.
Base Type	SystemID_Type

Message Structure		
Field Name	Field Type	Field Usage
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-78: MS UCI Extensions MA_CommAvailableEndpointsMT - SystemID UCI Extension Field

1.3.2.5 MA_CommTeamReportMT UCI Extension

MA_CommTeamReportMT	
Description	This message provides the status for the available endpoints within the communication network. See the XSD File for more info
Ext. Fields	MessageData : MA_CommTeamReportMDT AvailableEndpoint : MA_AvailableNetworkEndPointsType SystemID : SystemID_Type

Table A-1-79: MS UCI Extensions MA_CommTeamReportMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_CommTeamReportMDT	
Message Structure		
Field Name	Field Type	Field Usage
AvailableEndpoint	MA_AvailableNetworkEndPointsType	Repeated
Description	Lists the asset(s) or endpoint(s) part of the specific Group.	

Table A-1-80: MS UCI Extensions MA_CommTeamReportMT - MessageData UCI Extension Field

AvailableEndpoint		
Description	Lists the asset(s) or endpoint(s) part of the specific Group.	
Base Type	MA_AvailableNetworkEndPointsType	
Message Structure		
Field Name	Field Type	Field Usage
SystemID	SystemID_Type	Required
Description	Individual network endpoint's unique endpoint ID.	
EndpointLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-81: MS UCI Extensions MA_CommTeamReportMT - AvailableEndpoint UCI Extension Field

SystemID		
Description	Individual network endpoint's unique endpoint ID.	
Base Type	SystemID_Type	
Message Structure		
Field Name	Field Type	Field Usage
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-82: MS UCI Extensions MA_CommTeamReportMT - SystemID UCI Extension Field

1.3.2.6 MA_TaskMT UCI Extension

MA_TaskMT	
Description	This message specifies a Task that a System will perform against a specific entity or geographical location. Tasks include flight, collection, strike, electronic attack, directed energy, laser designation, cargo delivery and communication relay. They are abstracted to a level of C2 and autonomy derived from UCI operational vignettes. Tasks enable services, operators and other actors to do automated, dynamic, machine-to-machine allocation of Tasks to Systems and subsequent mission planning. If necessary, authorization of Users to Task Systems can be controlled using the Authorization message. See the XSD File for more info
Ext. Fields	TaskType : MA_TaskType TaskPlurality : MA_TaskPluralityEnum ConstraintID : MA_ConstraintID_Type HSA_CSA : MA_HSA_CSA_Type WaypointFollowing : MA_WaypointFollowingType CurveFollowing : MA_CurveControlType Speed : PathSegmentSpeedType Heading : MA_DirectionReferenceType SpeedChoice : MA_HSA_SpeedChoiceType SpeedValue : PathSegmentSpeedValueType CurveFitMethod : MA_CurveFitMethodEnum

Table A-1-83: MS UCI Extensions MA_TaskMT UCI Extension

TaskType	
Description	Identifies the type of this Task instance.
Base Type	MA_TaskType

Message Structure		
Field Name	Field Type	Field Usage
AirSample	AirSampleTaskType	Required
Description	Air sample includes direct sampling of the air (SAMPLE) and remote sensing with spectral analysis (SPECTROMETER) with the intent of detecting NBC events.	
AMTI	AMTI_TaskType	Required
Description	Indicates a Task to collect Air Moving Target Indicator (AMTI) data.	
AO	AO_TaskType	Required
Description	Indicates a Task to perform an optical emission such as laser designation.	
CAP	MA_CAP_TaskType	Required
Description	Indicates a Task to perform Combat Air Patrol.	
CargoDelivery	CargoDeliveryTaskType	Required
Description	Indicates a Task to transfer cargo between locations.	
COMINT	COMINT_TaskType	Required
Description	Indicates a Task to provide COMINT.	
CommRelay	CommRelayTaskType	Required
Description	Indicates a Task to provide communications relay support.	
CounterSpace	CounterSpaceTaskType	Required
Description	Indicates a Task to employ a CounterSpace capability.	
Escort	MA_EscortTaskType	Required
Description	Indicates a Task to escort another Entity or System.	
EA	EA_TaskType	Required
Description	Indicates a Task to provide electronic attack support to another System. It guides/constrains the EA System by specifying where it should fly, what it should protect and what the threat is.	
ESM	ESM_TaskType	Required
Description	Indicates a Task to collect ESM data.	
Flight	MA_FlightTaskType	Required
Description	Indicates a Task to effect the flight path/plan of the System.	
Jettison	MA_JettisonTaskType	Required
Description	Indicates a Task to a jettisonable weapon from the system.	
OrbitChange	OrbitChangeTaskType	Required
Description	Indicates a task to perform an orbit change via a spacecraft maneuver.	
OrbitalSurveillance	OrbitalSurveillanceTaskType	Required
Description	Indicates an Orbital Surveillance Task.	
OrbitalSurveillanceSensor	OrbitalSurveillanceSensorTaskType	Required
Description	Indicates a task to perform orbital surveillance sensor tasking.	

PO	PO_TaskType	Required
Description	Indicates a Task to collect Passive Optical data, imagery and video as well as perform PO search and track capabilities.	
Refuel	RefuelTaskType	Required
Description	Indicates a Task for one System to refuel another.	
SAR	SAR_TaskType	Required
Description	Indicates a Task to collect a Synthetic Aperture Radar (SAR) image.	
SMTI	SMTI_TaskType	Required
Description	Indicates a Task to collect Moving Target Indicator (MTI) data.	
Strike	StrikeTaskType	Required
Description	Indicates a Task to kinetically attack/strike, with a weapon that can be released from the System.	
SystemDeployment	SystemDeploymentTaskType	Required
Description	Indicates a task to perform a deployment or release of a system at a specified location.	
TacticalOrder	TacticalOrderTaskType	Required
Description	Indicates a task to perform a tactical order.	
WeatherRadar	WeatherRadarTaskType	Required
Description	Indicates a task to collect weather radar data.	

Table A-1-84: MS UCI Extensions MA_TaskMT - TaskType UCI Extension Field

TaskPlurality	
Description	Indicates whether the task is meant to be executed by a single entity or as a joint operation. A join operation implies forming a group prior to execution. A single entity could be a single system or a single pre-formed group.
Base Type	MA_TaskPluralityEnum
Message Structure	
Enumeration Literal	Description
UNSPECIFIED	Indicates the task plurality is unspecified, and should be defaulted or determined by the receiving system.
JOINT	Indicates the task is to be executed as a joint task. A group will be formed when this task is assigned.
SINGLE_ENTITY	Indicates the task is to be executed by a single entity. A single entity could be a single system or a single pre-existing group.

Table A-1-85: MS UCI Extensions MA_TaskMT - TaskPlurality UCI Extension Field

ConstraintID	
Description	Indicates any Constraints that apply to the Task (e.g FormationConstraint). A FormationConstraint implies that if a group is formed to execute this task, the referenced Constraint(s) should be applied to that group.
Base Type	MA_ConstraintID_Type

Message Structure		
Field Name	Field Type	Field Usage
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-86: MS UCI Extensions MA_TaskMT - ConstraintID UCI Extension Field

HSA_CSA		
Description	This message shall be used to provide the ability to command a new flight vector to the Platform.	
Base Type	MA_HSA_CSA_Type	
Message Structure		
Field Name	Field Type	Field Usage
Altitude	AltitudeReferenceType	Optional
Description	Altitude hold value to be achieved and reference to altitude measurement source.	
Speed	PathSegmentSpeedType	Optional
Description	Defines speed type (reference frame) for all speed related fields in this message.	
Direction	MA_DirectionChoiceType	Optional
Description	Defines heading or course type and reference to hold	

Table A-1-87: MS UCI Extensions MA_TaskMT - HSA_CSA UCI Extension Field

WaypointFollowing		
Description	This message shall be used to provide the ability to send a new waypoint path or segment for the Platform to follow.	
Base Type	MA_WaypointFollowingType	
Message Structure		
Field Name	Field Type	Field Usage
Route	MA_RouteType	Required
Description	Indicates the Paths and Segments that make up the Route to be flown.	

Table A-1-88: MS UCI Extensions MA_TaskMT - WaypointFollowing UCI Extension Field

CurveFollowing		
Description	Indicates the parameters for curve following control.	
Base Type	MA_CurveControlType	
Message Structure		
Field Name	Field Type	Field Usage
CurveSegments	MA_NURBS_PointType	Repeated

Description	Specifies the curve segments as a set of NURBS points.	
AppendCurve	EmptyType	Optional
Description	Specifies this curve will be appended to a previous curve command. This is only valid if the vehicle is currently executing a curve.	
EndOfCurveBehavior	MA_EndOfCurveBehaviorEnum	Optional
Description	Specifies the vehicle behavior at the completion of the curve.	

Table A-1-89: MS UCI Extensions MA_TaskMT - CurveFollowing UCI Extension Field

Speed		
Description	Defines speed type (reference frame) for all speed related fields in this message.	
Base Type	PathSegmentSpeedType	
Message Structure		
Field Name	Field Type	Field Usage
SpeedChoice	PathSegmentSpeedChoiceType	Optional
Description	Indicates the speed that the vehicle shall traverse the path segment. Speed can be represented as referenced speed (e.g., ground speed in m/s) or as a unitless Mach number.	
SpeedOptimization	SpeedOptimizationEnum	Optional
Description	Indicates that the System will vary its speed autonomously according to the specified optimization algorithm.	

Table A-1-90: MS UCI Extensions MA_TaskMT - Speed UCI Extension Field

Heading		
Description	Defines heading type and reference to hold.	
Base Type	MA_DirectionReferenceType	
Message Structure		
Field Name	Field Type	Field Usage
Value	AngleType	Required
Description	Indicates the heading or course value to hold.	
Reference	MA_HeadingReferenceEnum	Required
Description	Reference to type of heading or course given.	

Table A-1-91: MS UCI Extensions MA_TaskMT - Heading UCI Extension Field

SpeedChoice	
Description	Indicates the speed that the vehicle shall traverse the path segment. Speed can be represented as referenced speed (e.g., ground speed in m/s) or as a unitless Mach number.
Base Type	MA_HSA_SpeedChoiceType
Message Structure	

Field Name	Field Type	Field Usage
SpeedValue	PathSegmentSpeedValueType	Required
Description	Indicates speed in meters per second (m/s).	
MachValue	MachType	Required
Description	Indicates the speed reference for the associated speed value.	

Table A-1-92: MS UCI Extensions MA_TaskMT - SpeedChoice UCI Extension Field

SpeedValue		
Description	Indicates speed in meters per second (m/s).	
Base Type	PathSegmentSpeedValueType	
Message Structure		
Field Name	Field Type	Field Usage
Value	SpeedType	Required
Description	Indicates the speed along the PathSegment at the EndPoint.	
Reference	SpeedReferenceEnum	Required
Description	Indicates whether the given vehicle speed is reference to ground speed or true air speed.	

Table A-1-93: MS UCI Extensions MA_TaskMT - SpeedValue UCI Extension Field

CurveFitMethod	
Description	Method through which a curve should be fitted to the discretized points in ControlPoint.
Base Type	MA_CurveFitMethodEnum
Message Structure	
Enumeration Literal	Description
B_SPLINE	B-Spline (basis spline) fit.
BEZIER	Bezier curve fit.
CATMULL_ROM	Catmull-Rom spline fit.
NATURAL_CUBIC	Natural Cubic curve fit.

Table A-1-94: MS UCI Extensions MA_TaskMT - CurveFitMethod UCI Extension Field

1.3.2.7 MA_TaskStatusMT UCI Extension

MA_TaskStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	Plans : MA_PlansReferenceType OpPlanID : MA_OpPlanID_Type

ExecutingSystemOrPackage : MA_PackageSystemType
 MessageData : MA_TaskStatusMDT
 Traceability : MA_RequirementStatusTraceabilityType

Table A-1-95: MS UCI Extensions MA_TaskStatusMT UCI Extension

Plans		
Description	Indicates *Plans the Requirement is associated with.	
Base Type	MA_PlansReferenceType	
Message Structure		
Field Name	Field Type	Field Usage
MissionPlanID	MissionPlanID_Type	Repeated
Description	Indicates a reference to a MissionPlan.	
TaskPlanID	TaskPlanID_Type	Repeated
Description	Indicates a reference to a TaskPlan.	
OrbitPlanID	OrbitPlanID_Type	Repeated
Description	Indicates a reference to an OrbitPlan.	
OrbitActivityPlanID	OrbitActivityPlanID_Type	Repeated
Description	Indicates a reference to an OrbitActivityPlan.	
RoutePlanID	RoutePlanID_Type	Repeated
Description	Indicates a reference to a RoutePlan.	
RouteActivityPlanID	RouteActivityPlanID_Type	Repeated
Description	Indicates a reference to a RouteActivityPlan.	
ActivityPlanID	ActivityPlanID_Type	Repeated
Description	Indicates a reference to an ActivityPlan.	
EffectPlanID	EffectPlanID_Type	Repeated
Description	Indicates a reference to an EffectPlan.	
ActionPlanID	ActionPlanID_Type	Repeated
Description	Indicates a reference to an ActionPlan.	
ResponsePlanID	ResponsePlanID_Type	Repeated
Description	Indicates a reference to an ResponsePlan.	
OpPlanID	MA_OpPlanID_Type	Repeated
Description	Indicates a reference to an OpPlan.	

Table A-1-96: MS UCI Extensions MA_TaskStatusMT - Plans UCI Extension Field

OpPlanID	
Description	Indicates a reference to an OpPlan.
Base Type	MA_OpPlanID_Type

Message Structure		
Field Name	Field Type	Field Usage
Version	unsignedInt	Optional
Description	Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval.	
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-97: MS UCI Extensions MA_TaskStatusMT - OpPlanID UCI Extension Field

ExecutingSystemOrPackage		
Description	Indicates the ID of the System or package executing the Requirement; the System or package whose status on execution of the Requirement is reported in sibling elements. *Plans and Requirements must be managed such that only a single System or package executes a given Requirement.	
Base Type	MA_PackageSystemType	
Message Structure		
Field Name	Field Type	Field Usage
ALL	EmptyType	Required
Description	Indicates applies to ALL entities.	
NONE	EmptyType	Required
Description	Indicates applies to NO entities.	
PackageID	PackageID_Type	Required
Description	The ID of a Package.	
SystemID	SystemID_Type	Required
Description	The ID of an ACP or C2 Operator.	

Table A-1-98: MS UCI Extensions MA_TaskStatusMT - ExecutingSystemOrPackage UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_TaskStatusMDT	
Message Structure		
Field Name	Field Type	Field Usage
TaskID	TaskID_Type	Required
Description	Indicates the unique ID of the Task whose execution status is being reported.	

ExecutingSystemOrPackage	MA_PackageSystemType	Required
Description	Indicates the ID of the System or package executing the Requirement; the System or package whose status on execution of the Requirement is reported in sibling elements. *Plans and Requirements must be managed such that only a single System or package executes a given Requirement.	
ExecutionState	RequirementExecutionStateEnum	Required
Description	Identifies the state of the Requirement.	
ExecutionStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling State element. This element is only expected when the State is in a potentially undesirable state such as failed or completed with less than optimum results. It is strongly encouraged for transitions to failed or dropped.	
ActualTiming	RequirementTimingType	Repeated
Description	Indicates actual timing status for the Requirement. Allows for defining for each timing type (RequirementTimingEnum).	
PercentCompleted	PercentType	Optional
Description	Percentage of assigned Requirement completion. The percentage increments until the Requirement has COMPLETED or FAILED. A percent complete of 0 indicates the Requirement has not yet begun execution, and a failed Requirement may have any percentage between 1 - 99.	
Traceability	MA_RequirementStatusTraceabilityType	Optional
Description	Indicates traceability to the sources and implementation of the Requirement. During the life cycle of execution of a Requirement, the traceability can change/update. Traceability can vary depending on a given System's architecture. For example, traceability to only a CommandID is reasonable if the architecture includes [Capability]Commands without *Plans or [Capability]Activities. Both CommandID and ActivityID are reasonable for architectures where a single [Capability]Command could generate one or multiple [Capability]Activities, e.g. a SAR command could result in multiple activities to cover the area. ActivityID only is reasonable for architectures where [Capability]Activities can originate automatically without *Plans or [Capability]Commands.	
Metrics	MetricsType	Optional
Description	Specifies the actual metrics associated with the Requirements.	

Table A-1-99: MS UCI Extensions MA_TaskStatusMT - MessageData UCI Extension Field

Traceability	
Description	Indicates traceability to the sources and implementation of the Requirement. During the life cycle of execution of a Requirement, the traceability can change/update. Traceability can vary depending on a given System's architecture. For example, traceability to only a CommandID is reasonable if the architecture includes [Capability]Commands without *Plans or [Capability]Activities. Both CommandID and ActivityID are reasonable for

	architectures where a single [Capability]Command could generate one or multiple [Capability]Activities, e.g. a SAR command could result in multiple activities to cover the area. ActivityID only is reasonable for architectures where [Capability]Activities can originate automatically without *Plans or [Capability]Commands.	
Base Type	MA_RequirementStatusTraceabilityType	
Message Structure		
Field Name	Field Type	Field Usage
Plans	MA_PlansReferenceType	Optional
Description	Indicates *Plans the Requirement is associated with.	
PlannedActivityID	PlannedActivityID_Type	Repeated
Description	Indicates a specific Planned Activity the Requirement is associated with.	
CapabilityID	CapabilityID_Type	Repeated
Description	Identifies a specific Capability the Requirement is associated with.	
CommandID	CommandID_Type	Repeated
Description	Indicates the ID of an associated [Capability]Command.	
ActivityID	ActivityID_Type	Repeated
Description	Indicates the ID of an associated actual/executing/executed Activity.	

Table A-1-100: MS UCI Extensions MA_TaskStatusMT - Traceability UCI Extension Field

1.3.2.8 MA_TaskCommandStatusMT UCI Extension

MA_TaskCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_TaskCommandStatusMDT

Table A-1-101: MS UCI Extensions MA_TaskCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_TaskCommandStatusMDT	
Message Structure		
Field Name	Field Type	Field Usage
Activity	ResultingActivityType	Repeated
Description	Indicates an Activity that satisfies the associated Command. This element is required when the sibling State element indicates ACCEPTED.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	

CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-102: MS UCI Extensions MA_TaskCommandStatusMT - MessageData UCI Extension Field

1.3.2.9 MA_TaskCommandMT UCI Extension

MA_TaskCommandMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_TaskCommandMDT

Table A-1-103: MS UCI Extensions MA_TaskCommandMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_TaskCommandMDT	
Message Structure		
Field Name	Field Type	Field Usage
Command	MA_TaskCommandType	Repeated
Description	Indicates an instance of a Task Capability command. A Capability command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each command instance reports status via a separate, independent Capability CommandStatus message.	

Table A-1-104: MS UCI Extensions MA_TaskCommandMT - MessageData UCI Extension Field

1.3.2.10 MA_ActivityPlanMT UCI Extension

MA_ActivityPlanMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ActivityPlanMDT

Table A-1-105: MS UCI Extensions MA_ActivityPlanMT UCI Extension

MessageData

Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ActivityPlanMDT	
Message Structure		
Field Name	Field Type	Field Usage
ActivityPlanID	ActivityPlanID_Type	Required
Description	Indicates a unique ID assigned to the ActivityPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.	
PlanCommandID	ActivityPlanCommandID_ChoiceType	Optional
Description	Indicates the unique ID of the *PlanCommand which this ActivityPlan originated from.	
Plan	MA_ActivityPlanType	Required
Description	Indicates the details of the ActivityPlan.	
ForPlanningUseOnly	boolean	Required
Description	When TRUE, indicates the *Plan is a used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	
PlanInputs	ActivityPlanInputsType	Optional
Description	Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.	

Table A-1-106: MS UCI Extensions MA_ActivityPlanMT - MessageData UCI Extension Field

1.3.2.11 MA_MessagingSubscriptionSettingCommandMT UCI Extension

MA_MessagingSubscriptionSettingCommandMT	
Description	Command message to adjust the topic subscription settings for the secure messaging interface between ACP's on a network. See the XSD File for more info
Ext. Fields	MessageData : MA_MessagingSubscriptionSettingCommandMDT AddTopicSubscription : MA_TopicSubscriptionType ChangeTopicSubscription : MA_ChangeTopicSubscriptionType DataPayloadOriginatorID : SystemID_Type SubscriptionID : MA_SubscriptionID_Type SubscriptionFilterType : MA_SubscriptionFilterType SubscriptionID : MA_SubscriptionID_Type SubscriptionChangeType : MA_SubscriptionChangeTypeEnum

Table A-1-107: MS UCI Extensions MA_MessagingSubscriptionSettingCommandMT UCI Extension

MessageData		
Description	Message field for data for topic subscription additions and removals.	
Base Type	MA_MessagingSubscriptionSettingCommandMDT	
Message Structure		
Field Name	Field Type	Field Usage
AddTopicSubscription	MA_TopicSubscriptionType	Repeated
Description	Field indicates the topic and possible data originator(s) for the topic to be subscribed to.	
ChangeTopicSubscription	MA_ChangeTopicSubscriptionType	Repeated
Description	Field indicates a command to change the state of an existing message subscription.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-108: MS UCI Extensions MA_MessagingSubscriptionSettingCommandMT - MessageData UCI Extension Field

AddTopicSubscription		
Description	Field indicates the topic and possible data originator(s) for the topic to be subscribed to.	
Base Type	MA_TopicSubscriptionType	
Message Structure		
Field Name	Field Type	Field Usage
Topic	VisibleString1024Type	Required
Description	Indicates the topic name.	
DataPayloadOriginatorID	SystemID_Type	Repeated
Description	Indicates a data originator for data on this topic. To subscribe to messages on the specified topic from only a subset of originators, specify those originators using this field. If no DataPayloadOriginatorID values are specified then all data originators will be allowed.	
SubscriptionID	MA_SubscriptionID_Type	Required
Description	Unique ID that represents the subscription to a topic from a set of data originators.	
SubscriptionFilterType	MA_SubscriptionFilterType	Optional
Description	Specifies a filter for the data received via this topic. The SubscriptionFilterType can be used in addition to the Topic to provide further filtering of messaging within a specific Topic, or across all Topics. This allows a system to exclude messages in the Topic that do not meet its needs during a period of operation. Could be implemented as a receive filter or transmit filter.	

Priority	ComparableRankingType	Optional
Description	Indicates the priority of the subscription relative to others. Priority gives guidance on which subscriptions to favor when resources are limited.	

Table A-1-109: MS UCI Extensions MA_MessagingSubscriptionSettingCommandMT - AddTopicSubscription UCI Extension Field

ChangeTopicSubscription		
Description	Field indicates a command to change the state of an existing message subscription.	
Base Type	MA_ChangeTopicSubscriptionType	
Message Structure		
Field Name	Field Type	Field Usage
SubscriptionID	MA_SubscriptionID_Type	Required
Description	Unique ID that represents the subscription to a topic from a set of data originators.	
SubscriptionChangeType	MA_SubscriptionChangeTypeEnum	Required
Description	Indicates the type of requested subscription change.	

Table A-1-110: MS UCI Extensions MA_MessagingSubscriptionSettingCommandMT - ChangeTopicSubscription UCI Extension Field

DataPayloadOriginatorID		
Description	Indicates a data originator for data on this topic. To subscribe to messages on the specified topic from only a subset of originators, specify those originators using this field. If no DataPayloadOriginatorID values are specified then all data originators will be allowed.	
Base Type	SystemID_Type	
Message Structure		
Field Name	Field Type	Field Usage
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-111: MS UCI Extensions MA_MessagingSubscriptionSettingCommandMT - DataPayloadOriginatorID UCI Extension Field

SubscriptionID		
Description	Unique ID that represents the subscription to a topic from a set of data originators.	
Base Type	MA_SubscriptionID_Type	
Message Structure		
Field Name	Field Type	Field Usage

UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-112: MS UCI Extensions MA_MessagingSubscriptionSettingCommandMT - SubscriptionID UCI Extension Field

SubscriptionFilterType		
Description	Specifies a filter for the data received via this topic. The SubscriptionFilterType can be used in addition to the Topic to provide further filtering of messaging within a specific Topic, or across all Topics. This allows a system to exclude messages in the Topic that do not meet its needs during a period of operation. Could be implemented as a receive filter or transmit filter.	
Base Type	MA_SubscriptionFilterType	
Message Structure		
Field Name	Field Type	Field Usage
UCI_Message	QueryMessageType	Required
Description	Indicates a subscription aligned with specific UCI messages with optional filtering of the message.	

Table A-1-113: MS UCI Extensions MA_MessagingSubscriptionSettingCommandMT - SubscriptionFilterType UCI Extension Field

SubscriptionID		
Description	Unique ID that represents the subscription to a topic from a set of data originators.	
Base Type	MA_SubscriptionID_Type	
Message Structure		
Field Name	Field Type	Field Usage
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-114: MS UCI Extensions MA_MessagingSubscriptionSettingCommandMT - SubscriptionID UCI Extension Field

SubscriptionChangeType	
Description	Indicates the type of requested subscription change.
Base Type	MA_SubscriptionChangeTypeEnum
Message Structure	

Enumeration Literal	Description
ACTIVATE	Activate the referenced subscription.
DEACTIVATE	Deactivate the referenced subscription.
DELETE	Remove the referenced subscription.

Table A-1-115: MS UCI Extensions MA_MessagingSubscriptionSettingCommandMT - SubscriptionChangeType UCI Extension Field

1.3.2.12 MA_RoutePlanMT UCI Extension

MA_RoutePlanMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_RoutePlanMDT

Table A-1-116: MS UCI Extensions MA_RoutePlanMT UCI Extension

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_RoutePlanMDT	
Message Structure		
Field Name	Field Type	Field Usage
RoutePlanID	RoutePlanID_Type	Required
Description	Indicates a unique ID assigned to the RoutePlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.	
PlanCommandID	RoutePlanCommandID_ChoiceType	Optional
Description	Indicates the unique ID of the *PlanCommand which the RoutePlan originated from.	
Plan	MA_RoutePlanType	Required
Description	Contains the details of the route plan.	
ForPlanningUseOnly	boolean	Required
Description	When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	
PlanInputs	RoutePlanInputsType	Optional
Description	Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.	

Table A-1-117: MS UCI Extensions MA_RoutePlanMT - MessageData UCI Extension Field

1.3.2.13 MA_TaskPlanMT UCI Extension

MA_TaskPlanMT	
Description	See annotations in child elements and messages/elements that use this type for details.
	See the XSD File for more info
Ext. Fields	MessageData : MA_TaskPlanMDT

Table A-1-118: MS UCI Extensions MA_TaskPlanMT UCI Extension

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_TaskPlanMDT	
Message Structure		
Field Name	Field Type	Field Usage
TaskPlanID	TaskPlanID_Type	Required
Description	Indicates a unique ID assigned to the TaskPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.	
PlanCommandID	TaskPlanCommandID_ChoiceType	Optional
Description	Indicates the unique ID of the *PlanCommand which the TaskPlan originated from.	
Plan	MA_TaskPlanType	Required
Description	Contains the details of the task allocation plan.	
ForPlanningUseOnly	Boolean	Required
Description	When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	
PlanInputs	MA_TaskPlanInputsType	Optional
Description	Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.	

Table A-1-119: MS UCI Extensions MA_TaskPlanMT - MessageData UCI Extension Field

1.3.2.14 MA_TxDataPayloadCommandMT UCI Extension

MA_TxDataPayloadCommandMT	
Description	Command message to send a hexbinary data payload offboard to either a single or multiple network endpoints.
	See the XSD File for more info
Ext. Fields	MessageData : MA_TxDataPayloadCommandMDT

SpecificDestinationID : SystemID_Type

Table A-1-120: MS UCI Extensions MA_TxDataPayloadCommandMT UCI Extension

MessageData		
Description	Message field for data payload to be sent offboard.	
Base Type	MA_TxDataPayloadCommandMDT	
Message Structure		
Field Name	Field Type	Field Usage
DestinationRouting	MA_DestinationHandlingEnum	Required
Description	Indicates whether the message should be routed only to the listed SpecificDestinationIDs or to both the topic which corresponds to the listed MessageType and also to any SpecificDestinationIDs. The command is invalid if SPECIFIC_DESTINATION_ONLY is specified and no SpecificDestinationIDs are listed.	
EncodedPayload	hexBinary	Required
Description	Indicates the encoded data payload that is to be sent to the designated destination. This is an encoded binary message intended to be sent to another system (e.g. C2, autonomous platform, etc.) for decoding.	
MessageExpiry	DateTimeType	Optional
Description	Indicates the time at which the data payload message expires.	
Priority	ComparableRankingType	Required
Description	Indicates the precedence and priority levels of the particular data payloads being sent.	
SpecificDestinationID	SystemID_Type	Repeated
Description	Indicates a unique network endpoint that is identified as a destination for the encoded payload.	
Timestamp	DateTimeType	Optional
Description	Indicates the time which the encoded payload was generated.	
MessageType	MessageTypeEnum	Required
Description	Indicates the UCI type of message in the EncodedPayload.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-121: MS UCI Extensions MA_TxDataPayloadCommandMT - MessageData UCI Extension Field

SpecificDestinationID	
Description	Indicates a unique network endpoint that is identified as a destination for the encoded payload.

Base Type	SystemID_Type	
Message Structure		
Field Name	Field Type	Field Usage
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-122: MS UCI Extensions MA_TxDataPayloadCommandMT - SpecificDestinationID UCI Extension Field

1.3.2.15 MA_WEZ_RequestMT UCI Extension

MA_WEZ_RequestMT	
Description	This message is used to request the generation of a Weapon Engagement Zone (WEZ). The WEZ represents a less dynamic capability than the DLZ for air to air engagements and is not reliant on target state data to be produced. See the XSD File for more info
Ext. Fields	MessageData : MA_WEZ_RequestMDT WEZ_RequestData : MA_WEZ_RequestDataType WEZ_ChoiceType : MA_WEZ_ChoiceType StaticWEZ : MA_WEZ_StaticType PlannedStaticWEZ : MA_WEZ_PlannedStaticType PredictedWEZ : MA_WEZ_PredictedType PlannedPredictedWEZ : MA_WEZ_PlannedPredictedType WEZ_FieldOfView : MA_WEZ_FieldOfViewEnum PlannedReleaseConditions : MA_WEZ_StateDataType WEZ_FieldOfView : MA_WEZ_FieldOfViewEnum WEZ_FieldOfView : MA_WEZ_FieldOfViewEnum PlannedReleaseConditions : MA_WEZ_StateDataType PredictedTargetStateData : MA_WEZ_StateDataType WEZ_FieldOfView : MA_WEZ_FieldOfViewEnum PredictedTargetStateData : MA_WEZ_StateDataType

Table A-1-123: MS UCI Extensions MA_WEZ_RequestMT UCI Extension

MessageData	
Description	See annotations in child elements and messages/elements that use this type for details.
Base Type	MA_WEZ_RequestMDT

Message Structure		
Field Name	Field Type	Field Usage
SystemID	SystemID_Type	Required
Description	The ID of the system to perform the processing.	
ResultsInNativeMessage	boolean	Required
Description	If TRUE this element specifies that the response to this message should be in a MA_WEZ_MDT message. If FALSE the response to this message should be in the MA_WEZ_RequestStatusMDT.	
WEZ_RequestData	MA_WEZ_RequestDataType	Repeated
Description	Indicates the inputs needed for a specified weapon to return an array of capability ranges per azimuth from the ownship.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message.	
RequestState	RequestStateEnum	Required
Description	The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request.	

Table A-1-124: MS UCI Extensions MA_WEZ_RequestMT - MessageData UCI Extension Field

WEZ_RequestData		
Description	Indicates the inputs needed for a specified weapon to return an array of capability ranges per azimuth from the ownship.	
Base Type	MA_WEZ_RequestDataType	
Message Structure		
Field Name	Field Type	Field Usage
StoreInformation	WeaponeeringStoreType	Required
Description	Indicates Store Type and Category.	
WEZ_ChoiceType	MA_WEZ_ChoiceType	Required
Description	Indicates the type of WEZ that is being requested by the user.	
TargetPlatformInformation	PlatformIdentityType	Required
Description	Provides information about the Target Platform	

Table A-1-125: MS UCI Extensions MA_WEZ_RequestMT - WEZ_RequestData UCI Extension Field

WEZ_ChoiceType		
Description	Indicates the type of WEZ that is being requested by the user.	
Base Type	MA_WEZ_ChoiceType	
Message Structure		
Field Name	Field Type	Field Usage
StaticWEZ	MA_WEZ_StaticType	Required
Description	Indicates a target has a heading pointed at the ownship, and the same altitude and	

	velocity as the ownship in its current state, as reported by the vehicle interface. The service will subscribe to position data as reported by the vehicle interface for ownship state data.	
PlannedStaticWEZ	MA_WEZ_PlannedStaticType	Required
Description	Indicates a target has a heading pointed at the ownship and an altitude and velocity as described in the "PlannedReleaseConditions" field. The service will use the "PlannedReleaseConditions" field for the ownship state data.	
PredictedWEZ	MA_WEZ_PredictedType	Required
Description	Indicates a target has the heading, altitude and velocity as described by the "PredictedTargetStateData" field. The service will subscribe to position data as reported by the vehicle interface for ownship state data.	
PlannedPredictedWEZ	MA_WEZ_PlannedPredictedType	Required
Description	Indicates a target has the heading, altitude and velocity as described by the "PredictedTargetStateData" field. The service will use the "PlannedReleaseConditions" field for the ownship state data.	

Table A-1-126: MS UCI Extensions MA_WEZ_RequestMT - WEZ_ChoiceType UCI Extension Field

StaticWEZ		
Description	Indicates a target has a heading pointed at the ownship, and the same altitude and velocity as the ownship in its current state, as reported by the vehicle interface. The service will subscribe to position data as reported by the vehicle interface for ownship state data.	
Base Type	MA_WEZ_StaticType	
Message Structure		
Field Name	Field Type	Field Usage
WEZ_FieldOfView	MA_WEZ_FieldOfViewEnum	Required
Description	Identifies the region in radians for the WEZ by indicating the desired azimuth range around the ownship for which the capability ranges are being requested for. The ranges for these regions may be pre-calculated by the service prior to flight or calculated upon the request in flight.	

Table A-1-127: MS UCI Extensions MA_WEZ_RequestMT - StaticWEZ UCI Extension Field

PlannedStaticWEZ		
Description	Indicates a target has a heading pointed at the ownship and an altitude and velocity as described in the "PlannedReleaseConditions" field. The service will use the "PlannedReleaseConditions" field for the ownship state data.	
Base Type	MA_WEZ_PlannedStaticType	
Message Structure		
Field Name	Field Type	Field Usage
WEZ_FieldOfView	MA_WEZ_FieldOfViewEnum	Required
Description	Identifies the region in radians for the WEZ by indicating the desired azimuth range around the ownship for which the capability ranges are being requested for. The ranges for these regions may be pre-calculated by the service prior to flight or calculated upon the request in flight.	

PlannedReleaseConditions	MA_WEZ_StateDataType	Required
Description	Specifies the planned ownship position and state data for the WEZ.	

Table A-1-128: MS UCI Extensions MA_WEZ_RequestMT - PlannedStaticWEZ UCI Extension Field

PredictedWEZ		
Description	Indicates a target has the heading, altitude and velocity as described by the "PredictedTargetStateData" field. The service will subscribe to position data as reported by the vehicle interface for ownship state data.	
Base Type	MA_WEZ_PredictedType	
Message Structure		
Field Name	Field Type	Field Usage
WEZ_FieldOfView	MA_WEZ_FieldOfViewEnum	Required
Description	Identifies the region in radians for the WEZ by indicating the desired azimuth range around the ownship for which the capability ranges are being requested for. The ranges for these regions may be pre-calculated by the service prior to flight or calculated upon the request in flight.	
PredictedTargetStateData	MA_WEZ_StateDataType	Required
Description	Indicates predicted target state data for the WEZ.	

Table A-1-129: MS UCI Extensions MA_WEZ_RequestMT - PredictedWEZ UCI Extension Field

PlannedPredictedWEZ		
Description	Indicates a target has the heading, altitude and velocity as described by the "PredictedTargetStateData" field. The service will use the "PlannedReleaseConditions" field for the ownship state data.	
Base Type	MA_WEZ_PlannedPredictedType	
Message Structure		
Field Name	Field Type	Field Usage
WEZ_FieldOfView	MA_WEZ_FieldOfViewEnum	Required
Description	Identifies the region in radians for the WEZ by indicating the desired azimuth range around the ownship for which the capability ranges are being requested for. The ranges for these regions may be pre-calculated by the service prior to flight or calculated upon the request in flight.	
PlannedReleaseConditions	MA_WEZ_StateDataType	Required
Description	Specifies the planned ownship position and state data for the WEZ.	
PredictedTargetStateData	MA_WEZ_StateDataType	Required
Description	Indicates predicted target state data for the WEZ.	

Table A-1-130: MS UCI Extensions MA_WEZ_RequestMT - PlannedPredictedWEZ UCI Extension Field

WEZ_FieldOfView	
Description	Identifies the region in radians for the WEZ by indicating the desired azimuth range around the ownship for which the capability ranges are being requested for. The ranges for these regions may be pre-calculated by the service prior to

	flight or calculated upon the request in flight.
Base Type	MA_WEZ_FieldOfViewEnum
Message Structure	
Enumeration Literal	Description
FULL_WEZ	Represents a WEZ requested for the azimuths from 0 to 360 degrees around the ownship. Region is built out for azimuth range in radians with a default 5 degree step.
FRONT_WEZ	Represents a WEZ requested for the azimuths from 0 to 180 degrees off the nose of the ownship. Region is built out for azimuth range in radians with a default 5 degree step.
RIGHT_WEZ	Represents a WEZ requested for the azimuths from 0 to 180 degrees off the right wing of the ownship. Region is built out for azimuth range in radians with a default 5 degree step.
LEFT_WEZ	Represents a WEZ requested for the azimuths from 0 to 180s degree off the left wing of the ownship. Region is built out for azimuth range in radians with a default 5 degree step.

Table A-1-131: MS UCI Extensions MA_WEZ_RequestMT - WEZ_FieldOfView UCI Extension Field

PlannedReleaseConditions		
Description	Specifies the planned ownship position and state data for the WEZ.	
Base Type	MA_WEZ_StateDataType	
Message Structure		
Field Name	Field Type	Field Usage
Position	Point3D_Type	Required
Description	Notional System position at time of release.	
Velocity	Velocity2D_Type	Required
Description	Representation of the vehicle's velocity as directional components in True Air Speed.	
MagneticHeading	AngleType	Required
Description	Indicates the heading in relationship to magnetic north.	
Orientation	OrientationType	Optional
Description	Euler Angle Sequence describing the orientation of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle.	
WindVelocity	Velocity2D_Type	Optional
Description	This element represents wind speed at the release point.	

Table A-1-132: MS UCI Extensions MA_WEZ_RequestMT - PlannedReleaseConditions UCI Extension Field

WEZ_FieldOfView	
Description	Identifies the region in radians for the WEZ by indicating the desired azimuth range around the ownship for which the capability ranges are being requested for. The ranges for these regions may be pre-calculated by the service prior to flight or calculated upon the request in flight.

Base Type	MA_WEZ_FieldOfViewEnum
Message Structure	
Enumeration Literal	Description
FULL_WEZ	Represents a WEZ requested for the azimuths from 0 to 360 degrees around the ownship. Region is built out for azimuth range in radians with a default 5 degree step.
FRONT_WEZ	Represents a WEZ requested for the azimuths from 0 to 180 degrees off the nose of the ownship. Region is built out for azimuth range in radians with a default 5 degree step.
RIGHT_WEZ	Represents a WEZ requested for the azimuths from 0 to 180 degrees off the right wing of the ownship. Region is built out for azimuth range in radians with a default 5 degree step.
LEFT_WEZ	Represents a WEZ requested for the azimuths from 0 to 180s degree off the left wing of the ownship. Region is built out for azimuth range in radians with a default 5 degree step.

Table A-1-133: MS UCI Extensions MA_WEZ_RequestMT - WEZ_FieldOfView UCI Extension Field

WEZ_FieldOfView	
Description	Identifies the region in radians for the WEZ by indicating the desired azimuth range around the ownship for which the capability ranges are being requested for. The ranges for these regions may be pre-calculated by the service prior to flight or calculated upon the request in flight.
Base Type	MA_WEZ_FieldOfViewEnum
Message Structure	
Enumeration Literal	Description
FULL_WEZ	Represents a WEZ requested for the azimuths from 0 to 360 degrees around the ownship. Region is built out for azimuth range in radians with a default 5 degree step.
FRONT_WEZ	Represents a WEZ requested for the azimuths from 0 to 180 degrees off the nose of the ownship. Region is built out for azimuth range in radians with a default 5 degree step.
RIGHT_WEZ	Represents a WEZ requested for the azimuths from 0 to 180 degrees off the right wing of the ownship. Region is built out for azimuth range in radians with a default 5 degree step.
LEFT_WEZ	Represents a WEZ requested for the azimuths from 0 to 180s degree off the left wing of the ownship. Region is built out for azimuth range in radians with a default 5 degree step.

Table A-1-134: MS UCI Extensions MA_WEZ_RequestMT - WEZ_FieldOfView UCI Extension Field

PlannedReleaseConditions		
Description	Specifies the planned ownship position and state data for the WEZ.	
Base Type	MA_WEZ_StateDataType	
Message Structure		
Field Name	Field Type	Field Usage
Position	Point3D_Type	Required

Description	Notional System position at time of release.	
Velocity	Velocity2D_Type	Required
Description	Representation of the vehicle's velocity as directional components in True Air Speed.	
MagneticHeading	AngleType	Required
Description	Indicates the heading in relationship to magnetic north.	
Orientation	OrientationType	Optional
Description	Euler Angle Sequence describing the orientation of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle.	
WindVelocity	Velocity2D_Type	Optional
Description	This element represents wind speed at the release point.	

Table A-1-135: MS UCI Extensions MA_WEZ_RequestMT - PlannedReleaseConditions UCI Extension Field

PredictedTargetStateData		
Description	Indicates predicted target state data for the WEZ.	
Base Type	MA_WEZ_StateDataType	
Message Structure		
Field Name	Field Type	Field Usage
Position	Point3D_Type	Required
Description	Notional System position at time of release.	
Velocity	Velocity2D_Type	Required
Description	Representation of the vehicle's velocity as directional components in True Air Speed.	
MagneticHeading	AngleType	Required
Description	Indicates the heading in relationship to magnetic north.	
Orientation	OrientationType	Optional
Description	Euler Angle Sequence describing the orientation of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle.	
WindVelocity	Velocity2D_Type	Optional
Description	This element represents wind speed at the release point.	

Table A-1-136: MS UCI Extensions MA_WEZ_RequestMT - PredictedTargetStateData UCI Extension Field

WEZ_FieldOfView	
Description	Identifies the region in radians for the WEZ by indicating the desired azimuth range around the ownship for which the capability ranges are being requested for. The ranges for these regions may be pre-calculated by the service prior to flight or calculated upon the request in flight.
Base Type	MA_WEZ_FieldOfViewEnum
Message Structure	
Enumeration Literal	Description

FULL_WEZ	Represents a WEZ requested for the azimuths from 0 to 360 degrees around the ownship. Region is built out for azimuth range in radians with a default 5 degree step.
FRONT_WEZ	Represents a WEZ requested for the azimuths from 0 to 180 degrees off the nose of the ownship. Region is built out for azimuth range in radians with a default 5 degree step.
RIGHT_WEZ	Represents a WEZ requested for the azimuths from 0 to 180 degrees off the right wing of the ownship. Region is built out for azimuth range in radians with a default 5 degree step.
LEFT_WEZ	Represents a WEZ requested for the azimuths from 0 to 180s degree off the left wing of the ownship. Region is built out for azimuth range in radians with a default 5 degree step.

Table A-1-137: MS UCI Extensions **MA_WEZ_RequestMT - WEZ_FieldOfView** UCI Extension Field

PredictedTargetStateData		
Description	Indicates predicted target state data for the WEZ.	
Base Type	MA_WEZ_StateDataType	
Message Structure		
Field Name	Field Type	Field Usage
Position	Point3D_Type	Required
Description	Notional System position at time of release.	
Velocity	Velocity2D_Type	Required
Description	Representation of the vehicle's velocity as directional components in True Air Speed.	
MagneticHeading	AngleType	Required
Description	Indicates the heading in relationship to magnetic north.	
Orientation	OrientationType	Optional
Description	Euler Angle Sequence describing the orientation of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle.	
WindVelocity	Velocity2D_Type	Optional
Description	This element represents wind speed at the release point.	

Table A-1-138: MS UCI Extensions **MA_WEZ_RequestMT - PredictedTargetStateData** UCI Extension Field

1.3.2.16 MA_RouteActivityPlanMT UCI Extension

MA_RouteActivityPlanMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_RouteActivityPlanMDT

Table A-1-139: MS UCI Extensions **MA_RouteActivityPlanMT** UCI Extension

MessageData

Description	Indicates the data being conveyed by the message.	
Base Type	MA_RouteActivityPlanMDT	
Message Structure		
Field Name	Field Type	Field Usage
RouteActivityPlanID	RouteActivityPlanID_Type	Required
Description	Indicates a unique ID assigned to the RouteActivityPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.	
PlanCommandID	RouteActivityPlanCommandID_ChoiceType	Optional
Description	Indicates the unique ID of the *PlanCommand which the RouteActivityPlan originated from.	
Plan	MA_RouteActivityPlanType	Required
Description	Indicates the details of the RouteActivityPlan.	
ForPlanningUseOnly	boolean	Required
Description	When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	
PlanInputs	RouteActivityPlanInputsType	Optional
Description	Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.	

Table A-1-140: MS UCI Extensions MA_RouteActivityPlanMT - MessageData UCI Extension Field

1.3.2.17 MA_FaultMT UCI Extension

MA_FaultMT	
Description	This represents a fault for the system. Intended to be used by services such as a health monitoring service for centralized fault management and fault isolation of a system. Allows announcing a fault state change (raised or cleared) from a service as opposed to a capability or component of a subsystem. Also conveys a service relationship in addition to a subsystem relationship. Extended to include a modified Fault Severity Enumeration for Mission Autonomy. See the XSD File for more info
Ext. Fields	MessageData : MA_FaultMDT FaultInformation : MA_SubsystemFaultType Severity : MA_FaultSeverityEnum

Table A-1-141: MS UCI Extensions MA_FaultMT UCI Extension

MessageData	
Description	Reports a fault for the system.
Base Type	MA_FaultMDT

Message Structure		
Field Name	Field Type	Field Usage
FaultInformation	MA_SubsystemFaultType	Repeated
Description	Contains the fault information for the message.	

Table A-1-142: MS UCI Extensions MA_FaultMT - MessageData UCI Extension Field

FaultInformation		
Description	Contains the fault information for the message.	
Base Type	MA_SubsystemFaultType	
Message Structure		
Field Name	Field Type	Field Usage
FaultID	FaultID_Type	Required
Description	The unique ID for this fault. Every fault from a Subsystem, Capability, SupportCapability, Component, or Service should have a unique ID.	
Severity	MA_FaultSeverityEnum	Optional
Description	The severity of the fault. This enumeration is modified for Mission Autonomy adaptation.	
FaultState	FaultStateEnum	Optional
Description	The state of the fault.	
FaultData	FaultDataType	Repeated
Description	Additional fault data which contains key/value pairs with format and unit indicated for the data value. This data supports fault isolation and helps to minimize false alarms.	
DetectionTime	DateTimeType	Optional
Description	Indicates the time the fault was detected; it does not indicate the time the fault was transmitted.	
FaultCode	VisibleString256Type	Required
Description	Indicates the "code" or name of the fault.	
FaultCodeCount	int	Optional
Description	Indicates the number of occurrences of the associated FaultCode and a set FaultState.	
FaultDescription	VisibleString256Type	Optional
Description	Indicates a human readable description of the fault and/or its cause. This could be different than DescriptiveLabel in the sibling FaultID field. For example, DescriptiveLabel could be "Engine 1 overheat" and the FaultDescription could be "Lubrication oil" or "Exhaust nozzle" indicating the underlying cause of the fault.	
SubsystemID	SubsystemID_Type	Optional
Description	The Subsystem that the fault occurred within.	
ServiceID	ServiceID_Type	Optional
Description	The Service that this fault occurred within.	
CapabilityID	CapabilityID_Type	Repeated

Description	Indicates the unique ID of a Capability related to the fault.	
SupportCapabilityID	SupportCapabilityID_Type	Repeated
Description	Indicates the unique ID of a SupportCapability related to the fault.	
ComponentID	ComponentID_Type	Repeated
Description	Indicates the unique ID of a Component related to the fault. When multiple components are identified, it should be inferred that each Component has been individually determined to be related to the fault. In situations where the Subsystem has not yet been able to isolate the fault to individual components, the sibling AmbiguityGroup element should be used.	
AmbiguityGroup	SubsystemFaultAmbiguityGroupType	Repeated
Description	Indicates that a fault has been narrowed down to a set of components, but not yet fully isolated. Separate AmbiguityGroups can be provided when it is desirable to associate specific tests with a subset of the set of components.	

Table A-1-143: MS UCI Extensions MA_FaultMT - FaultInformation UCI Extension Field

Severity	
Description	The severity of the fault. This enumeration is modified for Mission Autonomy adaptation.
Base Type	MA_FaultSeverityEnum
Message Structure	
Enumeration Literal	Description
NOMINAL	A fault that carries the nominal level indicates that fault does not impact the current mission but may indicate something that should be checked in maintenance. Faults should be tagged as nominal if they are targeting maintenance. An example might be air filter hasn't been changed in 5000 hours or queue length reached 60% of max.
ADVISORY	A fault that is important but is not a Caution or Warning; this can be a configuration change, a condition that is contributing to a degraded aircraft or subsystem operation, or an anomaly that is being automatically controlled or corrected by a higher level system.
CAUTION	A fault that is indicative of a condition that, without a timely and correct response, could result in equipment damage or failure of mission.
WARNING	A fault that is indicative of a condition requiring immediate response or attention to prevent loss of aircraft, or failure of mission.
FAILED	A fault that carries a failed level indicates the failure of a component which may preclude the use of a capability. Faults should be tagged with failed if the component state will be marked faulted. Failed components may or may not prevent the completion of the mission.

Table A-1-144: MS UCI Extensions MA_FaultMT - Severity UCI Extension Field

1.3.2.18 MA_MessagingSubscriptionSettingCommandStatusMT UCI Extension

MA_MessagingSubscriptionSettingCommandStatusMT	
Description	Command status message in response to MA_MessagingSubscriptionSettingCommand.

	See the XSD File for more info
Ext. Fields	MessageData : MA_MessagingSubscriptionSettingCommandStatusMDT

Table A-1-145: MS UCI Extensions MA_MessagingSubscriptionSettingCommandStatusMT UCI Extension

MessageData		
Description	Command status message in response to MA_MessagingSubscriptionSettingCommand	
Base Type	MA_MessagingSubscriptionSettingCommandStatusMDT	
Message Structure		
Field Name	Field Type	Field Usage
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-146: MS UCI Extensions MA_MessagingSubscriptionSettingCommandStatusMT - MessageData UCI Extension Field

1.3.2.19 MA_RxDataPayloadMT UCI Extension

MA_RxDataPayloadMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_RxDataPayloadMDT

Table A-1-147: MS UCI Extensions MA_RxDataPayloadMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_RxDataPayloadMDT	
Message Structure		
Field Name	Field Type	Field Usage

RxDataPayloadID	MA_RxDataPayloadID_Type	Required
Description	Indicate the unique ID assigned to this DataPayload.	
DataPayloadOriginatorID	SystemID_Type	Required
Description	Indicates the unique network endpoint that is identified as the originator for the encoded payload. This field represents the original source of the payload, not the immediate sender.	
EncodedPayload	hexBinary	Required
Description	Represents the encoded data payload received by the DMS.	
Timestamp	DateTimeType	Required
Description	Indicates the time at which the encoded message was initially received.	
MessageType	MessageTypeEnum	Required
Description	Indicates the UCI type of message in the EncodedPayload.	
Priority	ComparableRankingType	Optional
Description	Indicates the precedence and priority levels of the particular data payloads being sent.	
MessageExpiry	DateTimeType	Optional
Description	Indicates the time at which the encoded message is expired.	
DestinationRouting	MA_DestinationHandlingEnum	Required
Description	Indicates whether the message was routed only to the listed SpecificDestinationIDs or to both the topic which corresponds to the listed MessageType and also to any SpecificDestinationIDs.	
SpecificDestinationID	SystemID_Type	Repeated
Description	Indicates a unique network endpoint that is identified as a destination for the encoded payload.	

Table A-1-148: MS UCI Extensions MA_RxDataPayloadMT - MessageData UCI Extension Field

1.3.2.20 MA_TxDataPayloadCommandStatusMT UCI Extension

MA_TxDataPayloadCommandStatusMT	
Description	Command status message in response to MA_TxDataPayloadCommand. See the XSD File for more info
Ext. Fields	MessageData : MA_TxDataPayloadCommandStatusMDT

Table A-1-149: MS UCI Extensions MA_TxDataPayloadCommandStatusMT UCI Extension

MessageData		
Description	Command status message in response to MA_TxDataPayloadCommand	
Base Type	MA_TxDataPayloadCommandStatusMDT	
Message Structure		
Field Name	Field Type	Field Usage

DestinationTransmissionStatus	MA_TransmissionStatusType	Repeated
Description	Each occurrence represents the transmission status of the message to a specific destination.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-150: MS UCI Extensions MA_TxDataPayloadCommandStatusMT - MessageData UCI Extension Field

1.3.2.21 MA_WEZ_MT UCI Extension

MA_WEZ_MT	
Description	This message represents a single Weapon Engagement Zone (WEZ). The WEZ represents a less dynamic capability than the DLZ for air to air engagements and is not reliant on target state data to be produced. See the XSD File for more info
Ext. Fields	MessageData : MA_WEZ_MDT WEZ_ID : MA_WEZ_ID_Type WEZ_Data : MA_WEZ_DataType RangeTurnAndClose : MA_WeaponRangeType RangeMinimum : MA_WeaponRangeType RangeTurnAndRun : MA_WeaponRangeType RangeProbabilityOfIntercept : MA_WeaponRangeType RangeMaxGuidance : MA_WeaponRangeType RangeOptimal : MA_WeaponRangeType RangeMaxAero : MA_WeaponRangeType

Table A-1-151: MS UCI Extensions MA_WEZ_MT UCI Extension

MessageData	
Description	This represents the characteristics that identify and describe a single Weapon Engagement Zone (WEZ).
Base Type	MA_WEZ_MDT

Message Structure		
Field Name	Field Type	Field Usage
WEZ_ID	MA_WEZ_ID_Type	Required
Description	This is an ID to uniquely identify the Weapon Engagement Zone (WEZ).	
WEZ_RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message.	
CapabilityID	CapabilityID_Type	Optional
Description	Indicates the weapon Capability instance that this WEZ is associated with.	
GeneratedByWeapon	boolean	Optional
Description	This element indicates if the WEZ was generated using payload algorithms.	
WEZ_Data	MA_WEZ_DataType	Required
Description	Indicates WEZ range calculations for air-to-air engagement assessments.	

Table A-1-152: MS UCI Extensions MA_WEZ_MT - MessageData UCI Extension Field

WEZ_ID		
Description	This is an ID to uniquely identify the Weapon Engagement Zone (WEZ).	
Base Type	MA_WEZ_ID_Type	
Message Structure		
Field Name	Field Type	Field Usage
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-153: MS UCI Extensions MA_WEZ_MT - WEZ_ID UCI Extension Field

WEZ_Data		
Description	Indicates WEZ range calculations for air-to-air engagement assessments.	
Base Type	MA_WEZ_DataType	
Message Structure		
Field Name	Field Type	Field Usage
RangeTurnAndClose	MA_WeaponRangeType	Repeated
Description	Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.Target turn-and-close maneuver	

	RangeTurnAndClose is not applicable to ballistic trajectory targets.	
RangeMinimum	MA_WeaponRangeType	Repeated
Description	Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers Note: ballistic trajectory velocity changes are not considered maneuvers.	
RangeTurnAndRun	MA_WeaponRangeType	Repeated
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.Target turn-and-run maneuver RangeTurnAndRun is not applicable to ballistic trajectory targets.	
RangeProbabilityOfIntercept	MA_WeaponRangeType	Repeated
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers Note: ballistic trajectory velocity changes are not considered maneuvers.	
RangeMaxGuidance	MA_WeaponRangeType	Repeated
Description	Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
RangeOptimal	MA_WeaponRangeType	Repeated

Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Optimal steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
RangeMaxAero	MA_WeaponRangeType	Repeated
Description	Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for: 1.Optimal steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
TimeToRelease	TimeType	Required
Description	The seconds time to activate for the proposed launch store type.	
TimeOfActivation	TimeType	Optional
Description	The time of weapon seeker activation, assuming no platform guidance or weapon errors, for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No post-launch target maneuvers	

Table A-1-154: MS UCI Extensions MA_WEZ_MT - WEZ_Data UCI Extension Field

RangeTurnAndClose		
Description	Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.Target turn-and-close maneuver RangeTurnAndClose is not applicable to ballistic trajectory targets.	
	MA_WeaponRangeType	
Message Structure		
Field Name	Field Type	Field Usage

Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-155: MS UCI Extensions MA_WEZ_MT - RangeTurnAndClose UCI Extension Field

RangeMinimum		
Description	Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers Note: ballistic trajectory velocity changes are not considered maneuvers.	
	Base Type	
MA_WeaponRangeType		
Message Structure		
Field Name	Field Type	Field Usage
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-156: MS UCI Extensions MA_WEZ_MT - RangeMinimum UCI Extension Field

RangeTurnAndRun		
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.Target turn-and-run maneuver RangeTurnAndRun is not applicable to ballistic trajectory targets.	
	Base Type	
MA_WeaponRangeType		
Message Structure		
Field Name	Field Type	Field Usage
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required

Description	Indicates the azimuth the slant range pertains to.
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Table A-1-157: MS UCI Extensions MA_WEZ_MT - RangeTurnAndRun UCI Extension Field

RangeProbabilityOfIntercept		
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers Note: ballistic trajectory velocity changes are not considered maneuvers.	
	Base Type	
MA_WeaponRangeType		
Message Structure		
Field Name	Field Type	Field Usage
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-158: MS UCI Extensions MA_WEZ_MT - RangeProbabilityOfIntercept UCI Extension Field

RangeMaxGuidance		
Description	Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
	MA_WeaponRangeType	
Message Structure		
Field Name	Field Type	Field Usage
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-159: MS UCI Extensions MA_WEZ_MT - RangeMaxGuidance UCI Extension Field

RangeOptimal

Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Optimal steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
Base Type	MA_WeaponRangeType	
Message Structure		
Field Name	Field Type	Field Usage
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-160: MS UCI Extensions MA_WEZ_MT - RangeOptimal UCI Extension Field

RangeMaxAero		
Description	Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for: 1.Optimal steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
	MA_WeaponRangeType	
Message Structure		
Field Name	Field Type	Field Usage
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-161: MS UCI Extensions MA_WEZ_MT - RangeMaxAero UCI Extension Field

1.3.2.22 MA_WEZ_RequestStatusMT UCI Extension

MA_WEZ_RequestStatusMT	
Description	This message provides status for a previously submitted MA_WEZ_Request.

	See the XSD File for more info
Ext. Fields	MessageData : MA_WEZ_RequestStatusMDT WeaponEngagementZone : MA_WEZ_MDT WEZ_ID : MA_WEZ_ID_Type WEZ_Data : MA_WEZ_DataType RangeTurnAndClose : MA_WeaponRangeType RangeMinimum : MA_WeaponRangeType RangeTurnAndRun : MA_WeaponRangeType RangeProbabilityOfIntercept : MA_WeaponRangeType RangeMaxGuidance : MA_WeaponRangeType RangeOptimal : MA_WeaponRangeType RangeMaxAero : MA_WeaponRangeType

Table A-1-162: MS UCI Extensions MA_WEZ_RequestStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_WEZ_RequestStatusMDT	
Message Structure		
Field Name	Field Type	Field Usage
WeaponEngagementZone	MA_WEZ_MDT	Repeated
Description	This element defines the response to a WEZ assessment request.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message corresponding to this status message.	
RequestProcessingState	RequestProcessingStateEnum	Required
Description	Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.	
RequestProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-163: MS UCI Extensions MA_WEZ_RequestStatusMT - MessageData UCI Extension Field

WeaponEngagementZone		
Description	This element defines the response to a WEZ assessment request.	
Base Type	MA_WEZ_MDT	
Message Structure		
Field Name	Field Type	Field Usage

WEZ_ID	MA_WEZ_ID_Type	Required
Description	This is an ID to uniquely identify the Weapon Engagement Zone (WEZ).	
WEZ_RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message.	
CapabilityID	CapabilityID_Type	Optional
Description	Indicates the weapon Capability instance that this WEZ is associated with.	
GeneratedByWeapon	boolean	Optional
Description	This element indicates if the WEZ was generated using payload algorithms.	
WEZ_Data	MA_WEZ_DataType	Required
Description	Indicates WEZ range calculations for air-to-air engagement assessments.	

Table A-1-164: MS UCI Extensions MA_WEZ_RequestStatusMT - WeaponEngagementZone UCI Extension Field

WEZ_ID		
Description	This is an ID to uniquely identify the Weapon Engagement Zone (WEZ).	
Base Type	MA_WEZ_ID_Type	
Message Structure		
Field Name	Field Type	Field Usage
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-165: MS UCI Extensions MA_WEZ_RequestStatusMT - WEZ_ID UCI Extension Field

WEZ_Data		
Description	Indicates WEZ range calculations for air-to-air engagement assessments.	
Base Type	MA_WEZ_DataType	
Message Structure		
Field Name	Field Type	Field Usage
RangeTurnAndClose	MA_WeaponRangeType	Repeated
Description	Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.Target turn-and-close maneuver RangeTurnAndClose is not applicable to ballistic trajectory targets.	

RangeMinimum	MA_WeaponRangeType	Repeated
Description	Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers Note: ballistic trajectory velocity changes are not considered maneuvers.	
RangeTurnAndRun	MA_WeaponRangeType	Repeated
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.Target turn-and-run maneuver RangeTurnAndRun is not applicable to ballistic trajectory targets.	
RangeProbabilityOfIntercept	MA_WeaponRangeType	Repeated
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers Note: ballistic trajectory velocity changes are not considered maneuvers.	
RangeMaxGuidance	MA_WeaponRangeType	Repeated
Description	Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
RangeOptimal	MA_WeaponRangeType	Repeated
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Optimal steering	

	2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
RangeMaxAero	MA_WeaponRangeType	Repeated
Description	Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for: 1.Optimal steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
TimeToRelease	TimeType	Required
Description	The seconds time to activate for the proposed launch store type.	
TimeOfActivation	TimeType	Optional
Description	The time of weapon seeker activation, assuming no platform guidance or weapon errors, for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No post-launch target maneuvers	

Table A-1-166: MS UCI Extensions MA_WEZ_RequestStatusMT - WEZ_Data UCI Extension Field

RangeTurnAndClose		
Description	Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.Target turn-and-close maneuver RangeTurnAndClose is not applicable to ballistic trajectory targets.	
	MA_WeaponRangeType	
Message Structure		
Field Name	Field Type	Field Usage
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required

Description	Indicates the azimuth the slant range pertains to.
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Table A-1-167: MS UCI Extensions MA_WEZ_RequestStatusMT - RangeTurnAndClose UCI Extension Field

RangeMinimum		
Description	Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers Note: ballistic trajectory velocity changes are not considered maneuvers.	
	MA_WeaponRangeType	
Message Structure		
Field Name	Field Type	Field Usage
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-168: MS UCI Extensions MA_WEZ_RequestStatusMT - RangeMinimum UCI Extension Field

RangeTurnAndRun		
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.Target turn-and-run maneuver RangeTurnAndRun is not applicable to ballistic trajectory targets.	
	MA_WeaponRangeType	
Message Structure		
Field Name	Field Type	Field Usage
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-169: MS UCI Extensions MA_WEZ_RequestStatusMT - RangeTurnAndRun UCI Extension Field

RangeProbabilityOfIntercept

Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers Note: ballistic trajectory velocity changes are not considered maneuvers.	
Base Type	MA_WeaponRangeType	
Message Structure		
Field Name	Field Type	Field Usage
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-170: MS UCI Extensions MA_WEZ_RequestStatusMT - RangeProbabilityOfIntercept UCI Extension Field

RangeMaxGuidance		
Description	Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
	MA_WeaponRangeType	
Message Structure		
Field Name	Field Type	Field Usage
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-171: MS UCI Extensions MA_WEZ_RequestStatusMT - RangeMaxGuidance UCI Extension Field

RangeOptimal	
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Optimal steering

	2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
Base Type	MA_WeaponRangeType	
Message Structure		
Field Name	Field Type	Field Usage
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-172: MS UCI Extensions MA_WEZ_RequestStatusMT - RangeOptimal UCI Extension Field

RangeMaxAero		
Description	Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for: 1.Optimal steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
	MA_WeaponRangeType	
Message Structure		
Field Name	Field Type	Field Usage
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-173: MS UCI Extensions MA_WEZ_RequestStatusMT - RangeMaxAero UCI Extension Field

1.3.2.23 MA_StrikeActivityMT UCI Extension

MA_StrikeActivityMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	ActiveWeaponHSA_CSA : MA_HSA_CSA_Type Direction : MA_DirectionChoiceType

	MunitionActive : MA_MunitionActiveType MessageData : MA_StrikeActivityMDT ActiveWeaponState : MA_ActiveWeaponStateEnum Activity : MA_StrikeActivityType Course : MA_DirectionReferenceType Heading : MA_DirectionReferenceType Reference : MA_HeadingReferenceEnum
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Table A-1-174: MS UCI Extensions MA_StrikeActivityMT UCI Extension

ActiveWeaponHSA_CSA		
Description	Provides the heading/course, speed, and altitude of the munition while the Activity is ACTIVE_*.	
Base Type	MA_HSA_CSA_Type	
Message Structure		
Field Name	Field Type	Field Usage
Altitude	AltitudeReferenceType	Optional
Description	Altitude hold value to be achieved and reference to altitude measurement source.	
Speed	PathSegmentSpeedType	Optional
Description	Defines speed type (reference frame) for all speed related fields in this message.	
Direction	MA_DirectionChoiceType	Optional
Description	Defines heading or course type and reference to hold	

Table A-1-175: MS UCI Extensions MA_StrikeActivityMT - ActiveWeaponHSA_CSA UCI Extension Field

Direction		
Description	Defines heading or course type and reference to hold	
Base Type	MA_DirectionChoiceType	
Message Structure		
Field Name	Field Type	Field Usage
Heading	MA_DirectionReferenceType	Optional
Description	Defines heading type and reference to hold.	
Course	MA_DirectionReferenceType	Optional
Description	Defines course type and reference to hold.	

Table A-1-176: MS UCI Extensions MA_StrikeActivityMT - Direction UCI Extension Field

MunitionActive	
Description	Data regarding the munition while it is active or in flight.
Base Type	MA_MunitionActiveType
Message Structure	

Field Name	Field Type	Field Usage
ActiveWeaponState	MA_ActiveWeaponStateEnum	Repeated
Description	Provides further munition state data detail if the Activity is ENABLED, ACTIVE_* or COMPLETED as reported by the ActivityState. This message can be omitted if ActivityState is DISABLED, FAILED or DELETED. List size for this element is based on "Select All That Apply" condition.	
ActiveWeaponHSA_CSA	MA_HSA_CSA_Type	Optional
Description	Provides the heading/course, speed, and altitude of the munition while the Activity is ACTIVE_*.	

Table A-1-177: MS UCI Extensions MA_StrikeActivityMT - MunitionActive UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_StrikeActivityMDT	
Message Structure		
Field Name	Field Type	Field Usage
SubsystemID	SubsystemID_Type	Required
Description	Indicates the Subsystem which hosts the Strike Capabilities whose Activities are given in this message.	
Activity	MA_StrikeActivityType	Repeated
Description	Indicates the dynamic operational characteristics of an Activity of a Strike Capability.	

Table A-1-178: MS UCI Extensions MA_StrikeActivityMT - MessageData UCI Extension Field

ActiveWeaponState	
Description	Provides further munition state data detail if the Activity is ENABLED, ACTIVE_* or COMPLETED as reported by the ActivityState. This message can be omitted if ActivityState is DISABLED, FAILED or DELETED. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_ActiveWeaponStateEnum
Message Structure	
Enumeration Literal	Description
TARGETED	Indicates that a target has been selected and the munition has not been released yet. Report with the ENABLED state generated by ActivityStateEnum.
REQUESTED	Indicates the munition has been requested for release. Report with the ENABLED state generated by ActivityStateEnum.
SEEKER_INACTIVE	Indicates the munition has been released but its local seeker has not been activated. Report with the ACTIVE_* state generated by ActivityStateEnum.
SEEKER_ACTIVE	Indicates the munition has been released and its local seeker has been activated. Report with the ACTIVE_* state generated by ActivityStateEnum.
FUZE_INACTIVE	Indicates the munition has been released but its fuze has not been activated. Report with the ACTIVE_* state generated by ActivityStateEnum.

FUZE_ACTIVE	Indicates the munition has been released and its fuze has been activated. Report with the ACTIVE_* state generated by ActivityStateEnum.
NOT_FULLY_AUTONOMOUS	Indicates the munition has been released not fully autonomous and thus not able to guide without datalink or target destination. Report with the ACTIVE_* state generated by ActivityStateEnum.
FULLY_AUTONOMOUS	Indicates the munition has been released and is fully autonomous and able guide without datalink or target destination. Report with the ACTIVE_* state generated by ActivityStateEnum.
DEFEATED	Indicates the munition has been released and but it has been out maneuvered by the assigned target. Report with the COMPLETED state generated by ActivityStateEnum.
TIMED_OUT	Indicates the munition flight has ended, and it did not intercept the assigned target. Report with the COMPLETED state generated by ActivityStateEnum.
SUCCESS	Indicates the munition flight has ended, and it intercepted the assigned target. Report with the COMPLETED state generated by ActivityStateEnum.

Table A-1-179: MS UCI Extensions **MA_StrikeActivityMT - ActiveWeaponState UCI Extension Field**

Activity		
Description	Indicates the dynamic operational characteristics of an Activity of a Strike Capability.	
Base Type	MA_StrikeActivityType	
Message Structure		
Field Name	Field Type	Field Usage
SelectedForKeyLoad	boolean	Required
Description	Key Load indicates the store has been put into a state where it is ready to receive cryptographic keys.	
WeaponArmed	boolean	Required
Description	When true, the weapon is in the armed state.	
ReadyForRelease	boolean	Required
Description	The store has been prepared for launch such that when released the store will function properly.	
SelectedForRelease	boolean	Required
Description	When Master Arm, Release Consent, and any LAR or vehicle Attitude parameters enforced by the carriage are all met, this store will be released.	
SelectedForJettison	boolean	Required
Description	Jettison indicates an ejection from the host system without powering or preparing the delivery vehicle. The store will be in an inert state. When Master Arm, Release Consent, and any LAR or vehicle attitude parameters enforced by the carriage are all met, this store will be jettisoned.	
AssignedTarget	GeoLocatedObjectType	Optional
Description	Indicates the Entity target assigned to the StrikeCapability.	
AO_Code	AO_CodeType	Optional
Description	Indicates the current/set AO/laser PRF and/or PIM code.	

LAR_ID	LAR_ID_Type	Optional
Description	Indicates the ID of the associated LAR and/or LAR_Report associated with the Strike capability.	
ConsentRequired	boolean	Optional
Description	Indicates whether (TRUE) or not (FALSE) the Activity requires consent to transition from the ENABLED state to an ACTIVE state. This element is required and shall be TRUE for Activities associated with "mother may I" Capabilities.	
ConsentState	ConsentEnum	Optional
Description	Indicates the state of consent for the Activity. This element is required for Activities which require Consent.	
KineticFootprint	EnduranceFootprintType	Optional
Description	The estimated footprint of where the store may fall if released from the current position.	
TargetLockAcquired	boolean	Optional
Description	Indicated whether (TRUE) or not (FALSE) a lock on target has been acquired.	
MunitionActive	MA_MunitionActiveType	Optional
Description	Data regarding the munition while it is active or in flight.	
TimeToRelease	DurationType	Optional
Description	Time interval before strike capability is released.	

Table A-1-180: MS UCI Extensions MA_StrikeActivityMT - Activity UCI Extension Field

Course		
Description	Defines course type and reference to hold.	
Base Type	MA_DirectionReferenceType	
Message Structure		
Field Name	Field Type	Field Usage
Value	AngleType	Required
Description	Indicates the heading or course value to hold.	
Reference	MA_HeadingReferenceEnum	Required
Description	Reference to type of heading or course given.	

Table A-1-181: MS UCI Extensions MA_StrikeActivityMT - Course UCI Extension Field

Heading		
Description	Defines heading type and reference to hold.	
Base Type	MA_DirectionReferenceType	
Message Structure		
Field Name	Field Type	Field Usage
Value	AngleType	Required
Description	Indicates the heading or course value to hold.	
Reference	MA_HeadingReferenceEnum	Required

Description	Reference to type of heading or course given.
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Table A-1-182: MS UCI Extensions MA_StrikeActivityMT - Heading UCI Extension Field

Reference	
Description	Reference to type of heading or course given.
Base Type	MA_HeadingReferenceEnum
Message Structure	
Enumeration Literal	Description
TRUE_NORTH	Heading value according to Geographic North.
MAGNETIC_NORTH	Heading value according to location where the planet's magnetic field points vertically downward.

Table A-1-183: MS UCI Extensions MA_StrikeActivityMT - Reference UCI Extension Field

1.3.2.24 MA_AMTI_CapabilityMT UCI Extension

MA_AMTI_CapabilityMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_AMTI_CapabilityMDT Capability : MA_AMTI_CapabilityType

Table A-1-184: MS UCI Extensions MA_AMTI_CapabilityMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_AMTI_CapabilityMDT	
Message Structure		
Field Name	Field Type	Field Usage
Capability	MA_AMTI_CapabilityType	Repeated
Description	Indicates an installed AMTI Capability, its configurable characteristics and its controllability.	
AvailableMDF	MDF_Type	Repeated
Description	Indicates a Mission Data File (MDF) that is available for use. All MDFs currently available for use should be listed. An MDF is a collection of data, organized into files, that is used to tailor, configure or otherwise affect the behavior of a Capability. MDFs can vary from mission to mission or even by phases of a single mission. They are generally developed in pre-mission planning and are unique for each Subsystem model. Their content is not standardized by UCI, but the mechanism for advertising their availability and activating them is. Their content is abstracted by the UCI concept of Capability.	

Table A-1-185: MS UCI Extensions MA_AMTI_CapabilityMT - MessageData UCI Extension Field

Capability

Description	Indicates an installed AMTI Capability, its configurable characteristics and its controllability.	
Base Type	MA_AMTI_CapabilityType	
Message Structure		
Field Name	Field Type	Field Usage
MT_IIRS_Level	NIIRS_Type	Repeated
Description	Indicates a Moving Target Indication Interpretability Rating Scale level supported by the Capability. See annotations of the underlying type for further details.	
ComponentFieldOfRegard	ComponentFieldOfRegardType	Repeated
Description	Fields of Regard for each component in the capability.	
CapabilityType	AMTI_CapabilityEnum	Required
Description	Indicates the Capability's first tier type in the AMTI taxonomy. For AMTI, the first tier is an indication of the physical area covered. See enumerated type annotations for further details.	
SubCapabilityType	AMTI_SubCapabilityEnum	Repeated
Description	Indicates a SubCapability of the AMTI Capability; the second tier in the taxonomy of AMTI. For AMTI, the second tier is waveform category. See enumeration annotations for further details. List size for this element is based on "Select All That Apply" condition. The selection of AUTO_SUBCAPABILITY_SELECT still requires the evaluation of all sibling attributes by the Capability.	
CapabilityOptions	AMTI_CapabilityOptionsType	Required
Description	Indicates the Capability's support for select optional elements in its accepted control interface messages. Support for these elements is expected to vary across Capability implementations but must be known in order to adapt and configure upstream Systems and Services which use the Capability.	
SupportedFrequencies	FrequencyRangeType	Repeated
Description	Indicates a frequency range used by the Capability.	
MessageOutput	AMTI_MessageOutputsEnum	Repeated
Description	Indicates a message that is an output of the Capability. See enumerated type annotations for details. List size for this element is based on "Select All That Apply" condition.	
AcceptedInterface	CapabilityControlInterfacesEnum	Repeated
Description	Indicates an accepted control interface for the Capability. See enumeration annotations for further details. List size for this element is based on "Select All That Apply" condition.	
CommsRequired	boolean	Optional
Description	Indicates whether (TRUE) or not (FALSE) communications are required to use the Capability. Omitted is equivalent to FALSE. For example, an unmanned system with no onboard storage would require communications to disseminate the outputs of the Capability.	
CapabilityID	CapabilityID_Type	Required
Description	Indicates the unique ID of this Capability. This ID is used to invoke the Capability via Command messages, report its activity via Activity messages, etc.	

DomainPairing	EnvironmentPairingEnum	Repeated
Description	Indicates the domain/environment which the Capability is employed from and the "target" domain/environment. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.	
ProductOutput	ProductOutputType	Repeated
Description	Indicates a Product that can be an output of the Capability.	
PackageOperation	PackageOperationEnum	Optional
Description	Indicates if and how the operation of the Capability can be coordinated with a partner Capability of another System in a multi-System Package. Omitting this element is equivalent to the enumerate NO_PACKAGE_OPERATION. The actual Package configuration including per-Capability partnering between Systems is given in the PackageStatus message. For example, a System that can perform cooperative geolocation could advertise with this element in the ESM_Capability message. A System that can navigate in a swarm with other Systems could advertise with this element in the FlightCapability message.	

Table A-1-186: MS UCI Extensions MA_AMTI_CapabilityMT - Capability UCI Extension Field

1.3.2.25 MA_MessagingSubscriptionStatusMT UCI Extension

MA_MessagingSubscriptionStatusMT	
Description	This message reports the active message subscriptions which may be activated, deactivated, or deleted via MA_MessagingSubscriptionSettingCommand. See the XSD File for more info
Ext. Fields	MessageData : MA_MessagingSubscriptionStatusMDT

Table A-1-187: MS UCI Extensions MA_MessagingSubscriptionStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_MessagingSubscriptionStatusMDT	
Message Structure		
Field Name	Field Type	Field Usage
TopicSubscription	MA_TopicSubscriptionType	Repeated
Description	Indicates details of an existing topic subscription.	

Table A-1-188: MS UCI Extensions MA_MessagingSubscriptionStatusMT - MessageData UCI Extension Field

1.3.2.26 MA_MissionPlanValidationCommandMT UCI Extension

MA_MissionPlanValidationCommandMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	Inputs : MA_MissionPlanValidationInputsType

	OpLineCategory : MA_OpLineCategoryEnum
	TaskType : MA_TaskTypeEnum
	CapabilityCommand : MA_CapabilityTypeEnum
	PlanPart : MA_PlanPartsType
	TaskPlan : MA_TaskPlanPartsType
	ActivityPlan : MA_ActivityPlanPartsType
	OpPlan : MA_OpPlanPartsType
	RouteActivityPlan : MA_ActivityPlanPartsType
	OpZoneCategory : MA_OpZoneCategoryEnum
	OpVolumeCategory : MA_OpZoneCategoryEnum
	OrbitActivityPlan : MA_ActivityPlanPartsType
	MessageData : MA_MissionPlanValidationCommandMDT

Table A-1-189: MS UCI Extensions MA_MissionPlanValidationCommandMT UCI Extension

Inputs		
Description	Indicates input parameters for performing MissionPlan validation.	
Base Type	MA_MissionPlanValidationInputsType	
Message Structure		
Field Name	Field Type	Field Usage
PlanID	MissionPlanID_Type	Required
Description	Indicates the unique ID of the MissionPlan to be validated.	
PlanType	PlanTypeEnum	Repeated
Description	Indicates which sub-*Plan types to include in MissionPlan validation. If omitted, all sub-*Plans should be validated. List size for this element is based on "Select All That Apply" condition.	
PlanPart	MA_PlanPartsType	Optional
Description	Indicates which part or parts to include in MissionPlan validation. If omitted, all Parts should be validated.	
ValidateSubordinateMissionPlans	boolean	Required
Description	Indicates whether (TRUE) or not (FALSE) subordinate MissionPlans should be validated along with the parent MissionPlan given by the sibling PlanID.	
PlanningProcessID	PlanningProcessID_Type	Required
Description	ID of the Planning Process to be used for validation.	
ModifyToValidate	boolean	Required
Description	Indicates whether (TRUE) or not (FALSE) portions of the associated plan, that are found to be invalid, should be modified to make the plan valid.	
ValidationWindow	TimeWindowType	Optional
Description	Indicates the time window, within the time period of the Plan, to validate. If omitted, the entire duration of the Plan should be validated. The child Duration element, when the child Range element is omitted, indicates a time window	

	beginning at the start time of the Plan and ending at the relative Duration. Otherwise, the child Duration element is relative to the Range element if left open ended. If the child Range element is fully specified, the child Duration element should be omitted.	
WeatherAreaData	WeatherAreaDataType	Optional
Description	This element represents an override of the existing weather data available through UCI weather messages.	

Table A-1-190: MS UCI Extensions MA_MissionPlanValidationCommandMT - Inputs UCI Extension Field

OpLineCategory	
Description	Indicates an OpLine type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpLineCategoryEnum
Message Structure	
Enumeration Literal	Description
GENERAL_LINE	A general line where the function must be described by other procedures.
FORWARD_EDGE_OF_BATTLE_AREA	The foremost limits of a series of areas in which ground combat units are deployed, excluding the areas in which the covering or screening forces are operating.
GUN_TARGET_LINE	Straight line from the gun or guns to the target. Altitude, if included, is the apex of the projectile.
CORRIDOR	Line with width describing the passageway designated for travel by friendly aircraft.
HOSTILE_BOUNDARY	Line that indicates a limit beyond which exists a hostile environment.
BUFFER_ZONE_BOUNDARY	Line beyond which friendly aircraft usually do not fly in peacetime.
LOW_LEVEL_TRANSIT_ROUTE	Line describing the center line of a route through the forward missile engagement zone that must be flown by friendly aircraft to ensure they are not designated hostile by defense systems.
TACTICAL_ACTION_LINE	Line positioned in hostile airspace which may be used in conjunction with other criteria for hostile determination.
FIRE_SUPPORT_COORDINATION_LINE	Line established by the appropriate ground commander to ensure coordination of fire not under his command but which may affect current tactical operations. Attacks against ground targets behind this line must be coordinated with the appropriate ground commander.
FORWARD_LINE_OF_OWN_TROOPS	Line defining the forward position of own troops within a ground area of responsibility, that does not usually coincide with the forward edge of battle area.
LINE_OF_COMMUNICATIONS	Line the route that connects an operating military unit with its supply base.
JETWAY	Line with width that defines areas where commercial aircraft are flying through.
DEPARTURE_RADIAL	Line the defines the departure radial when performing Carrier admin departures.
RECOVERY_RADIAL	Line the defines the recovery radial that is used to define the recovery marshal. May be used when performing Carrier admin recoveries.

FINAL_BEARING	Line the defines the final bearing when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.
BRC	Base Recovery Course defines the aircraft carrier heading when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.

Table A-1-191: MS UCI Extensions MA_MissionPlanValidationCommandMT - OpLineCategory UCI Extension Field

TaskType	
Description	Indicates a Task type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_TaskTypeEnum
Message Structure	
Enumeration Literal	Description
AIR_SAMPLE	Indicates the task is to sample and analyze the air around the vehicle.
AMTI	Indicates the task is to track air targets using an Airborne Moving Target Indicator.
AO	Indicates the task is to employ an Active Optical capability.
CARGO_DELIVERY	Indicates the task is to deliver cargo.
CAP	Indicates the task is to execute a Combat Air Patrol.
COMINT	Indicates the task is to employ a communications intelligence capability.
COMM_RELAY	Indicates the task is to relay communications from other battlespace participants.
COUNTER_SPACE	Indicates the task is to employ a Counter Space Capability.
EA	Indicates the task is to employ an electronic attack capability.
ESCORT	Indicates the task is to escort another entity.
ESM	Indicates the task is to employ an Electronic Support Measure.
FLIGHT	Indicates the task is to fly.
ORBIT_CHANGE	Indicates a task to perform an orbit change via a spacecraft maneuver.
ORBITAL_SURVEILLANCE	Indicates a task to perform orbital surveillance.
ORBITAL_SURVEILLANCE_SENSOR	Indicates a task to perform orbital surveillance via sensor collection capabilities.
PO	Indicates the task is to employ a Passive Optical capability.
REFUEL	Indicates the task is to refuel another System or Entity.
SAR	Indicates the task is to employ a Synthetic Aperture Radar capability.
SMTI	Indicates the task is to track surface targets using a Surface Moving Target Indicator.
STRIKE	Indicates the task is to strike a target.
SYSTEM_DEPLOYMENT	Indicates the task is to deploy a system.
TACTICAL_ORDER	Indicates the task is to perform a tactical order.

WEATHER_RADAR	Indicates the task is to collect weather data.
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Table A-1-192: MS UCI Extensions MA_MissionPlanValidationCommandMT - TaskType UCI Extension Field

CapabilityCommand	
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_CapabilityTypeEnum
Message Structure	
Enumeration Literal	Description
ACTION	Indicates an action capability.
AIR_SAMPLE	Indicates an air sampling capability.
AMTI	Indicates an airborne moving target indication (AMTI) capability.
AO	Indicates an active optical capability.
CAP	Indicates a cargo delivery capability.
CARGO_DELIVERY	Indicates a cargo delivery capability.
COMINT	Indicates a Communications Intelligence (COMINT) capability.
COMM_RELAY	Indicates a communications relay capability.
COMM_TERMINAL	Indicates a communications terminal capability
COUNTER_SPACE	Indicates a Counter Space Capability.
EA	Indicates an electronic attack (EA) capability.
EFFECT	Indicates an effect capability.
ESM	Indicates an electronic support measures (ESM) capability.
FLIGHT	Indicates a capability for waypoint-based flight tasks/commands.
IFF	Indicates an Identification Friend or Foe (IFF) capability.
OPAQUE	Indicates a Opaque Capability
ORBIT_CHANGE	Indicates this space system has the ability to change orbits.
ORBITAL_SURVEILLANCE	Indicates a task to perform Orbital Surveillance.
ORBITAL_SURVEILLANCE_SENSOR	Indicates an Orbital Surveillance Sensor Capability.
PO	Indicates a passive optical capability.
RADAR_ALTIMETER	Indicates a Radar Altimeter capability
REFUEL	Indicates a capability to refuel another asset.
RESPONSE	Indicates an response capability.
SAR	Indicates a synthetic aperture radar (SAR) capability.
SMTI	Indicates a surface moving target indication (SMTI) capability.
STRIKE	Indicates a strike capability.
SYSTEM_DEPLOYMENT	Indicates a deployment capability.

TACTICAL_ORDER	Indicates a tactical order capability.
TASK	Indicates a task capability.
WEATHER_RADAR	Indicates a weather radar capability.

Table A-1-193: MS UCI Extensions MA_MissionPlanValidationCommandMT - CapabilityCommand UCI Extension Field

PlanPart		
Description	Indicates which part or parts to include in MissionPlan validation. If omitted, all Parts should be validated.	
Base Type	MA_PlanPartsType	
Message Structure		
Field Name	Field Type	Field Usage
ActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable ActivityPlan parts.	
RoutePlan	RoutePlanPartsType	Optional
Description	Indicates applicable RoutePlan parts.	
RouteActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable RouteActivityPlan parts.	
OrbitPlan	OrbitPlanPartsType	Optional
Description	Indicates applicable OrbitPlan parts.	
OrbitActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable OrbitActivityPlan parts.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-194: MS UCI Extensions MA_MissionPlanValidationCommandMT - PlanPart UCI Extension Field

TaskPlan	
Description	Indicates applicable TaskPlan parts.
Base Type	MA_TaskPlanPartsType
Message Structure	

Field Name	Field Type	Field Usage
TaskType	MA_TaskTypeEnum	Repeated
Description	Indicates a Task type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	

Table A-1-195: MS UCI Extensions MA_MissionPlanValidationCommandMT - TaskPlan UCI Extension Field

ActivityPlan		
Description	Indicates applicable ActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	Field Type	Field Usage
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-196: MS UCI Extensions MA_MissionPlanValidationCommandMT - ActivityPlan UCI Extension Field

OpPlan

Description	Indicates applicable OpPlan parts.	
Base Type	MA_OpPlanPartsType	
Message Structure		
Field Name	Field Type	Field Usage
OpPointCategory	OpPointCategoriesType	Repeated
Description	Indicates an OpPoint type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpLineCategory	MA_OpLineCategoryEnum	Repeated
Description	Indicates an OpLine type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpZoneCategory	MA_OpZoneCategoryEnum	Repeated
Description	Indicates an OpZone type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpVolumeCategory	MA_OpZoneCategoryEnum	Repeated
Description	Indicates an OpVolume type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpRouting	EmptyType	Optional
Description	Indicates an OpRouting is applicable to the associated *Plan. List size for this element is based on there being no category type for OpRouting objects	

Table A-1-197: MS UCI Extensions **MA_MissionPlanValidationCommandMT - OpPlan UCI Extension Field**

RouteActivityPlan		
Description	Indicates applicable RouteActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	Field Type	Field Usage
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask,	

	ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-198: MS UCI Extensions MA_MissionPlanValidationCommandMT - RouteActivityPlan UCI Extension Field

OpZoneCategory	
Description	Indicates an OpZone type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpZoneCategoryEnum
Message Structure	
Enumeration Literal	Description
AIRBORNE_EARLY_WARNING	A point at which AEW aircraft are or will be orbiting or maneuvering.
COMBAT_AIR_PATROL	A point at which CAP aircraft are or will be orbiting or maneuvering.
CONSTRAINED_ENTRY_EXIT	Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.
CONTAMINATED	Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.
DANGER	A specified area above, below, or within which there may be a potential danger.
DEFENDED_AREA	An area the source tn is capable of defending against ballistic missiles, etc. (i.e. the source tn is operational with ready weapons and has designated the area for defense).
DIRECTED_SEARCH_AREA	Indicates the zone is a directed search area.
EOB	Indicates the zone is the boundary for an electronic order of battle request.
EXERCISE	An area for military maneuver or simulated wartime operation involving planning, preparation and execution.
FIGHTER_ENGAGE_AOR	Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.
FILTER_AREA	Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the

	OpZoneFilterAreaPET choice in OpZoneCategoryType.
GENERAL_HAZARD	Indicates the zone is a hazard area that doesn't have a specified operational purpose.
GENERAL_STATION	A general zone describing an air station that must be amplified by voice or predetermined procedures.
GROUND_AOR	Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.
HOSTILE_TACTICAL_AREA	Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.
HOSTILE_WEAPON	Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.
IMPACT_POINT	The real or expected point that a device has or will penetrate or contact a surface.
JAMMING	Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.
KEEP_IN	Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.
KILL	Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.
MISSILE_ENGAGEMENT	Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.
MISSILE_LAUNCH_POINT	A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.
MISSION_AREA	Indicates the zone is an area in which a mission will take place.
NAMED_AREA_OF_INTEREST	An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.
NO_COMM	Indicates the zone is an area where communications use is restricted.
NO_FIRE	Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.
NO_FLY	Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.
NO_IMAGE	Indicates the zone is an area where image collection is restricted.
ORBIT_FIGURE_EIGHT	A figure eight pattern within a rectangle that the aircraft is or will be tracking.
ORBIT_RACE_TRACK	An oval within a rectangle that the aircraft is or will be tracking.
ORBIT_RANDOM_CLOSED	A rectangle in which the aircraft will track randomly.
REFERENCE	Indicates the zone is an area that doesn't have a specified operational purpose.
SEARCH	An area where a search is being or will be conducted. normally associated with a SAR mission.

SEPARATION	Indicates an update in reservation volume for an individual vehicle.
SHORAD	Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.
SPEED_LIMITS	An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.
SUBMARINE_PATROL_AREA	A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.
TARGET_AREA_OF_INTEREST	An engagement area usually along an enemy mobility corridor.
VEHICLE_CONFIGURATION	Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.
WEAPON_RESTRICTION	Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType.
WEATHER_CONDITIONS	Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.

Table A-1-199: MS UCI Extensions MA_MissionPlanValidationCommandMT - OpZoneCategory UCI Extension Field

OpVolumeCategory	
Description	Indicates an OpVolume type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpZoneCategoryEnum
Message Structure	
Enumeration Literal	Description
AIRBORNE_EARLY_WARNING	A point at which AEW aircraft are or will be orbiting or maneuvering.
COMBAT_AIR_PATROL	A point at which CAP aircraft are or will be orbiting or maneuvering.
CONSTRAINED_ENTRY_EXIT	Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.
CONTAMINATED	Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.
DANGER	A specified area above, below, or within which there may be a potential danger.
DEFENDED_AREA	An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).
DIRECTED_SEARCH_AREA	Indicates the zone is a directed search area.
EOB	Indicates the zone is the boundary for an electronic order of battle request.
EXERCISE	An area for military maneuver or simulated wartime operation involving planning,

	preparation and execution.
FIGHTER_ENGAGE_AOR	Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.
FILTER_AREA	Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.
GENERAL_HAZARD	Indicates the zone is a hazard area that doesn't have a specified operational purpose.
GENERAL_STATION	A general zone describing an air station that must be amplified by voice or predetermined procedures.
GROUND_AOR	Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.
HOSTILE_TACTICAL_AREA	Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.
HOSTILE_WEAPON	Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.
IMPACT_POINT	The real or expected point that a device has or will penetrate or contact a surface.
JAMMING	Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.
KEEP_IN	Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.
KILL	Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.
MISSILE_ENGAGEMENT	Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.
MISSILE_LAUNCH_POINT	A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.
MISSION_AREA	Indicates the zone is an area in which a mission will take place.
NAMED_AREA_OF_INTEREST	An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.
NO_COMM	Indicates the zone is an area where communications use is restricted.
NO_FIRE	Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.
NO_FLY	Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.
NO_IMAGE	Indicates the zone is an area where image collection is restricted.
ORBIT_FIGURE_EIGHT	A figure eight pattern within a rectangle that the aircraft is or will be tracking.

ORBIT_RACE_TRACK	An oval within a rectangle that the aircraft is or will be tracking.
ORBIT_RANDOM_CLOSED	A rectangle in which the aircraft will track randomly.
REFERENCE	Indicates the zone is an area that doesn't have a specified operational purpose.
SEARCH	An area where a search is being or will be conducted. normally associated with a SAR mission.
SEPARATION	Indicates an update in reservation volume for an individual vehicle.
SHORAD	Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.
SPEED_LIMITS	An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.
SUBMARINE_PATROL_AREA	A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.
TARGET_AREA_OF_INTEREST	An engagement area usually along an enemy mobility corridor.
VEHICLE_CONFIGURATION	Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.
WEAPON_RESTRICTION	Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType.
WEATHER_CONDITIONS	Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.

Table A-1-200: MS UCI Extensions MA_MissionPlanValidationCommandMT - OpVolumeCategory UCI Extension Field

OrbitActivityPlan		
Description	Indicates applicable OrbitActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	Field Type	Field Usage
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	

CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-201: MS UCI Extensions MA_MissionPlanValidationCommandMT - OrbitActivityPlan UCI Extension Field

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_MissionPlanValidationCommandMDT	
Message Structure		
Field Name	Field Type	Field Usage
Inputs	MA_MissionPlanValidationInputsType	Required
Description	Indicates input parameters for performing MissionPlan validation.	
ForPlanningUseOnly	boolean	Required
Description	When TRUE, indicates any *Plan modified is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	
RequestFrequencyType	RepeatEnum	Required
Description	Indicates whether this request for validation is a request for a single reply, or a request for periodic validation with a corresponding request to cease periodic validation.	
RequestFrequencyPeriod	DurationType	Optional
Description	If requesting periodic validation, this field specifies the amount of time that should elapse between the periodic validation, i.e. the repetition time interval of the validation.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	

CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-202: MS UCI Extensions MA_MissionPlanValidationCommandMT - MessageData UCI Extension Field

1.3.2.27 MA_MissionPlanValidationCommandStatusMT UCI Extension

MA_MissionPlanValidationCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_MissionPlanValidationCommandStatusMDT

Table A-1-203: MS UCI Extensions MA_MissionPlanValidationCommandStatusMT UCI Extension

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_MissionPlanValidationCommandStatusMDT	
Message Structure		
Field Name	Field Type	Field Usage
MissionPlanValidationID	MissionPlanValidationID_Type	Optional
Description	MissionPlanValidation message associated with this CommandStatus.	
CommandStatus	ProcessingStatusEnum	Optional
Description	Indicates the status of the processing of the command after the command has reached a RECEIVED state in the prior state machine.	
CompletionStatus	CompletionStatusType	Optional
Description	Indicates actual and predicted metrics for completion of the command.	
IncompleteItem	IncompleteProcessingType	Repeated
Description	Indicates items associated with command processing that failed to complete. For example if a particular required Capability wasn't available, it could be referenced here. This element/type is highly flexible. It can reference almost any message/concept in UCI as a planning issue. The wide range of messages/concepts it can reference makes it 1) difficult to use as a basis for software/services to resolve the issue but 2) an excellent general method for processing algorithms to convey issues to operators. This element shouldn't be used to report processing issues/results that are more directly indicated by other processing status elements. For example, for planning processing, other elements focus specifically on per-Requirement allocation results therefore this element shouldn't be used for per-Requirement allocation results.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required

Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-204: MS UCI Extensions MA_MissionPlanValidationCommandStatusMT - MessageData UCI Extension Field

1.3.2.28 MA_SystemNotificationMT UCI Extension

MA_SystemNotificationMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_SystemNotificationMDT NotificationDetail : CannotComplyType NotificationDetail : CannotComplyType

Table A-1-205: MS UCI Extensions MA_SystemNotificationMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_SystemNotificationMDT	
Message Structure		
Field Name	Field Type	Field Usage
SystemSubjectIDs	MA_PackageSystemType	Required
Description	Indicates the subject of the Notification, the thing which the Notification is about.	
SystemPerspective	NotificationPerspectiveEnum	Repeated
Description	Indicates a perspective (characteristic, behavior and/or change) of the Subject that is the basis of the Notification. List size for this element is based on "Select All That Apply" condition.	
AssociatedMessage	AssociatedMessageType	Repeated
Description	Indicates a message that is associated with the Notification.	
NotificationDetail	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the unsuccessful processing of the sibling AssociatedMessage element. This element is only expected when the Associated Message processing is not successful	
NotificationID	NotificationID_Type	Required

Description	Indicates the unique ID of the Notification.	
NotificationState	NotificationStateEnum	Required
Description	Indicates the current state within an enumerated state machine that characterizes the life cycle of a notification.	
Timestamp	DateTimeType	Required
Description	The time at which the Notification was originally triggered (ex: event time).	
Source	NotificationSourceType	Required
Description	Indicates the originating source of the Notification.	
Severity	NotificationSeverityEnum	Required
Description	Indicates the overall severity of the notification. Informational notifications are meant to provide informational context of events. Indicates the existence of a hazardous condition requiring immediate action to prevent loss of life, equipment damage, or abortion of the mission. Critical notifications indicate the need for immediate attention.	
AppliesTo	SubjectType	Repeated
Description	Indicates an Asset that the Notification applies to and/or is affected by. Optional when the Notification characterizes the Subject without regard for Assets it might affect. Omission of this element doesn't necessarily mean the Notification isn't applicable to an Asset.	
NotificationNarrative	VisibleString1024Type	Optional
Description	Human readable message for display to an operator.	

Table A-1-206: MS UCI Extensions MA_SystemNotificationMT - MessageData UCI Extension Field

NotificationDetail		
Description	Indicates details regarding the reason for and/or a contributing factor to the unsuccessful processing of the sibling AssociatedMessage element. This element is only expected when the Associated Message processing is not successful	
Base Type	CannotComplyType	
Message Structure		
Field Name	Field Type	Field Usage
Reason	CannotComplyEnum	Required
Description	Indicates the reason why an action (Task, [Capability]Command, [SupportingCapability]Command, [Capability]Activity, etc.) was rejected, interrupted, unallocated, failed and/or generally can't be completed satisfactorily. See enumeration annotations for further details.	
Description	VisibleString1024Type	Optional
Description	Indicates a human-readable text description of the reason for the "can't comply".	
AssociatedID	ID_Type	Optional
Description	Indicates an ID corresponding to another object that is associated with or the cause of the "can't comply".	

Table A-1-207: MS UCI Extensions MA_SystemNotificationMT - NotificationDetail UCI Extension Field

NotificationDetail		
Description	Indicates details regarding the reason for and/or a contributing factor to the unsuccessful processing of the sibling AssociatedMessage element. This element is only expected when the Associated Message processing is not successful	
Base Type	CannotComplyType	
Message Structure		
Field Name	Field Type	Field Usage
Reason	CannotComplyEnum	Required
Description	Indicates the reason why an action (Task, [Capability]Command, [SupportingCapability]Command, [Capability]Activity, etc.) was rejected, interrupted, unallocated, failed and/or generally can't be completed satisfactorily. See enumeration annotations for further details.	
Description	VisibleString1024Type	Optional
Description	Indicates a human-readable text description of the reason for the "can't comply".	
AssociatedID	ID_Type	Optional
Description	Indicates an ID corresponding to another object that is associated with or the cause of the "can't comply".	

Table A-1-208: MS UCI Extensions MA_SystemNotificationMT - NotificationDetail UCI Extension Field

1.3.2.29 MA_EntityMT UCI Extension

MA_EntityMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	ObjectState : ObjectStateEnum MessageData : MA_EntityMDT

Table A-1-209: MS UCI Extensions MA_EntityMT UCI Extension

ObjectState	
Description	Indicates the state of the internal software object that the data in the message is based upon. The state of the software object is determined by the publisher. The publisher could be the original/master publisher or an intermediary re-publisher such as a persistence/data manager. This element is not a commanded CRUD operation. It is not an indication of the state of an instance of a message. It is not a mechanism for partial, incremental, or fragmentary publication of a message. It is a signal from a publisher to subscribers. For REMOVED state, in particular, some subscribers will mirror the publisher and treat the message as if the object no longer exists, but others, such as mission recorders, might not (in order to support retaining historical records of activity, for example).
Base Type	ObjectStateEnum

Message Structure	
Enumeration Literal	Description
NEW	Indicates the internal software object corresponding to this message currently exists and this message is the first publication corresponding to the object.
UPDATED	Indicates the internal software object corresponding to this message currently exists and has been updated since the last publication of the message; this publication of the message may contain updates.
REMOVED	Indicates the internal software object corresponding to this message no longer exists; there should be no further publications of this message corresponding to the object.

Table A-1-210: MS UCI Extensions MA_EntityMT - ObjectState UCI Extension Field

MessageData		
Description	Defines the characteristics of this Entity.	
Base Type	MA_EntityMDT	
Message Structure		
Field Name	Field Type	Field Usage
EntityID	EntityID_Type	Required
Description	This element represents the unique identifier of the entity.	
CreationTimestamp	DateTimeSigmaType	Required
Description	Indicates the time at which the Entity was initially detected and/or created.	
Source	EntitySourceType	Required
Description	Indicates the System and other details regarding the Source of the Entity. Summarizes the creation and correlation ancestry of the Entity. A given Entity has a single source but could be the result of fusion of multiple Entity children.	
EntityStatus	EntityStatusEnum	Required
Description	Indicates the status of an Entity. This includes early status (POTENTIAL, TENTATIVE), active status (CONFIRMED), inactive status (DROP_PENDING), end status (DROPPED, DESTROYED), or unknown status (MISSING, UNKNOWN, GHOST).	
OperationalStatus	OB_OperationalStatusEnum	Optional
Description	Indicates the degree to which the real world asset which the Entity represents is ready to perform the overall operational mission or missions for which it was organized and equipped. An Entity with this element specified will generally have an xOB contributor reference which can used to query for additional xOB data that isn't duplicated in Entity. Thus, for example, when an enum value of "O" is used refer to available xOB OperationalStatus elements.	
Identity	EntityIdentityType	Required
Description	This element represents the identity of the Entity. In UCI, Entity and therefore this element are abstracted to a level comparable to MIL-STD-6016 J3.X messages. In particular, the identity taxonomy used in this element is derived directly from MIL-STD-6016. It is left to applications using UCI to enforce compatibility with the portions of MIL-STD-6016 referenced herein.	
SiteEntityID	EntityID_Type	Repeated

	Description	Indicates the unique ID of an Entity that is a part of this Site Entity. When omitted, the Entity is not a Site. When one or more are given, this Entity is a Site.	
Mobility		MobilityEnum	Optional
	Description	Indicates the mobility of the Entity. Omission is equivalent to the enumerate UNDETERMINED.	
Kinematics		KinematicsType	Optional
	Description	Indicates the current kinematic information for the Entity. "Current" is subjective and varies widely by application. For example, some Service might only consider directly sensed kinematic information as current and always report extrapolated data in the sibling EstimatedKinematics element. Another Service might have "staleness" criteria which allows current kinematics to be extrapolated for some period of time before being considered stale. When the staleness criteria are met, that Service might then continue to extrapolate but report the continued extrapolation in the sibling EstimatedKinematics element while freezing this element in a "last-known-good" kinematic state.	
EstimatedKinematics		KinematicsType	Optional
	Description	This element indicates extrapolated or otherwise estimated kinematics for the Entity.	
DownLocation		Point2D_Type	Optional
	Description	When present, indicates the Entity is in a "down" state (damaged, broken or otherwise not operating) and the last known location is given. The child Timestamp element, when given, indicates the time at which the Entity entered the down state.	
PlatformStatus		PlatformStatusType	Optional
	Description	General status information about the asset.	
WeaponState		MA_WeaponStateType	Optional
	Description	State of the weapon flyout.	
SignalSummary		EntitySignalSummaryType	Repeated
	Description	Indicates a summary of an RF Signal from the Entity.	
PulseDataID		PulseDataID_Type	Repeated
	Description	Indicates the unique ID of a PulseData collection associated with the Entity.	
MeasurementID		MeasurementID_Type	Repeated
	Description	Indicates the unique ID of a Measurement within an Observation Measurement Report associated with the Entity.	
Strength		StrengthType	Optional
	Description	Specifies the strength for the Entity. When omitted, a single object/entity should be assumed.	
ActivityAgainst		ActivityAgainstType	Repeated
	Description	This element indicates an activity, that isn't otherwise represented by UCI Tasks, Capability commands, etc. being performed on/against the Entity.	
ActivityBy		ActivityByType	Repeated
	Description	This element indicates an activity, that isn't otherwise represented by UCI Tasks, Capability commands, etc. being performed by the Entity.	
Endurance		EnduranceType	Optional

Description	Indicates the Entity's remaining endurance.	
VoiceControl	VoiceControlType	Optional
Description	This type contains the connection data needed to communicate with a vehicle using voice radios.	
AssociatedID	ID_Type	Repeated
Description	This element indicates an association (not a fusion or derivation relationship) between this Entity and another UCI message or object. For example, an Entity could be associated with an OpPoint. Data content of the Entity and the associated message/object is maintained independently.	
RemoveInfo	EntityRemoveInfoType	Optional
Description	Indicates supplementary information regarding an Entity whose ObjectState indicates REMOVED.	
CapabilityID	CapabilityID_Type	Repeated
Description	Indicates a capability of the Entity.	
OrbitalKinematics	OrbitalKinematicsChoiceType	Optional
Description	Indicates the current single-position orbital kinematic information for the Entity in the form of a single Cartesian state vector of position and velocity that together with its time (epoch) uniquely determine the trajectory of an orbiting body in space. Requires selection of the frame of reference and should, in general, only be propagated using the same model used for the orbit determination. Optionally includes a covariance matrix that contains information about the uncertainties associated with the solution of the element corrections. "Current" is subjective and varies widely by application. For example, some Service might only consider directly sensed kinematic information as current and always report extrapolated data in the sibling EstimatedOrbitalKinematics element. Another Service might have "staleness" criteria which allows current kinematics to be extrapolated for some period of time before being considered stale. When the staleness criteria are met, that Service might then continue to extrapolate but report the continued extrapolation in the sibling EstimatedOrbitalKinematics element while freezing this element in a "last-known-good" kinematic state.	
OrbitalKinematicsParameters	OrbitalSingleVectorParametersType	Optional
Description	Indicates the parameters used in the Propagation Model for the Entity. If the OrbitalKinematics are relative to another object (Reference Object), these parameters only relate to this entity. If it is necessary to capture the parameters for the Reference Object, those parameters should be captured in the Entity message for that Reference Object, and the Reference Object's EntityID should be specified in the ReferenceAsset element of the sibling OrbitalKinematics.	
EstimatedOrbitalKinematics	OrbitalKinematicsChoiceType	Optional
Description	This element indicates extrapolated or otherwise estimated single-position orbital kinematics for the Entity.	

Table A-1-211: MS UCI Extensions MA_EntityMT - MessageData UCI Extension Field

1.3.2.30 MA_GatewayActivityMT UCI Extension

MA_GatewayActivityMT	
Description	See the annotation in the associated message for an overall description of the message and this type.

	See the XSD File for more info
Ext. Fields	ObjectState : ObjectStateEnum MessageData : MA_GatewayActivityMDT

Table A-1-212: MS UCI Extensions MA_GatewayActivityMT UCI Extension

ObjectState	
Description	Indicates the state of the internal software object that the data in the message is based upon. The state of the software object is determined by the publisher. The publisher could be the original/master publisher or an intermediary re-publisher such as a persistence/data manager. This element is not a commanded CRUD operation. It is not an indication of the state of an instance of a message. It is not a mechanism for partial, incremental, or fragmentary publication of a message. It is a signal from a publisher to subscribers. For REMOVED state, in particular, some subscribers will mirror the publisher and treat the message as if the object no longer exists, but others, such as mission recorders, might not (in order to support retaining historical records of activity, for example).
Base Type	ObjectStateEnum
Message Structure	
Enumeration Literal	Description
NEW	Indicates the internal software object corresponding to this message currently exists and this message is the first publication corresponding to the object.
UPDATED	Indicates the internal software object corresponding to this message currently exists and has been updated since the last publication of the message; this publication of the message may contain updates.
REMOVED	Indicates the internal software object corresponding to this message no longer exists; there should be no further publications of this message corresponding to the object.

Table A-1-213: MS UCI Extensions MA_GatewayActivityMT - ObjectState UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_GatewayActivityMDT	
Message Structure		
Field Name	Field Type	Field Usage
GatewayServiceID	ServiceID_Type	Required
Description	This element represents the ServiceID of the Gateway Service that sent this message.	
Activity	MA_GatewayActivityType	Repeated
Description	Indicates the dynamic operational characteristics of an Activity of a GatewayCapability.	

Table A-1-214: MS UCI Extensions MA_GatewayActivityMT - MessageData UCI Extension Field

1.3.2.31 MA_GatewayCommandMT UCI Extension

MA_GatewayCommandMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_GatewayCommandMDT

Table A-1-215: MS UCI Extensions MA_GatewayCommandMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_GatewayCommandMDT	
Message Structure		
Field Name	Field Type	Field Usage
GatewayServiceID	ServiceID_Type	Required
Description	This element represents the ServiceID of the Gateway Service that sent this message.	
Command	MA_GatewayCommandType	Repeated
Description	Indicates an instance of a Waveform Capability command. A Capability command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each command instance reports status via a separate, independent Capability CommandStatus message.	

Table A-1-216: MS UCI Extensions MA_GatewayCommandMT - MessageData UCI Extension Field

1.3.2.32 MA_GatewayCommandStatusMT UCI Extension

MA_GatewayCommandStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_GatewayCommandStatusMDT

Table A-1-217: MS UCI Extensions MA_GatewayCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_GatewayCommandStatusMDT	
Message Structure		
Field Name	Field Type	Field Usage
Activity	ResultingActivityType	Repeated

Description	Indicates an Activity that satisfies the associated Command. This element is required when the sibling State element indicates ACCEPTED.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-218: MS UCI Extensions **MA_GatewayCommandStatusMT - MessageData UCI Extension Field**

1.3.2.33 MA_DataDisseminationPlanOverrideRequestMT UCI Extension

MA_DataDisseminationPlanOverrideRequestMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_DataDisseminationPlanOverrideRequestMDT

Table A-1-219: MS UCI Extensions **MA_DataDisseminationPlanOverrideRequestMT UCI Extension**

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_DataDisseminationPlanOverrideRequestMDT	
Message Structure		
Field Name	Field Type	Field Usage
AllocatedToSystemIDs	MA_PackageSystemType	Repeated
Description	Indicates the unique ID of the system(s) or package that is the recipient of this command.	
AllocatedToServiceID	ServiceID_Type	Optional
Description	Indicates the unique ID of the service that is the recipient of this command.	
Request	MA_DataDisseminationPlanOverrideRequestType	Repeated
Description	Indicates an instance of a data plan override request. A data plan override request message can have multiple request instances to allow a sequence of related requests in a single coherent message. Each request instance reports status via a separate, independent DataPlanOverrideRequestStatus message.	

Table A-1-220: MS UCI Extensions **MA_DataDisseminationPlanOverrideRequestMT - MessageData UCI**

Extension Field

1.3.2.34 MA_DataDisseminationPlanOverrideRequestStatusMT UCI Extension

MA_DataDisseminationPlanOverrideRequestStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_DataDisseminationPlanOverrideRequestStatusMDT

Table A-1-221: MS UCI Extensions MA_DataDisseminationPlanOverrideRequestStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_DataDisseminationPlanOverrideRequestStatusMDT	
Message Structure		
Field Name	Field Type	Field Usage
PlanID	MA_DataDisseminationPlanID_Type	Required
Description	Indicates the unique ID of the DataDisseminationPlan that this override modification status applies to.	
EffectivityActivationStatus	DataPlanEffectivityActivationStatusType	Repeated
Description	This element indicates the status of pre-mission planned Data-Link gateway settings. It is used to indicate details about specific planned effectivity activations.	
OperatorModificationStatus	OperatorGatewayDesignationStatusType	Repeated
Description	This element indicates the status of operator designations. It is used to indicate details about specific designation configurations.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message corresponding to this status message.	
RequestProcessingState	RequestProcessingStateEnum	Required
Description	Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.	
RequestProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-222: MS UCI Extensions MA_DataDisseminationPlanOverrideRequestStatusMT - MessageData UCI Extension Field

1.3.2.35 MA_DataDisseminationPlanActivationCommandMT UCI Extension

MA_DataDisseminationPlanActivationCommandMT

Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_DataDisseminationPlanActivationCommandMDT

Table A-1-223: MS UCI Extensions MA_DataDisseminationPlanActivationCommandMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_DataDisseminationPlanActivationCommandMDT	
Message Structure		
Field Name	Field Type	Field Usage
PlanID	MA_DataDisseminationPlanID_Type	Required
Description	The DataDisseminationPlan that this command pertains to.	
CommandedState	PlanActivationCommandEnum	Required
Description	Indicates the activation to perform on the associated DataDisseminationPlan. See enumerate annotations for further details.	
Applicability	MA_PackageSystemType	Repeated
Description	The System or Package IDs, that this message applies to. A lead is able to respond with ACCEPTED for the Package or individual Systems as needed.	
EffectivityActivation	MA_DataDisseminationPlanEffectivityActivationType	Repeated
Description	Indicates an Effectivity of the DataDisseminationPlan to activate.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-224: MS UCI Extensions MA_DataDisseminationPlanActivationCommandMT - MessageData UCI Extension Field

1.3.2.36 MA_DataDisseminationPlanActivationCommandStatusMT UCI Extension

MA_DataDisseminationPlanActivationCommandStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_DataDisseminationPlanActivationCommandStatusMDT

Table A-1-225: MS UCI Extensions MA_DataDisseminationPlanActivationCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_DataDisseminationPlanActivationCommandStatusMDT	
Message Structure		
Field Name	Field Type	Field Usage
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-226: MS UCI Extensions MA_DataDisseminationPlanActivationCommandStatusMT - MessageData UCI Extension Field

1.3.2.37 MA_DataDisseminationPlanActivationStatusMT UCI Extension

MA_DataDisseminationPlanActivationStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_DataDisseminationPlanActivationStatusMDT

Table A-1-227: MS UCI Extensions MA_DataDisseminationPlanActivationStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_DataDisseminationPlanActivationStatusMDT	
Message Structure		
Field Name	Field Type	Field Usage
PlanID	MA_DataDisseminationPlanID_Type	Required
Description	The ID of the DataDisseminationPlan this status pertains to.	
ActivationState	PlanActivationStateEnum	Required
Description	Indicates the activation state of the associated DataDisseminationPlan.	

StateReason	MA_CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling ActivationState element.	
EffectivityActivationState	MA_DataDisseminationPlanEffectivityStateType	Repeated
Description	Indicates the activation state of an Effectivity of the associated DataDisseminationPlan.	
OverrideModificationState	MA_OverriderModificationStatusType	Repeated
Description	This element indicates the status of operator designations. It is used to indicate details about specific designation configurations.	

Table A-1-228: MS UCI Extensions MA_DataDisseminationPlanActivationStatusMT - MessageData UCI Extension Field

1.3.2.38 MA_DataDisseminationPlanMT UCI Extension

MA_DataDisseminationPlanMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	ObjectState : ObjectStateEnum MessageData : MA_DataDisseminationPlanMDT

Table A-1-229: MS UCI Extensions MA_DataDisseminationPlanMT UCI Extension

ObjectState	
Description	Indicates the state of the internal software object that the data in the message is based upon. The state of the software object is determined by the publisher. The publisher could be the original/master publisher or an intermediary re-publisher such as a persistence/data manager. This element is not a commanded CRUD operation. It is not an indication of the state of an instance of a message. It is not a mechanism for partial, incremental, or fragmentary publication of a message. It is a signal from a publisher to subscribers. For REMOVED state, in particular, some subscribers will mirror the publisher and treat the message as if the object no longer exists, but others, such as mission recorders, might not (in order to support retaining historical records of activity, for example).
Base Type	ObjectStateEnum
Message Structure	
Enumeration Literal	Description
NEW	Indicates the internal software object corresponding to this message currently exists and this message is the first publication corresponding to the object.
UPDATED	Indicates the internal software object corresponding to this message currently exists and has been updated since the last publication of the message; this publication of the message may contain updates.
REMOVED	Indicates the internal software object corresponding to this message no longer exists; there should be no further publications of this message corresponding to the object.

Table A-1-230: MS UCI Extensions MA_DataDisseminationPlanMT - ObjectState UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_DataDisseminationPlanMDT	
Message Structure		
Field Name	Field Type	Field Usage
PlanID	MA_DataDisseminationPlanID_Type	Required
Description	Indicates the unique ID of the DataDisseminationPlan.	
AssociatedPlanID	MA_DataDisseminationPlanID_Type	Repeated
Description	Indicates the unique ID of a DataDisseminationPlan that is a parent to this one.	
Applicability	MA_PlanApplicabilityType	Repeated
Description	Indicates the System or Systems which have a direct relationship with the DataDisseminationPlan. See annotations for the underlying type and its child elements for details. If Applicability is not defined then it indicates this DataDisseminationPlan applies to ALL systems.	
IER	MA_CommIER_Planned_Type	Repeated
Description	Indicates a planned Information Exchange Requirement (IER). An IER can reflect the information needs of local or remote Systems or third party Systems coordinating on behalf of other Systems. The planned assignment of an IER to comms resources is given in the sibling Effectivity element.	
IER_DesiredService	MA_CommIER_DesiredServicesType	Repeated
Description	Defines the desired service constraints for an IER (Information Exchange Requirement), allowing multi-constrained solutions. All child elements are optional; when omitted software should apply no explicit bound for that metric.	
RateGroup	MA_CommRateGroupType	Repeated
Description	Indicates a RateGroup that can be associated with an IER to configure its transmit rate.	
Effectivity	MA_CommDataDisseminationEffectivityType	Repeated
Description	Indicates a DataDisseminationPlan "effectivity set" of items that are used together.	
Vector	MA_CommVectorPlannedType	Repeated
Description	Indicates a planned Vector. A vector describes a logical connection between nodes initiating on a specific NetworkLink (comms resource).	
DistributionGroup	MA_DistributionGroupType	Repeated
Description	Indicates a grouping of tiles or node groups and associated ip address.	
DistributionGroupMembership	MA_DistributionGroupMembershipType	Repeated
Description	Indicates a system's membership to a multicast group.	
Gateway	MA_GatewayReferenceType	Repeated
Description	Indicates a planned Gateway local to the system for which this DataDisseminationPlan is applicable to, with a summary of its Capability to be used for dissemination of IERs in the DataDisseminationPlan.	

Trigger	MA_CommTriggerType	Repeated
Description	Signifies available triggers which would automatically invoke this DataDisseminationPlan or parts within this DataDisseminationPlan. These Triggers that can be associated with activation of an Effectivity by using the TriggerID.	

Table A-1-231: MS UCI Extensions MA_DataDisseminationPlanMT - MessageData UCI Extension Field